

Project Deliverable 2 – Documentation and Coding

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Section A

Question 1

2.1) Introduction

The prototype consists of the main website, which facilitates online e-commerce website elements such as user registration, creating listings and performing transactions between buyers and vendors, as well as the admin website, where the admin users can view and update the website's database information.

This is accomplished by utilising html files for forms that are static, such as account registration and login, and php files can be utilised to process database information and for forms that are required to be dynamic to function appropriately, like the main page of the main website where listings should be visible and deleted listings should be removed. Moreover, JavaScript can be utilised to improve the user experience by providing quick input validation before inputted data is submitted. In addition, the css file is also utilised to style the html elements and ensure that html elements can change to best suit the size of the devices accessing the website, like desktop, tablet and mobiles.

The main objective of the creation of this website is to create a C2C e-commerce website that encourages community interaction through features such as a user rating system and the ability for buyers and vendors to chat with each other.

2.2) Prototyping

a)

Account registration page:

Desktop

Marketplace Return to the store front

Register an account

First name: Last name:

Username: Phone Number:

Email:

Address:

Password:

Confirm password:

Marketplace Return to the store front

Address:

Password:

Confirm password:

☐ Notify me of any promotions on the Marketplace platform.

☐ You have read and agreed to our user agreement and privacy policy.

Register

Already have an account? [Log in](#)

MarketplaceReturn to the store front

Register an account

First name:

Last name:

Username:

Phone Number:

Email:

Address:

Password:

Confirm password:

☐ Notify me of any promotions on the Marketplace platform.

☐ You have read and agreed to our user agreement and privacy policy.

Register

Already have an account? [Log in](#)

*Marketplace*Return to the store front

Register an account

First name:

Last name:

Username:

Phone Number:

Email:

Address:

Password:

Confirm password:

Email:

Address:

Password:

Confirm password:

- ☐ Notify me of any promotions on the Marketplace platform.
- ☐ You have read and agreed to our user agreement and privacy policy.

Register

Already have an account? [Log in](#)

Mobile portrait

Marketplace

Return to the store front

Register an account

First name:

Last name:

Username:

Phone Number:

Email:

Address:

Password:

Confirm password:

☐

 Notify me of any promotions on the Marketplace platform.

☐

 You have read and agreed to our user agreement and privacy policy.

Register

Already have an account? [Log in](#)

|||

○

<

Mobile landscape

Marketplace

Return to the store front

Register an account

First name:

Last name:

Username:

Phone Number:

Email:

<

○

|||

Address:

Password:

Confirm password:

- ☐ Notify me of any promotions on the Marketplace platform.
- ☐ You have read and agreed to our user agreement and privacy policy.

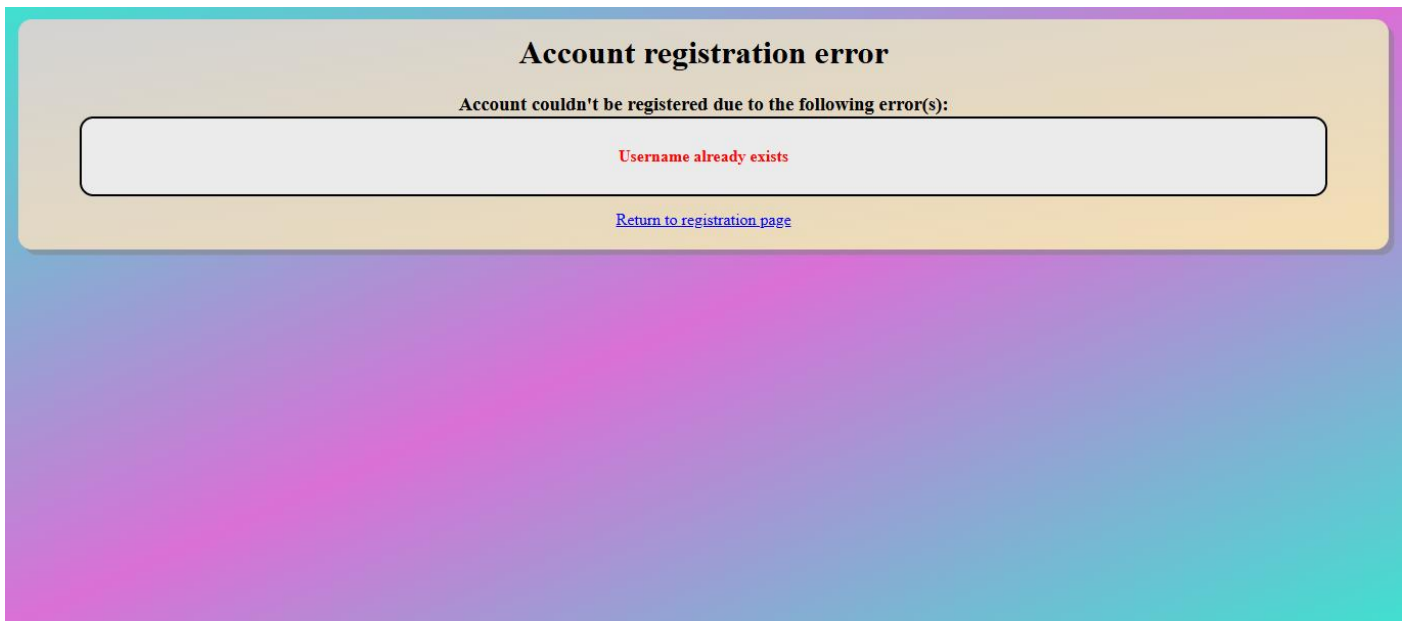
- ☐ Notify me of any promotions on the Marketplace platform.
- ☐ You have read and agreed to our user agreement and privacy policy.

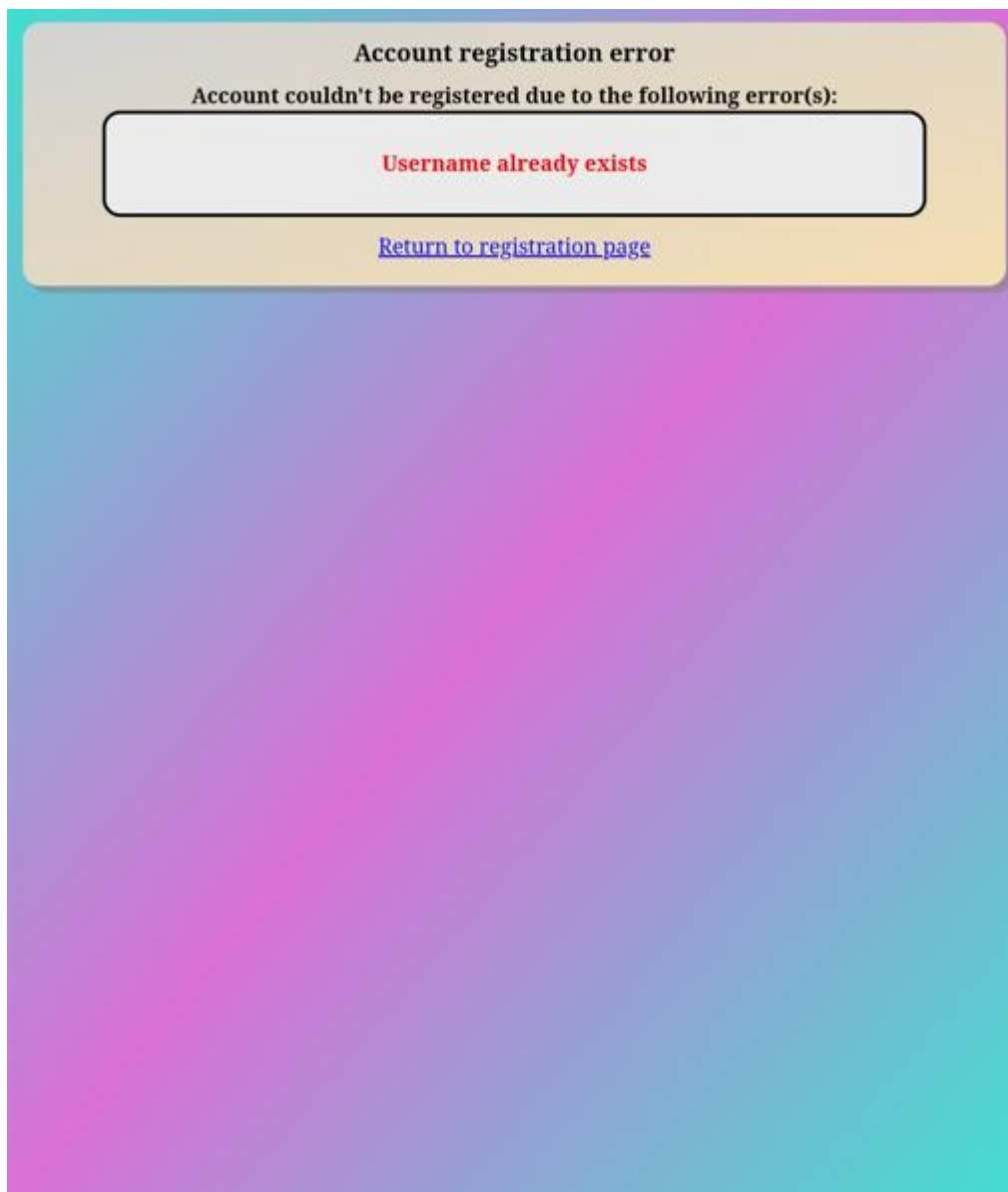
Register

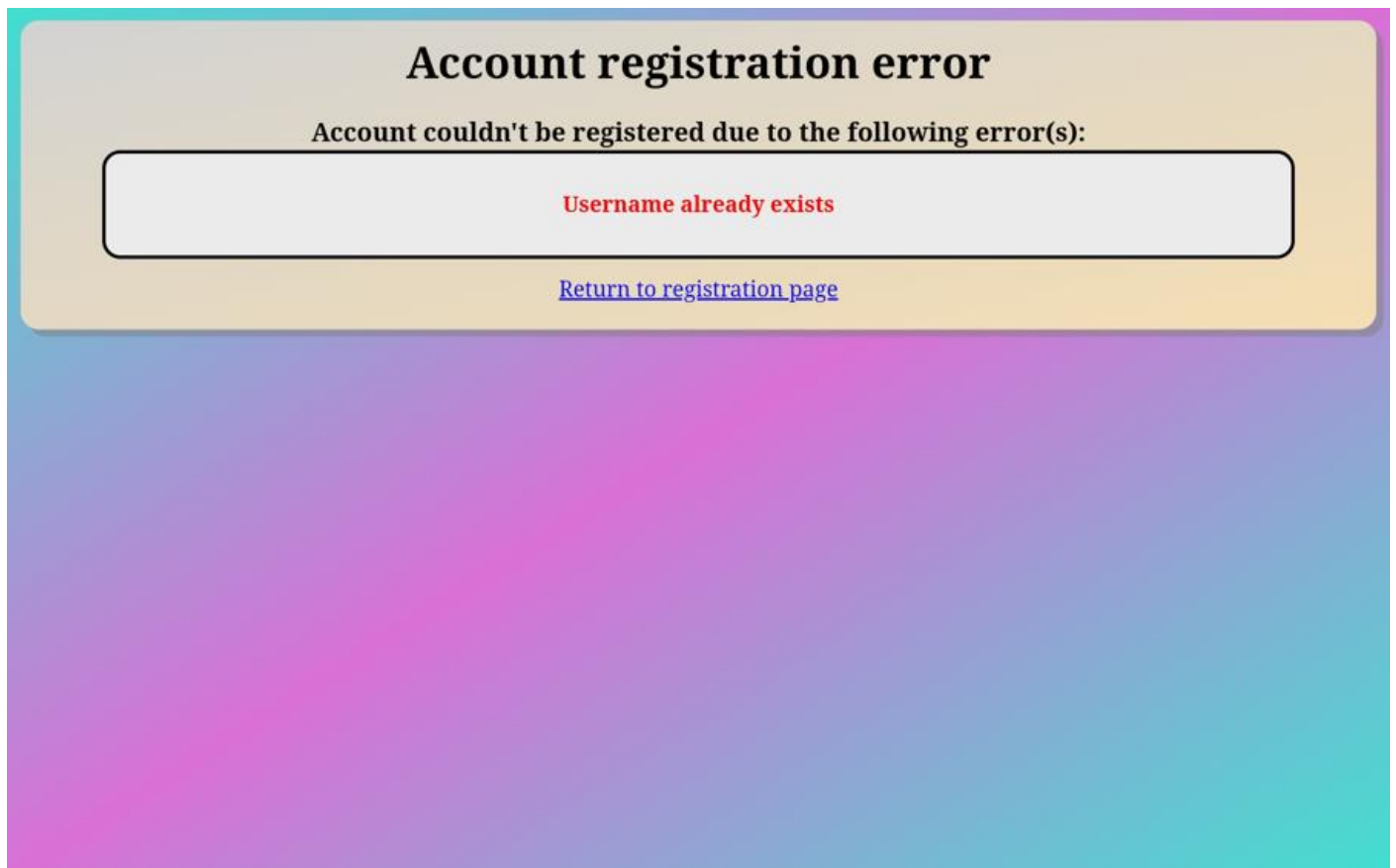
Already have an account? [Log in](#)

Registration error page:

Desktop



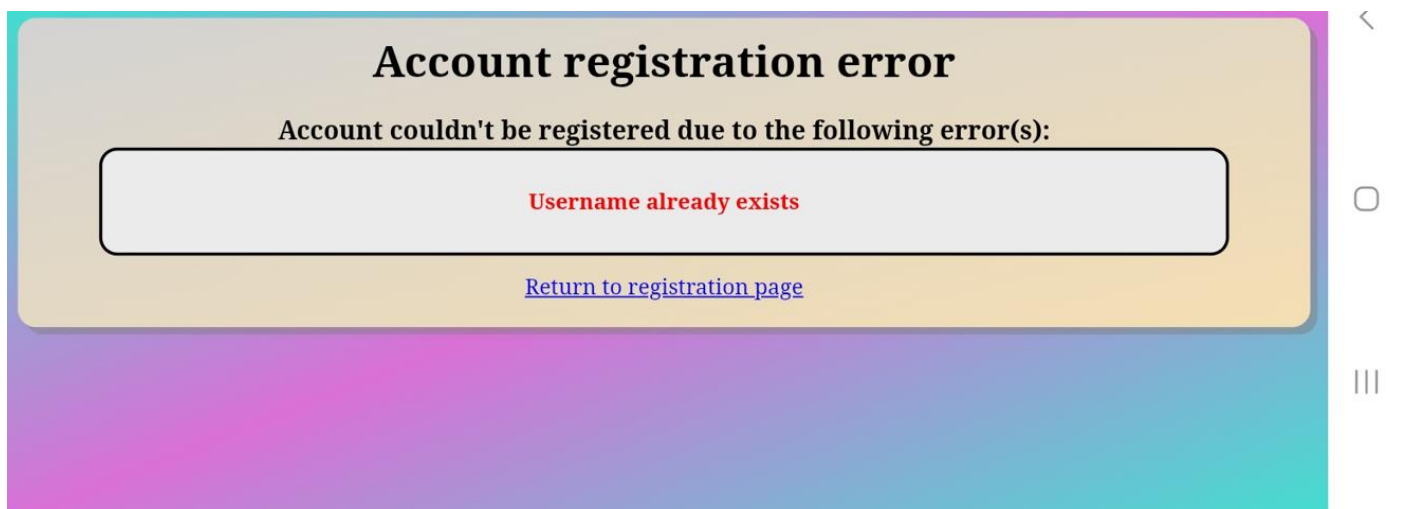




Mobile portrait

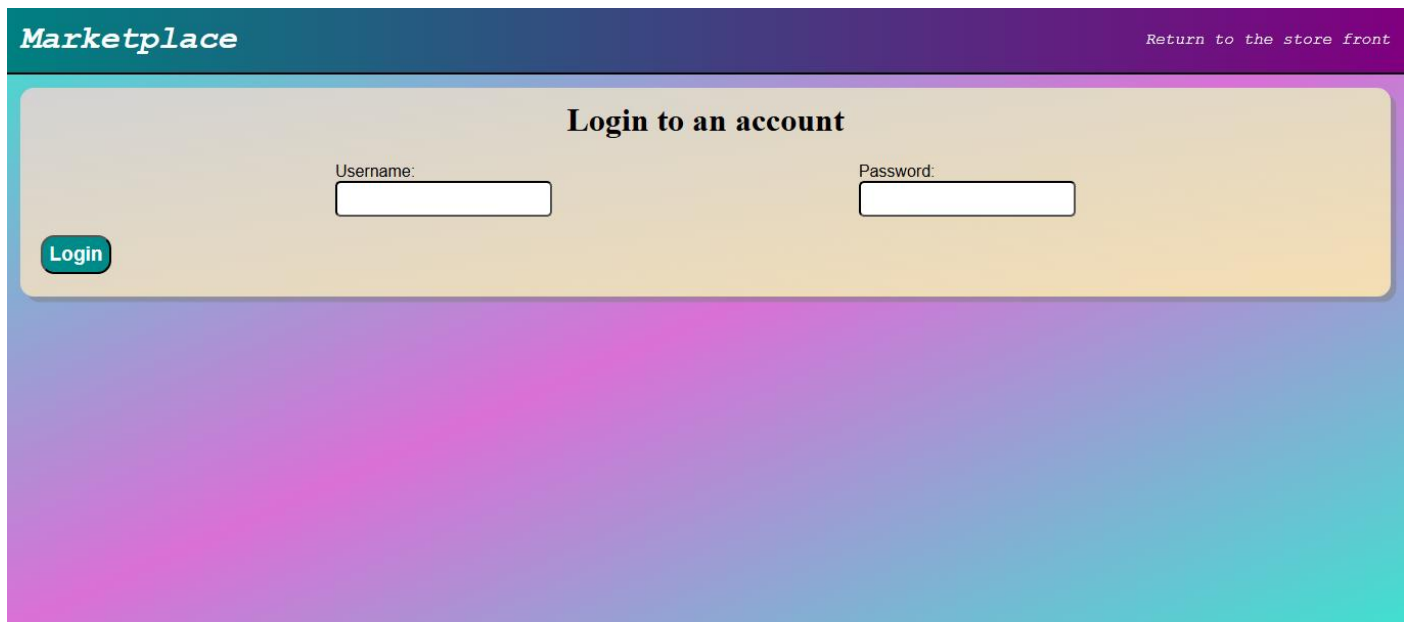


Mobile landscape



Account login page:

Desktop



The image shows a desktop mockup of an account login page. At the top, there is a dark purple header bar. On the left side of this bar, the word "Marketplace" is written in a white, italicized serif font. On the right side, the text "Return to the store front" is written in a small, white, italicized sans-serif font. Below the header, the main content area has a light beige background. Centered at the top of this area is the heading "Login to an account" in a bold, black, sans-serif font. Below this heading, there are two input fields. The first is labeled "Username:" and the second is labeled "Password:". Both labels are in a small, black, sans-serif font. The input fields are white rectangles with thin black borders. To the left of the "Username:" input field, there is a small, rounded rectangular button with a teal background and the word "Login" in white, bold, sans-serif font. The entire page is set against a background with a vertical gradient from purple on the left to teal on the right.

Marketplace

Return to the store front

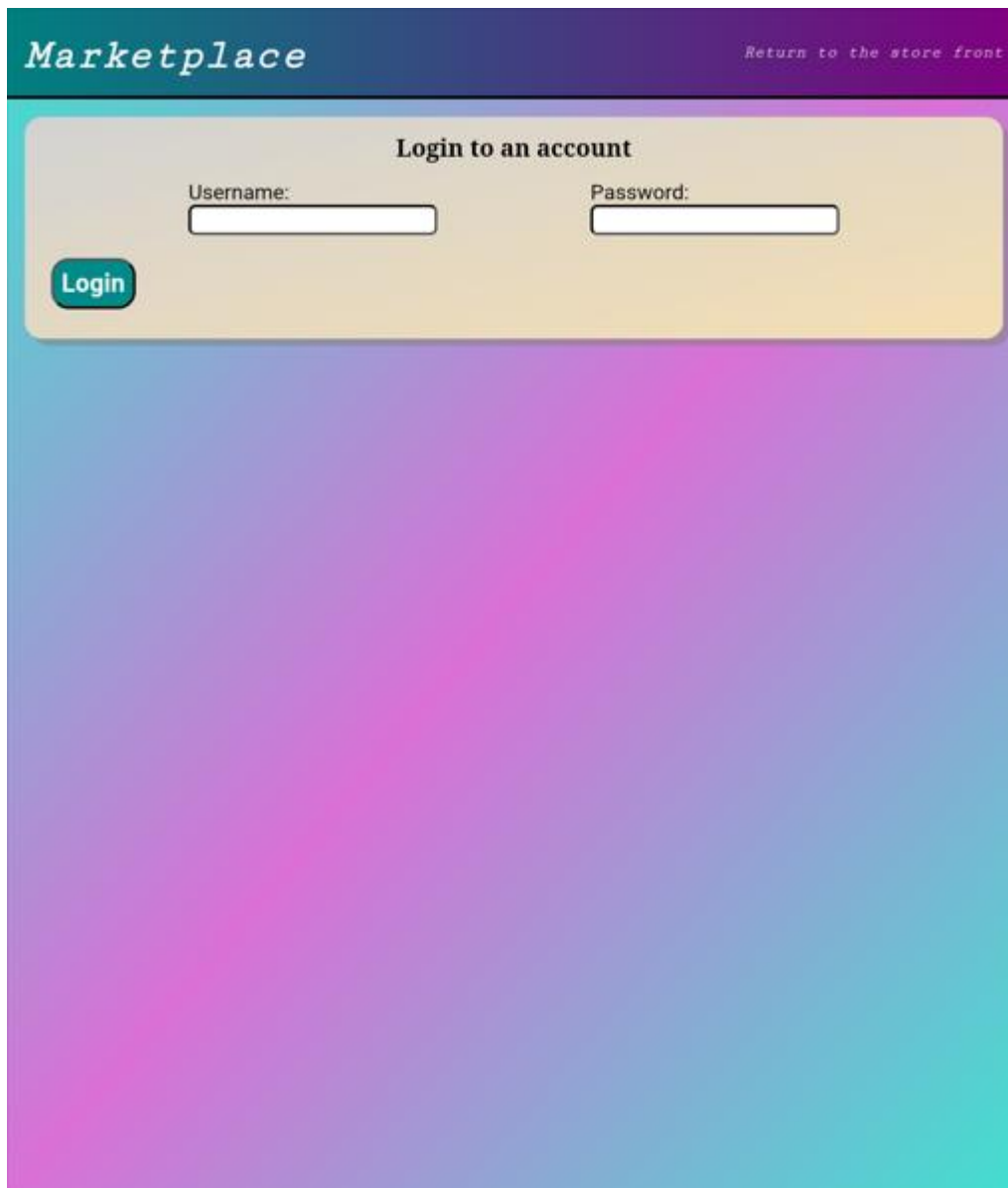
Login to an account

Username:

Password:

Login

Tablet portrait



The image shows a tablet portrait view of a login page. At the top, a dark purple header bar contains the word "Marketplace" in a white, stylized font on the left, and a link "Return to the store front" in a smaller, lighter purple font on the right. Below the header is a light beige rounded rectangle containing the login form. The form is titled "Login to an account" in bold black text. It features two input fields: "Username:" on the left and "Password:" on the right, both with white text and empty text boxes. A green "Login" button is positioned to the left of the input fields. The background of the tablet screen is a vibrant, multi-colored gradient transitioning from purple at the top to blue and green at the bottom.

Marketplace [Return to the store front](#)

Login to an account

Username:

Password:

Login

*Marketplace*Return to the store front

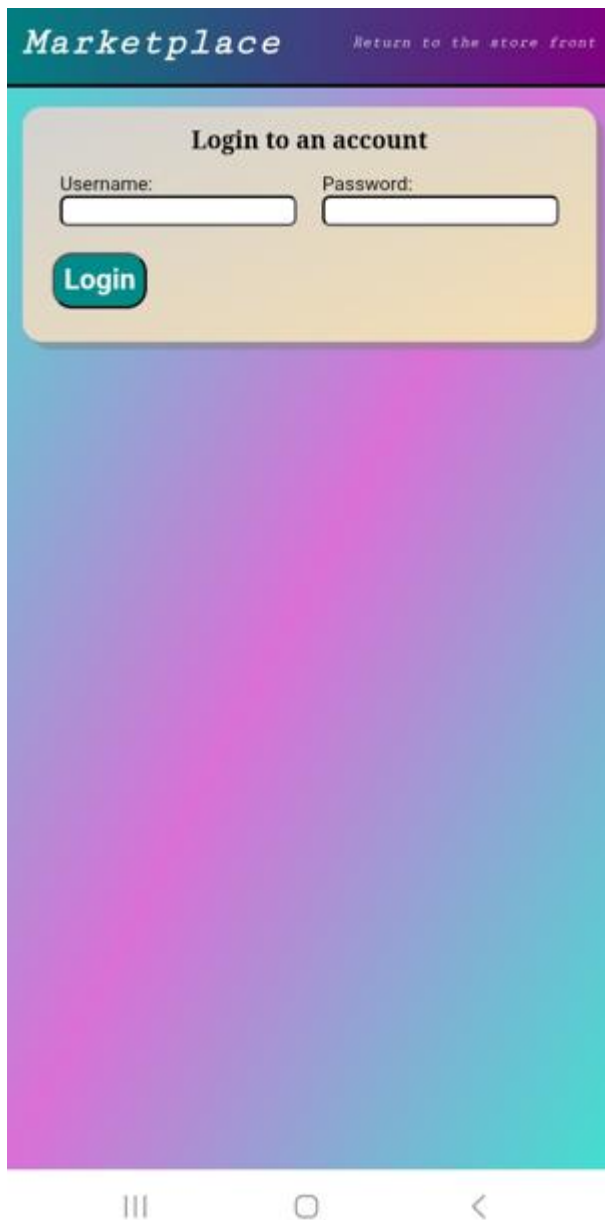
Login to an account

Username:

Password:

Login

Mobile portrait



The image shows a mobile portrait view of a login screen. At the top, there is a dark purple header bar with the word "Marketplace" in a light blue, monospace-style font on the left, and a link "Return to the store front" in a smaller, lighter font on the right. Below the header is a light beige rounded rectangle containing the login form. The form has the title "Login to an account" at the top. Below the title are two labels: "Username:" and "Password:". Each label is followed by a white rectangular input field. Below the input fields is a green rounded button with the word "Login" in white. The background of the screen is a vertical gradient from purple at the top to blue at the bottom. At the very bottom, there are three small, faint icons: a hamburger menu icon (three horizontal lines), a square icon, and a back arrow icon.

Marketplace *Return to the store front*

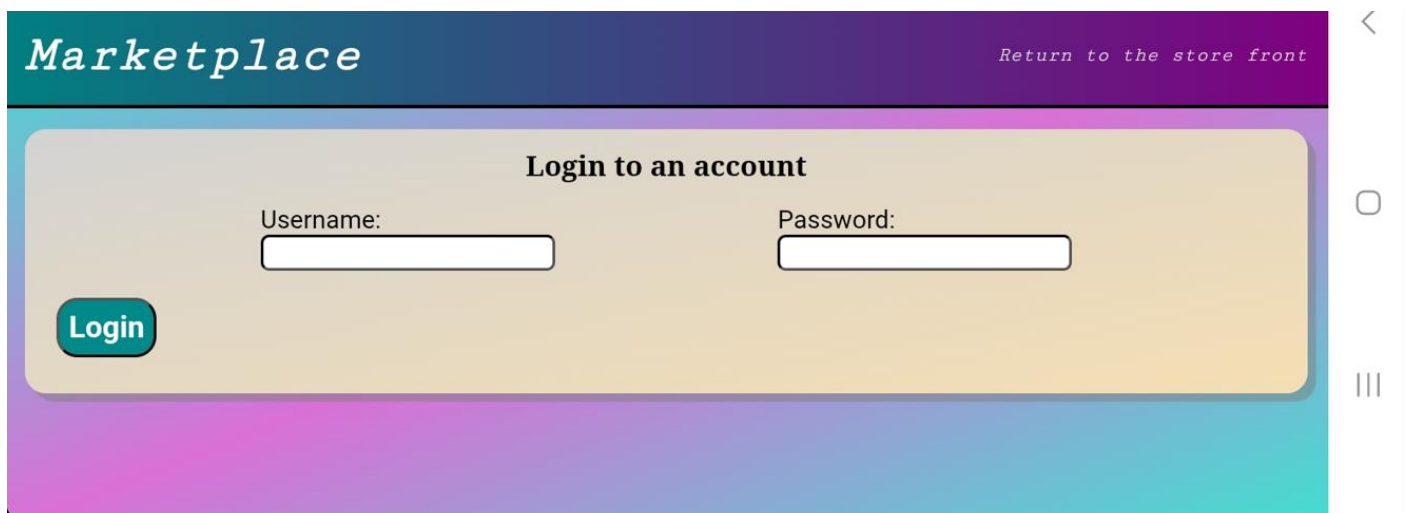
Login to an account

Username:

Password:

Login

Mobile landscape



The image shows a mobile landscape view of the same login screen. The layout is wider. The header bar is the same, but the "Return to the store front" link is now on the right side of the header. The login form is a light beige rounded rectangle that takes up most of the screen width. The title "Login to an account" is centered at the top of the form. Below the title, the "Username:" and "Password:" labels are positioned above their respective input fields. The "Login" button is now on the left side of the form. The background gradient is the same. On the right side of the screen, there are three small, faint icons: a back arrow icon, a square icon, and a hamburger menu icon.

Marketplace *Return to the store front*

Login to an account

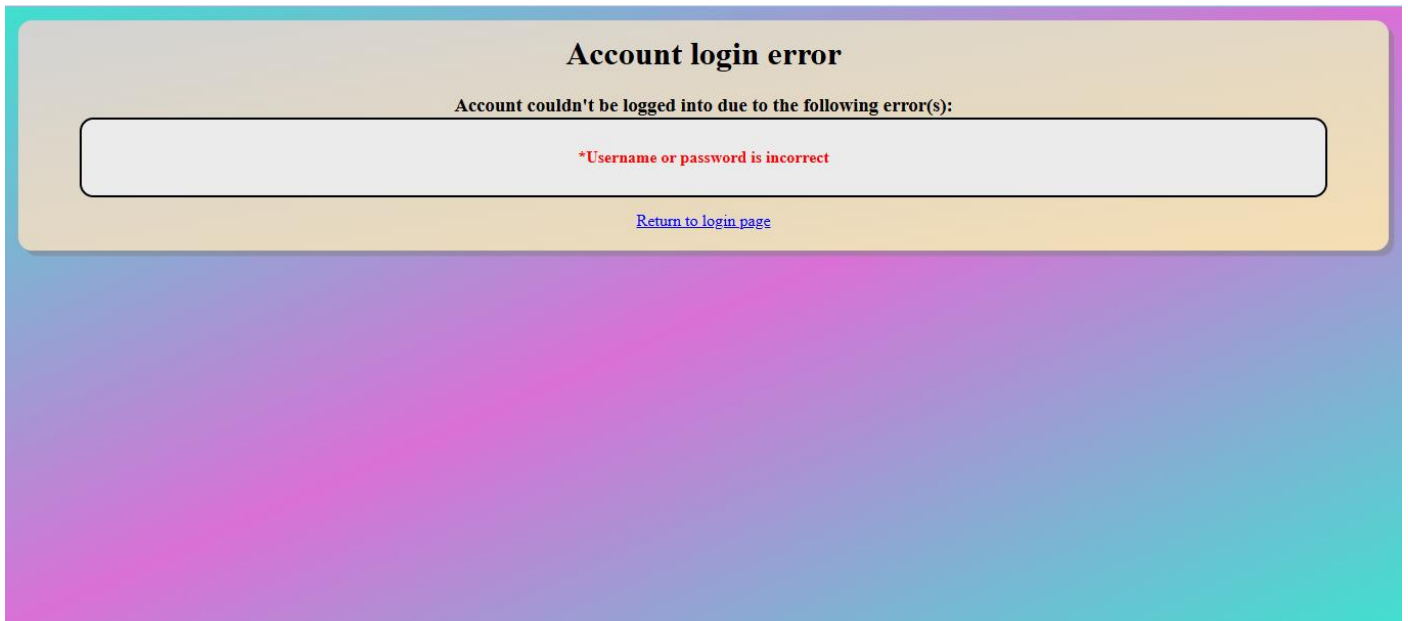
Username:

Password:

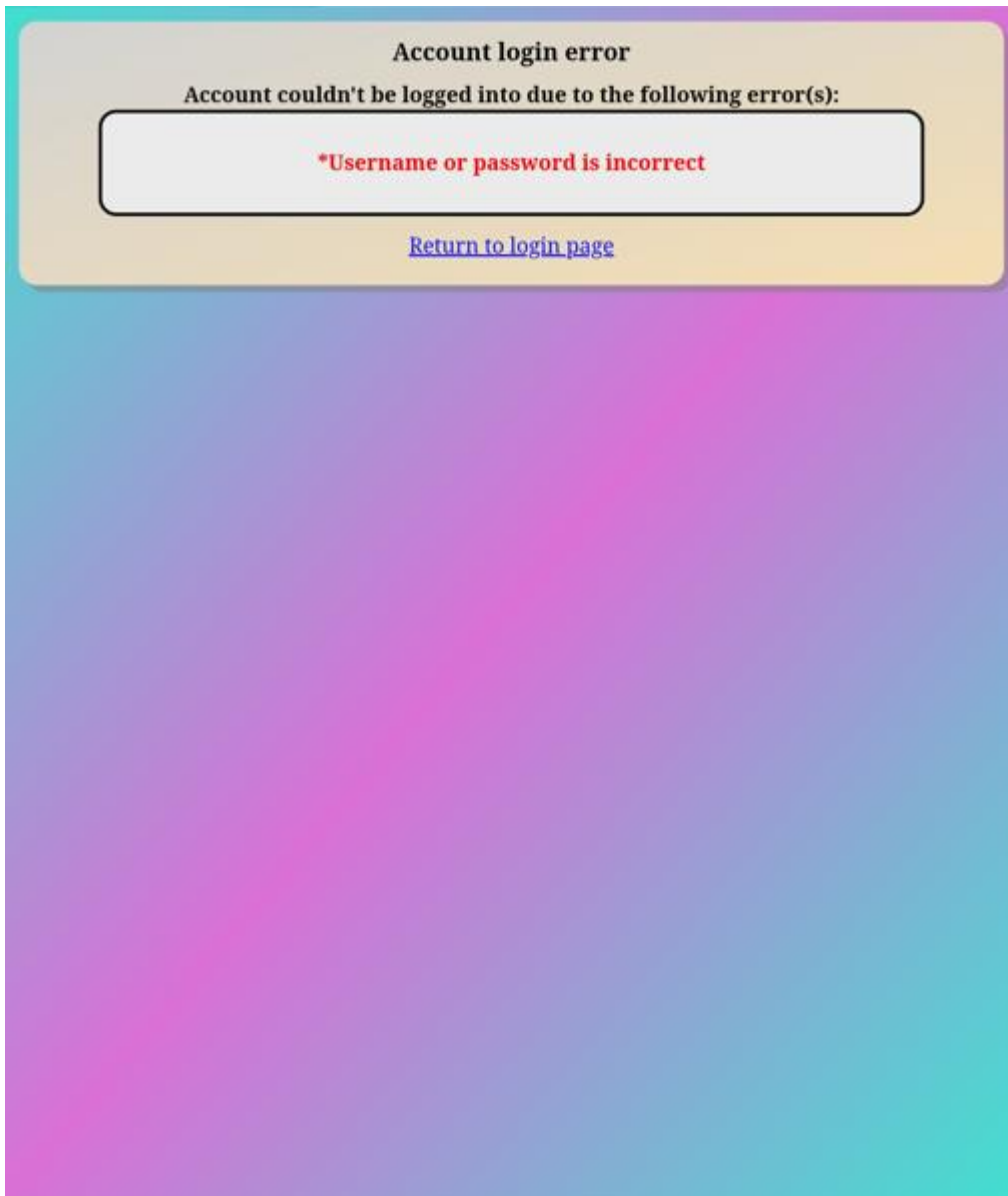
Login

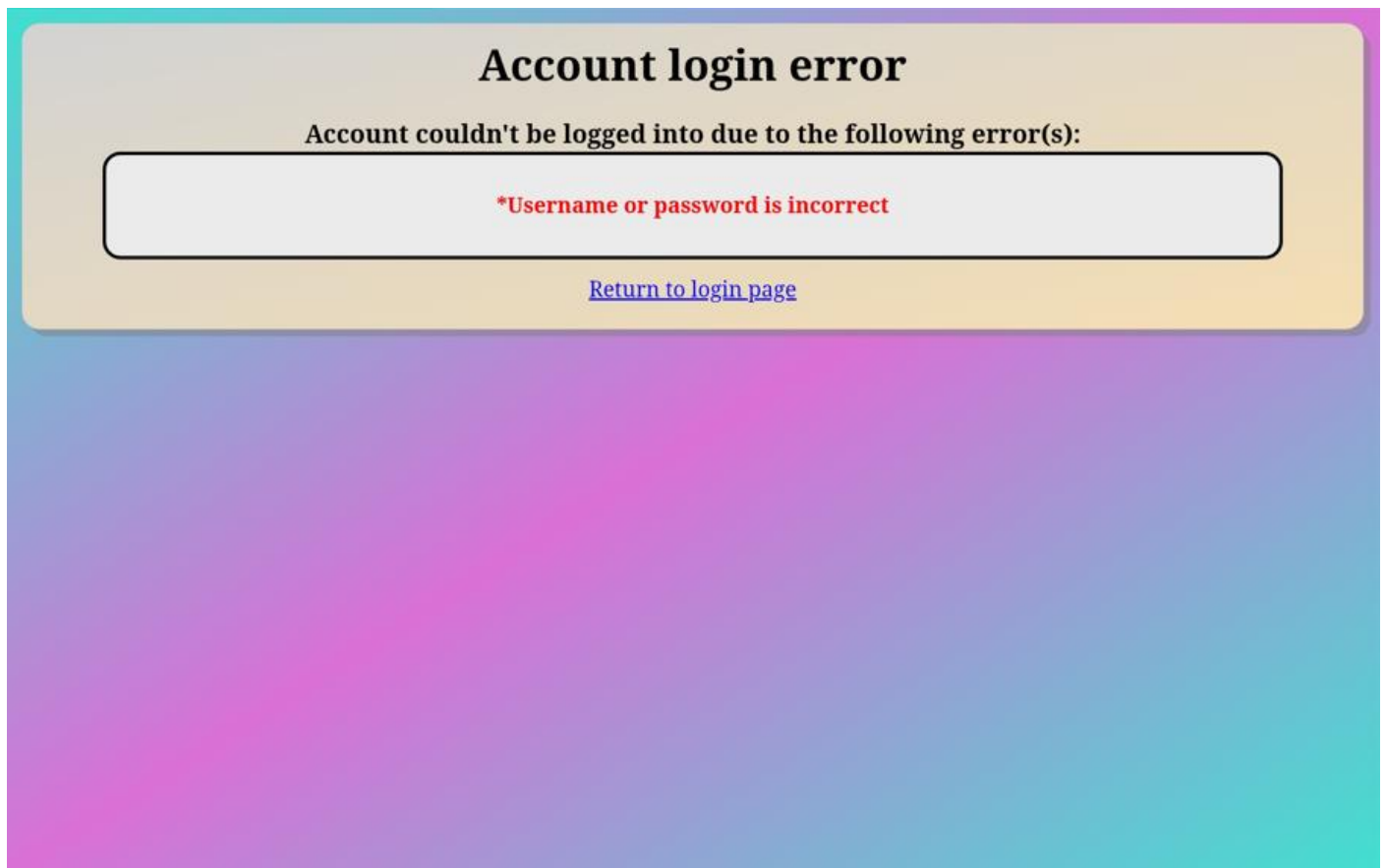
Login error page:

Desktop



Tablet portrait

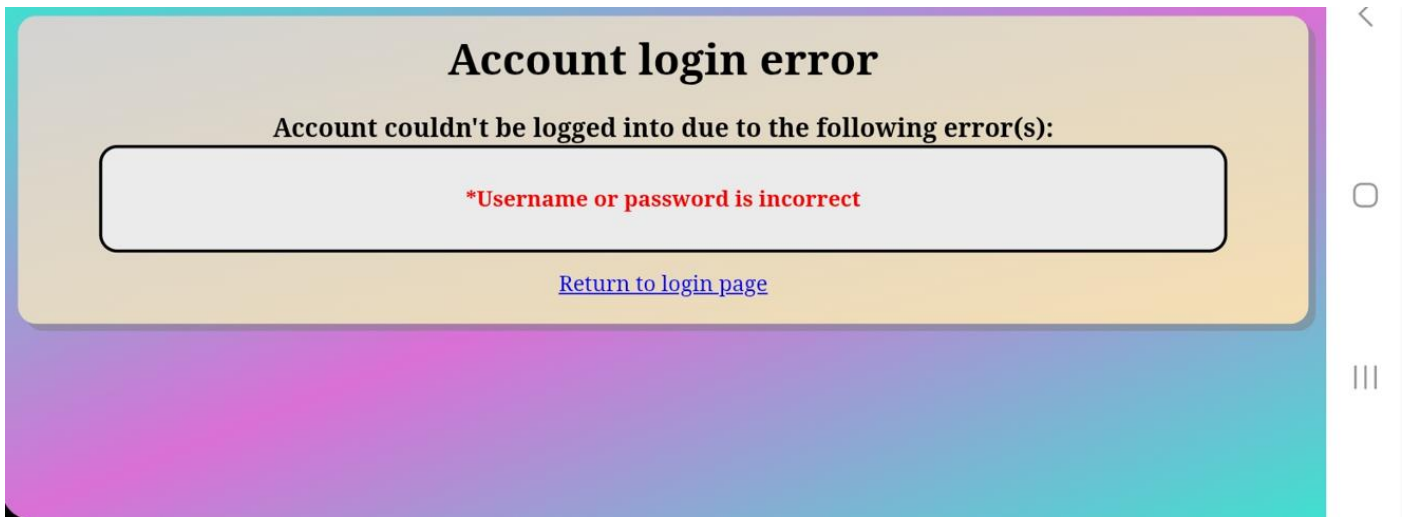




Mobile portrait

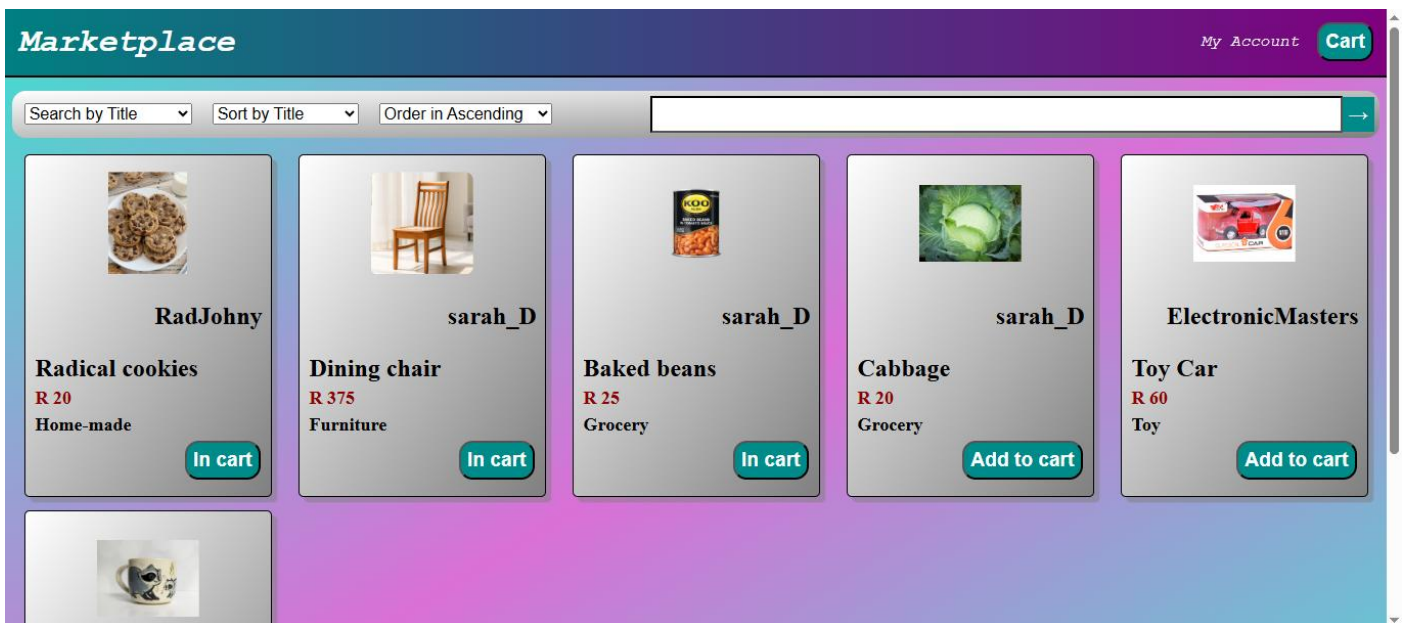


Mobile landscape

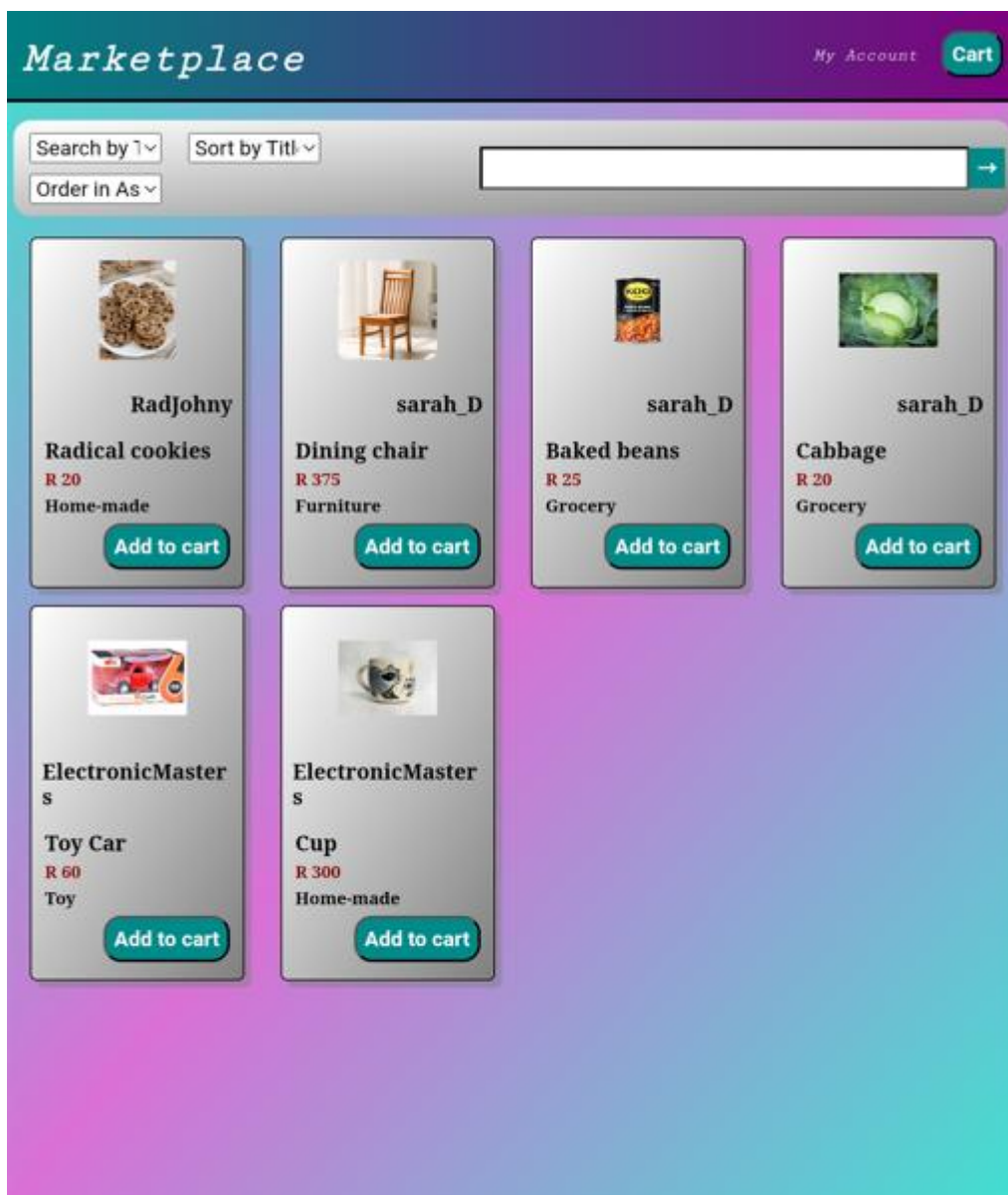


Main page:

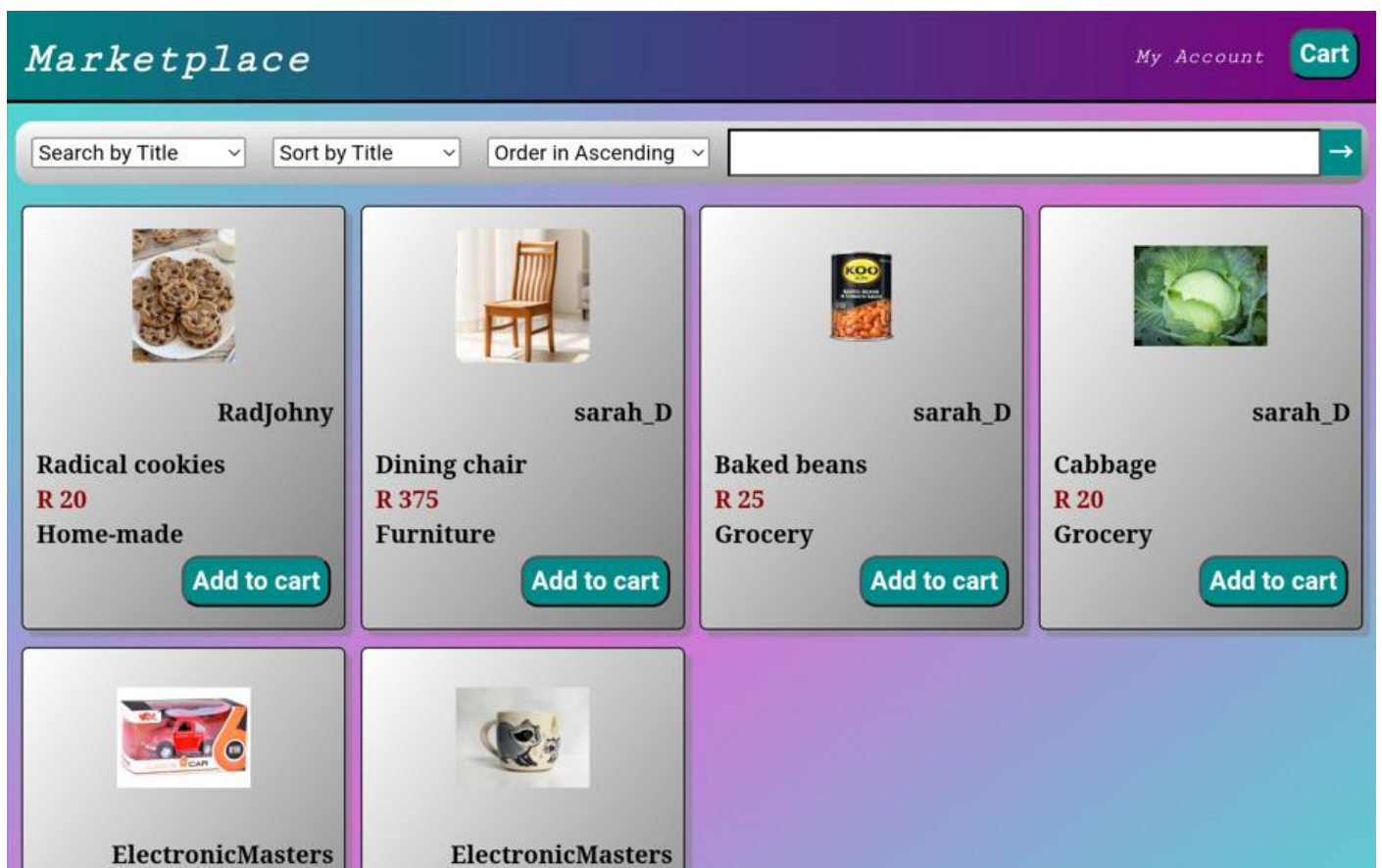
Desktop



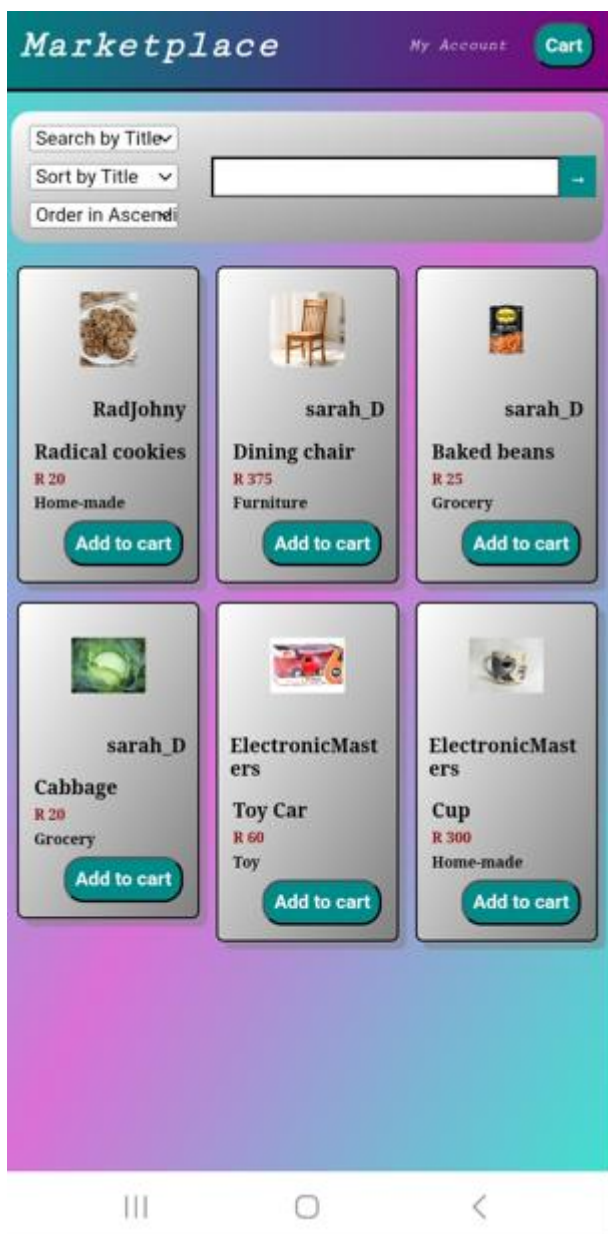
Tablet portrait



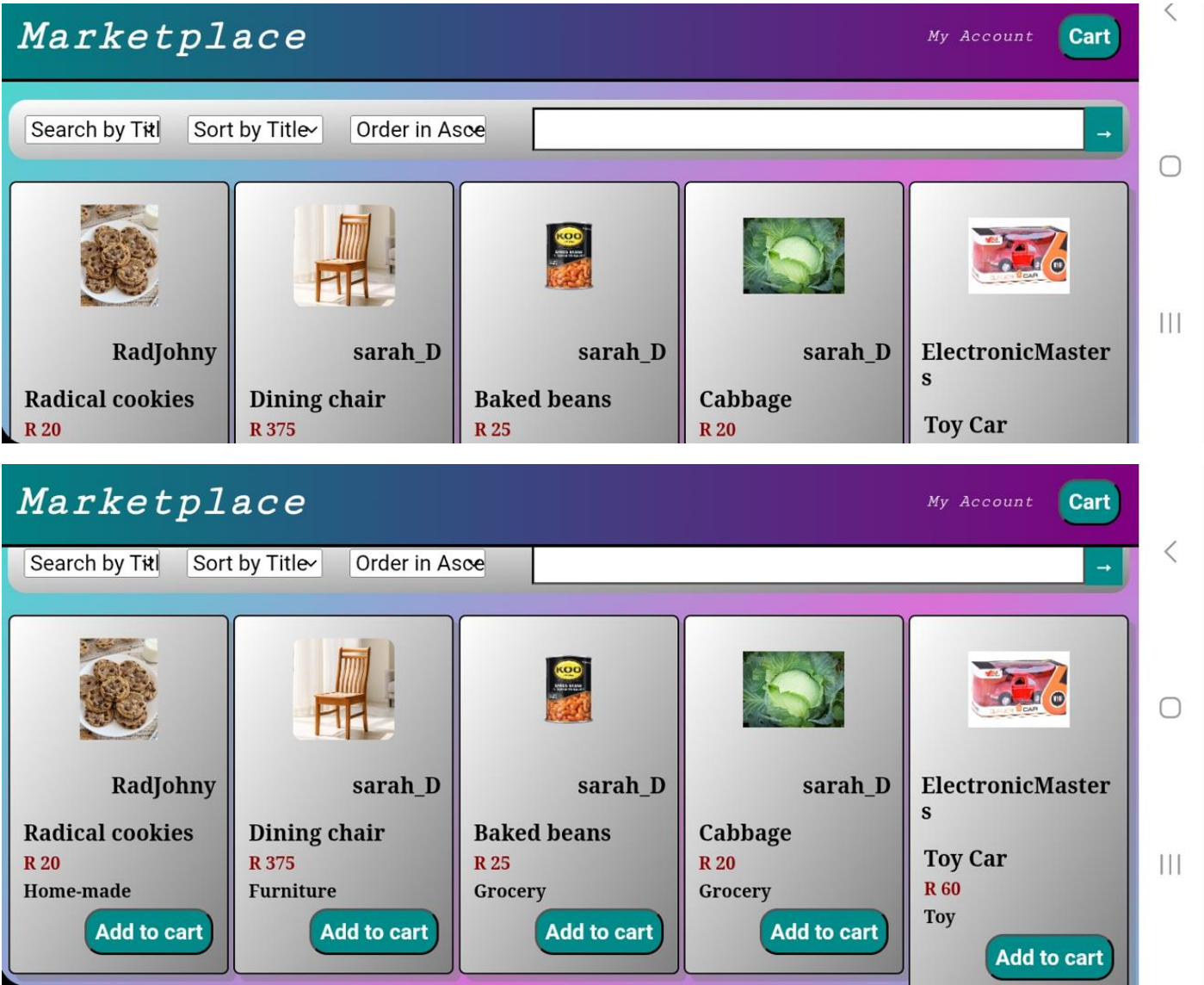
Tablet landscape



Mobile portrait



Mobile landscape



Product page:

Desktop



Baked beans

sarah_D

0 vendor rating

R25

17 left in stock

Description

Healthy, canned, baked beans.

[In cart](#)

Products in the same category



Tablet portrait



Baked beans

sarah_D

0 vendor rating

R25

17 left in stock

Description

Healthy, canned, baked beans.

Add to cart

Products in the same category



sarah_D

0

Cabbage

R 20

Grocery

Add to cart



Baked beans

sarah_D

0 vendor rating

R25

17 left in stock

Description

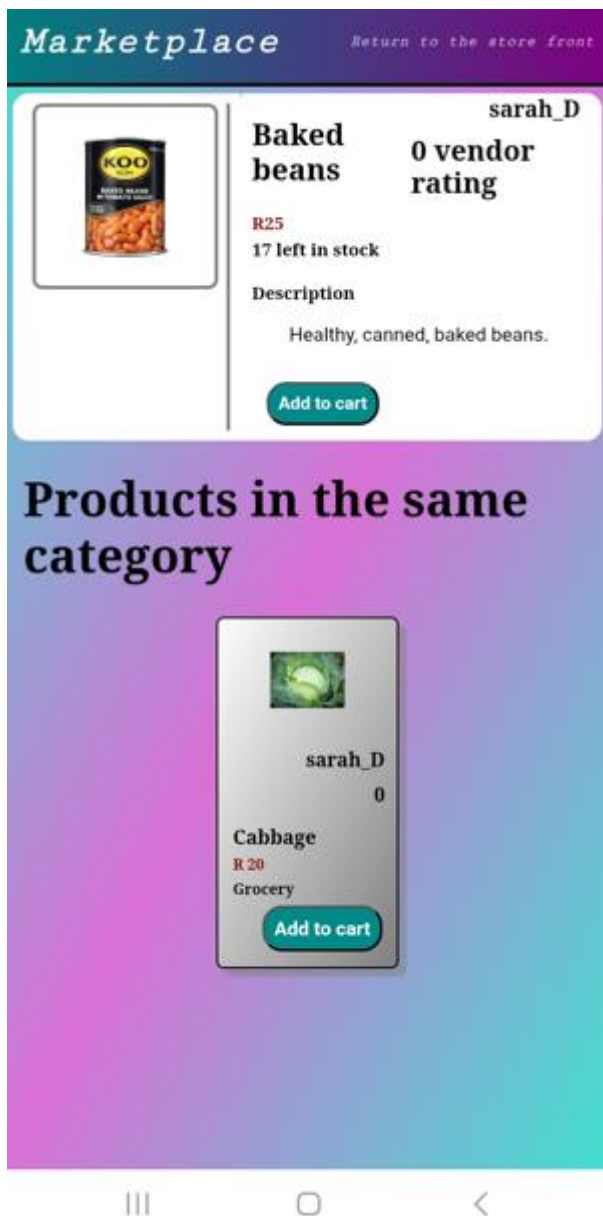
Healthy, canned, baked beans.

[Add to cart](#)

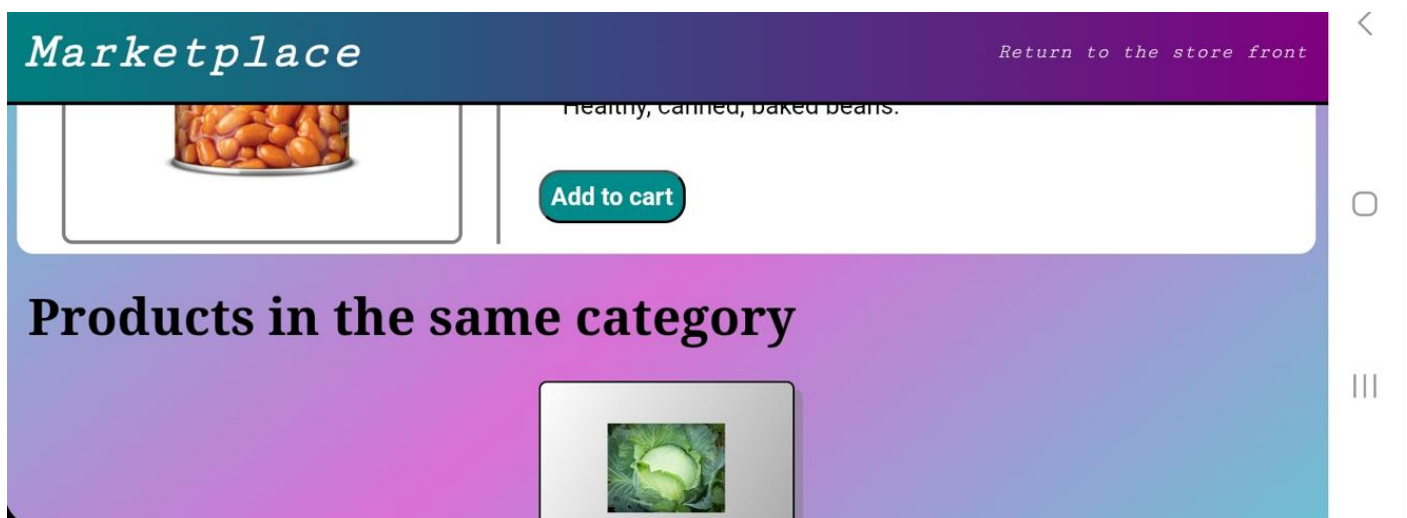
Products in the same category



Mobile portrait



Mobile landscape



Create listing page:

Desktop

MarketplaceReturn to the store front

Listing details

Product categories:

Listing's image:
 No file chosen

Create listing

*Marketplace*Return to the store front

Listing details

Title

Price

Stock

Description(Optional)

Product categories:

Listing's image:

Home-made ▾

Choose file

No file chosen

Create listing

*Marketplace*Return to the store front

Listing details

Product categories:

Listing's image:
 No file chosen

Mobile portrait

The image shows a mobile application interface in portrait orientation. At the top, there is a dark purple header bar with the word "Marketplace" in a white, stylized font on the left and a link "Return to the store front" in a smaller white font on the right. Below the header is a light beige rounded rectangle containing the "Listing details" form. The form has three input fields for "Title", "Price", and "Stock" arranged horizontally. Below these is a larger text area for "Description(Optional)". At the bottom of the form, there are two sections: "Product categories:" with a dropdown menu currently showing "Home-made", and "Listing's image:" with a "Choose File" button and the text "No file chosen". A green rounded button labeled "Create listing" is positioned at the bottom left of the form. The background of the screen is a vibrant, multi-colored gradient. At the very bottom, there is a white bar with three standard Android navigation icons: a square, a circle, and a triangle.

Marketplace *Return to the store front*

Listing details

Title Price Stock

Description(Optional)

Product categories: Listing's image:

Home-made ▾ Choose File No file chosen

Create listing

Mobile landscape

The image shows the same mobile application interface but in landscape orientation. The layout is stretched horizontally. The header bar remains at the top. The "Listing details" form is now wider, with the "Title", "Price", and "Stock" input fields still arranged horizontally but with more space between them. The "Description(Optional)" text area is also wider. The "Product categories:" dropdown and "Listing's image:" section are positioned further to the right. The "Create listing" button is now at the bottom left of the form. The background gradient is the same. On the right side of the screen, there is a vertical white bar containing three navigation icons: a triangle, a circle, and a square, which are the standard Android navigation icons rotated 90 degrees clockwise.

Marketplace *Return to the store front*

Listing details

Title Price Stock

Description(Optional)

Product categories: Listing's image:

Home-made ▾ Choose File No file chosen

Marketplace

[Return to the store front](#)

Product categories: Listing's image: No file chosen

Listing error page:

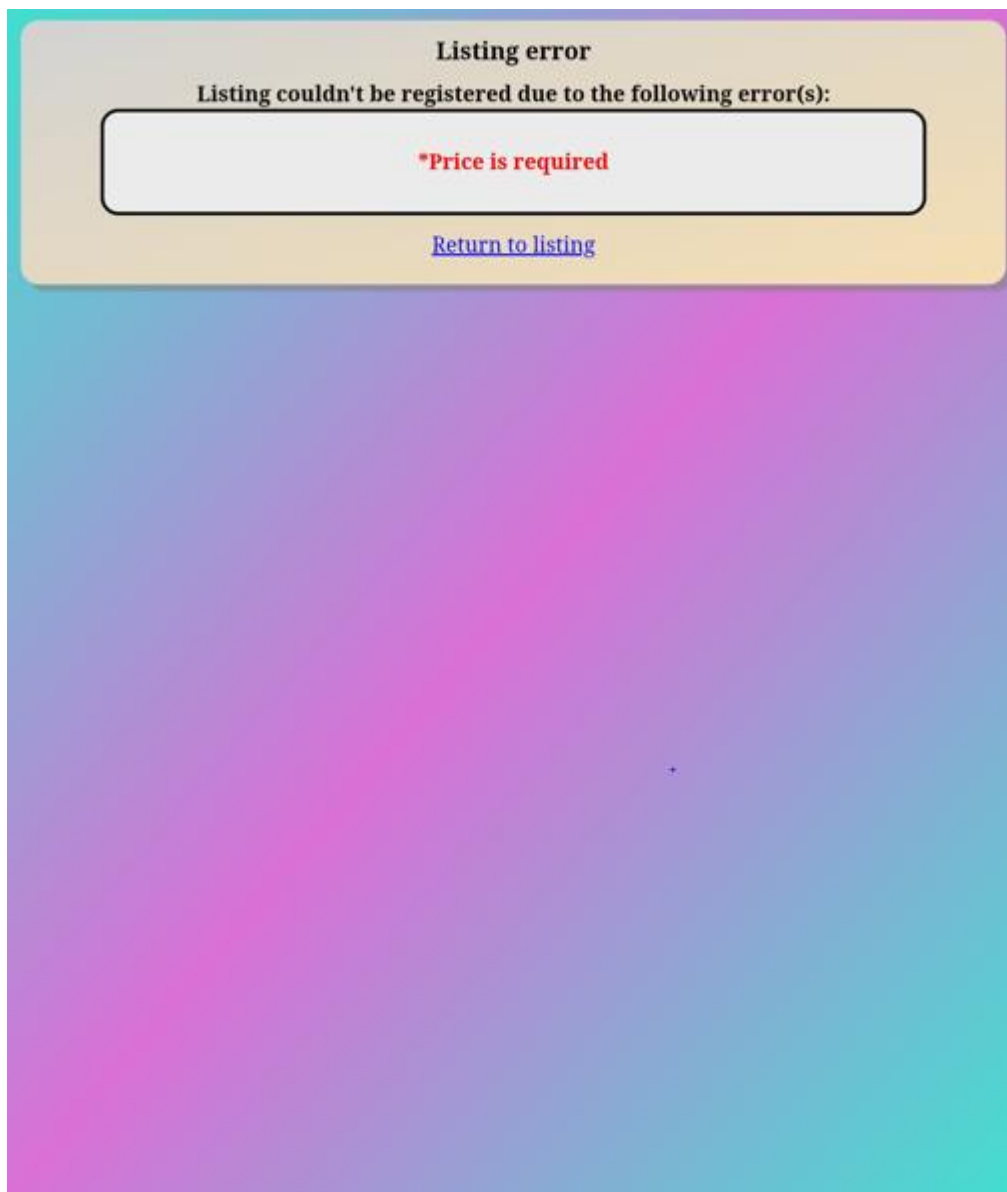
Desktop

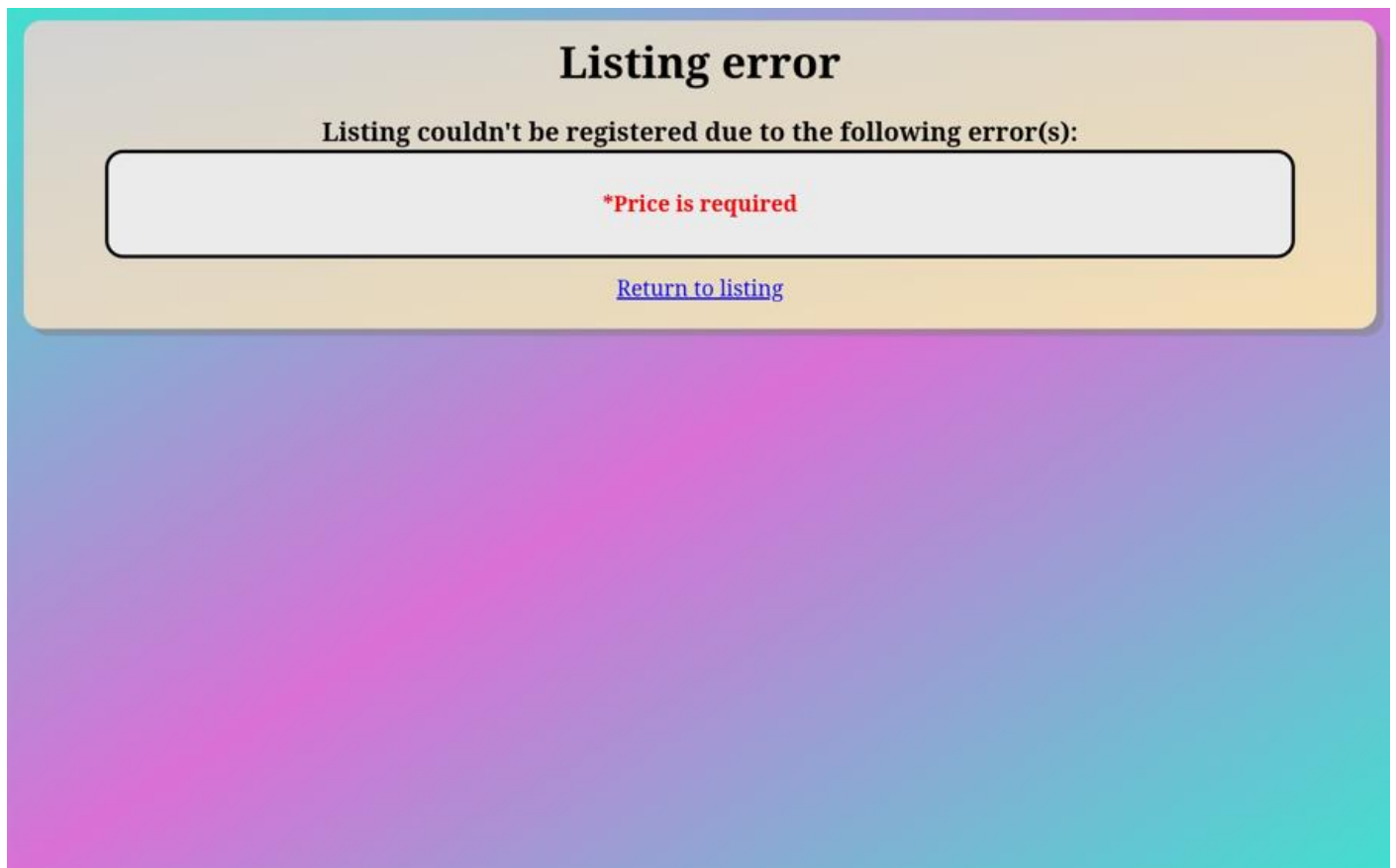
Listing error

Listing couldn't be registered due to the following error(s):

*Price is required

[Return to listing](#)

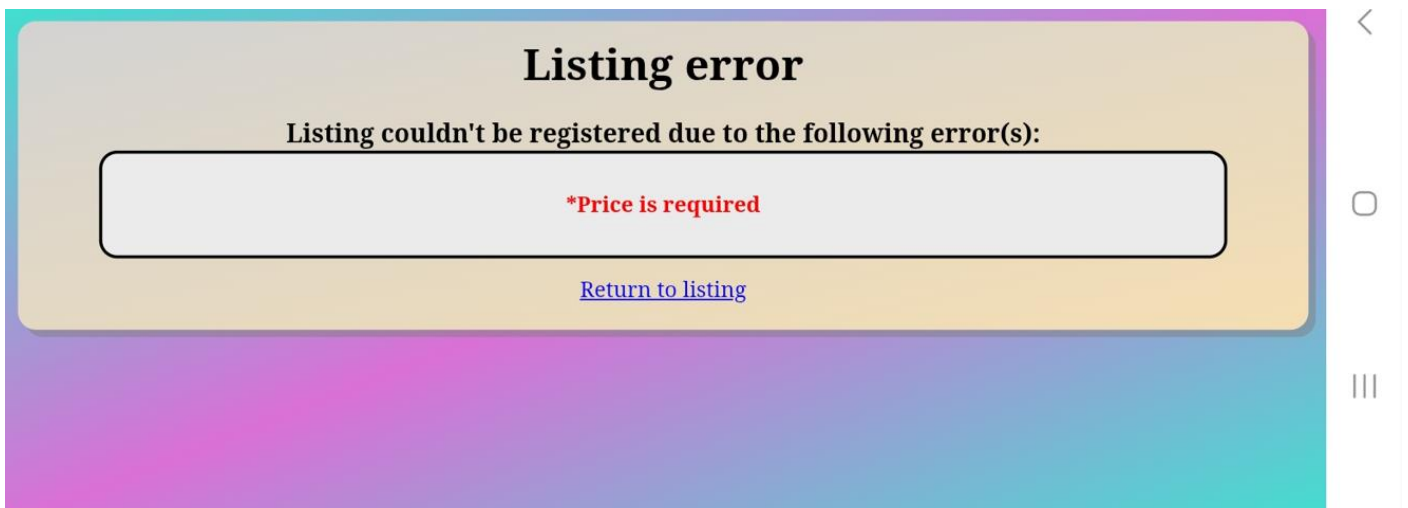




Mobile portrait



Mobile landscape



Cart page:

Desktop

MarketplaceReturn to the store front

Cart

RadJohnny
Total cost: R40
Delivery address:

Update address
Your approval: ☒ Vendor approval: ☒
Reject address
Chat with vendor
Cancel order delivery
Complete payment and delivery

Radical cookies

Price: R 40
Quantity:

Current funds: R 4500
Voucher amount (in Rands):

Redeem voucher

MarketplaceReturn to the store front

Radical cookies

Price: R 40
Quantity:
Update quantity
Remove

sarah_D
Total cost: R400
Delivery address:

Update address
Your approval: ☒ Vendor approval: ☐
Reject address

Marketplace

Return to the store front

Cart

RadJohny

Total cost: R40

Delivery address:

Update address

Your approval: ☒ Vendor approval: ☒

Reject address

Chat with vendor

Cancel order delivery

Complete payment and delivery

Radical cookies

Price: R 40



Quantity:

Update quantity

Remove

sarah_D

Total cost: R400

Delivery address:

Update address

Your approval: ☒ Vendor approval: ☐

Reject address

Chat with vendor

Cancel order delivery

Current funds: R 4500

Voucher amount (in Rands):

Redeem voucher

MarketplaceReturn to the store front

Cart

RadJohny
Total cost: R40

Delivery address:

Update address

Your approval: ☒ Vendor approval: ☒

Reject address

Chat with vendor

Cancel order delivery

Complete payment and delivery

Radical cookies


Price: R 40
Quantity:

Update quantity

Current funds: R 4500
Voucher amount (in Rands):

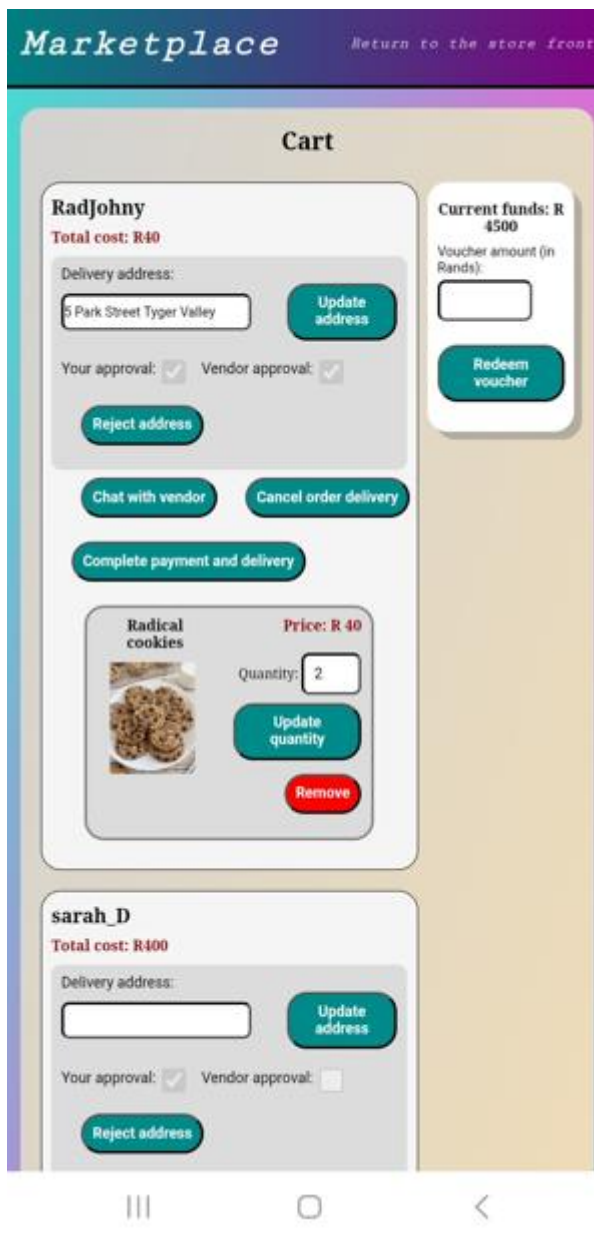
Redeem voucher

[Complete payment and delivery](#)**Radical cookies****Price: R 40**Quantity: [Update quantity](#)[Remove](#)**sarah_D****Total cost: R400**

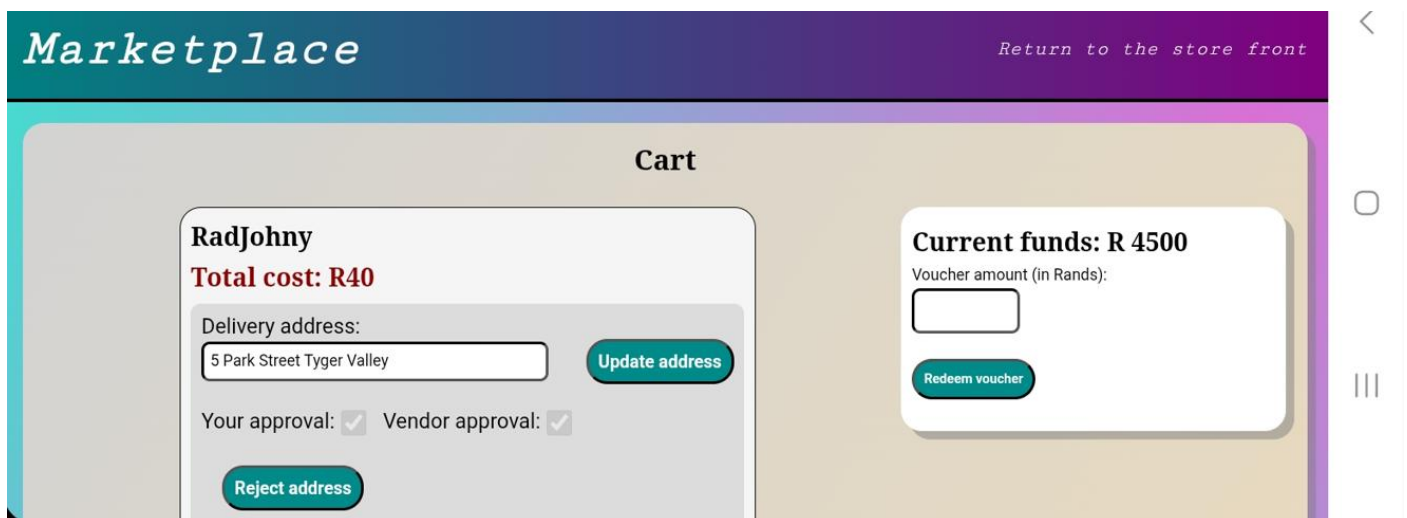
Delivery address:

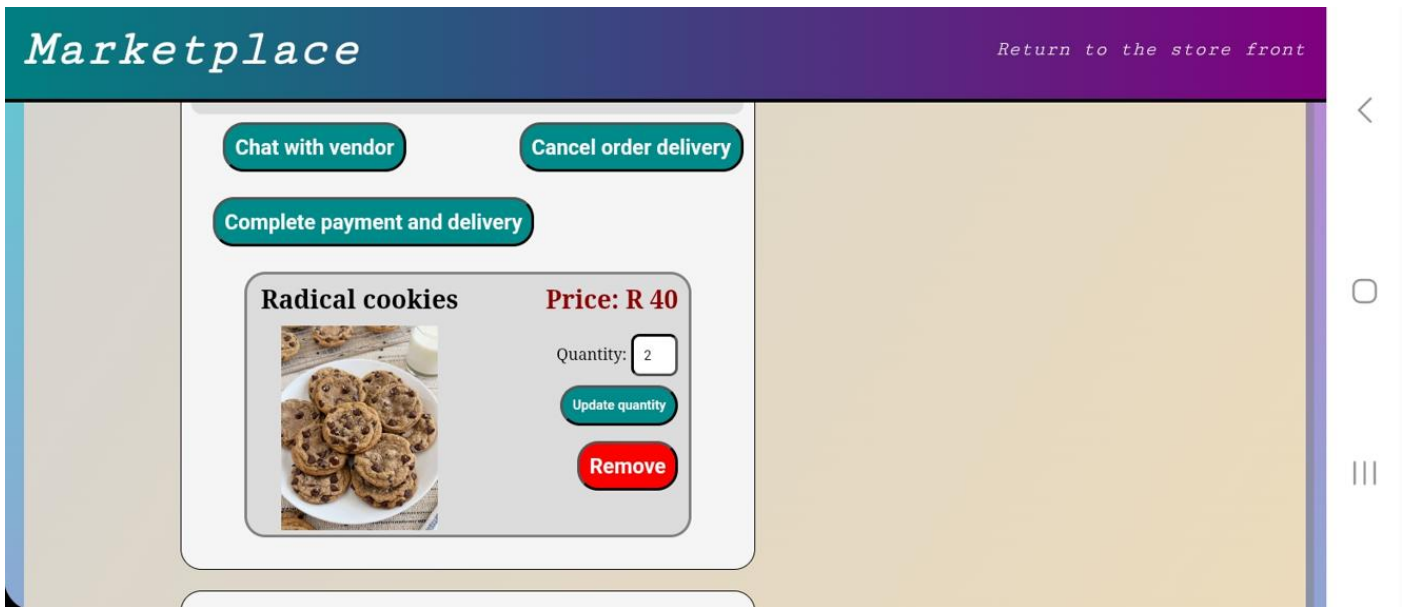
[Update address](#)Your approval: ☒ Vendor approval: ☐[Reject address](#)[Chat with vendor](#)[Cancel order delivery](#)[Complete payment and delivery](#)

Mobile portrait



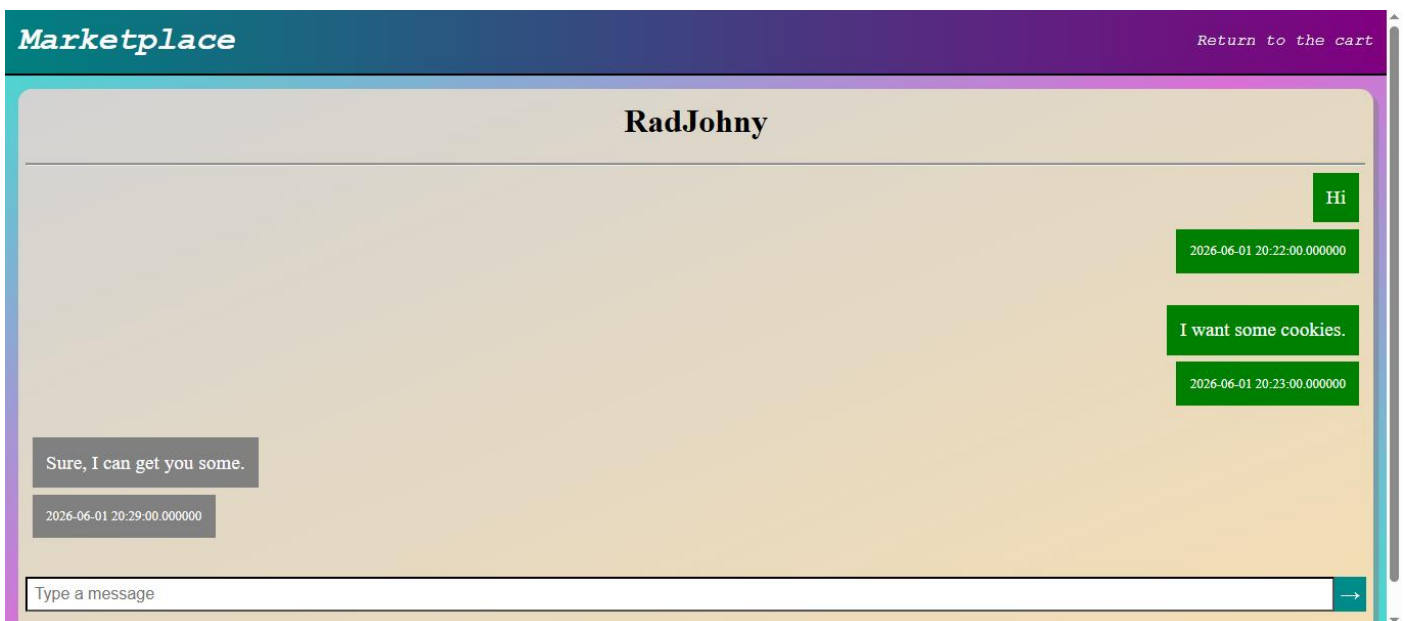
Mobile landscape

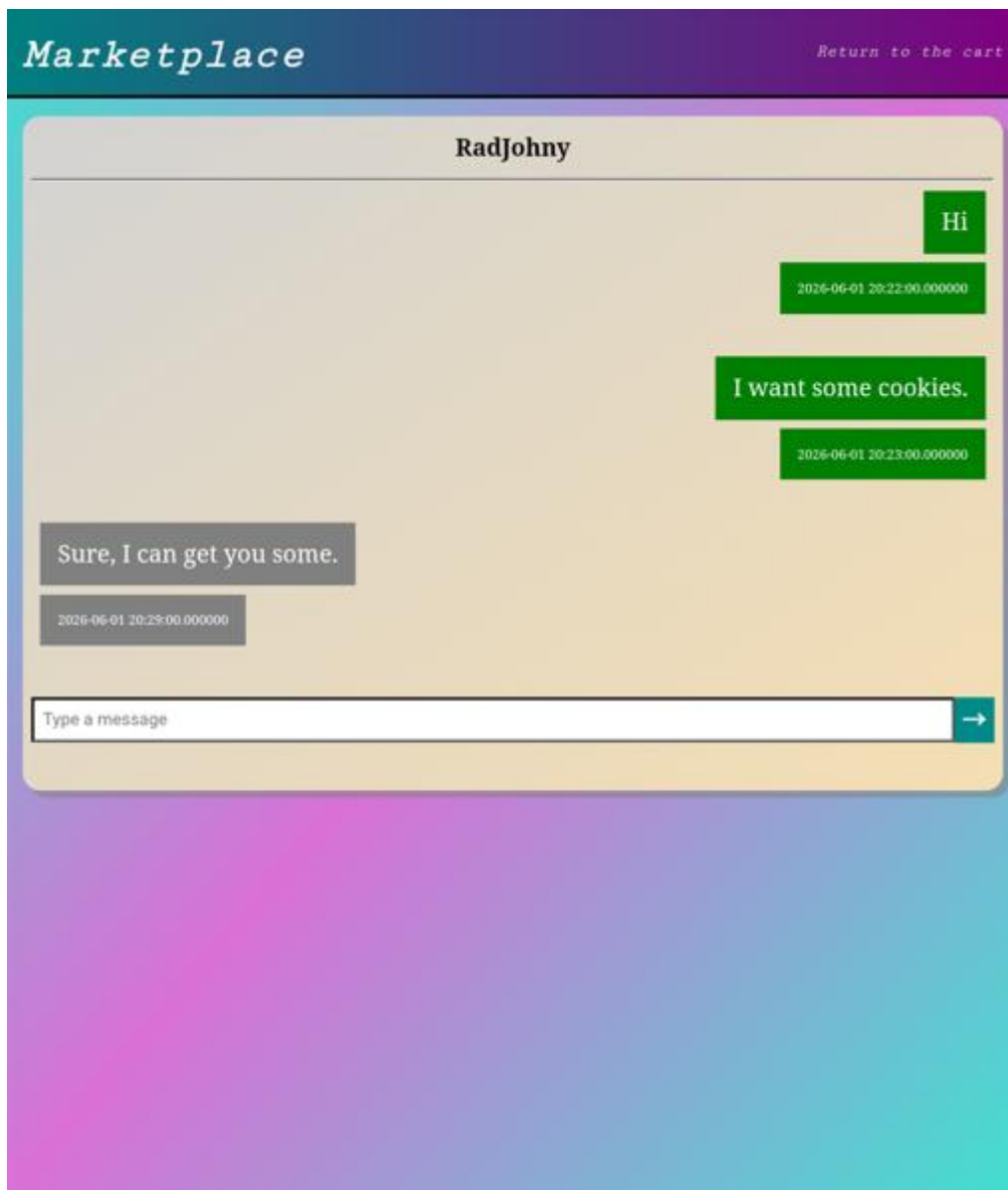


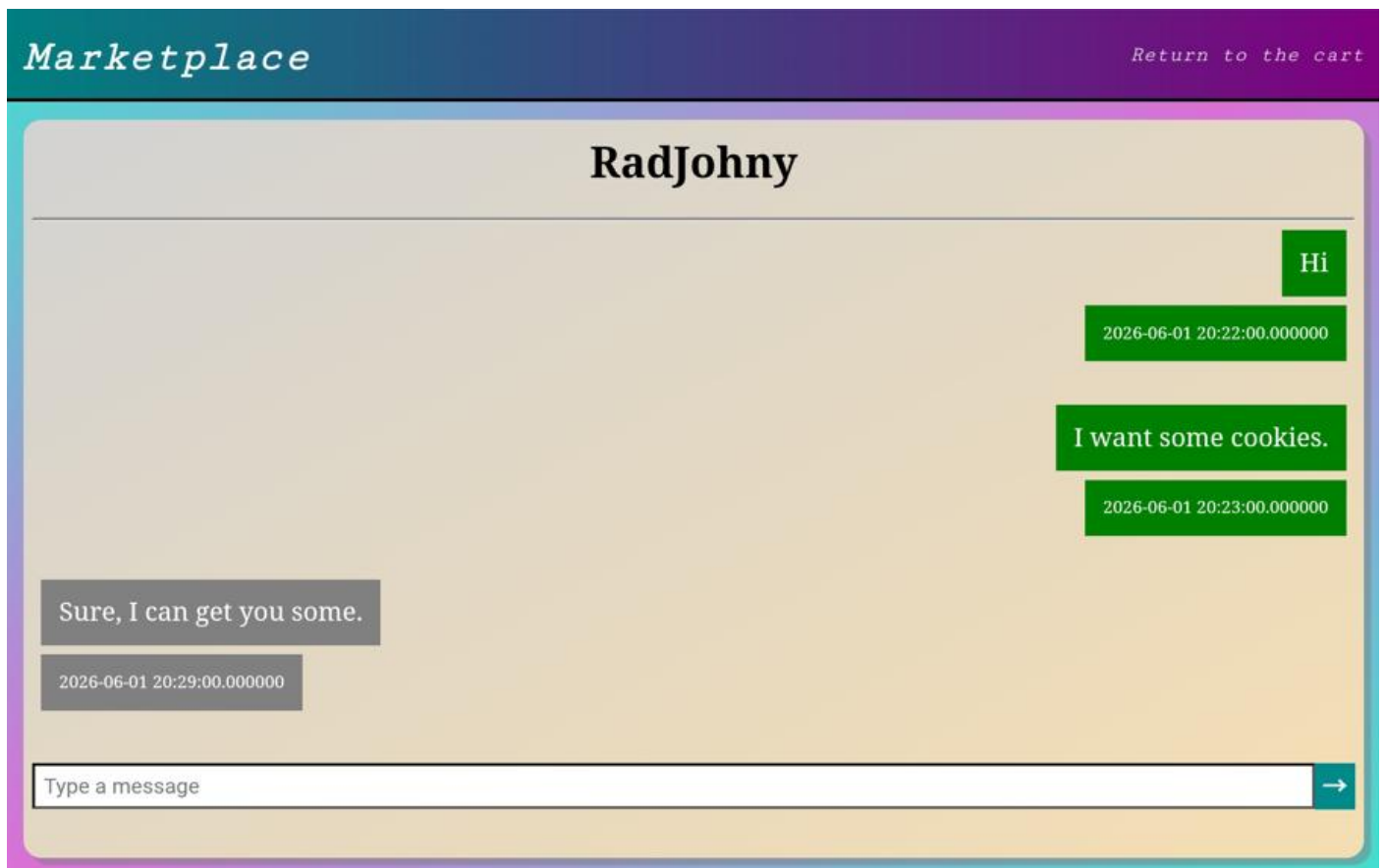


Delivery message page:

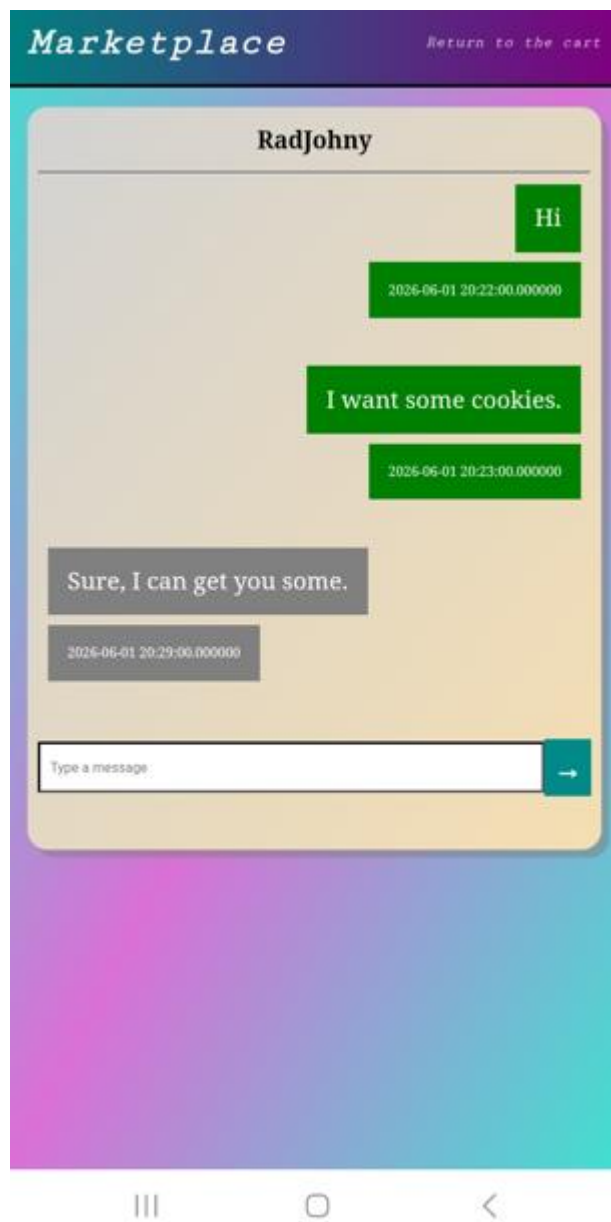
Desktop



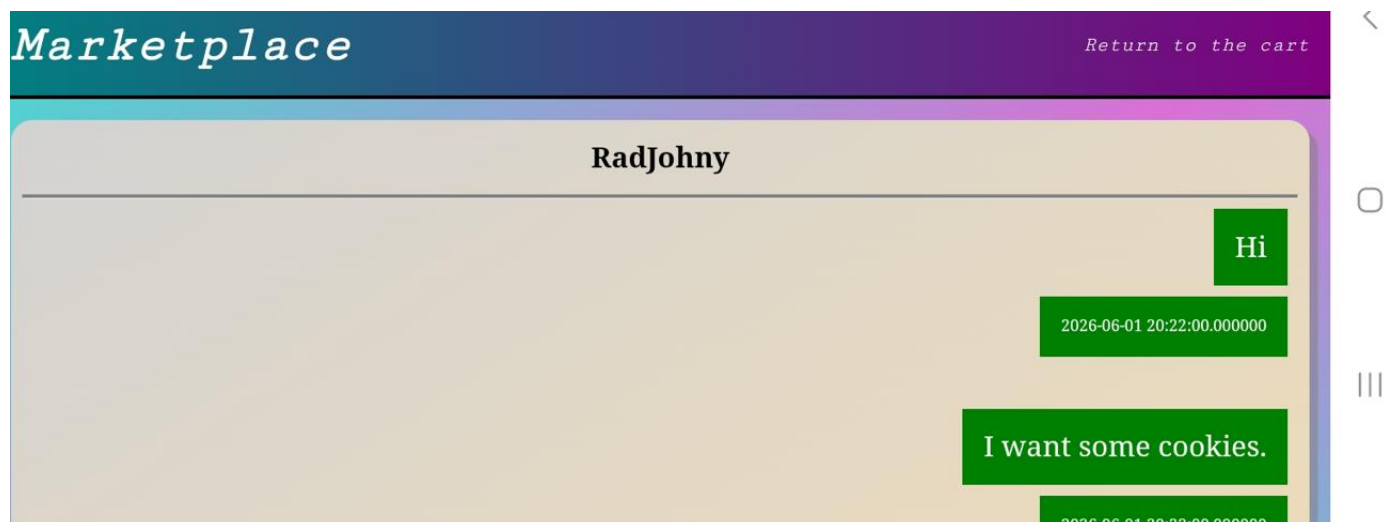


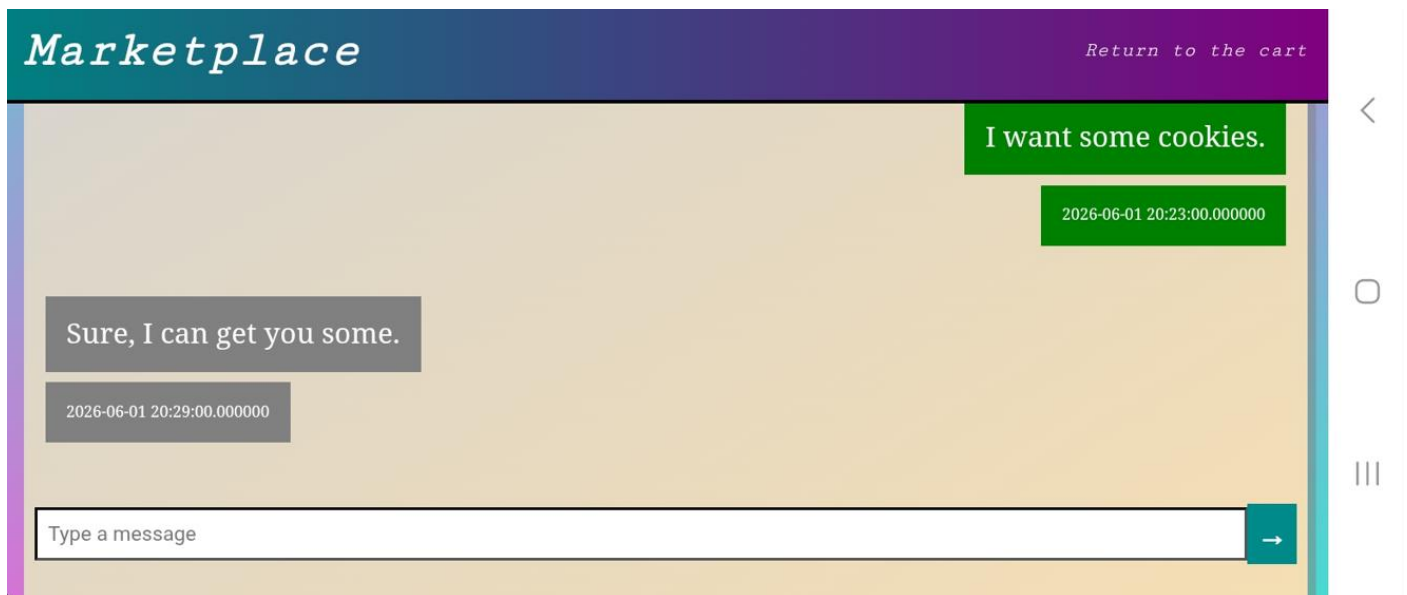


Mobile portrait



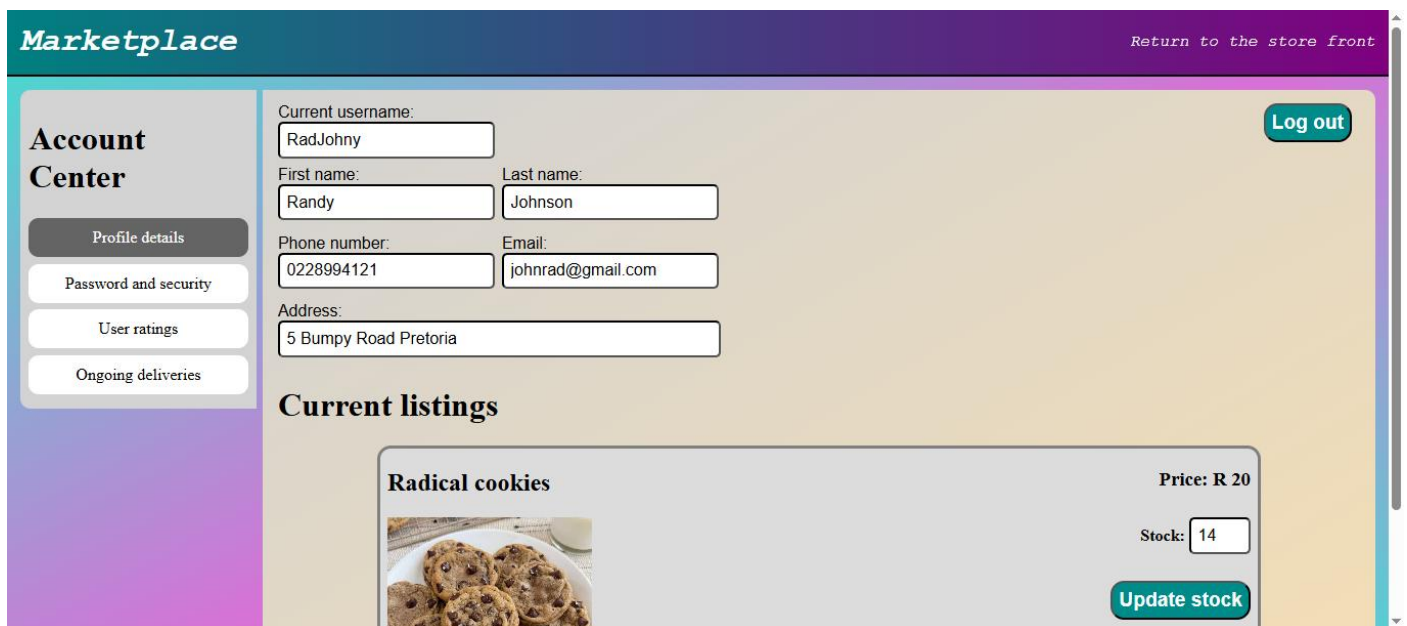
Mobile landscape





Account centre (Profile details) page:

Desktop



Marketplace

Return to the store front

Profile details

Password and security

User ratings

Ongoing deliveries

Phone number:

0228994121

Email:

johnrad@gmail.com

Address:

5 Bumpy Road Pretoria

Current listings

Radical cookies



Price: R 20

Stock: 14

Update stock

Remove listing

Tablet portrait

Marketplace

Return to the store front

Account Center

Profile details

Password and security

User ratings

Ongoing deliveries

Current username:

Rad.Johny

Log out

First name:

Randy

Last name:

Johnson

Phone number:

0228994121

Email:

johnrad@gmail.com

Address:

5 Bumpy Road Pretoria

Current listings

Radical cookies



Price: R 20

Stock: 14

Update stock

Remove listing

MarketplaceReturn to the store front

Account Center

Profile details

Password and security

User ratings

Ongoing deliveries

Current username:

First name:

Last name:

Phone number:

Email:

Address:

Current listings

Radical cookies



Price: R 20

Stock:

Update stock

Remove listing

48

Mobile portrait

Marketplace

Return to the store front

Account Center

Profile details

Password and security

User ratings

Ongoing deliveries

Current username:

Rad.Johny

Log out

First name:

Randy

Last name:

Johnson

Phone number:

0228994121

Email:

johnrad@gmail.com

Address:

5 Bumpy Road Pretoria

Current listings

Radical cookies



Price: R 20

Stock: 14

Update stock

Remove listing

Mobile landscape

Marketplace

Return to the store front

Account Center

Profile details

Password and security

Current username:

Rad.Johny

Log out

First name:

Randy

Last name:

Johnson

Phone number:

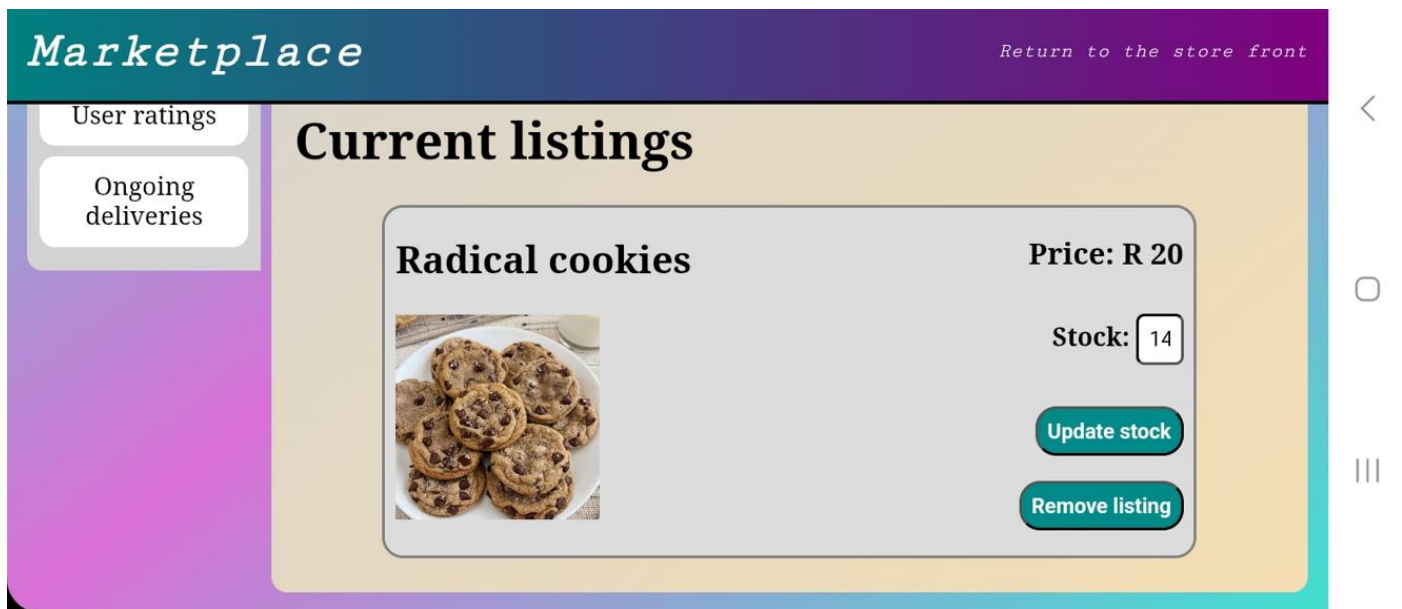
0228994121

Email:

johnrad@gmail.com

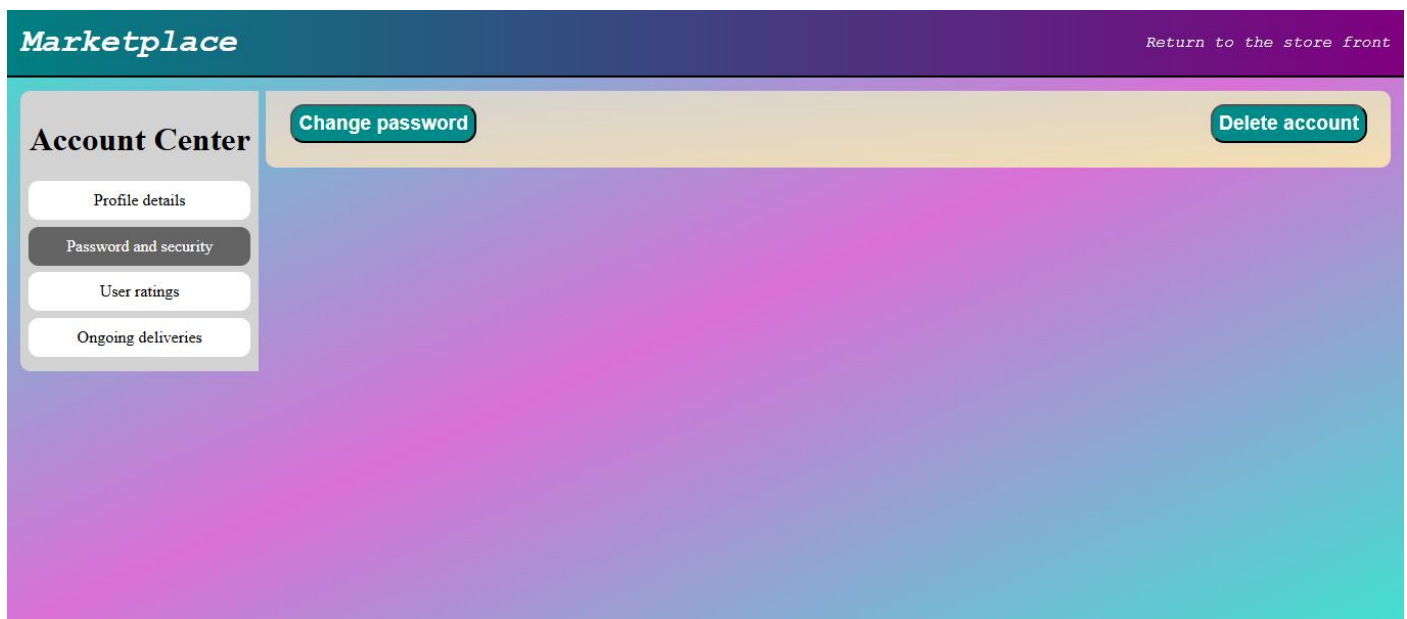
Address:

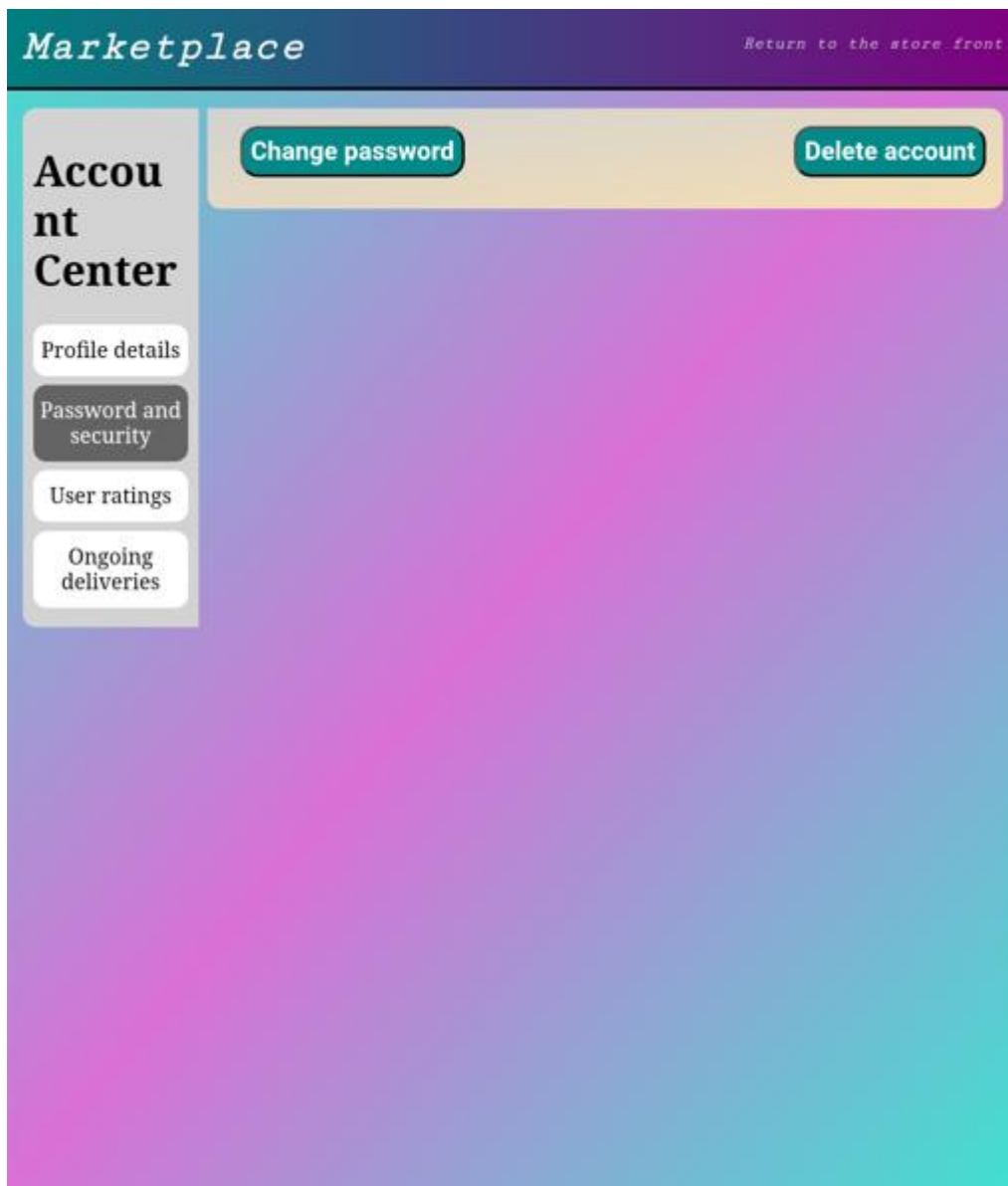
5 Bumpy Road Pretoria

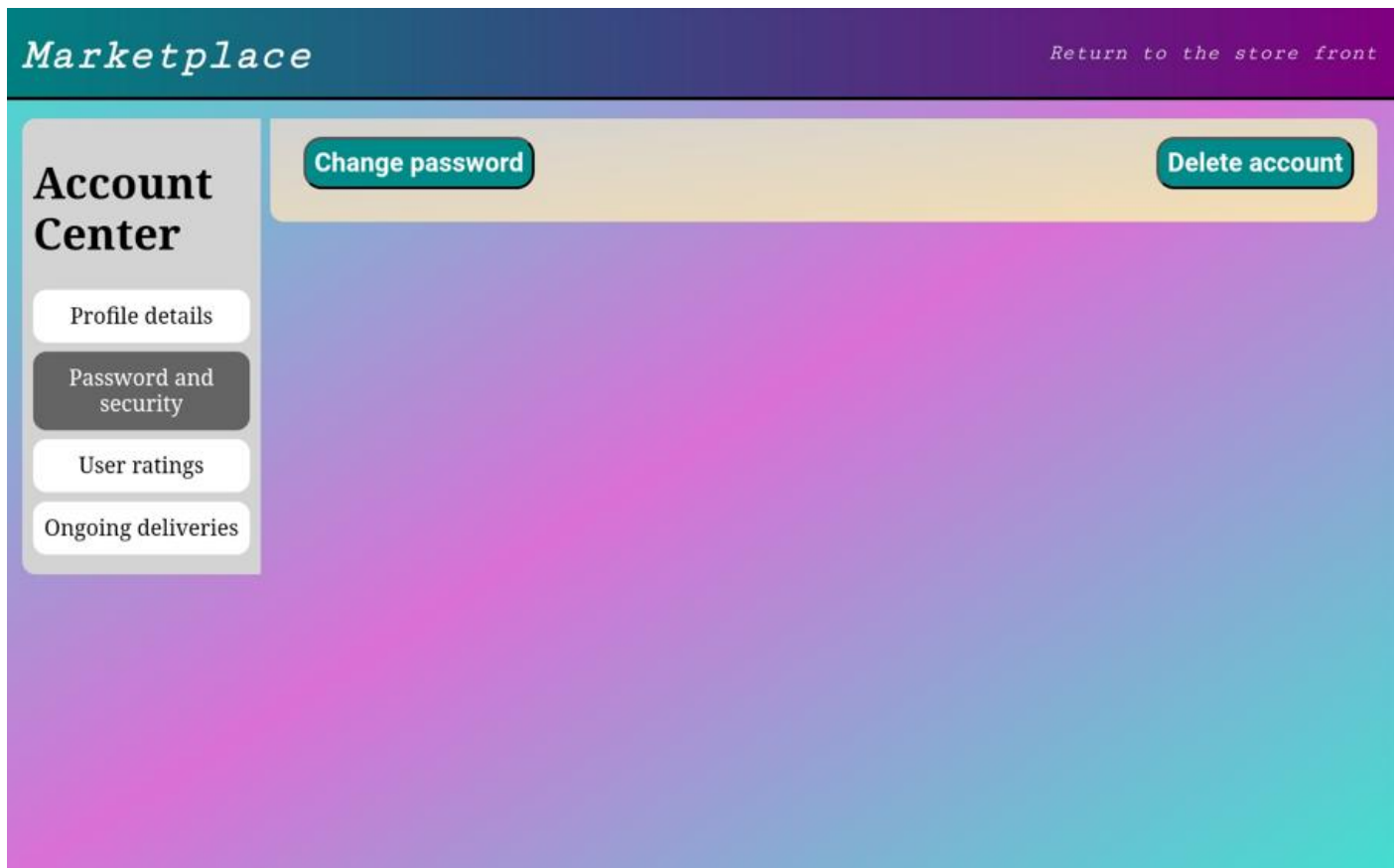


Account centre (Password and security) page:

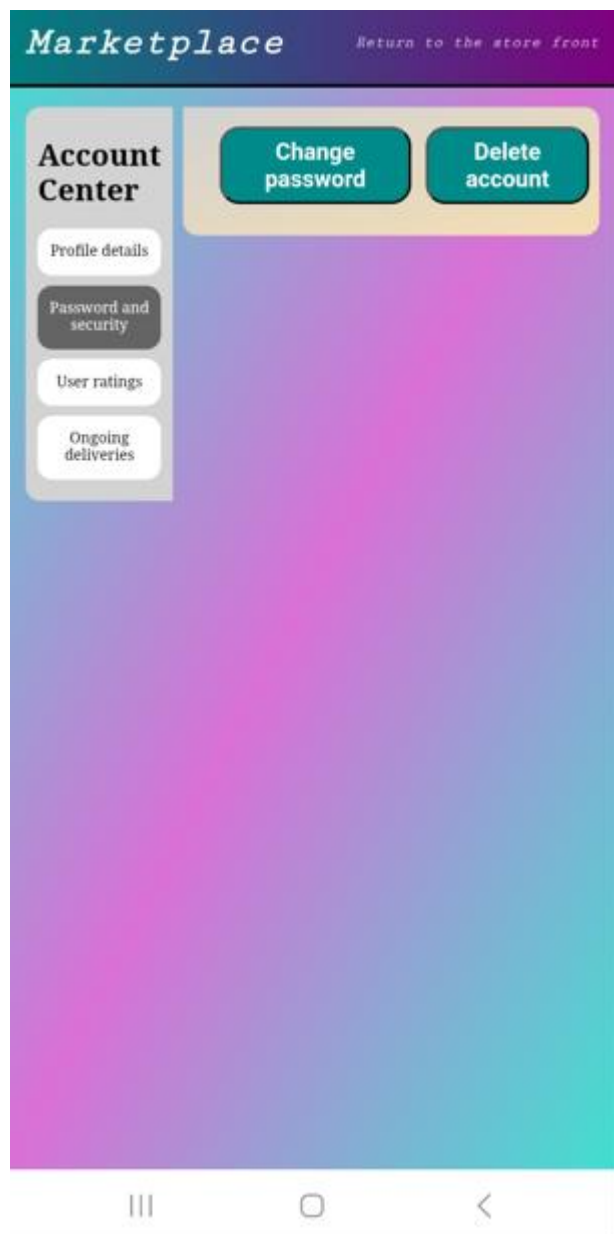
Desktop



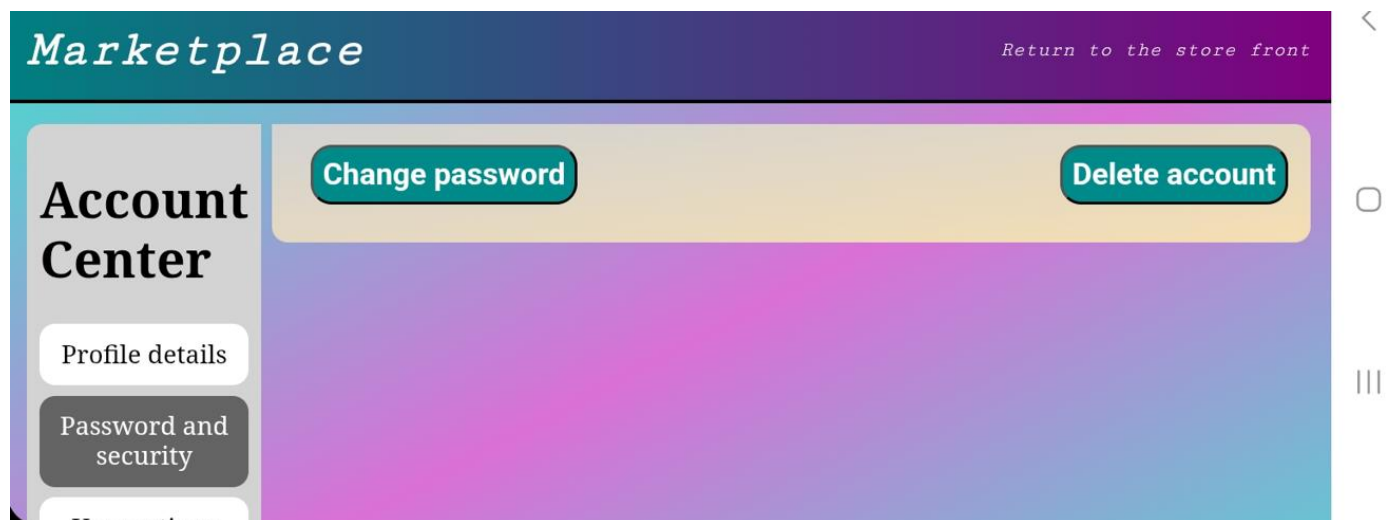




Mobile portrait

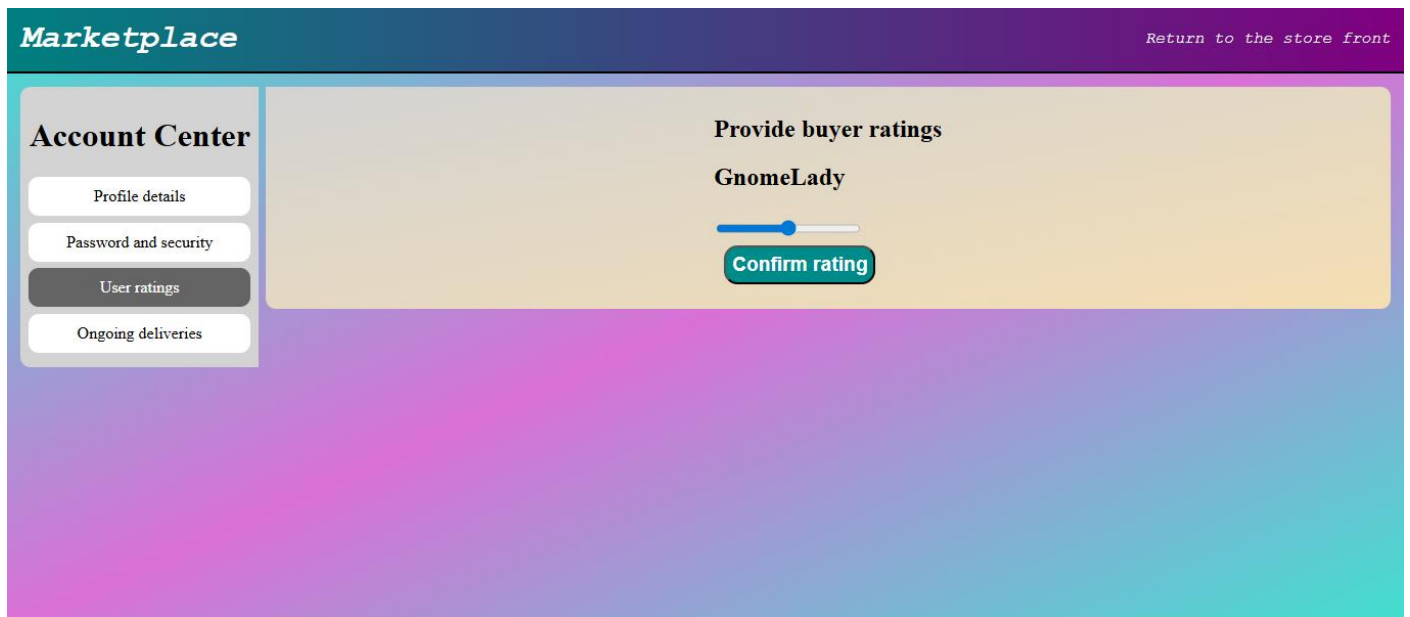


Mobile landscape

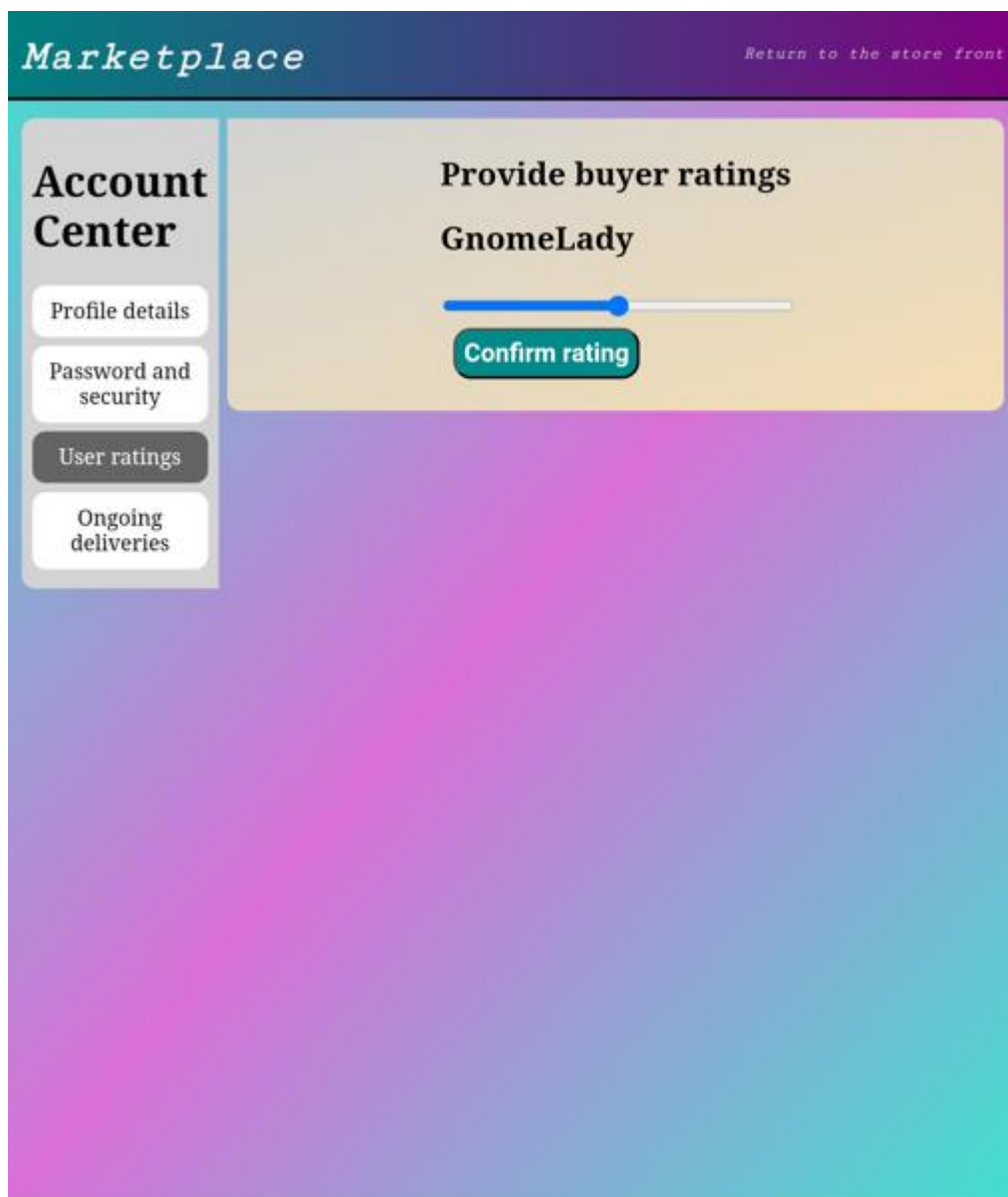


Account centre (User ratings) page:

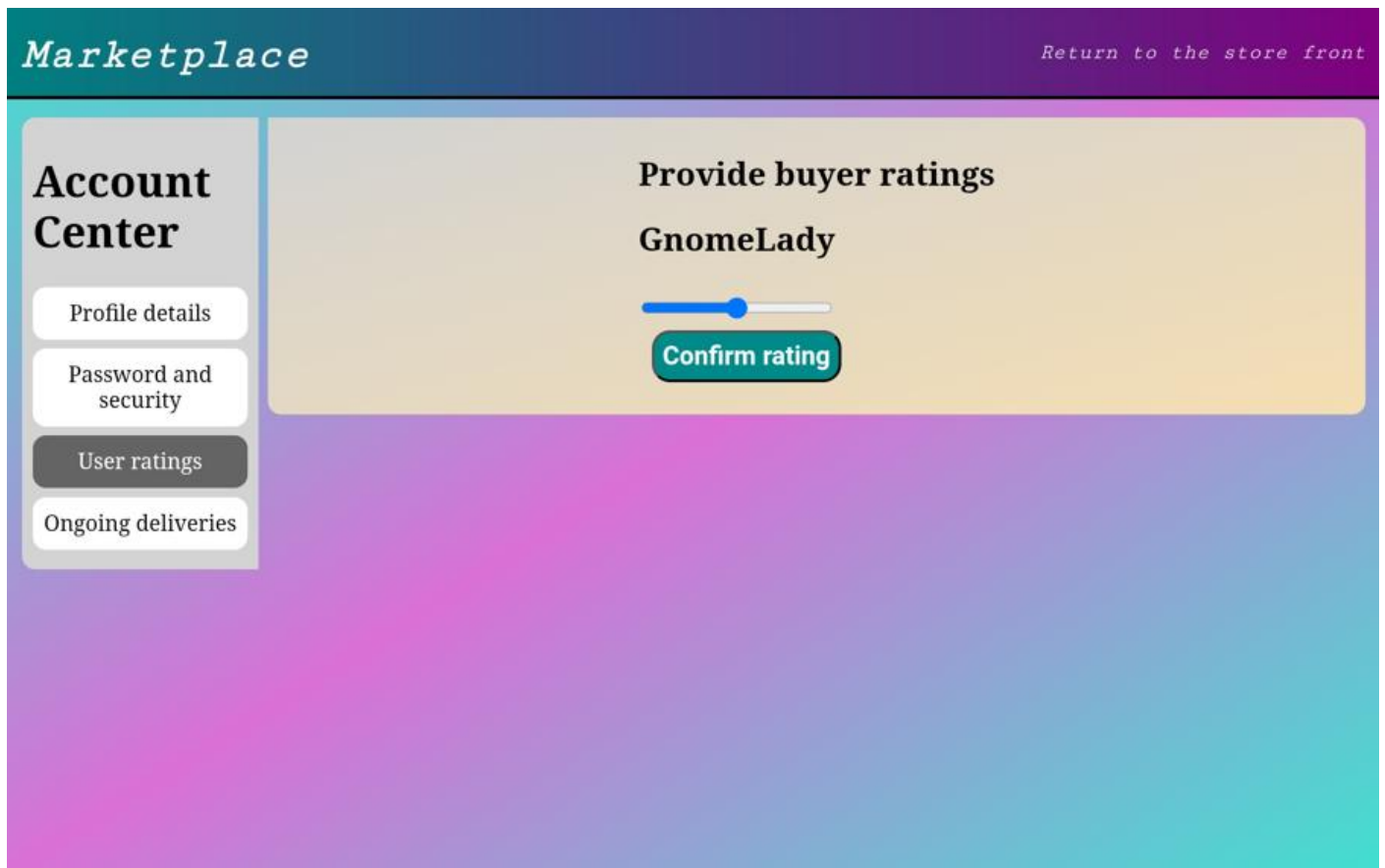
Desktop



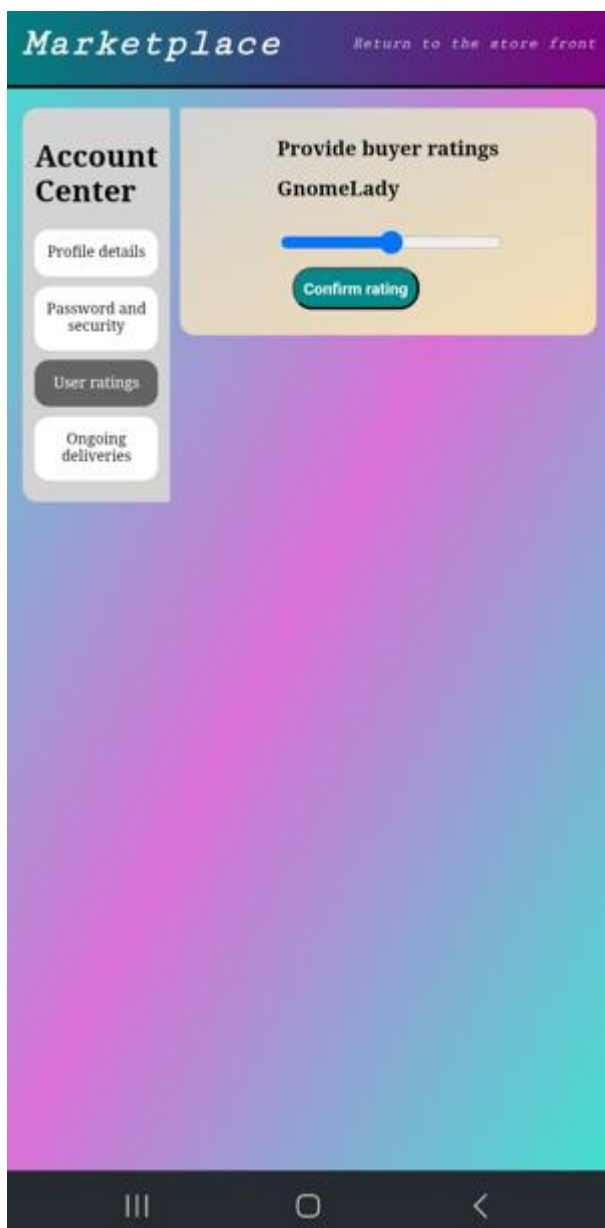
Tablet portrait



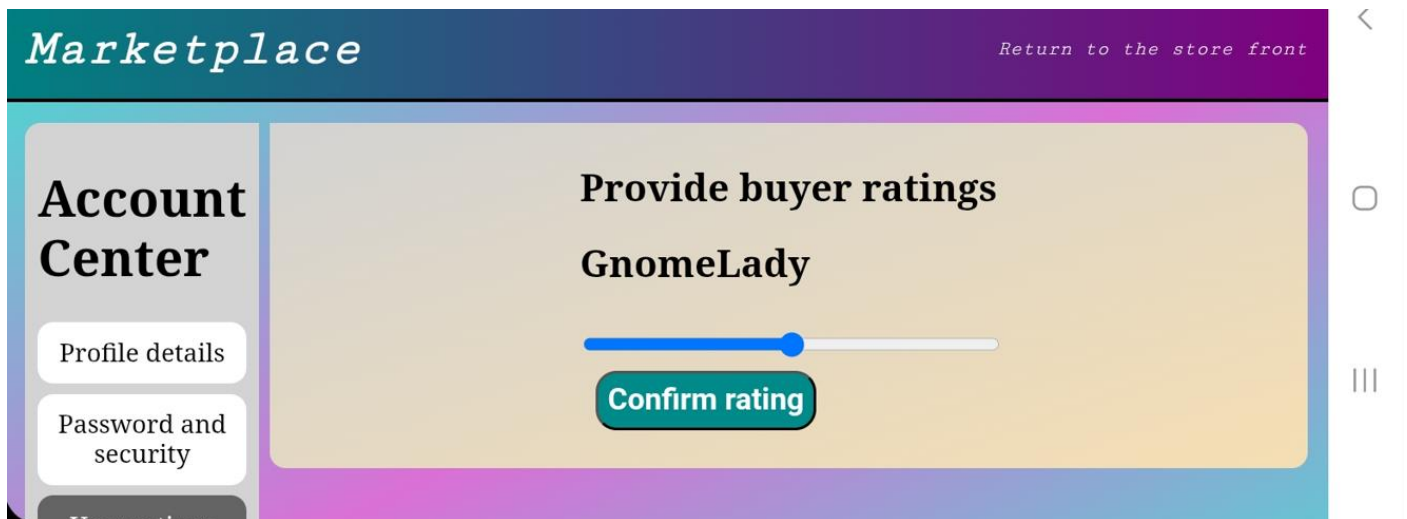
Tablet landscape



Mobile portrait

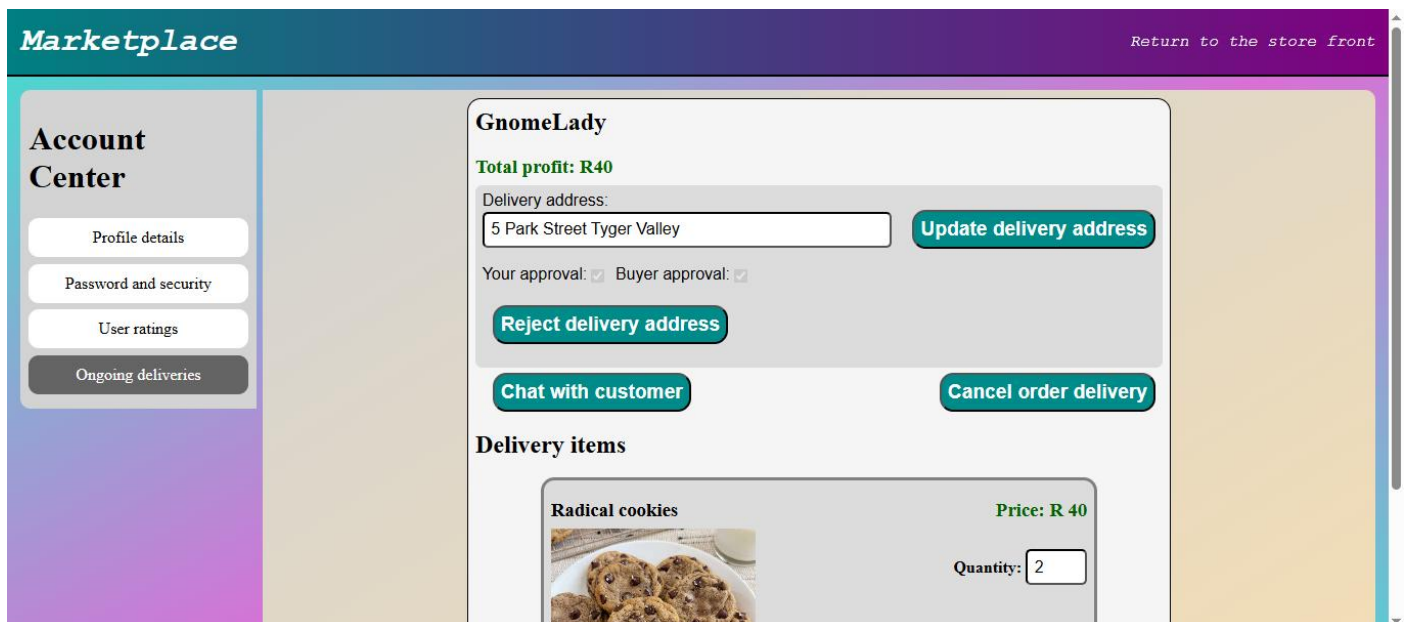


Mobile landscape



Account centre (Ongoing deliveries) page:

Desktop



[Password and security](#)[User ratings](#)[Ongoing deliveries](#)Your approval: ☒ Buyer approval: ☒[Reject delivery address](#)[Chat with customer](#)[Cancel order delivery](#)**Delivery items****Radical cookies****Price: R 40**Quantity:

Tablet portrait

Account Center

[Profile details](#)[Password and security](#)[User ratings](#)[Ongoing deliveries](#)

GnomeLady

Total profit: R40

Delivery address:

[Update delivery address](#)

Your approval:



Buyer approval:

[Reject delivery address](#)[Chat with customer](#)[Cancel order delivery](#)

Delivery items

Radical cookies**Price: R 40**Quantity:

MarketplaceReturn to the store front

Account Center

- Profile details
- Password and security
- User ratings
- Ongoing deliveries

GnomeLady

Total profit: R40

Delivery address: **Update delivery address**

Your approval: ☒ Buyer approval: ☒

Reject delivery address

Chat with customer **Cancel order delivery**

Delivery items

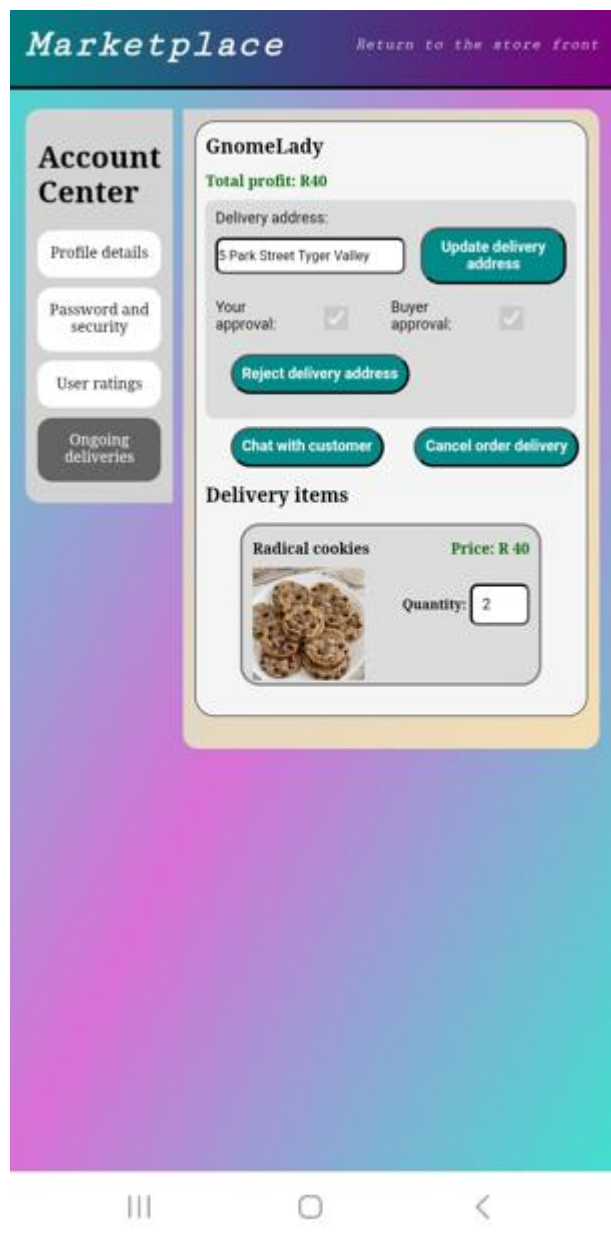
Radical cookies



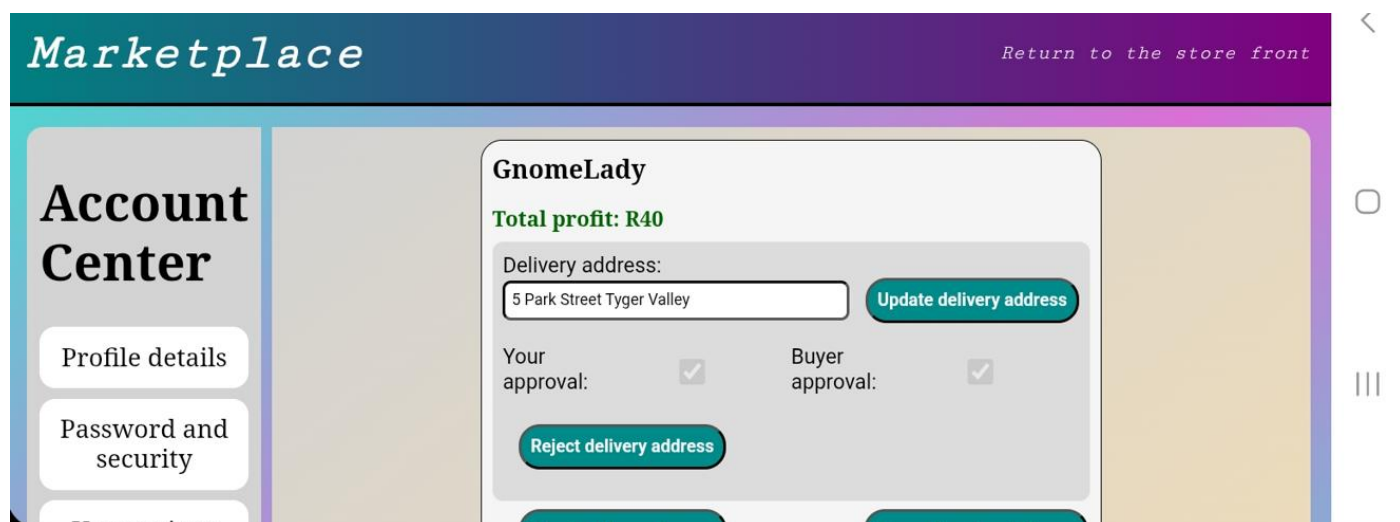
Price: R 40

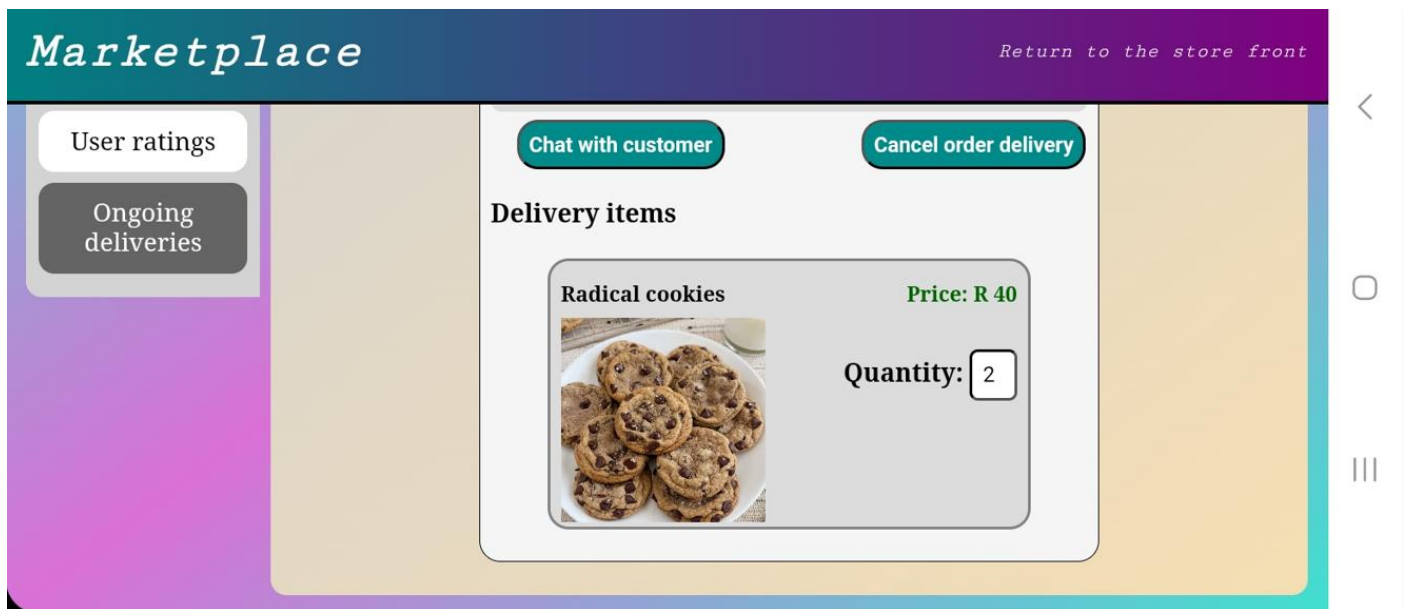
Quantity:

Mobile portrait



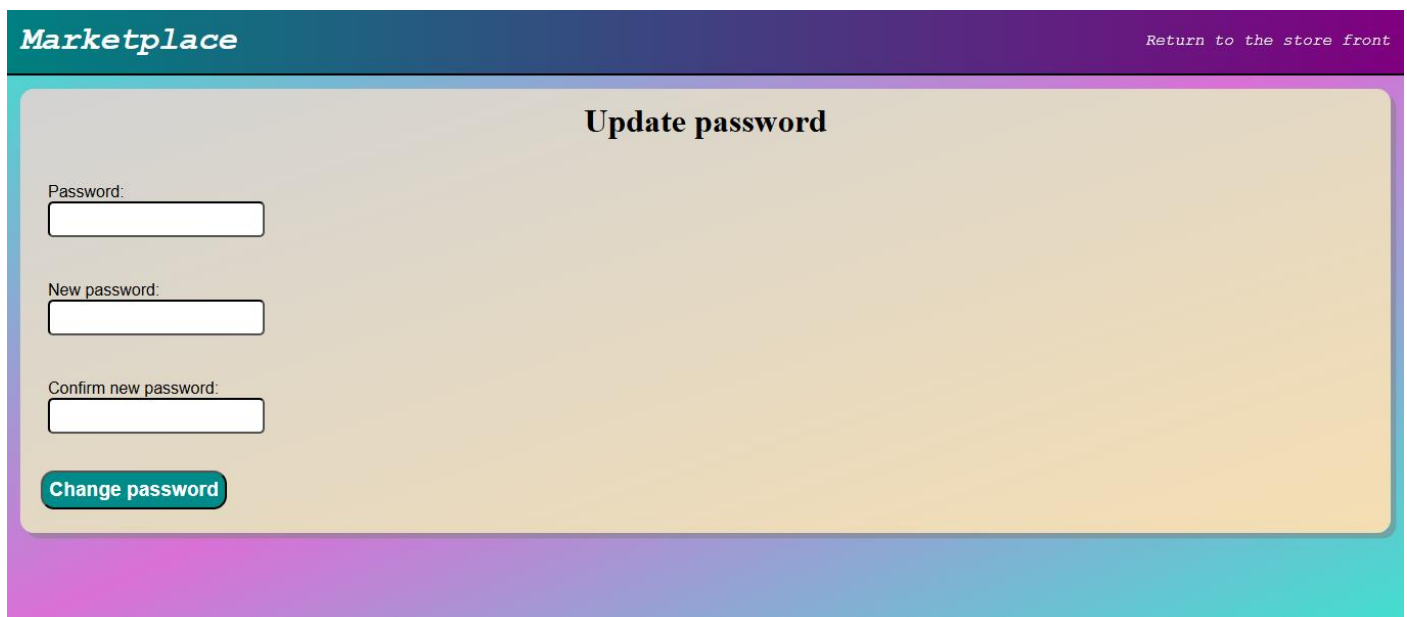
Mobile landscape





Change password page:

Desktop



Marketplace[Return to the store front](#)

Update password

Password:

New password:

Confirm new password:

Change password

*Marketplace*Return to the store front

Update password

Password:

New password:

Confirm new password:

Change password

Mobile portrait

The image shows a mobile application interface in portrait orientation. At the top, there is a dark purple header bar with the word "Marketplace" in a light blue, monospace-style font on the left, and the text "Return to the store front" in a smaller, lighter purple font on the right. Below the header is a large, light beige rounded rectangle with a subtle drop shadow, titled "Update password" in bold black text. Inside this rectangle are three input fields: "Password:", "New password:", and "Confirm new password:", each followed by a white rectangular text box. Below these fields is a green rounded button with the text "Change password" in white. The background of the screen is a vibrant gradient of purple, pink, and blue. At the bottom, there is a dark grey Android-style navigation bar with three icons: a square, a circle, and a triangle.

Mobile landscape

The image shows the same mobile application interface in landscape orientation. The header bar remains at the top with "Marketplace" and "Return to the store front". The "Update password" form is now wider, with the title centered. The input fields and the "Change password" button are still present. The background gradient is the same. On the right side of the screen, there is a vertical white bar containing three navigation icons: a back arrow at the top, a square in the middle, and a hamburger menu icon at the bottom.

Marketplace

Return to the store front

Password:

New password:

Confirm new password:

Change password

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b)

Admin centre (Admin database) page:

Desktop

Admin center

Admin database

Buyer database

Vendor database

Listing database

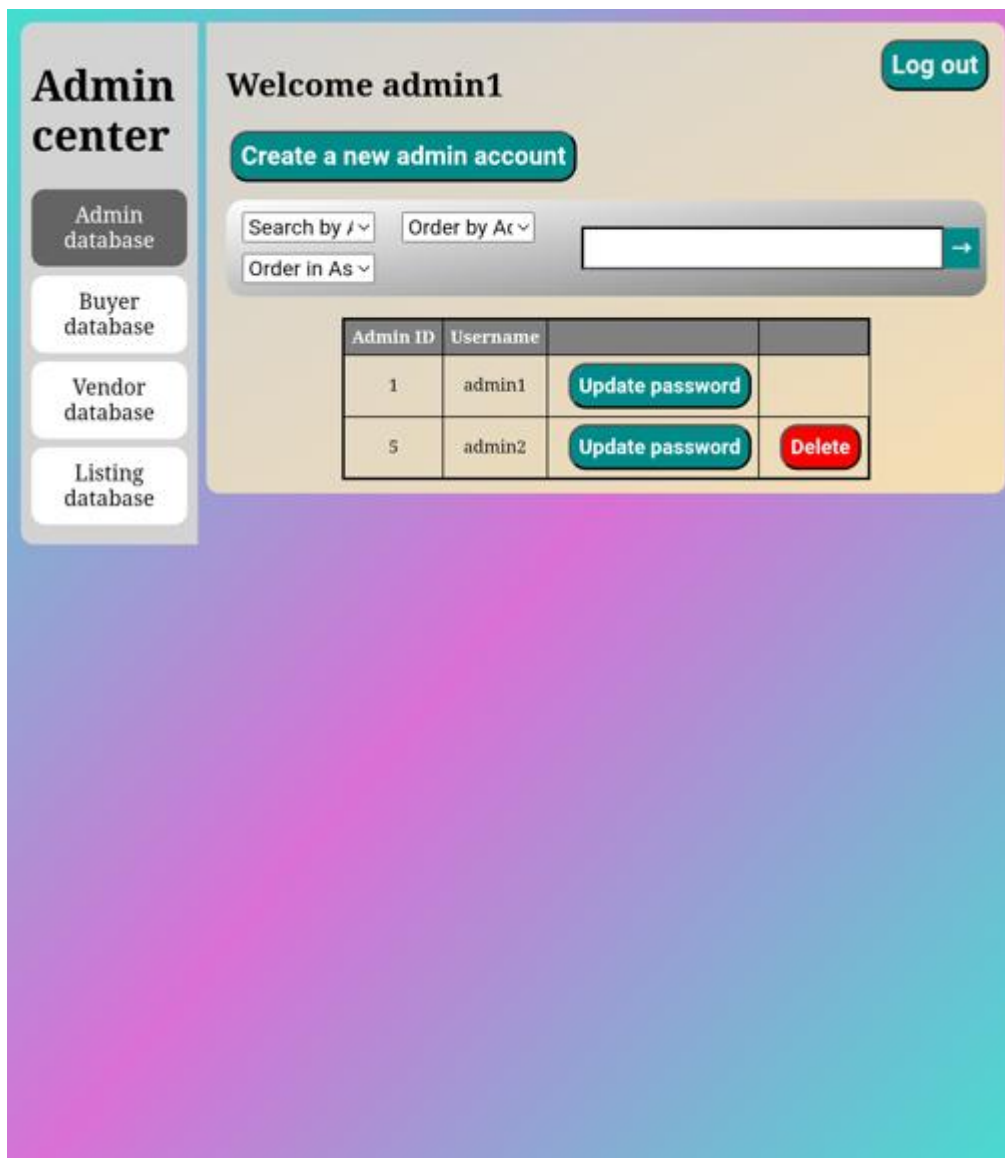
Welcome admin1

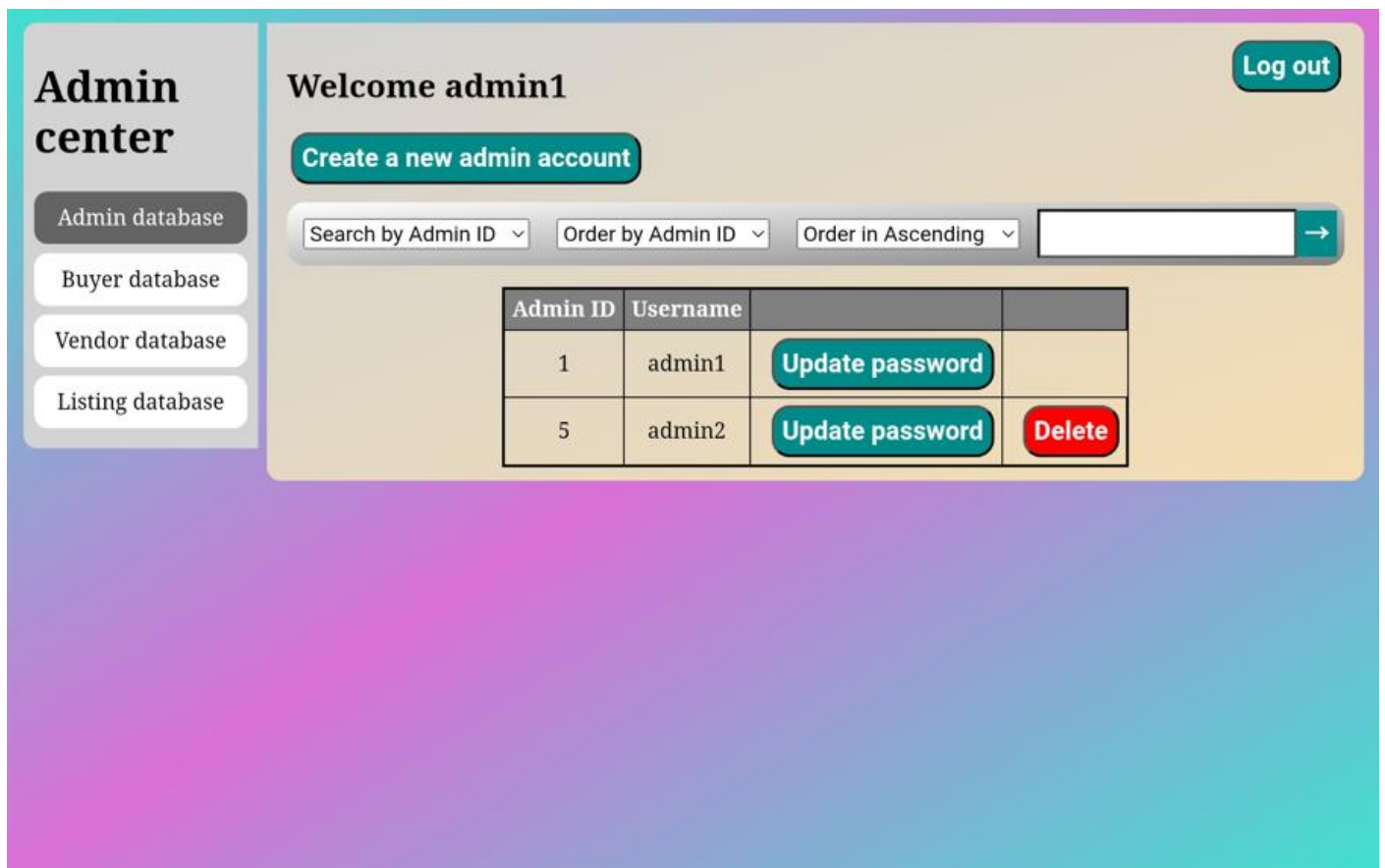
Log out

Create a new admin account

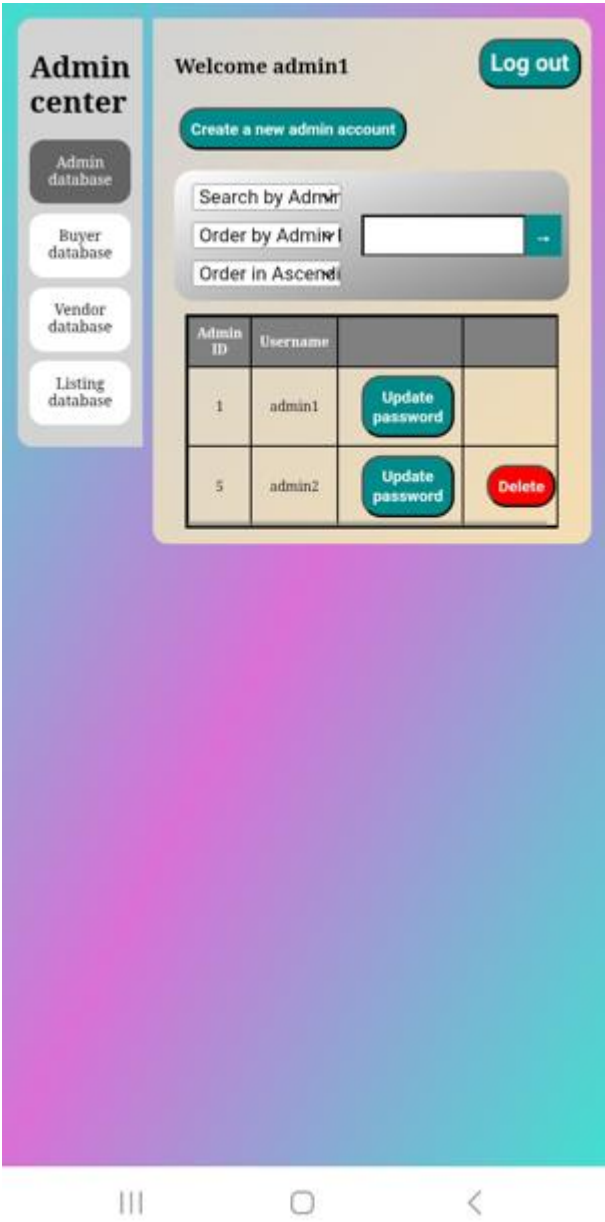
Search by Admin ID ▼ Order by Admin ID ▼ Order in Ascending ▼ →

Admin ID	Username		
1	admin1	Update password	
5	admin2	Update password	Delete

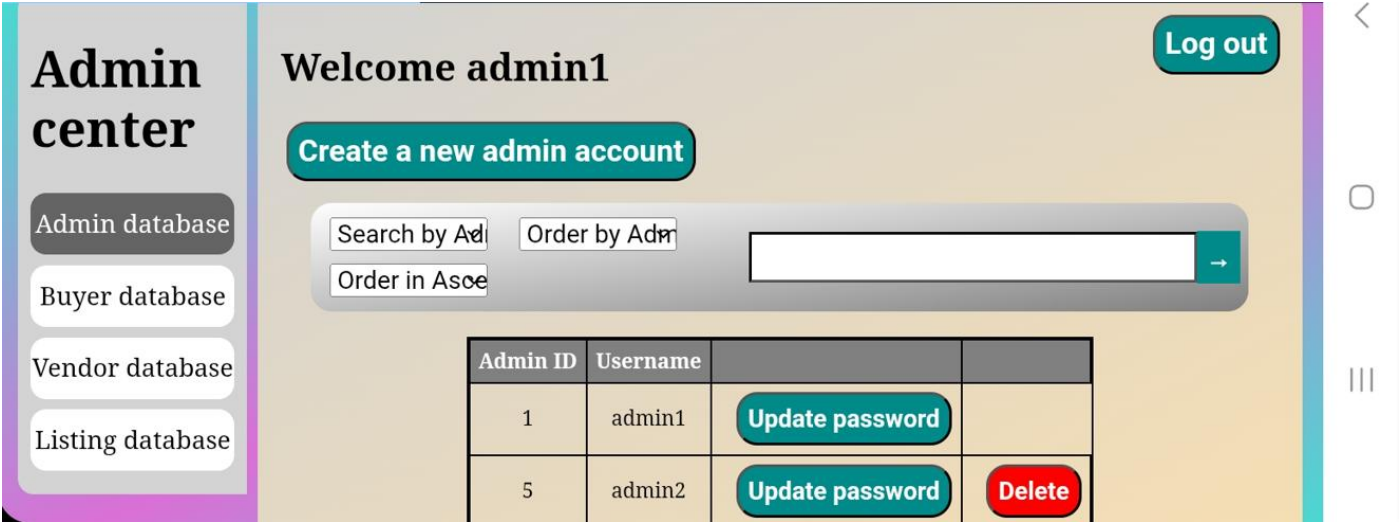




Mobile portrait



Mobile landscape



Admin centre (Buyer database) page:

Desktop

Admin center

Admin database

Buyer database

Vendor database

Listing database

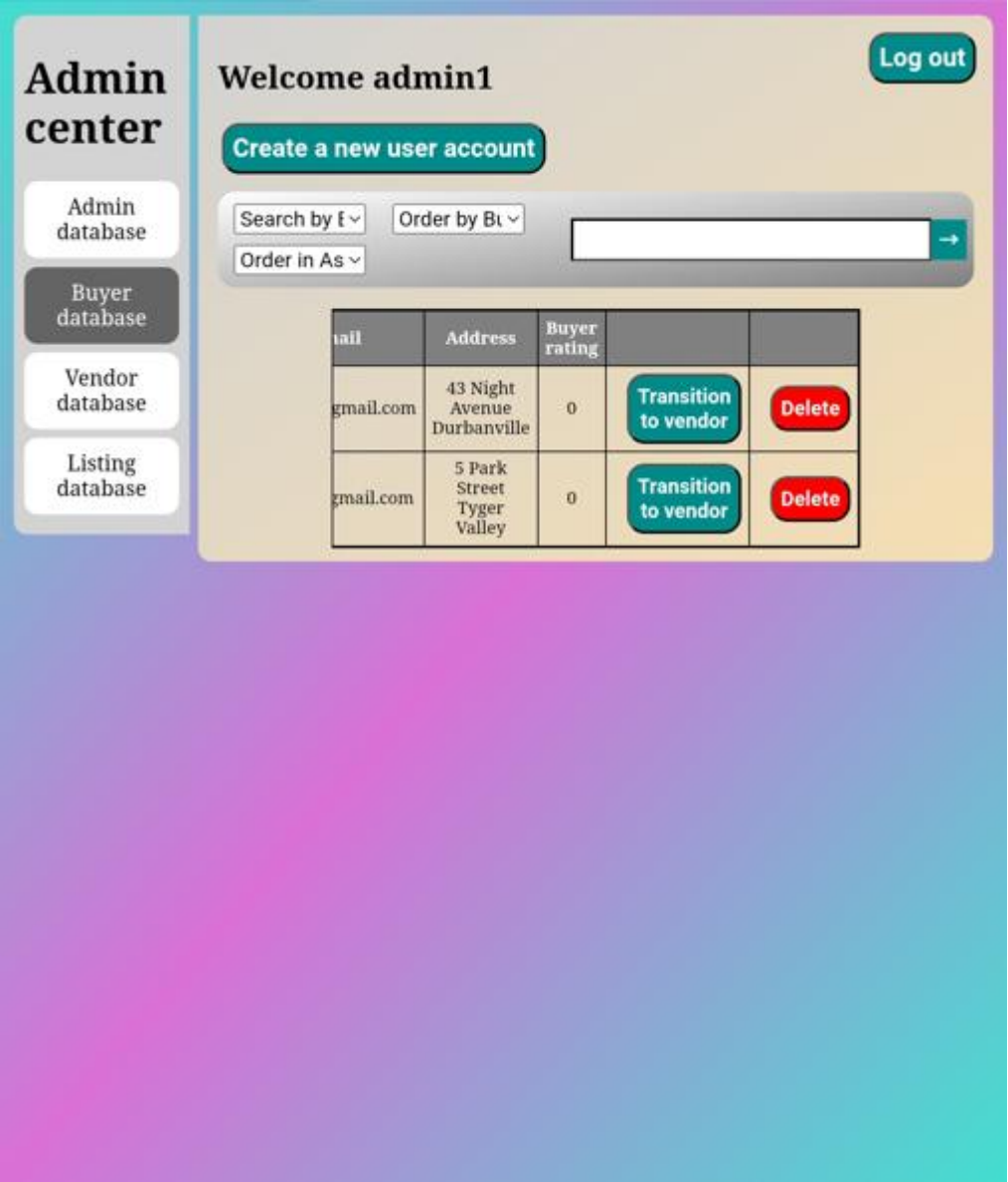
Welcome admin1

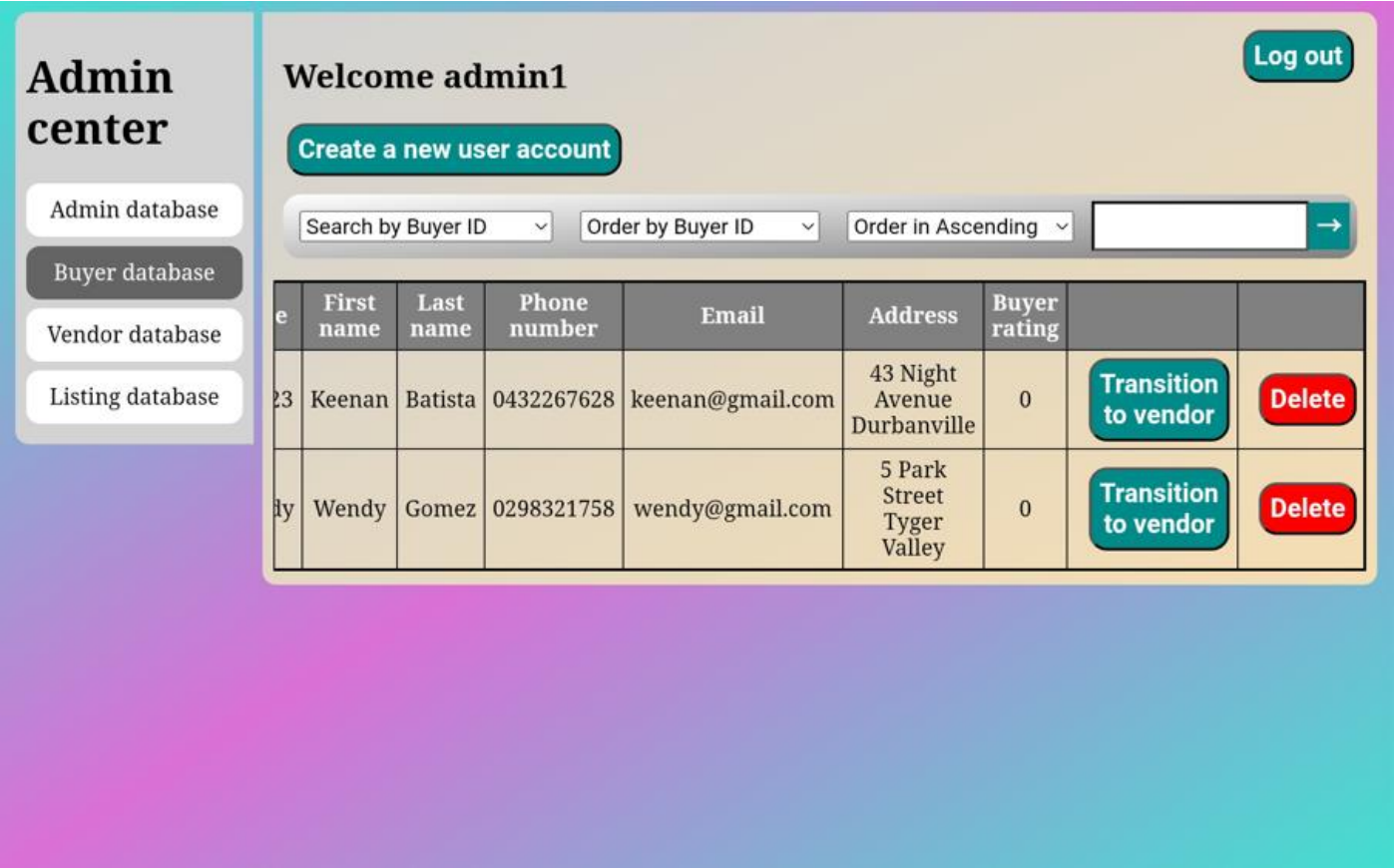
Log out

Create a new user account

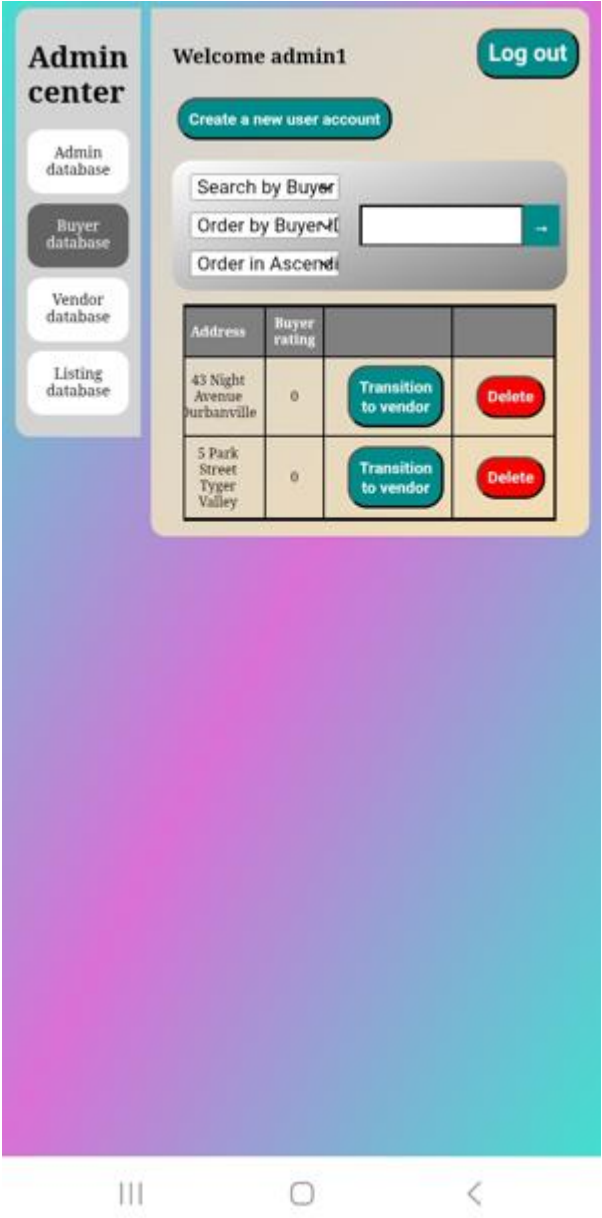
Search by Buyer ID Order by Buyer ID Order in Ascending

Username	First name	Last name	Phone number	Email	Address	Buyer rating		
oolGuy123	Keenan	Batista	0432267628	keenana@gmail.com	43 Night Avenue Durbanville	0	Transition to vendor	Delete
gnomeLady	Wendy	Gomez	0298321758	wendy@gmail.com	5 Park Street Tyger Valley	0	Transition to vendor	Delete

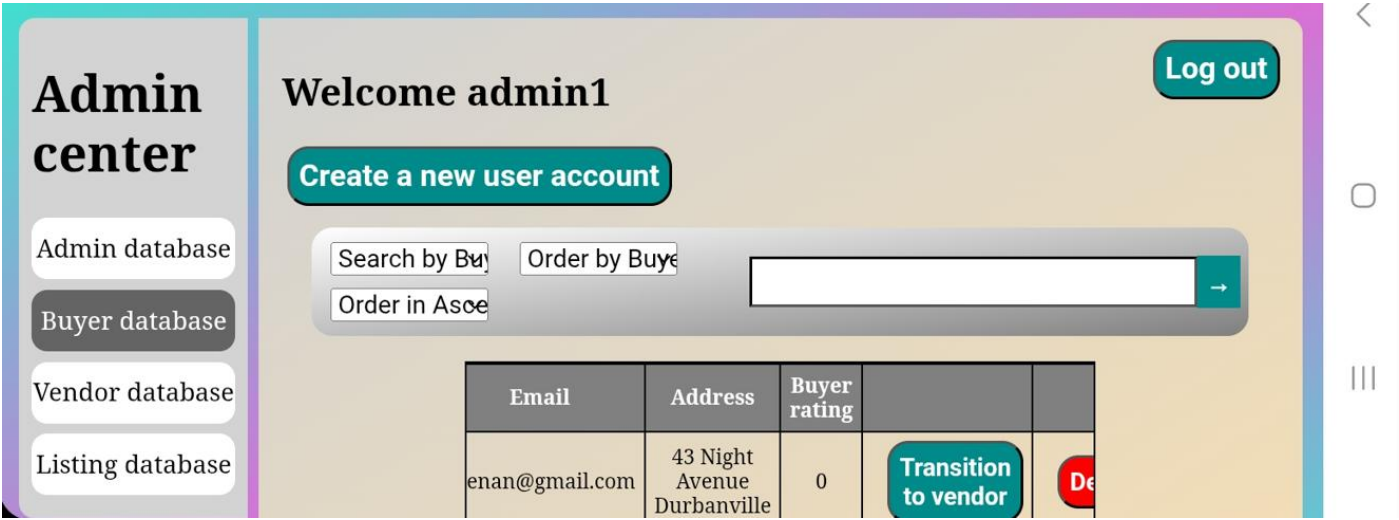




Mobile portrait



Mobile landscape



Admin centre (Vendor database) page:

Desktop

Admin center

Admin database

Buyer database

Vendor database

Listing database

Welcome admin1

Log out

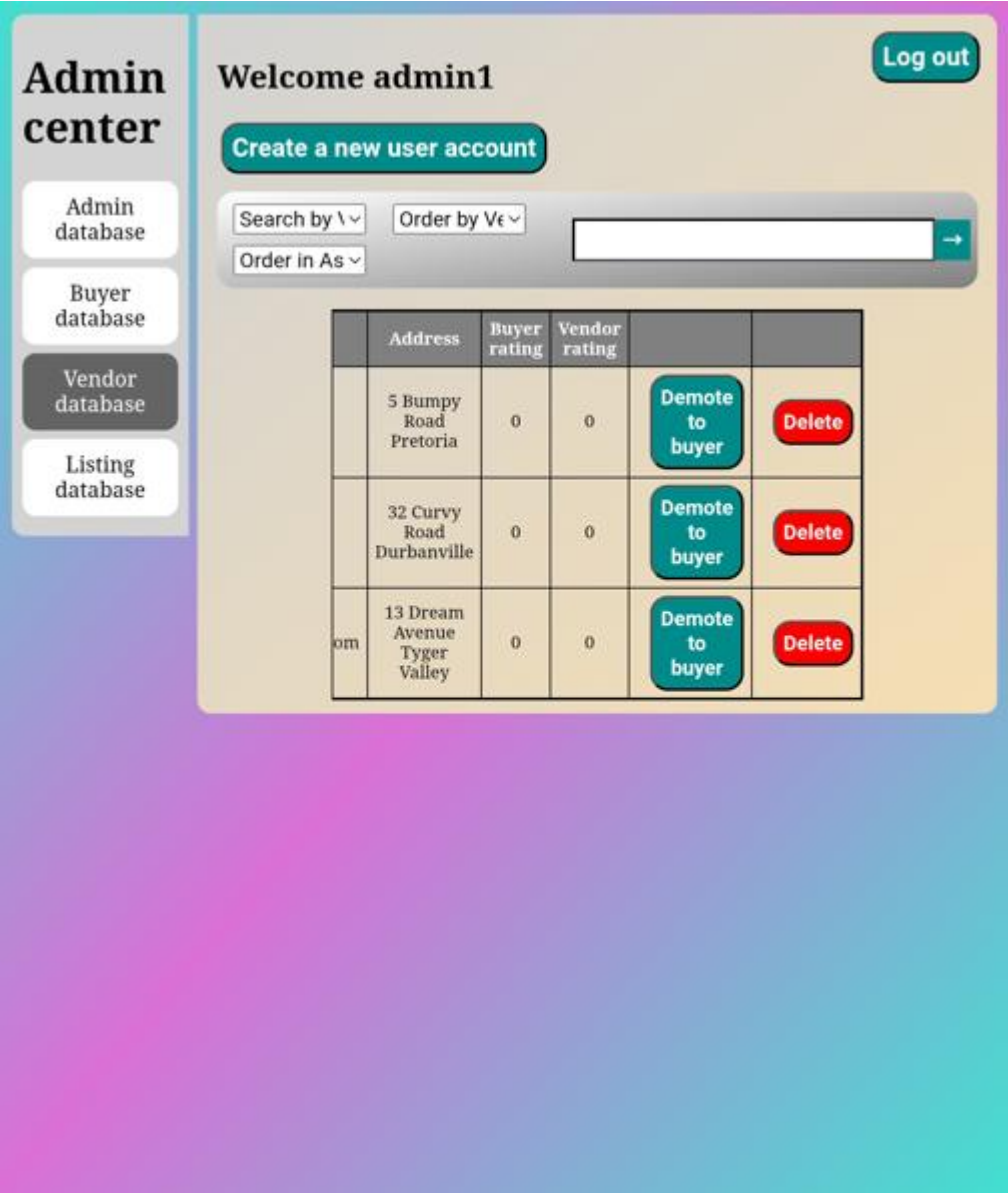
Create a new user account

Search by Vendor ID

Order by Vendor ID

Order in Ascending

	Last name	Phone number	Email	Address	Buyer rating	Vendor rating		
ty	Johnson	0228994121	johnrad@gmail.com	5 Bumpy Road Pretoria	0	0	Demote to buyer	Delete
h	Dickinson	0954879964	sarah@gmail.com	32 Curvy Road Durbanville	0	0	Demote to buyer	Delete
to	Downey	0341785456	electronicmasters@gmail.com	13 Dream Avenue Tyger Valley	0	0	Demote to buyer	Delete



Admin center

Admin database

Buyer database

Vendor database

Listing database

Welcome admin1

Log out

Create a new user account

Search by Vendor ID

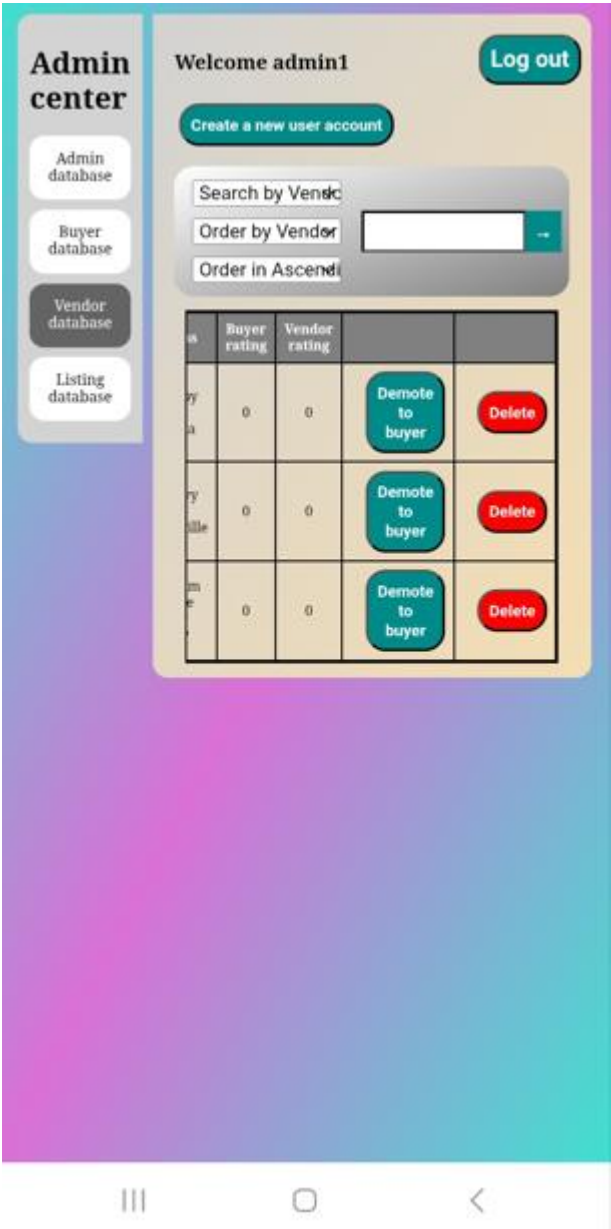
Order by Vendor ID

Order in Ascending

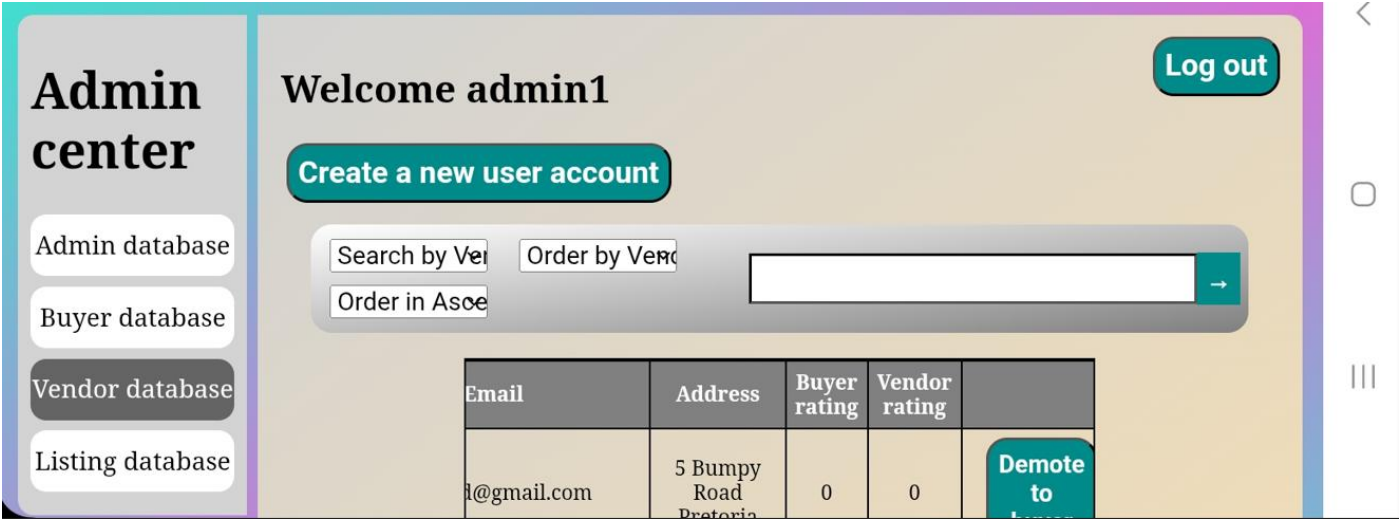
→

	Phone number	Email	Address	Buyer rating	Vendor rating		
n	0228994121	johnrad@gmail.com	5 Bumpy Road Pretoria	0	0	Demote to buyer	Delete
on	0954879964	sarah@gmail.com	32 Curvy Road Durbanville	0	0	Demote to buyer	Delete
y	0341785456	electronicmasters@gmail.com	13 Dream Avenue Tyger Valley	0	0	Demote to buyer	Delete

Mobile portrait



Mobile landscape



Admin centre (Listing database) page:

Desktop

Admin center

Admin database

Buyer database

Vendor database

Listing database

Welcome admin1

Log out

Search by Listing ID

Order by Listing ID

Order in Ascending

Listing ID	Title	Price	Description	Stock	Image	Vendor ID	Vendor username	
101	Radical cookies	20	Chocolate chip cookies for radical people. Freshly baked and home-made.	14		39	RadJohny	Delete
102	Dining chair	375	A well-made, antique, dining chair.	5		40	sarah_D	Delete

Admin center

Admin database

Buyer database

Vendor database

Listing database

Welcome admin1

Log out

Search by I: v

Order by Li: v

→

Order in As v

Listing ID	Title	Price	Description	Stock	Image
101	Radical cookies	20	Chocolate chip cookies for radical people. Freshly baked and home-made.	14	
102	Dining chair	375	A well-made, antique, dining chair.	5	
103	Baked beans	25	Healthy, canned, baked beans.	17	
					

Admin center

Admin database

Buyer database

Vendor database

Listing database

Welcome admin1

Log out

Search by Listing ID ▾

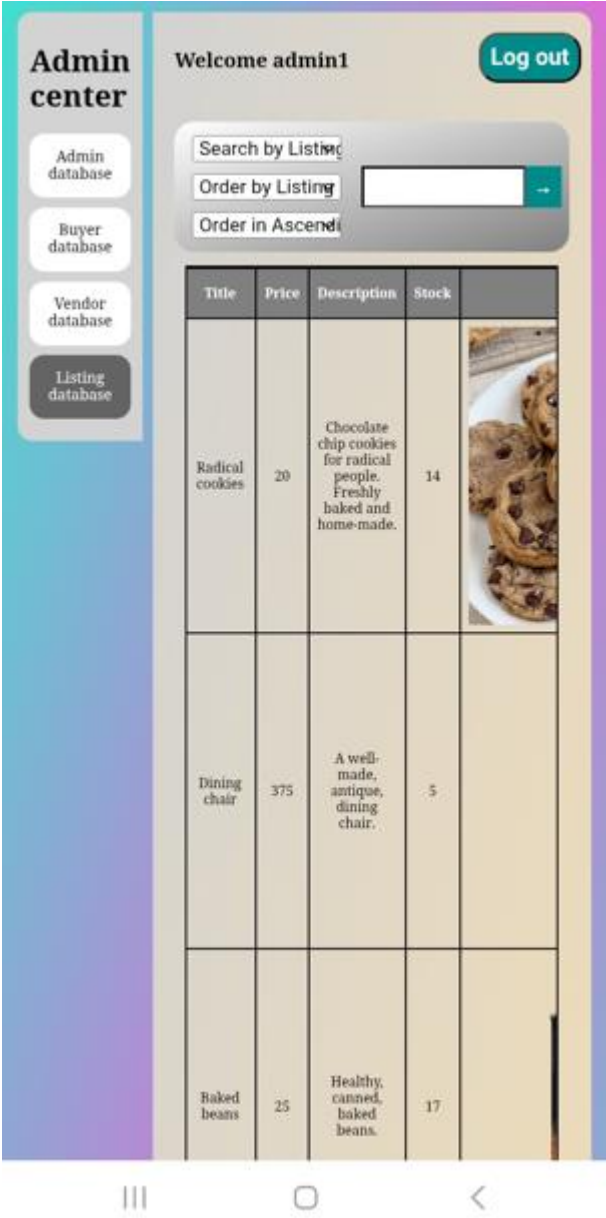
Order by Listing ID ▾

Order in Ascending ▾

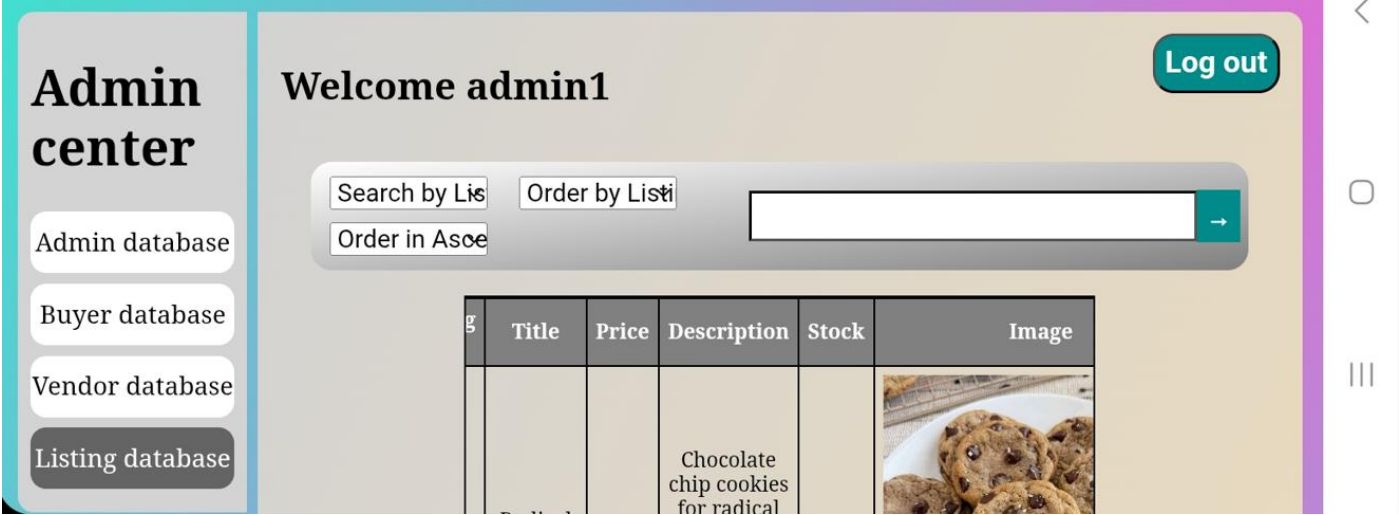
→

Listing ID	Title	Price	Description	Stock	Image	Vendor ID	Vendor username
101	Radical cookies	20	Chocolate chip cookies for radical people. Freshly baked and home-made.	14		39	RadJohny
102	Dining chair	375	A well-made, antique, dining chair.	5		40	sarah_D

Mobile portrait

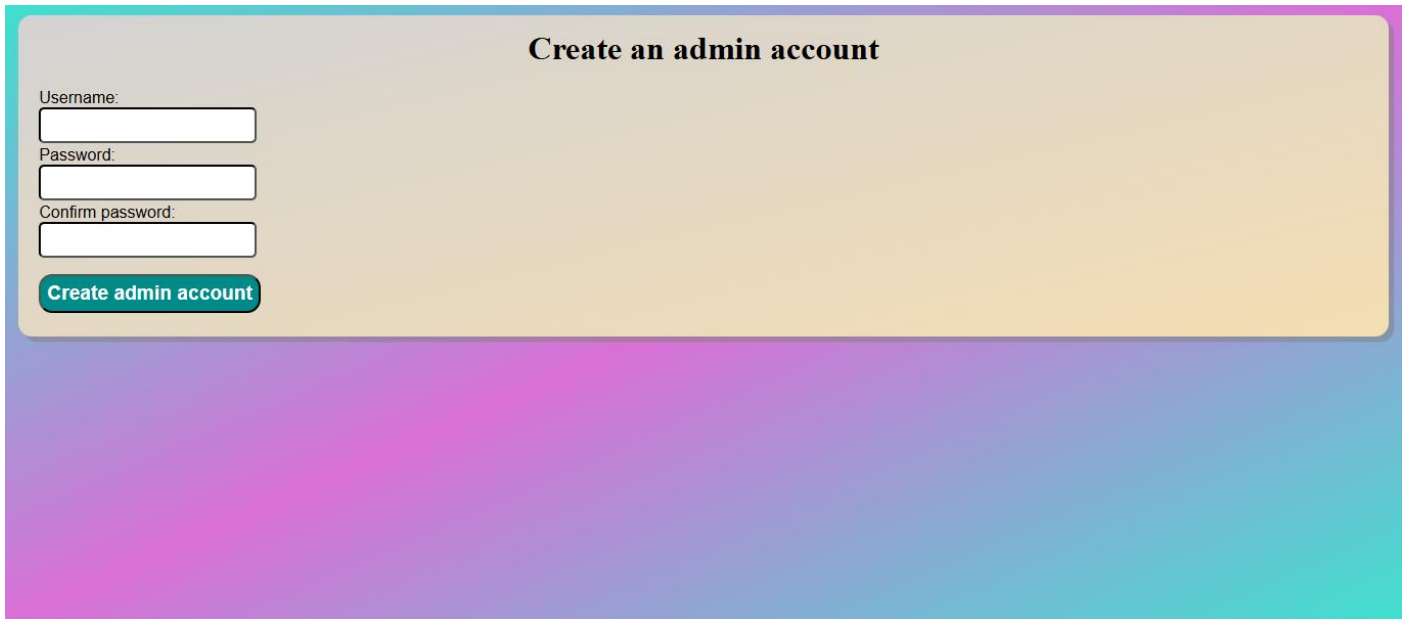


Mobile landscape



Admin creation page:

Desktop

A desktop mockup of an admin creation page. The page has a light beige header area with the title "Create an admin account" in bold black text. Below the title, on the left side, are three stacked input fields with labels "Username:", "Password:", and "Confirm password:". At the bottom left of the header area is a teal button with the text "Create admin account". The background of the page is a vibrant gradient of purple, pink, and blue.

Create an admin account

Username:

Password:

Confirm password:

Create admin account

Tablet portrait

The image shows a tablet in portrait orientation with a colorful gradient background transitioning from purple at the top to blue at the bottom. At the top, there is a light beige rectangular box with rounded corners and a thin purple border. Inside this box, the title "Create an admin account" is centered at the top in a bold, black font. Below the title, on the left side, are three labels: "Username:", "Password:", and "Confirm password:", each followed by a white rectangular input field with a thin black border. The input fields are stacked vertically. Below the input fields, there is a green button with rounded corners and a black border, containing the text "Create admin account" in white. The rest of the tablet screen is filled with the purple-to-blue gradient background.

Create an admin account

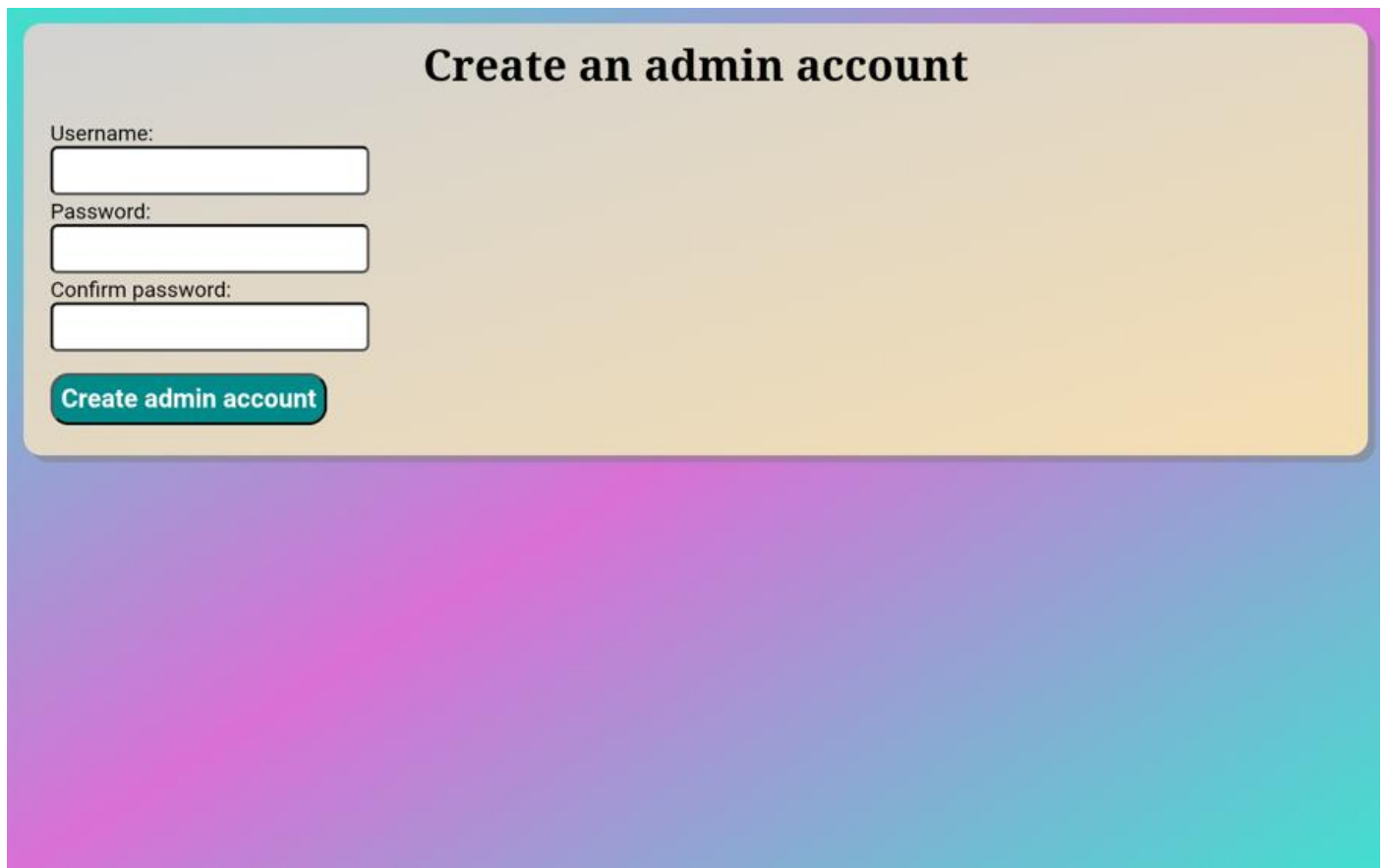
Username:

Password:

Confirm password:

Create admin account

Tablet landscape



The image shows a tablet landscape with a colorful gradient background transitioning from purple on the left to teal on the right. A light beige rounded rectangle is positioned in the upper portion of the screen. Inside this rectangle, the title "Create an admin account" is centered at the top in a bold, black, sans-serif font. Below the title, on the left side, are three vertically stacked input fields. Each field is preceded by a label: "Username:", "Password:", and "Confirm password:". The input fields are white with thin black borders. At the bottom left of the beige rectangle is a teal button with rounded corners and a thin black border, containing the text "Create admin account" in white, bold, sans-serif font.

Create an admin account

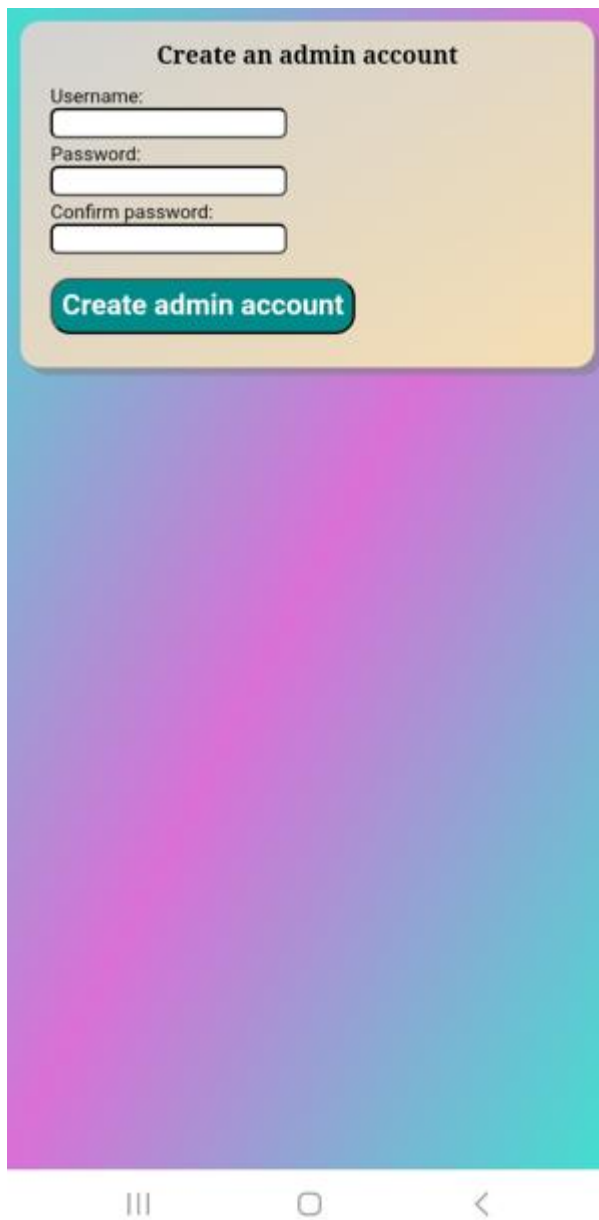
Username:

Password:

Confirm password:

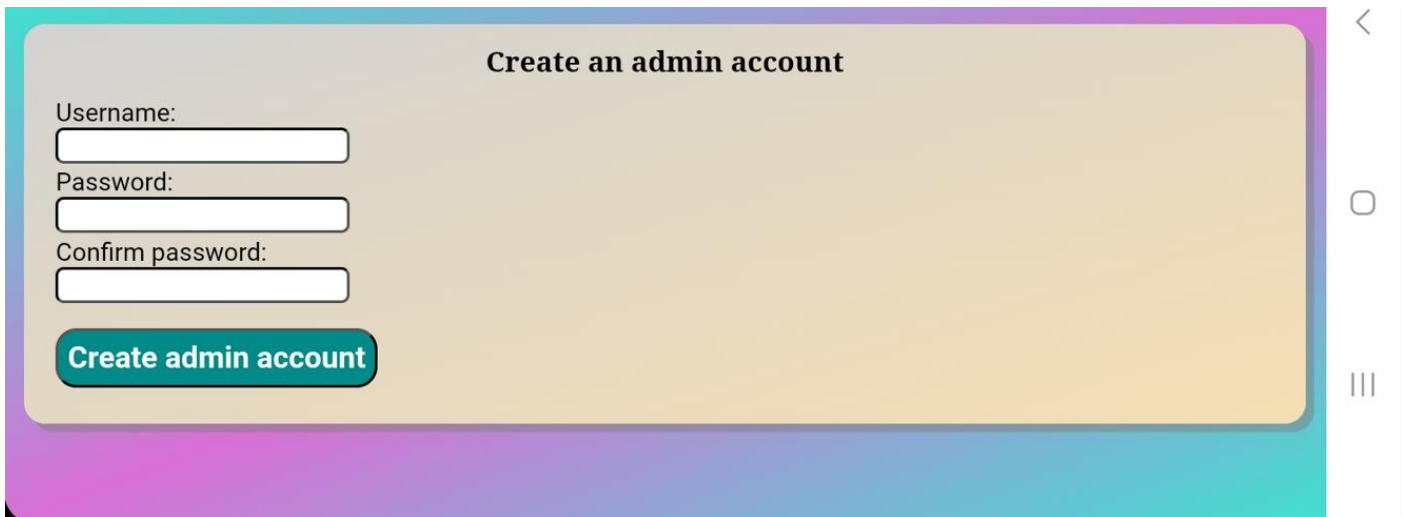
Create admin account

Mobile portrait



The image shows a mobile application interface in portrait orientation. At the top, there is a light beige rounded rectangle containing the title "Create an admin account" in bold black text. Below the title are three input fields: "Username:", "Password:", and "Confirm password:", each followed by a white rectangular text box. At the bottom of this beige area is a green rounded button with the text "Create admin account" in white. The background of the entire screen is a vertical gradient from purple at the top to teal at the bottom. At the very bottom of the screen, there are three small, faint icons: a hamburger menu icon (three horizontal lines), a square icon, and a back arrow icon.

Mobile landscape



The image shows a mobile application interface for creating an admin account. It features a light beige rounded rectangle on a colorful gradient background. On the right side of the screen, there are three vertical navigation icons: a back arrow, a home circle, and a menu hamburger icon. The form itself has the title "Create an admin account" at the top. Below the title are three input fields labeled "Username:", "Password:", and "Confirm password:". At the bottom of the form is a teal button with the text "Create admin account".

Create an admin account

Username:

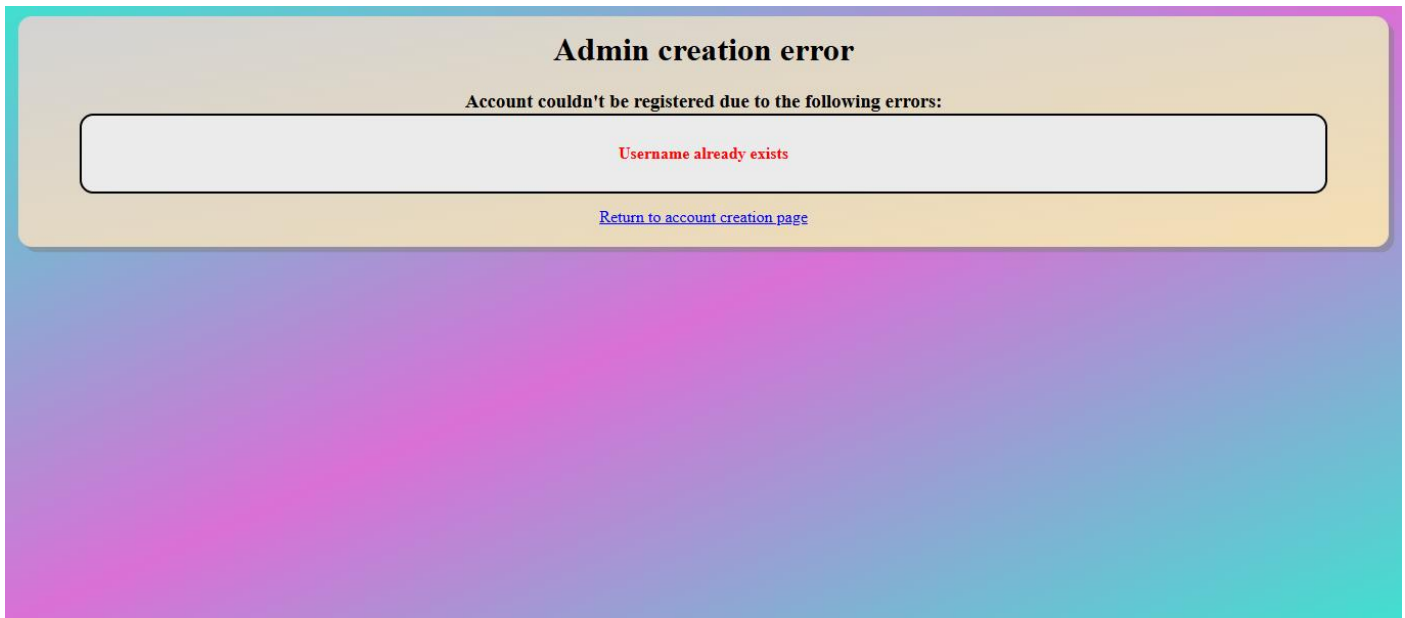
Password:

Confirm password:

Create admin account

Admin creation error page:

Desktop



The image shows a desktop application interface for an admin creation error. It features a light beige rounded rectangle on a colorful gradient background. The title "Admin creation error" is centered at the top. Below the title, the text "Account couldn't be registered due to the following errors:" is displayed. A light gray rounded rectangle contains the error message "Username already exists" in red. At the bottom of the beige area is a blue link that says "Return to account creation page".

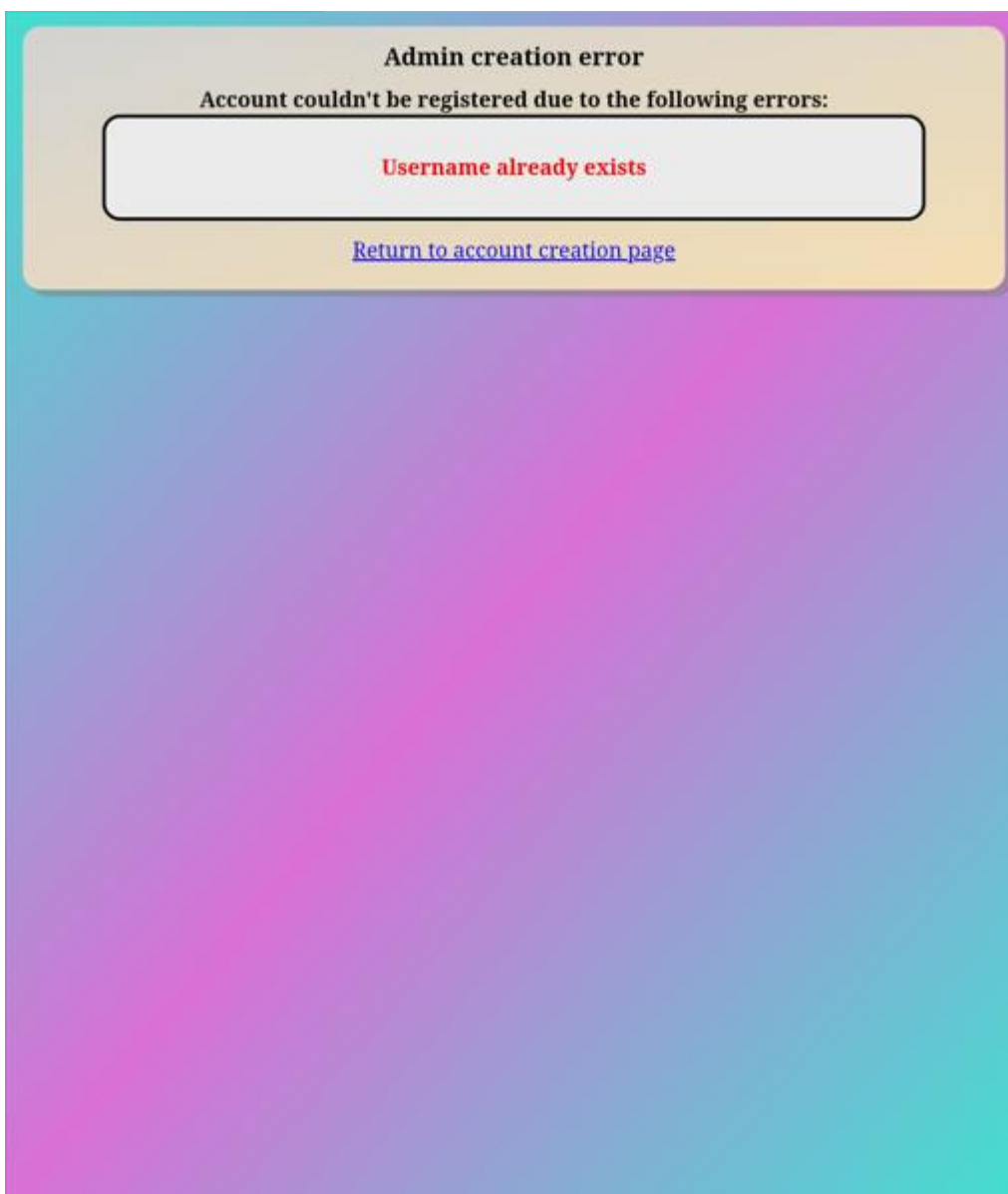
Admin creation error

Account couldn't be registered due to the following errors:

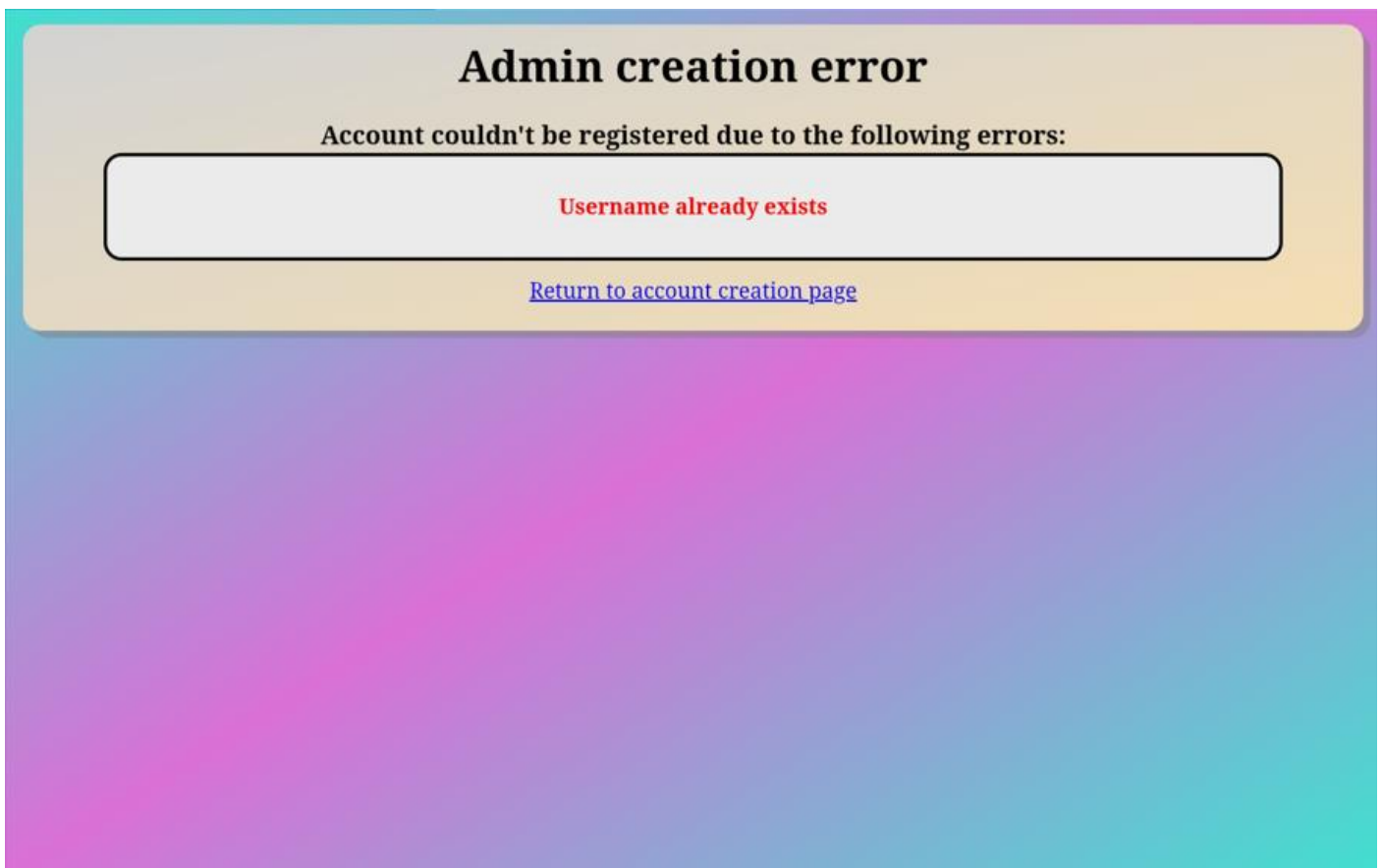
Username already exists

[Return to account creation page](#)

Tablet portrait



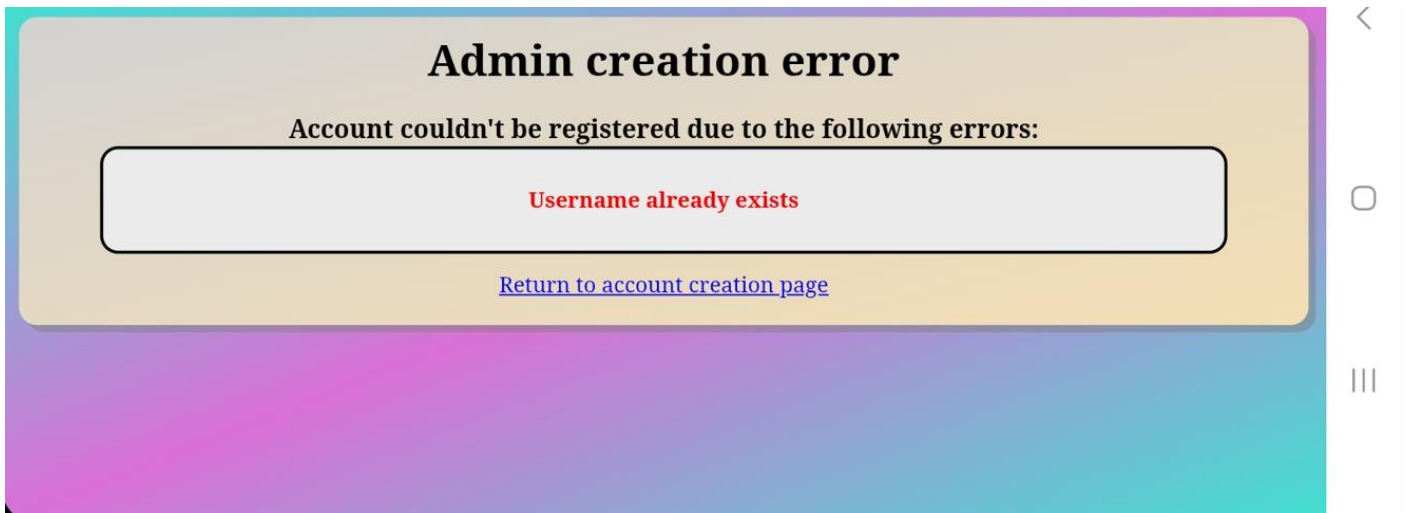
Tablet landscape



Mobile portrait

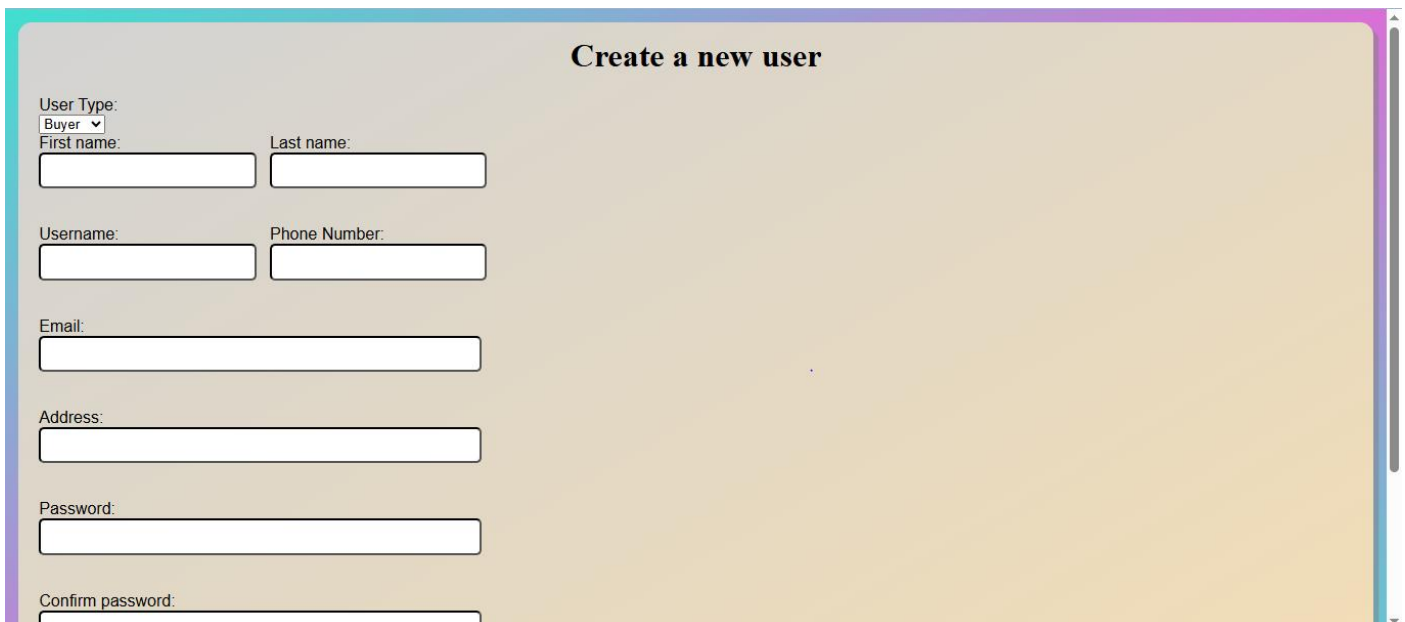


Mobile landscape



User creation page:

Desktop

A screenshot of a desktop application interface for creating a new user. The background is a gradient of purple, blue, and teal. A light beige rounded rectangle contains the title "Create a new user" in bold black font. Below the title, there are several form fields: "User Type:" with a dropdown menu showing "Buyer", "First name:" and "Last name:" with text input fields, "Username:" and "Phone Number:" with text input fields, "Email:" with a text input field, "Address:" with a text input field, "Password:" with a text input field, and "Confirm password:" with a text input field. A vertical scrollbar is visible on the right side of the form area.

Username: Phone Number:

Email:

Address:

Password:

Confirm password:

☐ Enable email notifications.

Register

Tablet portrait

Create a new user

User Type:
Buyer

First name:

Last name:

Username:

Phone Number:

Email:

Address:

Password:

Confirm password:

☐ Enable email notifications.

Register

Tablet landscape

Create a new user

User Type:

Buyer ▾

First name:

Last name:

Username:

Phone Number:

Email:

Address:

Password:

Confirm password:

User Type:

Buyer ▾

First name:

Last name:

Username:

Phone Number:

Email:

Address:

Password:

Confirm password:

☐

Enable email notifications.

Register

Mobile portrait

The form is titled "Create a new user" and is displayed in a portrait orientation. It features a light beige background with a colorful gradient border. The form fields are arranged vertically: "User Type:" with a dropdown menu set to "Buyer", "First name:" and "Last name:" (two separate input fields), "Username:" and "Phone Number:" (two separate input fields), "Email:" (one wide input field), "Address:" (one wide input field), "Password:" (one wide input field), and "Confirm password:" (one wide input field). Below the input fields is a checkbox labeled "Enable email notifications." and a green "Register" button. At the bottom of the screen, there is a navigation bar with three icons: a hamburger menu, a home button, and a back arrow.

Create a new user

User Type:
Buyer ▾

First name: Last name:

Username: Phone Number:

Email:

Address:

Password:

Confirm password:

☐ Enable email notifications.

Register

Mobile landscape

The form is titled "Create a new user" and is displayed in a landscape orientation. The layout is adjusted to fit the wider screen. The "User Type:" dropdown is at the top left. Below it, "First name:" and "Last name:" are side-by-side. "Username:" and "Phone Number:" are also side-by-side. "Email:" is a single wide field. "Address:" is not visible in this view. "Password:" and "Confirm password:" are also side-by-side. The "Enable email notifications." checkbox and the "Register" button are at the bottom left. On the right side of the screen, there is a vertical navigation bar with three icons: a back arrow, a home button, and a hamburger menu.

Create a new user

User Type:
Buyer ▾

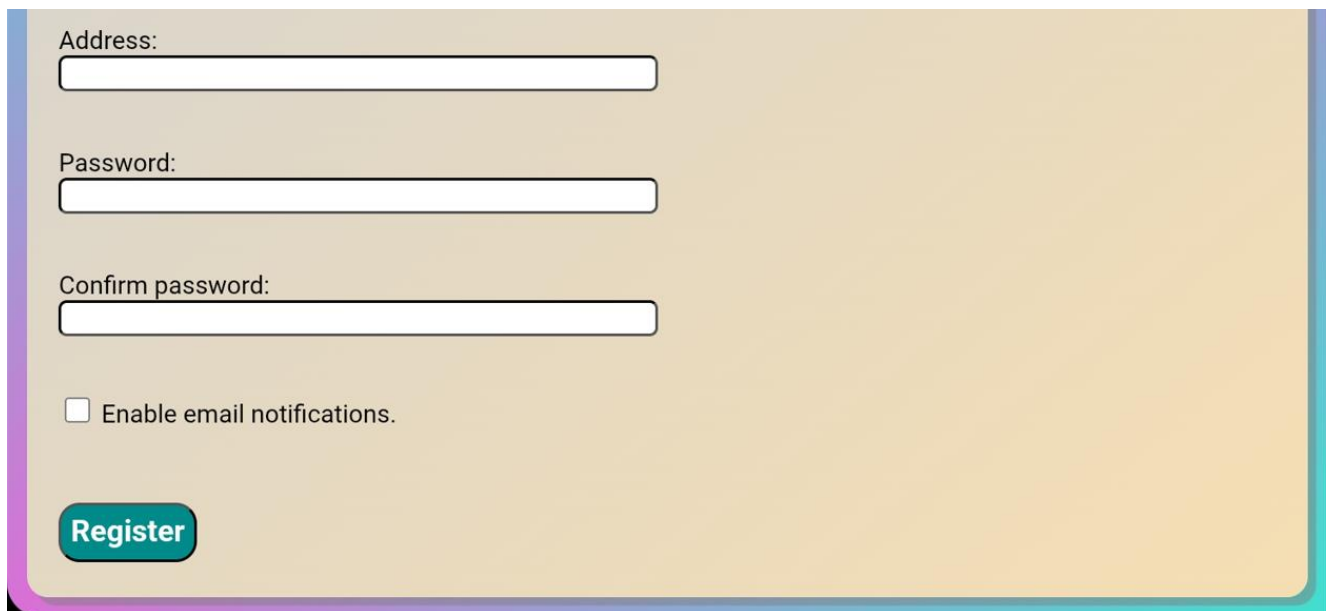
First name: Last name:

Username: Phone Number:

Email:

☐ Enable email notifications.

Register



A user registration form with a light beige background and a colorful gradient border. It includes three input fields for 'Address:', 'Password:', and 'Confirm password:'. Below these is a checkbox labeled 'Enable email notifications.' and a green 'Register' button. On the right side, there is a vertical navigation bar with icons: a back arrow, a home icon, and a menu icon.

Address:

Password:

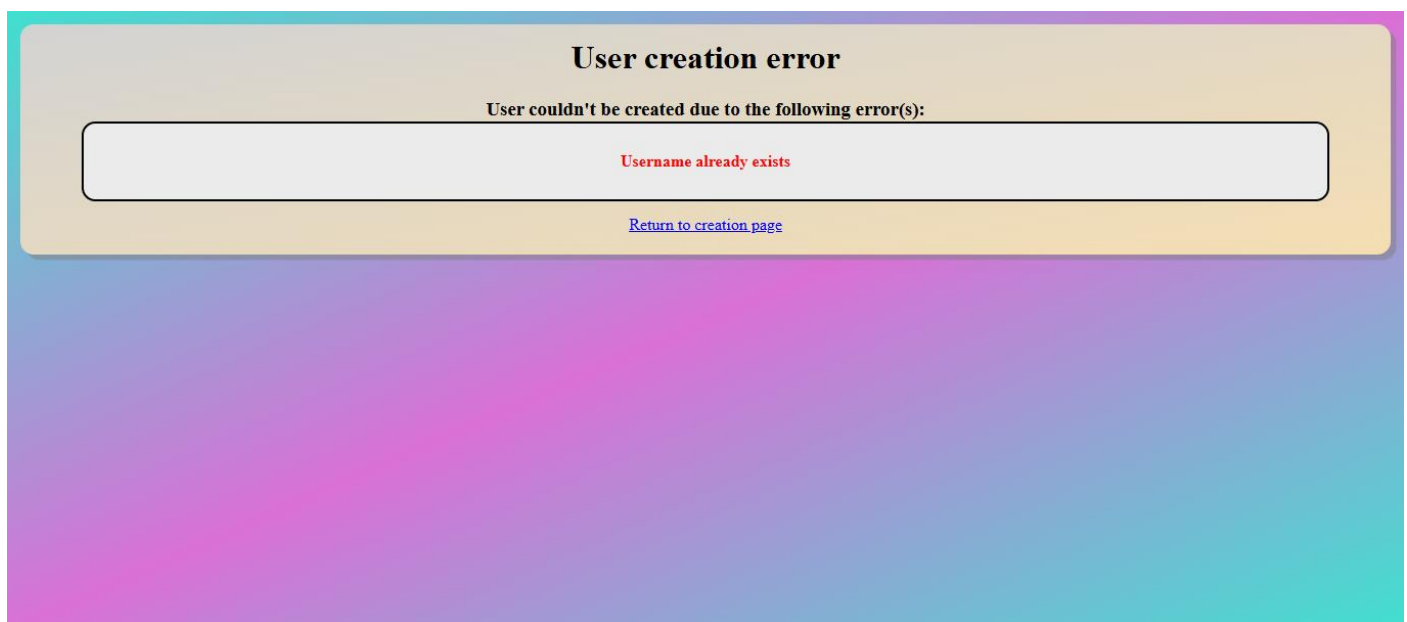
Confirm password:

☐ Enable email notifications.

Register

User creation error page:

Desktop



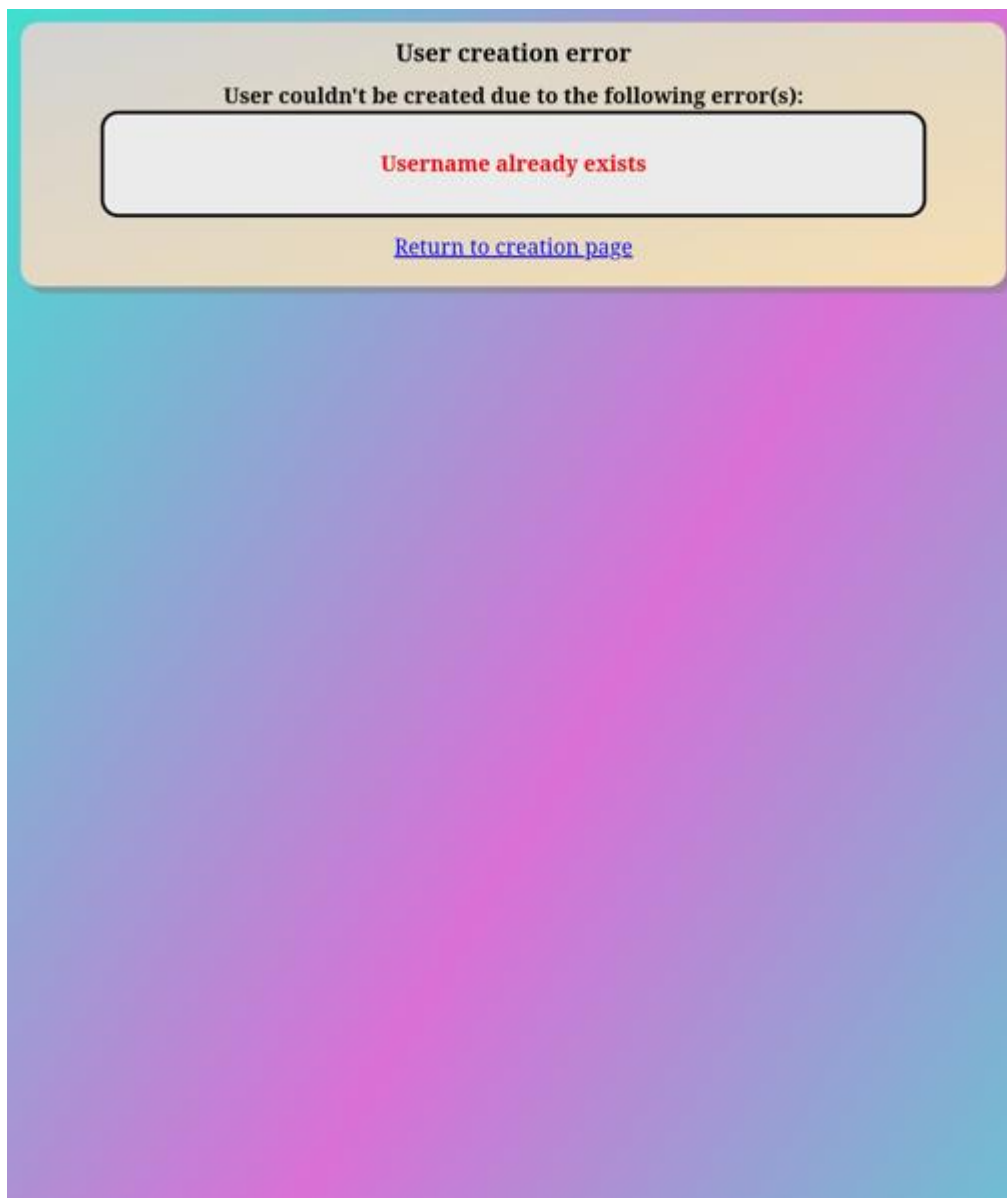
A desktop view of a user creation error page. It features a light beige header area with the title 'User creation error' and a message 'User couldn't be created due to the following error(s):'. Below this is a large white box containing the error message 'Username already exists' in red. At the bottom of the header area is a blue link 'Return to creation page'. The background of the page is a colorful gradient.

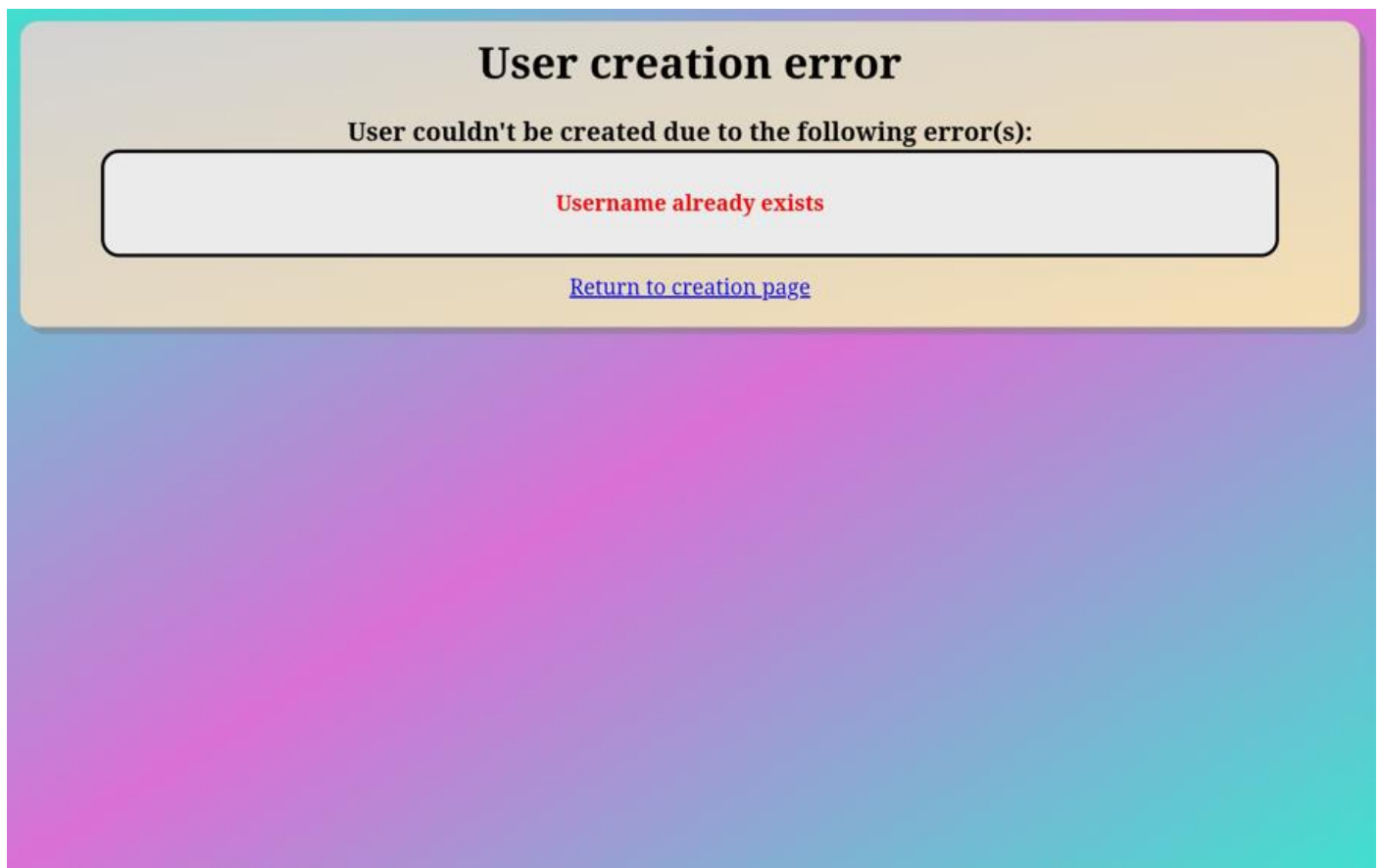
User creation error

User couldn't be created due to the following error(s):

Username already exists

[Return to creation page](#)

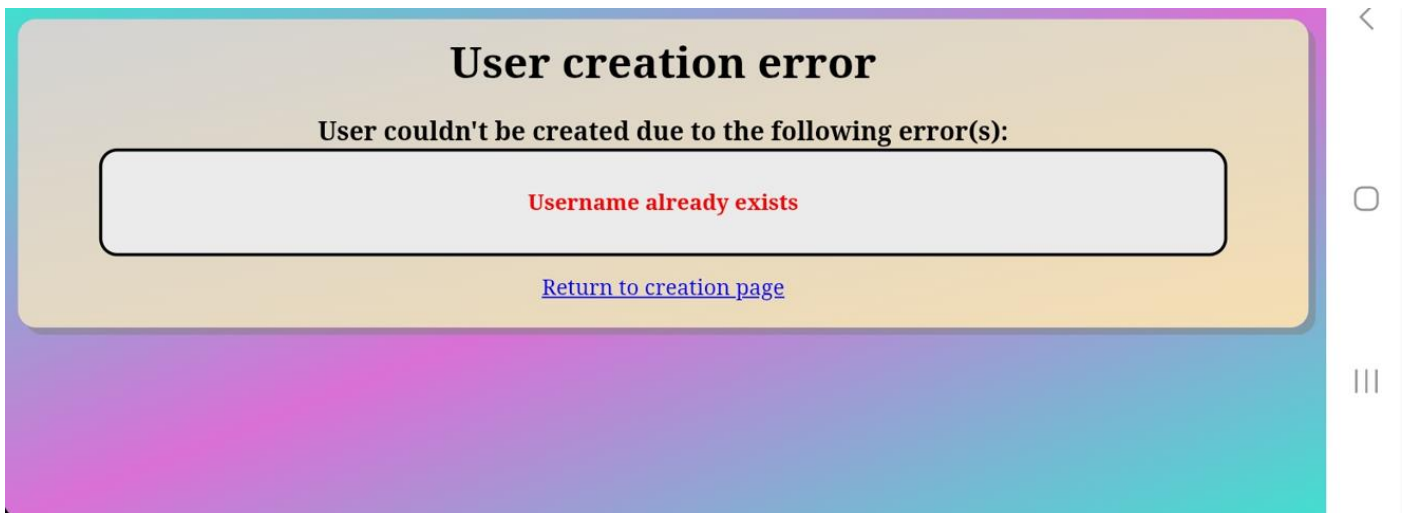




Mobile portrait

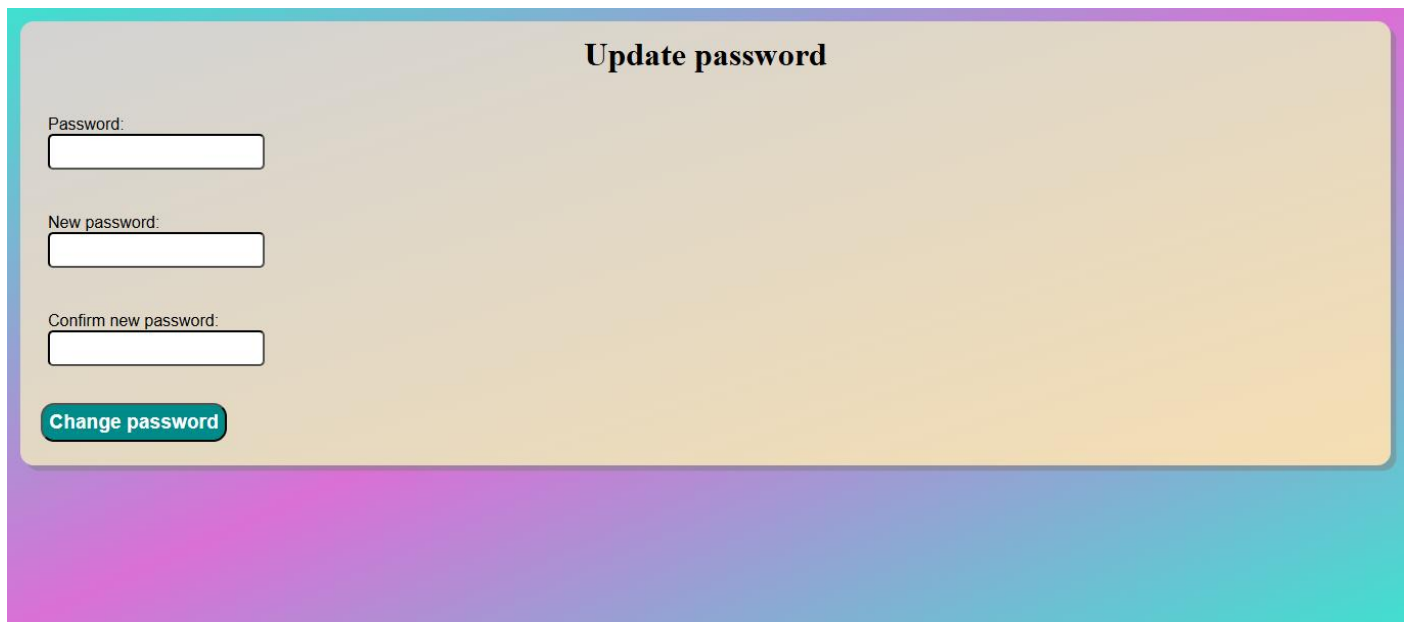


Mobile landscape



Change password page:

Desktop

The image shows a desktop view of a 'Change password' page. The page has a light beige background with a subtle gradient. At the top right, the title 'Update password' is displayed in a bold, black, sans-serif font. On the left side, there are three input fields stacked vertically. Each field is preceded by a label: 'Password:', 'New password:', and 'Confirm new password:'. The input fields are white with a thin black border. Below the input fields, there is a green button with the text 'Change password' in white. The entire form area is enclosed in a thin, multi-colored border that transitions from purple on the left to blue on the right. The bottom of the page features a wide, horizontal gradient bar that matches the bottom border of the form area, transitioning from purple on the left to blue on the right.

Update password

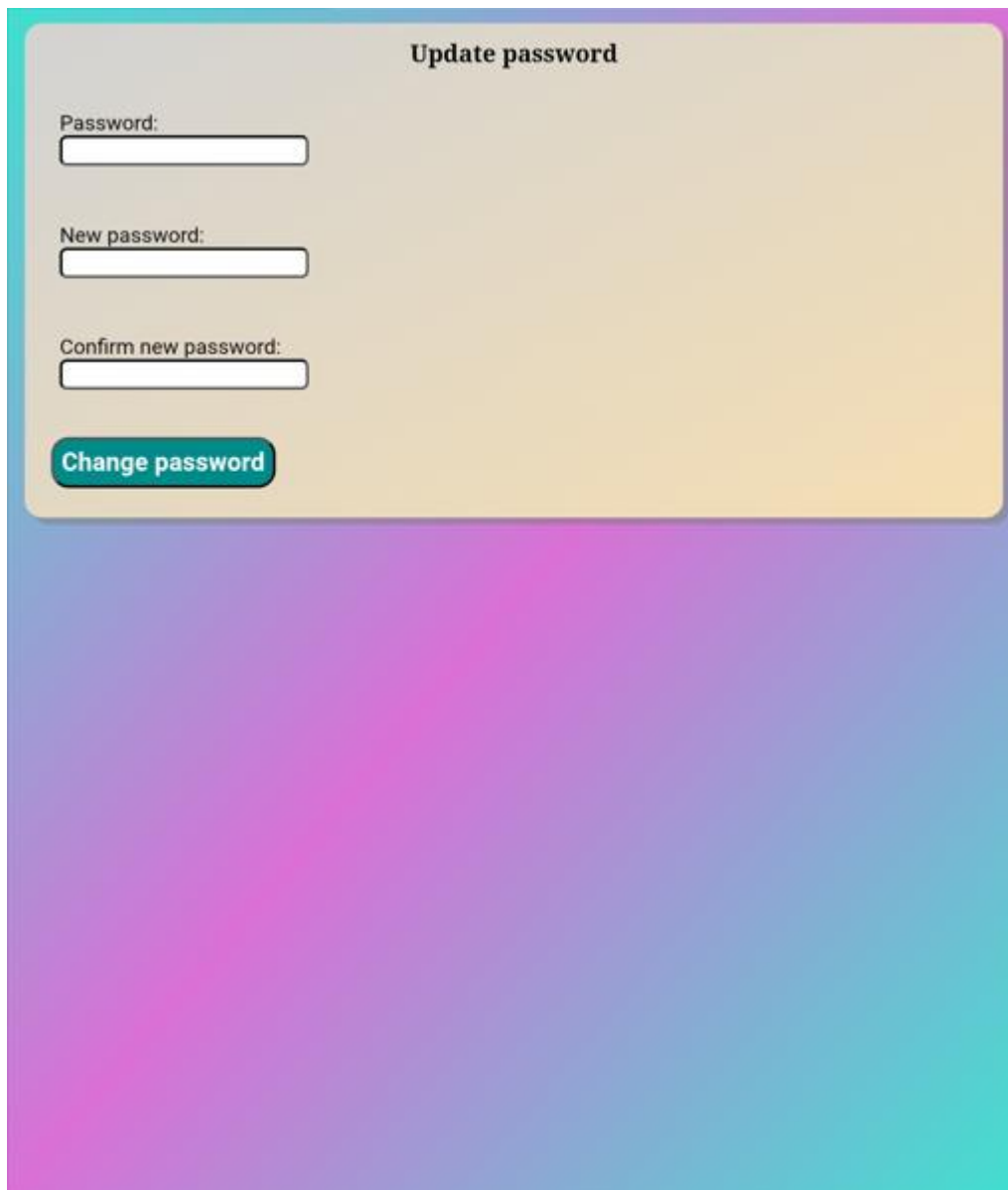
Password:

New password:

Confirm new password:

Change password

Tablet portrait



The image shows a tablet in portrait orientation displaying a password update form. The form is contained within a light beige rounded rectangle at the top of the screen. It features three input fields for 'Password:', 'New password:', and 'Confirm new password:', each with a white background and a thin black border. Below these fields is a teal button with white text that says 'Change password'. The background of the tablet screen is a vibrant, multi-colored gradient transitioning from purple at the top to blue and green at the bottom.

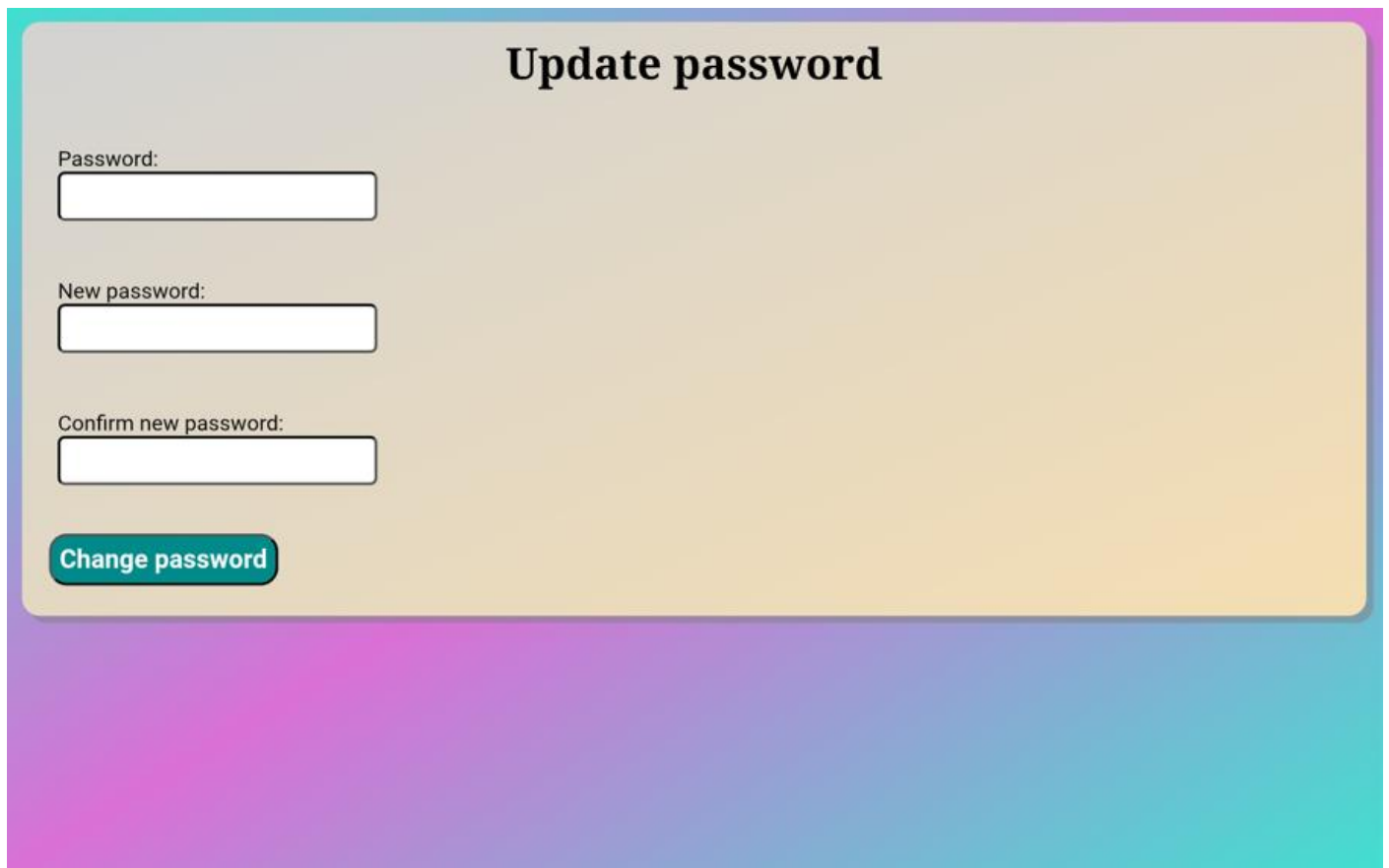
Update password

Password:

New password:

Confirm new password:

Change password



The image shows a mobile application interface in tablet landscape. A light beige card with rounded corners is centered on a background with a purple-to-teal gradient. The card has the title 'Update password' in bold black text at the top right. On the left side of the card, there are three text labels: 'Password:', 'New password:', and 'Confirm new password:'. Each label is followed by a white rectangular input field with a thin black border. Below these fields is a teal button with rounded corners and the text 'Change password' in white. The overall design is clean and modern.

Update password

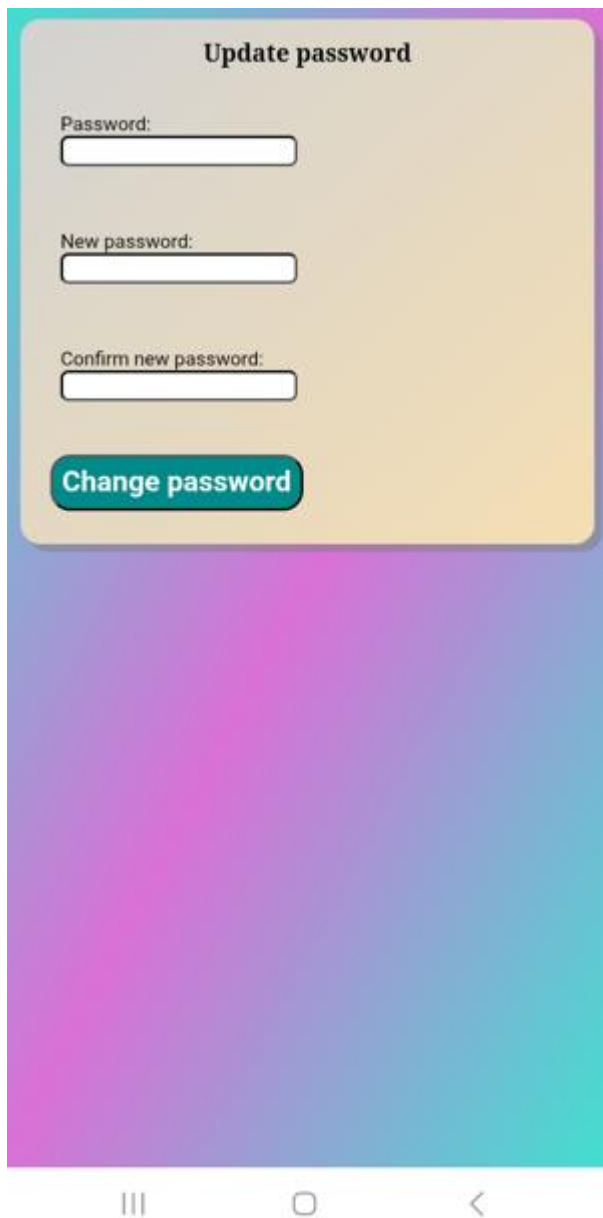
Password:

New password:

Confirm new password:

Change password

Mobile portrait



The image shows a mobile application interface in portrait orientation. At the top, there is a light beige rounded rectangle with the title "Update password" in bold black text. Below the title are three input fields, each with a label to its left: "Password:", "New password:", and "Confirm new password:". Each label is in a small, dark grey font. The input fields are white with a thin black border. Below the input fields is a green rounded rectangle with the text "Change password" in white, bold font. The background of the screen is a vibrant gradient of purple, pink, and blue. At the bottom of the screen, there are three standard Android navigation icons: a square, a circle, and a triangle.

Update password

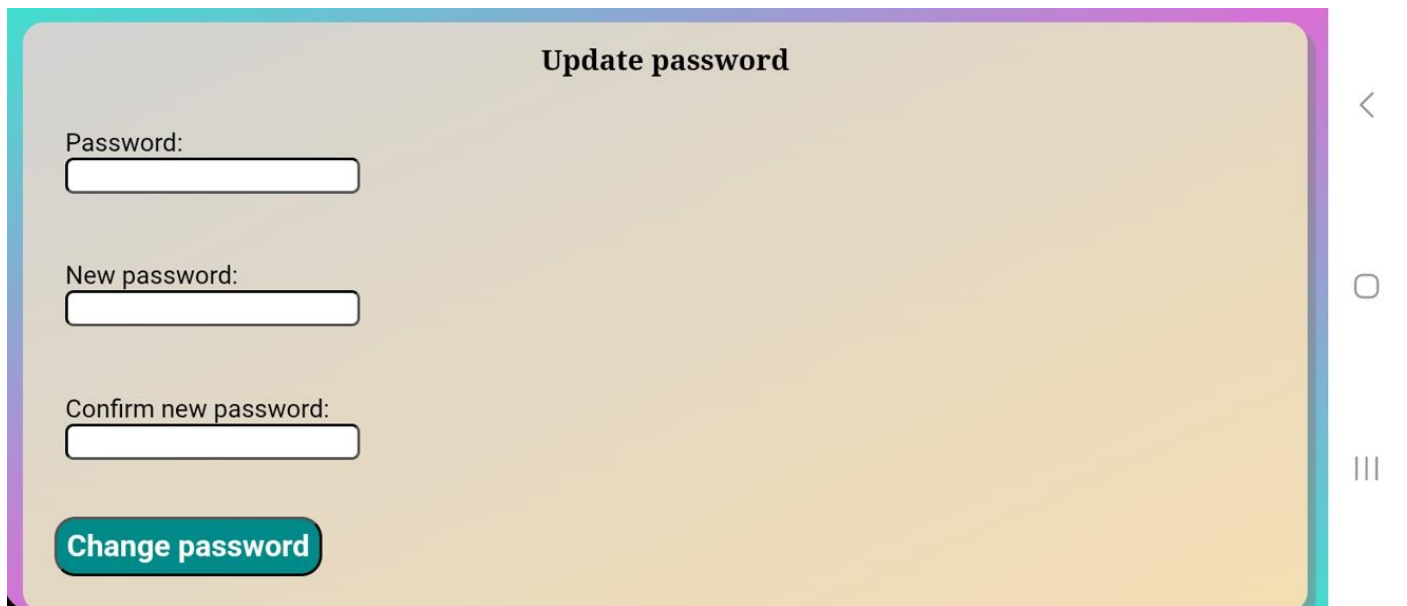
Password:

New password:

Confirm new password:

Change password

Mobile landscape



The image shows a mobile application interface in landscape orientation. The main content area has a light beige background and is titled "Update password" in bold black text at the top center. On the left side, there are three text input fields, each with a label to its left: "Password:", "New password:", and "Confirm new password:". Below these fields is a green button with white text that says "Change password". To the right of the main content area is a vertical sidebar with a white background. It contains three icons: a back arrow at the top, a square icon in the middle, and a hamburger menu icon (three horizontal lines) at the bottom.

Update password

Password:

New password:

Confirm new password:

Change password

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2.3) Designing

a)

Class name: User	
Description: The super class for all users in the system.	
Responsibilities: Login to account Change password	Collaborators:
Attributes: Username (string) Password (string)	
Relationships: Generalisation (a-kind-of): Admin, Non-AdminUser Aggregation: Other associations:	

Class name: Admin	
Description: Manages the administrative side of the system, such as updating user information.	
Responsibilities: Create new users Change user type Change user password	Collaborators: Buyer, Vendor
Attributes: adminID (Integer)	
Relationships: Generalisation (a-kind-of): User Aggregation: Other associations: Buyer, Vendor	

Class name: Non-admin user	
Description: Is the sub class of the User super class.	
Responsibilities: Delete account Add products to delivery Remove products from delivery Send a message Rate a vendor	Collaborators: User
Attributes: firstName (string) lastName (string) phoneNum (integer) email (string) address (string) buyerRating (float) eWalletAmount (float) notify (boolean)	
Relationships: Generalisation (a-kind-of): User, Buyer, Vendor Aggregation: Other associations: Listing	

Class name: Buyer	
Description: A user that buys products from vendors.	
Responsibilities: Register an account Register as a vendor	Collaborators: User, Non-AdminUser
Attributes: buyerID (integer)	
Relationships: Generalisation (a-kind-of): Non-AdminUser Aggregation: Other associations: Delivery, DeliveryMessage, CartItem, UserRatingHistory	

Class name: Vendor	
Description: A user that creates listings and conducts business with buyers and other vendors.	
Responsibilities: Create a listing Remove a listing Rate a buyer	Collaborators: Buyer, vendor
Attributes: vendorID (integer) vendorRating (integer)	
Relationships: Generalisation (a-kind-of): Non-AdminUser Aggregation: Other associations: Listing, Delivery, DeliveryMessage, CartItem, UserRatingHistory	

Class name: Listing	
Description: A product that is being sold by a vendor.	
Responsibilities:	Collaborators: Vendor
Attributes: listingID (integer) title (string) price (float) description (string) category (string) image (string)	
Relationships: Generalisation (a-kind-of): Aggregation: Other associations: Vendor, Delivery, CartItem	

Class name: CartItem	
Description: An individual product and it's details in a specific delivery.	
Responsibilities:	Collaborators: Listing, Delivery
Attributes: cartItemID (integer) quantity (integer)	
Relationships:	

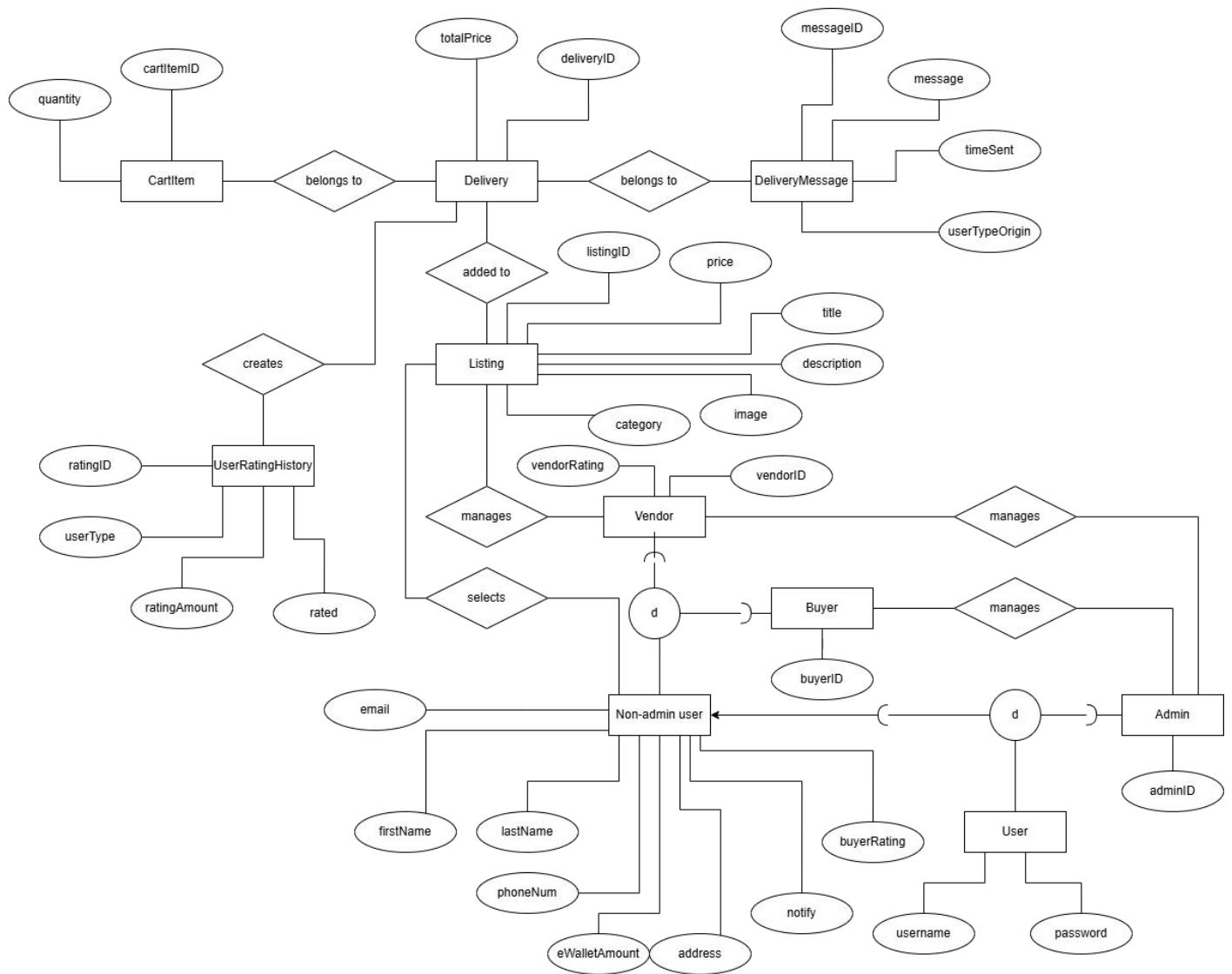
Generalisation (a-kind-of): Aggregation: Delivery Other associations: Listing	
---	--

Class name: DeliveryMessage	
Description: Messages send between both users associated with a specific delivery.	
Responsibilities:	Collaborators: Delivery, Buyer, Vendor
Attributes: messageID (integer) message (string) timeSent (date) userTypeOrigin (string)	
Relationships: Generalisation (a-kind-of): Aggregation: Delivery Other associations: Buyer, Vendor	

Class name: Delivery	
Description: A delivery of products between the buyer and vendor.	
Responsibilities: Update a product's quantity Pay for delivery	Collaborators: CartItem, Buyer, Vendor
Attributes: deliveryID (integer) totalPrice (float)	
Relationships: Generalisation (a-kind-of): Aggregation: CartItem Other associations: DeliveryMessage, Buyer, Vendor	

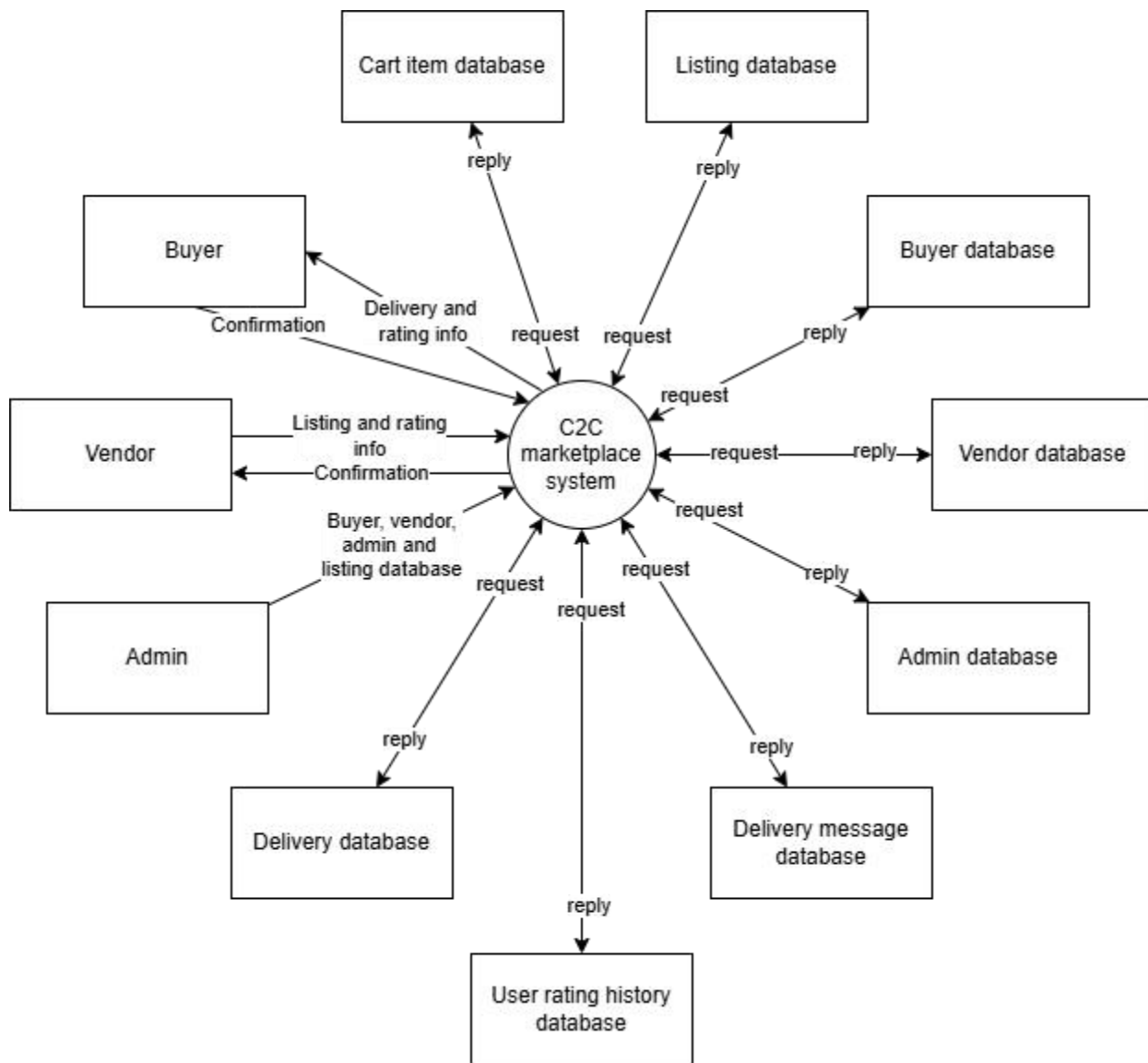
Class name: UserRatingHistory	
Description: A class that stores all user ratings between buyers and vendors.	
Responsibilities:	Collaborators: Delivery, Buyer, vendor
Attributes: ratingID (integer) rated (boolean) ratingAmount (Integer) ratingType (String) userType (String)	
Relationships: Generalisation (a-kind-of): Aggregation: Other associations: Delivery, Buyer, Vendor	

b)



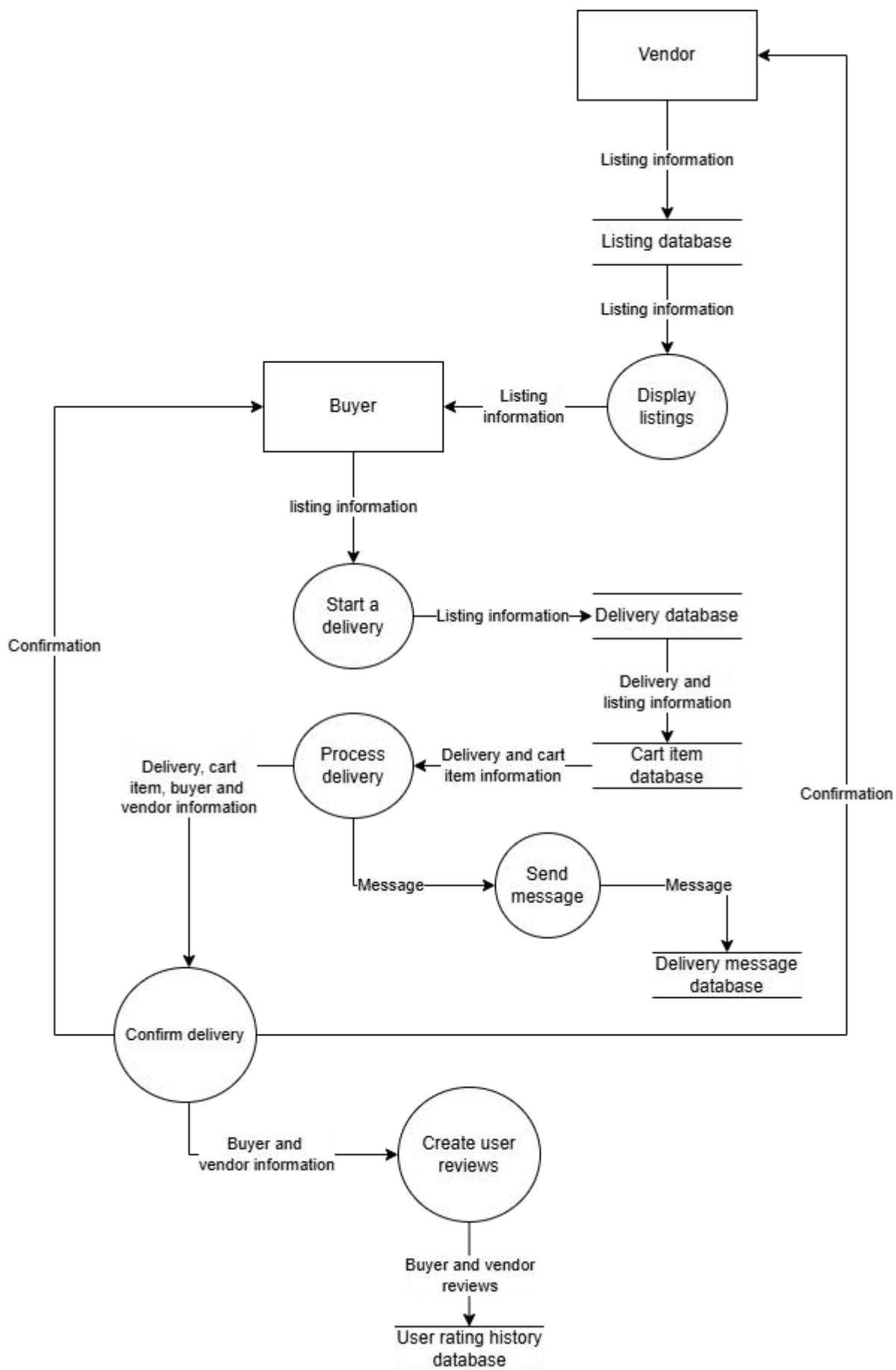
(Patel, 2026)

c)



(Reddy, 2024)

d)

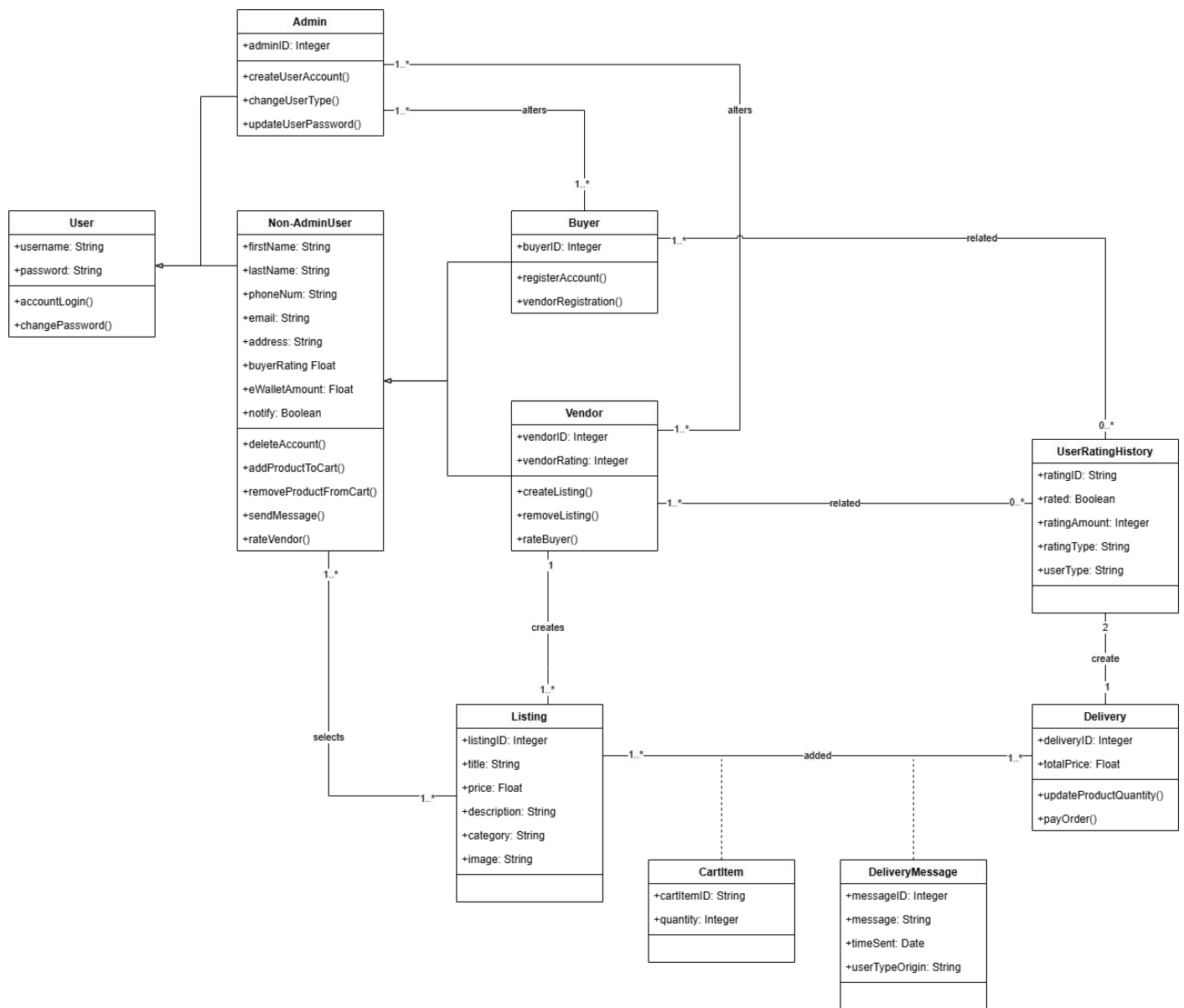


(Patni, 2025)

e)



f)



2.4) Coding

b)

accountRegistration.php

```
<?php
echo "<link rel='stylesheet' href='../style.css'>";
echo "<div class='form-accountDetails'>";
$success = false;
$connectionActive = false;
session_start();

if ($_SERVER["REQUEST_METHOD"] == "POST") {
    mysqli_report(MYSQLI_REPORT_ERROR | MYSQLI_REPORT_STRICT);
    $connection = null;

    try {
        $connection = new mysqli("localhost", "root", "", "c2cMarketplacedb");
        mysqli_autocommit($connection, false);
        mysqli_begin_transaction($connection);
        $connectionActive = true;

        $firstName = $_POST["fName"];
        $lastName = $_POST["lName"];
        $username = $_POST["username"];
        $phoneNum = $_POST["phoneNum"];
        $email = $_POST["email"];
        $address = $_POST["address"];
        $password = $_POST["password"];
        $conPassword = $_POST["conPassword"];
        $notify = isset($_POST["notify"]);
        $userAgreement = isset($_POST["userAgreement"]);

        $errors = [];

        if (empty($firstName)) {
            $errors[] = "First name is required";
        }
        if (empty($lastName)) {
            $errors[] = "Last name is required";
        }
        if (empty($username)) {
            $errors[] = "Username is required";
        }
        else{
            $sqlUsernameB = mysqli_execute_query($connection, "SELECT username FROM buyer
WHERE username = ?", [$username]);
            if (mysqli_num_rows($sqlUsernameB) > 0){
                $errors[] = "Username already exists";
            }
            else{
```

```

        $sqlUsernameV = mysqli_execute_query($connection, "SELECT username FROM
vendor WHERE username = ?", [$username]);
        if (mysqli_num_rows($sqlUsernameV) > 0){
            $errors[] = "Username already exists";
        }
        else{
            $sqlUsernameA = mysqli_execute_query($connection, "SELECT username
FROM admin WHERE username = ?", [$username]);
            if (mysqli_num_rows($sqlUsernameA) > 0){
                $errors[] = "Username already exists";
            }
        }
    }

    if (empty($phoneNum)) {
        $errors[] = "Phone number is required";
    }

    if (empty($email)) {
        $errors[] = "Email is required";
    } else if (!filter_var($email, FILTER_VALIDATE_EMAIL)) {
        $errors[] = "Invalid email";
    }

    if (empty($address)) {
        $errors[] = "Address is required";
    }

    if (empty($password)) {
        $errors[] = "Password is required";
    } else if (strlen($password) < 8) {
        $errors[] = "Password must be at least 8 or more characters in length";
    }

    if (empty($conPassword)) {
        $errors[] = "Password must be repeated for confirmation";
    } else if ($password != $conPassword) {
        $errors[] = "Password and confirmation password don't match";
    }

    if (empty($userAgreement)) {
        $errors[] = "You must agree to the user agreement and privacy policy if you
wish to use the marketplace";
    }

    if (empty($errors)) {
        $notifyChecked = 0;
        if (!empty($notify)){
            $notifyChecked = 1;
        }
    }

```

```

        mysqli_execute_query($connection, "INSERT INTO buyer (firstName, lastName,
username, phoneNum, email, address, password, buyerRating, eWalletAmount, notify) VALUES
(?, ?, ?, ?, ?, ?, ?, '0', '0', ?)", [$firstName, $lastName, $username, $phoneNum, $email,
$address, password_hash($password, PASSWORD_DEFAULT), $notifyChecked]);
        $userDetails = mysqli_fetch_array(mysqli_execute_query($connection, "SELECT
buyerID FROM buyer WHERE username = ?", [$username]));

        mysqli_commit($connection);
        $success = true;
        $_SESSION["message"] = "Account registered successfully";
        $_SESSION["userID"] = $userDetails["buyerID"];
        $_SESSION["firstName"] = $firstName;
        $_SESSION["lastName"] = $lastName;
        $_SESSION["username"] = $username;
        $_SESSION["phoneNum"] = $phoneNum;
        $_SESSION["email"] = $email;
        $_SESSION["address"] = $address;
        $_SESSION["userType"] = "buyer";
    } else {
        echo "<h1>Account registration error</h1>";
        echo "<h3>Account couldn't be registered due to the following error(s):</h3>";
        echo "<div class='form-errorBoard error'>";
            foreach ($errors as $error) {
                echo "<h4>$error</h4>";
            }
        echo "</div>";
        echo "<a href='accountRegistration.html' class='form-returnLink'>Return to
registration page</a>";
    }
}
catch(Exception $error){
    echo "<h1>Account registration error</h1>";
    echo "<h3>Account couldn't be registered due to the following error(s):</h3>";
    echo "<div class='form-errorBoard error'>";
        echo "<h4>Error: ". $error->getMessage() . "</h4>";
    echo "</div>";
    echo "<a href='accountRegistration.html' class='form-returnLink'>Return to
registration page</a>";
}

    if ($connectionActive == true){
        mysqli_rollback($connection);
    }
}

    if ($connection != null){
        $connection->close();
    }
} else {
    $success = true;
    $_SESSION["message"] = "Error: POST error occurred";
}
echo "</div>";

```

```

if ($success == true){
    header('Location:mainPage.php');
}
?>

```

(php, n.d.) (Code, 2024) (PHP.Watch, n.d.) (W3Schools, n.d.) (Zeljka, 2017) (W3Schools, n.d.)

A php file that validates and inserts the inputted user data from the accountRegistration.html into the buyer database.

accountLogin.php

```

<?php
echo "<link rel='stylesheet' href='style.css'>";
echo "<div class='form-accountDetails'>";
$userType = null;
$success = false;
session_start();

if ($_SERVER["REQUEST_METHOD"] == "POST") {
    mysqli_report(MYSQLI_REPORT_ERROR | MYSQLI_REPORT_STRICT);
    $connection = null;

    try{
        $connection = new mysqli("localhost", "root", "", "c2cMarketplacedb");
        $username = $_POST["username"];
        $password = $_POST["password"];
        $errors = [];

        if (empty($username)) {
            $errors[] = "Username is required";
        }
        if (empty($password)) {
            $errors[] = "Password is required";
        }

        if (!empty($username) && !empty($password)) {
            $sqlUsernameB = mysqli_execute_query($connection, "SELECT * FROM buyer WHERE
username = ?", [$username]);

            if (mysqli_num_rows($sqlUsernameB) <= 0) {
                $sqlUsernameV = mysqli_execute_query($connection, "SELECT * FROM vendor
WHERE username = ?", [$username]);

                if (mysqli_num_rows($sqlUsernameV) <= 0) {
                    $sqlUsernameA = mysqli_execute_query($connection, "SELECT * FROM admin
WHERE username = ?", [$username]);

                    if (mysqli_num_rows($sqlUsernameA) <= 0) {
                        $errors[] = "Username or password is incorrect";
                    }
                }
            }
        }
    }
}

```

```

        else{
            $userDetails = mysqli_fetch_array($sqlUsernameA);
            if (password_verify($password, $userDetails["password"])){
                $userType = "admin";
            }
            else{
                $errors[] = "Username or password is incorrect";
            }
        }
    }
    else{
        $userDetails = mysqli_fetch_array($sqlUsernameV);
        if (password_verify($password, $userDetails["password"])){
            $userType = "vendor";
        }
        else{
            $errors[] = "Username or password is incorrect";
        }
    }
}
else{
    $userDetails = mysqli_fetch_array($sqlUsernameB);
    if (password_verify($password, $userDetails["password"])){
        $userType = "buyer";
    }
    else{
        $errors[] = "Username or password is incorrect";
    }
}
}

if (empty($errors)) {
    $userDetails = mysqli_fetch_array(mysqli_execute_query($connection, "SELECT *
FROM $userType WHERE username = ?", [$username]));

    unset($_SESSION["userID"]);
    unset($_SESSION["firstName"]);
    unset($_SESSION["lastName"]);
    unset($_SESSION["username"]);
    unset($_SESSION["phoneNum"]);
    unset($_SESSION["email"]);
    unset($_SESSION["address"]);
    unset($_SESSION["userType"]);

    $_SESSION["userID"] = $userDetails[$userType."ID"];
    $_SESSION["username"] = $userDetails["username"];
    $_SESSION["userType"] = $userType;

    if ($userType != "admin"){
        $_SESSION["firstName"] = $userDetails["firstName"];
        $_SESSION["lastName"] = $userDetails["lastName"];
        $_SESSION["phoneNum"] = $userDetails["phoneNum"];
    }
}

```

```

        $_SESSION["email"] = $userDetails["email"];
        $_SESSION["address"] = $userDetails["address"];
    }

    $_SESSION["message"] = "Account logged in successfully";
    $success = true;
}
else {
    echo "<h1>Account login error</h1>";
    echo "<h3>Account couldn't be logged into due to the following error(s):</h3>";
    echo "<div class='form-errorBoard error'>";
    foreach ($errors as $error) {
        echo "<h4>*$error</h4>";
    }
    echo "</div>";
    echo "<a href='accountLogin.html' class='form-returnLink'>Return to login
page</a>";
}
}
catch(Exception $error){
    echo "<h1>Account login error</h1>";
    echo "<h3>Account couldn't be logged into due to the following error(s):</h3>";
    echo "<div class='form-errorBoard error'>";
    echo "<h4>*Error: ". $error->getMessage() . "</h4>";
    echo "</div>";
    echo "<a href='accountLogin.html' class='form-returnLink'>Return to login
page</a>";
}
if ($connection != null){
    $connection->close();
}
} else {
    $_SESSION["message"] = "Error: POST error occured";
}
echo "</div>";

if ($success == true){
    if ($userType != "admin"){
        header('Location:mainWebsite/mainPage.php');
    }
    else{
        header('Location:adminWebsite/adminDatabase.php');
    }
}
?>

```

A php file that validates and inserts the login information and determines what type of user the user is, such as a buyer vendor or admin.

accountLogout.php

```

<?php
session_start();

if ($_SERVER["REQUEST_METHOD"] == "POST") {
    unset($_SESSION["userID"]);
    unset($_SESSION["firstName"]);
    unset($_SESSION["lastName"]);
    unset($_SESSION["username"]);
    unset($_SESSION["phoneNum"]);
    unset($_SESSION["email"]);
    unset($_SESSION["address"]);
    unset($_SESSION["userType"]);
    $_SESSION["message"] = "Successfully logged out of account";
}

header('Location:mainWebsite/mainPage.php');
?>

```

A php form that logs the user out of their account by removing the corresponding user SESSION data.

mainPage.php

```

<?php
session_start();
$accountLink = "accountRegistration.html";
$sqlStatement = "SELECT * FROM listing";

if (isset($_SESSION["userID"]) && !empty($_SESSION["userID"])){
    $accountLink = "accountCenter(ProfileDetails).php";
}

echo "<html lang='en'>";
echo "<head>";
    echo "<meta charset='UTF-8'>";
    echo "<meta name='viewport' content='width=device-width, initial-scale=1.0'>";
    echo "<title>Marketplace</title>";
    echo "<link rel='stylesheet' href='../style.css'>";
echo "</head>";

if (isset($_SESSION["searchStatement"]) && !empty($_SESSION["searchStatement"])){
    $sqlStatement = $_SESSION["searchStatement"];
    unset($_SESSION["searchStatement"]);
}

if (isset($_SESSION["message"]) && !empty($_SESSION["message"])){
    echo "<div id='message' name='message' class='msg'>";
        echo "<p>". $_SESSION["message"] . "</p>";
        echo "<button onclick='closeMessage()'>X</button>";
    echo "</div>";
    unset($_SESSION["message"]);
}

```

```

    echo "<script src='../script.js'></script>";
}

echo "<header>";
echo "<h1><a href='mainPage.php'>Marketplace</a></h1>";
echo "<nav>";
    echo "<div class='form-mainHeading'>";
        if (isset($_SESSION["userType"])) {
            if ($_SESSION["userType"] == "vendor") {
                echo "<div class='form-account'>";
                    echo "<a href='createListing.html'>Create listing</a>";
                echo "</div>";
            }
        }
        echo "<div class='form-account'>";
            echo "<a href=' $accountLink ' >My Account</a>";
        echo "</div>";

        if (isset($_SESSION["userID"])){
            echo "<a href='cart.php'><button>Cart</button></a>";
        }
        else{
            echo "<a href='cart.php'><button disabled>Cart</button></a>";
        }

    echo "</div>";
echo "</nav>";
echo "</header>";

echo "<body>";
echo "<form action='../searchDatabase.php' method='POST'>";
    echo "<div class='form-searchBar'>";
        echo "<div class='form-searchBarItem'>";
            echo "<input type='hidden' id='isAdmin' name='isAdmin' value='false'>";
            echo "<select id='infoPriority' name='infoPriority'>";
                echo "<option value='title'>Search by Title</option>";
                echo "<option value='price'>Search by Price</option>";
                echo "<option value='category'>Search by Category</option>";
            echo "</select>";

            echo "<select id='orderItem' name='orderItem'>";
                echo "<option value='title'>Sort by Title</option>";
                echo "<option value='price'>Sort by Price</option>";
                echo "<option value='category'>Sort by Category</option>";
            echo "</select>";

            echo "<select id='adPriority' name='adPriority'>";
                echo "<option value='ASC'>Order in Ascending</option>";
                echo "<option value='DESC'>Order in Descending</option>";
                echo "<option value='None'>None</option>";
            echo "</select>";
        echo "</div>";

```

```

        echo "<div class='form-inputAndBtn'>";
        echo "<input type='search' id='searchBar' name='searchBar'>";
        echo "<button type='submit' name='tableBtn' id='tableBtn'
value='listing'>&rarr;</button>";
        echo "</div>";
    echo "</div>";
echo "</form>";

echo "<div class='form-productsGridContainer'>";
mysqli_report(MYSQLI_REPORT_ERROR | MYSQLI_REPORT_STRICT);
$connection = null;

try{
    $connection = new mysqli("localhost", "root", "", "c2cMarketplacedb");
    if ($listingRecords = mysqli_execute_query($connection, $sqlStatement)){
        if(mysqli_num_rows($listingRecords) > 0){
            while($recordDetails = mysqli_fetch_array($listingRecords)){
                $vendorDetails =
mysqli_fetch_array(mysqli_execute_query($connection, "SELECT * FROM vendor WHERE vendorID
= ?", [$recordDetails["vendorID"]]));
                echo "<div class='form-productsGridItem'>";
                echo "<div class='form-productsGridItemImage'>";
                echo "<form action='productPage.php' method='POST'>";
                echo "<button type='submit' id='formBtn'
name='formBtn' value='". $recordDetails["listingID"] ." '><img src='".
$recordDetails["image"] ." ' alt='Product.png'></button>";
                echo "</form>";
                echo "</div>";

                echo "<div class='form-productsGridVendor'>";
                echo "<h2>". $vendorDetails["username"] ."</h2>";
                echo "</div>";

                echo "<h3>". $recordDetails["title"] ."</h3>";
                echo "<h4 class='form-cost'>R ".
$recordDetails["price"] ."</h4>";
                echo "<h4>". $recordDetails["category"] ."</h4>";

                echo "<form action='addToCart.php' method='POST'>";
                if (isset($_SESSION["userID"])){
                    $deliveryDetails =
mysqli_fetch_array(mysqli_execute_query($connection, "SELECT * FROM delivery WHERE
userType = ? AND userID = ? AND vendorID = ?", [$_SESSION["userType"], $_SESSION["userID"],
$recordDetails["vendorID"]]));
                    if ($recordDetails["vendorID"] == $_SESSION["userID"]
&& $_SESSION["userType"] == 'vendor'){
                        echo "<button name='productBtn' id='productBtn'
value='-2'>Your listing</button>";
                    }
                    else{
                        if (!isset($deliveryDetails["deliveryID"])){

```

```

                                echo "<input type='hidden' name='vendorID'
id='vendorID' value='". $recordDetails["vendorID"] ."'>";
                                echo "<button type='submit' name='productBtn'
id='productBtn' value='". $recordDetails["listingID"] ."'>Add to cart</button>";
                                }
                                else{
                                    if ($cartItems =
mysqli_execute_query($connection, "SELECT * FROM cartItem WHERE deliveryID = ? AND
listingID = ?", [$deliveryDetails["deliveryID"], $recordDetails["listingID"]])){
                                        if(mysqli_num_rows($cartItems) <= 0){
                                            echo "<input type='hidden'
name='vendorID' id='vendorID' value='". $recordDetails["vendorID"] ."'>";
                                            echo "<button type='submit'
name='productBtn' id='productBtn' value='". $recordDetails["listingID"] ."'>Add to
cart</button>";
                                                }
                                            else{
                                                echo "<button type='submit'
name='productBtn' id='productBtn' value='-1'>In cart</button>";
                                            }
                                        }
                                    }
                                }
                            }
                        }
                    }
                }
            }
        }
    }
    catch(Exception $error){
        echo "<h2>An error occured</h2>";
    }

    if ($connection != null){
        $connection->close();
    }
    echo "</div>";
echo "</body>";
echo "</html>";
?>

```

A php form that dynamically displays all listings from the listing database and enables users to add it to their cart.

productPage.php

```

<?php
session_start();
if ($_SERVER["REQUEST_METHOD"] == "POST") {
    $listingID = $_POST["formBtn"];
    echo "<html lang='en'>";
    echo "<head>";
        echo "<meta charset='UTF-8'>";
        echo "<meta name='viewport' content='width=device-width, initial-scale=1.0'>";
        echo "<title>Marketplace</title>";
        echo "<link rel='stylesheet' href='../style.css'>";
    echo "</head>";

    echo "<header>";
        echo "<h1><a href='mainPage.php'>Marketplace</a></h1>";
        echo "<div class='form-mainHeading'>";
            echo "<div class='form-account'>";
                echo "<a href='mainPage.php'>Return to the store front</a>";
            echo "</div>";
        echo "</div>";
    echo "</header>";
    mysqli_report(MYSQLI_REPORT_ERROR | MYSQLI_REPORT_STRICT);
    $connection = null;

    try{
        $connection = new mysqli("localhost", "root", "", "c2cMarketplacedb");
        echo "<body>";
        $listingCategory = null;
        echo "<div class='form-productPage'>";
            if ($listingDetails = mysqli_fetch_array(mysqli_execute_query($connection,
"SELECT * FROM listing WHERE listingID = ?", [$listingID])){
                $listingCategory = $listingDetails["category"];
                $vendorDetails = mysqli_fetch_array(mysqli_execute_query($connection,
"SELECT * FROM vendor WHERE vendorID = ?", [$listingDetails["vendorID"]]));

                echo "<div class='form-productPageLeft vr'>";
                    echo "<img src='". $listingDetails["image"] .' alt='Product.png'>";
                echo "</div><hr>";

                echo "<div class='form-productPageRight'>";
                    echo "<div class='form-productPageMainHeading'>";
                        echo "<h1>". $listingDetails["title"] . "</h1>";
                        echo "<div class='form-productsGridVendor'>";
                            echo "<h2>". $vendorDetails["username"] . "</h2>";
                            echo "<h3>". $vendorDetails["vendorRating"] . " vendor
rating</h3>";
                        echo "</div>";
                    echo "</div>";

                    echo "<h3 class='form-cost'>R". $listingDetails["price"] . "</h3>";
                    echo "<h3>". $listingDetails["inStock"] . " left in stock</h3><br>";
                    echo "<h3>Description</h3>";
                    echo "<p>". $listingDetails["description"] . "</p><br>";
            }
        }
    }

```

```

        echo "<form action='addToCart.php' method='POST'>";
        if (isset($_SESSION["userID"])){
            $deliveryDetails =
mysqli_fetch_array(mysqli_execute_query($connection, "SELECT * FROM delivery WHERE
userType = ? AND userID = ? AND vendorID = ?", [$_SESSION["userType"], $_SESSION["userID"],
$listingDetails["vendorID"]]));
            if ($listingDetails["vendorID"] == $_SESSION["userID"] &&
$_SESSION["userType"] == 'vendor'){
                echo "<button name='productBtn' id='productBtn' value='-
2'>Your listing</button>";
            }
            else{
                if (!isset($deliveryDetails["deliveryID"])){
                    echo "<input type='hidden' name='vendorID'
id='vendorID' value='". $listingDetails["vendorID"] .'>";
                    echo "<button type='submit' name='productBtn'
id='productBtn' value='". $listingDetails["listingID"] .'>Add to cart</button>";
                }
                else{
                    if ($cartItems = mysqli_execute_query($connection,
"SELECT * FROM cartItem WHERE deliveryID = ? AND listingID = ?",
[$deliveryDetails["deliveryID"], $listingDetails["listingID"]])){
                        if(mysqli_num_rows($cartItems) <= 0){
                            echo "<input type='hidden' name='vendorID'
id='vendorID' value='". $listingDetails["vendorID"] .'>";
                            echo "<button type='submit' name='productBtn'
id='productBtn' value='". $listingDetails["listingID"] .'>Add to cart</button>";
                        }
                        else{
                            echo "<button type='submit' name='productBtn'
id='productBtn' value='-1'>In cart</button>";
                        }
                    }
                }
            }
        }
        else{
            echo "<button disabled>Add to cart</button>";
        }
        echo "</form>";
        echo "</div>";
    }
    echo "</div>";

    if ($relatedRecords = mysqli_execute_query($connection, "SELECT * FROM listing
WHERE category = ? AND NOT listingID = ?", [$listingCategory, $listingID])){
        if(mysqli_num_rows($relatedRecords) > 0){
            echo "<h1 class='form-moveLeft'>Products in the same category</h1>";
            echo "<div class='form-productsGridContainer'>";
            while($recordDetails = mysqli_fetch_array($relatedRecords)){

```

```

        $vendorDetails =
mysqli_fetch_array(mysqli_execute_query($connection, "SELECT * FROM vendor WHERE vendorID
= ?", [$recordDetails["vendorID"]]));
        echo "<div class='form-productsGridItem'>";
            echo "<div class='form-productsGridItemImage'>";
                echo "<form action='productPage.php' method='POST'>";
                    echo "<button type='submit' id='formBtn'
name='formBtn' value='". $recordDetails["listingID"] .'><img src='".
$recordDetails["image"] ." alt='Product.png'></button>";
                    echo "</form>";
                echo "</div>";

            echo "<div class='form-productsGridVendor'>";
                echo "<h2>". $vendorDetails["username"] . "</h2>";
                echo "<h3>". $vendorDetails["vendorRating"] . "</h3>";
            echo "</div>";

            echo "<h3>". $recordDetails["title"] . "</h3>";
            echo "<h4 class='form-cost'>R ".
$recordDetails["price"] . "</h4>";
            echo "<h4>". $recordDetails["category"] . "</h4>";

            echo "<form action='addToCart.php' method='POST'>";
                if (isset($_SESSION["userID"])){
                    $deliveryDetails =
mysqli_fetch_array(mysqli_execute_query($connection, "SELECT * FROM delivery WHERE
userType = ? AND userID = ? AND vendorID = ?", [$_SESSION["userType"], $_SESSION["userID"],
$recordDetails["vendorID"]]));
                    if ($recordDetails["vendorID"] == $_SESSION["userID"]
&& $_SESSION["userType"] == 'vendor'){
                        echo "<button name='productBtn' id='productBtn'
value='-2'>Your listing</button>";
                    }
                    else{
                        if (!isset($deliveryDetails["deliveryID"])){
                            echo "<input type='hidden' name='vendorID'
id='vendorID' value='". $recordDetails["vendorID"] .'>";
                            echo "<button type='submit' name='productBtn'
id='productBtn' value='". $recordDetails["listingID"] .'>Add to cart</button>";
                        }
                        else{
                            if ($cartItems =
mysqli_execute_query($connection, "SELECT * FROM cartItem WHERE deliveryID = ? AND
listingID = ?", [$deliveryDetails["deliveryID"], $recordDetails["listingID"]])){
                                if(mysqli_num_rows($cartItems) <= 0){
                                    echo "<input type='hidden'
name='vendorID' id='vendorID' value='". $recordDetails["vendorID"] .'>";
                                    echo "<button type='submit'
name='productBtn' id='productBtn' value='". $recordDetails["listingID"] .'>Add to
cart</button>";
                                }
                                else{

```

```

                                echo "<button type='submit'
name='productBtn' id='productBtn' value='-1'>In cart</button>";
                                }
                            }
                        }
                    }
                }
            }
        }
    }
    else{
        echo "<button disabled>Add to cart</button>";
    }
    echo "</form>";
    echo "</div>";
}
echo "</div>";
}
}
echo "</body>";
}
catch(Exception $error){
    echo "<h2>An error occurred</h2>";
}

echo "</html>";
}
else{
    echo "<h1>An error occurred</h1>";
}
?>

```

A php form that displays the details of a listing as well as recommending other listings that are in the same category.

addToCart.php

```

<?php
session_start();
$connectionActive = false;
if ($_SERVER["REQUEST_METHOD"] == "POST") {
    $listingID = $_POST["productBtn"];
    $vendorID = $_POST["vendorID"];
    mysqli_report(MYSQLI_REPORT_ERROR | MYSQLI_REPORT_STRICT);
    $connection = null;

    if ($listingID >= 0){
        try{
            $connection = new mysqli("localhost", "root", "", "c2cMarketplacedb");
            mysqli_autocommit($connection, false);
            mysqli_begin_transaction($connection);
            $connectionActive = true;
            $deliveryDetails = null;

```

```

        $deliveryExists = mysqli_execute_query($connection, "SELECT * FROM delivery
WHERE userType = ? AND userID = ? AND vendorID = ?", [$_SESSION["userType"],
$_SESSION["userID"], $vendorID]);

        if (mysqli_num_rows($deliveryExists) > 0){
            $deliveryDetails = mysqli_fetch_array($deliveryExists);
        }
        else{
            mysqli_execute_query($connection, "INSERT INTO delivery (totalCost,
deliveryAddress, userApproval, vendorApproval, userType, userID, vendorID) VALUES
(0, '', 0, 0, ?, ?, ?)", [$_SESSION["userType"], $_SESSION["userID"], $vendorID]);
            $deliveryDetails = mysqli_fetch_array(mysqli_execute_query($connection,
"SELECT * FROM delivery WHERE userType = ? AND userID = ? AND vendorID = ?",
[$_SESSION["userType"], $_SESSION["userID"], $vendorID]));
        }
        mysqli_execute_query($connection, "INSERT INTO cartItem (quantity, listingID,
deliveryID) VALUES (1, ?, ?)", [$listingID, $deliveryDetails["deliveryID"]]);

        $listingDetails = mysqli_fetch_array(mysqli_execute_query($connection, "SELECT
* FROM listing WHERE listingID = ?", [$listingID]));
        mysqli_execute_query($connection, "UPDATE delivery SET totalCost = ? WHERE
deliveryID = ?", [$deliveryDetails["totalCost"] + $listingDetails["price"],
$deliveryDetails["deliveryID"]]);
        $_SESSION["message"] = "Product added to cart successfully";

        mysqli_commit($connection);
    }
    catch(Exception $error){
        $_SESSION["message"] = "Error: ". $error->getMessage();
        if ($connectionActive == true){
            mysqli_rollback($connection);
        }
    }
}

if ($connection != null){
    $connection->close();
}
} else {
    $_SESSION["message"] = "Error: POST error occurred";
}

if ($listingID == -1){
    header('Location:cart.php');
}
else if ($listingID == -2){
    header('Location:accountCenter(ProfileDetails).php');
}
else{
    header('Location:mainPage.php');
}
?>

```

A php file that adds the listing to a delivery associated with the listing's vendor and if a delivery with the vendor doesn't exist then a new delivery is created.

cart.php

```
<?php
session_start();
if (!isset($_SESSION["userID"]) && empty($_SESSION["userID"])){
    header('Location:mainPage.php');
}
$totalCost = 0;
$totalItems = 0;
$cartItems = null;
$connectionActive = false;

echo "<html lang='en'>";
echo "<head>";
    echo "<meta charset='UTF-8'>";
    echo "<meta name='viewport' content='width=device-width, initial-scale=1.0'>";
    echo "<title>Cart</title>";
    echo "<link rel='stylesheet' href='../style.css'>";
echo "</head>";

if (isset($_SESSION["message"]) && !empty($_SESSION["message"])){
    echo "<div id='message' name='message' class='msg'>";
        echo "<p>". $_SESSION["message"] . "</p>";
        echo "<button onclick='closeMessage()'>X</button>";
    echo "</div>";
    unset($_SESSION["message"]);
    echo "<script src='../script.js'></script>";
}

echo "<header>";
    echo "<h1><a href='mainPage.php'>Marketplace</a></h1>";
    echo "<div class='form-mainHeading'>";
        echo "<div class='form-account'>";
            echo "<a href='mainPage.php'>Return to the store front</a>";
        echo "</div>";
    echo "</div>";
echo "</header>";

echo "<body>";
    echo "<div class='form-accountDetails'>";
        echo "<h1>Cart</h1>";
        mysqli_report(MYSQLI_REPORT_ERROR | MYSQLI_REPORT_STRICT);
        $connection = null;

        try{
            $connection = new mysqli("localhost", "root", "", "c2cMarketplacedb");
            $connectionActive = true;
```

```

        echo "<div class='form-cartFlex'>";
        echo "<div class='form-cartFlexItemLeft'>";
            $userDetails = mysqli_fetch_array(mysqli_execute_query($connection,
"SELECT * FROM ". $_SESSION["userType"] ." WHERE ". $_SESSION["userType"] ."ID = ?",
[$_SESSION["userID"]]]));
            $deliveries = mysqli_execute_query($connection, "SELECT * FROM
delivery WHERE userType = ? AND userID = ?",[$_SESSION["userType"], $_SESSION["userID"]]);

            if(mysqli_num_rows($deliveries) > 0){
                while($deliveryDetails = mysqli_fetch_array($deliveries)){
                    $vendorDetails =
mysqli_fetch_array(mysqli_execute_query($connection, "SELECT * FROM vendor WHERE vendorID
= ?", [$deliveryDetails["vendorID"]]]));
                    $cartItems = mysqli_execute_query($connection, "SELECT * FROM
cartItem WHERE deliveryID = ?", [$deliveryDetails["deliveryID"]]);
                    echo "<div class='form-deliveryForms'>";
                        echo "<h2>". $vendorDetails["username"] . "</h2>";
                        echo "<h3 class='form-cost'>Total cost: R".
$deliveryDetails["totalCost"] . "</h3>";

                        echo "<div class='form-
deliveryFormsAddress'>";
                            echo "<form action='updateDeliveryLocation.php'
method='POST'>";
                                echo "<label for='address'>Delivery address:
</label><br>";
                                echo "<input type='hidden' id='userType'
name='userType' value='user'>";
                                echo "<div class='form-deliveryFormsAddressFlex'>";
                                    echo "<input type='text' name='address'
id='address' value='". $deliveryDetails["deliveryAddress"] . "'>";
                                    echo "<button type='submit' name='formBtn'
id='formBtn' value='". $deliveryDetails["deliveryID"] . "'>Update address</button>";
                                    echo "</div>";
                                echo "</form>";

                                echo "<div class='form-deliveryFormsAddressFlex'>";
                                    echo "<label for='userApproval'>Your approval:
</label><br>";
                                    if ($deliveryDetails["userApproval"] == 0){
                                        echo "<input type='checkbox'
name='userApproval' id='userApproval' disabled>";
                                    }
                                    else{
                                        echo "<input type='checkbox'
name='userApproval' id='userApproval' checked disabled>";
                                    }

                                    echo "<label for='vendorApproval'>Vendor approval:
</label><br>";
                                    if ($deliveryDetails["vendorApproval"] == 0){

```

```

        echo "<input type='checkbox'
name='vendorApproval' id='vendorApproval' disabled>";
    }
    else{
        echo "<input type='checkbox'
name='vendorApproval' id='vendorApproval' checked disabled>";
    }
    echo "</div><br>";

    echo "<form action='approveDeliveryLocation.php'
method='POST'>";

        echo "<input type='hidden' id='userType'
name='userType' value='user'>";

        echo "<input type='hidden' id='currentApproval'
name='currentApproval' value='". $deliveryDetails["userApproval"] ."'>";
        if ($deliveryDetails["userApproval"] == 0){
            echo "<button type='submit' name='formBtn'
id='formBtn' value='". $deliveryDetails["deliveryID"] ."'>Approve address</button>";
        }
        else{
            echo "<button type='submit' name='formBtn'
id='formBtn' value='". $deliveryDetails["deliveryID"] ."'>Reject address</button>";
        }
        echo "</form>";
    echo "</div>";

    echo "<div class='form-settingOptionsSideItems'>";
    echo "<form action='deliveryMessages.php'
method='POST'>";

        echo "<input type='hidden' id='deliveryID'
name='deliveryID' value='". $deliveryDetails["deliveryID"] ."'>";
        echo "<input type='hidden' id='otherUserID'
name='otherUserID' value='". $deliveryDetails["vendorID"] ."'>";
        echo "<button type='submit' name='formBtn'
id='formBtn' value='user'>Chat with vendor</button>";
        echo "</form>";

        echo "<form action='../deleteRecord.php'
method='POST'>";

        echo "<input type='hidden' id='userType'
name='userType' value='". $_SESSION["userType"] ."'>";
        echo "<input type='hidden' id='recordType'
name='recordType' value='delivery'>";

        echo "<button type='submit' name='formBtn'
id='formBtn' value='". $deliveryDetails["deliveryID"] ."'>Cancel order delivery</button>";
        echo "</form>";
    echo "</div>";

    echo "<form action='purchaseCartItems.php' method='POST'>";
    if ($deliveryDetails["userApproval"] == 1 &&
$deliveryDetails["vendorApproval"] == 1){

```

```

        echo "<input type='hidden' id='total' name='total'
value='". $deliveryDetails["totalCost"] ."'>";
        echo "<input type='hidden' id='currentAmount'
name='currentAmount' value='". $userDetails["eWalletAmount"] ."'>";
        echo "<button type='submit' name='paymentBtn'
id='paymentBtn' value='". $deliveryDetails["deliveryID"] ."'>Complete payment and
delivery</button>";
    }
    else{
        echo "<button disabled>Complete payment and
delivery</button>";
    }
    echo "</form>";

    while($cartItem = mysqli_fetch_array($cartItems)){
        $productItems =
mysqli_fetch_array(mysqli_execute_query($connection, "SELECT * FROM listing WHERE
listingID = ?", [$cartItem["listingID"]]));

        echo "<div class='form-listingList'>";
        echo "<div class='form-cartLeftItem'>";
            echo "<h3>". $productItems["title"] ."</h3>";
            echo "<img src='". $productItems["image"] ."'
alt='Product.png'>";

        echo "</div>";

        echo "<div class='form-cartRightItem'>";
            echo "<h3 class='form-cost'>Price: R ".
$productItems["price"] * $cartItem["quantity"] ."</h3>";

            echo "<div class='form-quantityBtn'>";
                echo "<form action='updateQuantity.php'
method='POST'>";

                    echo "<input type='hidden'
id='deliveryID' name='deliveryID' value='". $deliveryDetails["deliveryID"] ."'>";
                    echo "<h4>Quantity: </h4><input
type='number' name='quantity' id='quantity' value='". $cartItem["quantity"] ."' min=1
max='". $productItems["inStock"] ."'>";

                    echo "<button type='submit'
name='updateBtn' id='updateBtn' value='". $cartItem["cartItemID"] ."'>Update
quantity</button>";

                echo "</form>";
            echo "</div>";

            echo "<form action='removeFromCart.php'
method='POST'>";

                echo "<input type='hidden' id='deliveryID'
name='deliveryID' value='". $cartItem["deliveryID"] ."'>";
                echo "<button type='submit' class='form-
redBtn' name='removeBtn' id='removeBtn' value='".
$cartItem["cartItemID"] ."'>Remove</button>";
            echo "</form>";

```

```

        echo "</div>";
        echo "</div>";
    }
    echo "</div>";
}
}
echo "</div>";

echo "<div class='form-cartFlexItemRight'>";
    echo "<div class='form-checkout'>";
        echo "<form action='redeemVoucher.php' method='POST'
enctype='multipart/form-data' onsubmit='return validateVoucher()'>";
        echo "<h3>Current funds: R ".
$userDetails["eWalletAmount"] . "</h3>";
        echo "<label for='fundInc'>Voucher amount (in Rands):</input>";
        echo "<div class='form-error'>";
            echo "<input type='number' id='fundInc' name='fundInc'
min=25 max=10000>";
            echo "<small class='error' id='fundError'></small>";
        echo "</div>";
        echo "<button type='submit' name='fundBtn' id='fundBtn'>Redeem
voucher</button>";
        echo "</form>";
    echo "</div>";
    echo "</div>";
    echo "</div>";
}
catch(Exception $error){
    echo "<h2>". $error->getMessage() . "</h2>";
}

if ($connection != null){
    $connection->close();
}
echo "<script src='../script.js'></script>";
echo "</div>";
echo "</body>";
echo "</html>";
?>

```

A php form that enables users to view and alter the details of their deliveries as well as redeem vouchers to add money to their e-wallet.

redeemVoucher.php

```

<?php
session_start();
$success = true;

if ($_SERVER["REQUEST_METHOD"] == "POST") {
    $fundInc = $_POST["fundInc"];

```

```

mysqli_report(MYSQLI_REPORT_ERROR | MYSQLI_REPORT_STRICT);
$connection = null;
$connectionActive = false;

try{
    $connection = new mysqli("localhost", "root", "", "c2cMarketplacedb");
    mysqli_autocommit($connection, false);
    mysqli_begin_transaction($connection);
    $connectionActive = true;

    if (empty($fundInc)){
        $_SESSION["message"] = "Voucher amount is required";
        $success = false;
    }
    else if ($fundInc < 25 || $fundInc > 10000){
        $_SESSION["message"] = "Invalid voucher amount! The voucher amount can only be
between R25 and R10000";
        $success = false;
    }

    if ($success == true) {
        $userDetails = mysqli_fetch_array(mysqli_execute_query($connection, "SELECT *
FROM ". $_SESSION["userType"] ." WHERE ". $_SESSION["userType"] ."ID = ?",
[$_SESSION["userID"]]));
        $totalMoney = $userDetails["eWalletAmount"] + $fundInc;
        mysqli_execute_query($connection, "UPDATE ". $_SESSION["userType"] ." SET
eWalletAmount = ? WHERE ". $_SESSION["userType"] ."ID = ?", [$totalMoney,
$_SESSION["userID"]]);

        $_SESSION["message"] = "Voucher redeemed successfully";
        mysqli_commit($connection);
    }
}
catch(Exception $error){
    $_SESSION["message"] = "Error: ". $error->getMessage();
    if ($connectionActive == true){
        mysqli_rollback($connection);
    }
}

if ($connection != null){
    $connection->close();
}
} else {
    $_SESSION["message"] = "Error: POST error occured";
}

header('Location:cart.php');
?>

```

A php file that adds money to the user's e-wallet.

deliveryMessages.php

```
<?php
session_start();
if ($_SERVER["REQUEST_METHOD"] == "POST") {
    $_SESSION["deliveryID"] = $_POST["deliveryID"];
    $_SESSION["otherUserID"] = $_POST["otherUserID"];
    $_SESSION["userTypeOrigin"] = $_POST["formBtn"];
}

$deliveryID = $_SESSION["deliveryID"];
$otherUserID = $_SESSION["otherUserID"];
$userType = $_SESSION["userTypeOrigin"];

if ((!isset($_SESSION["userID"]) && empty($_SESSION["userID"])) ||
(!isset($_SESSION["deliveryID"]) && empty($_SESSION["deliveryID"]))) {
    unset($_SESSION["deliveryID"]);
    unset($_SESSION["otherUserID"]);
    unset($_SESSION["userTypeOrigin"]);
    header('Location:mainPage.php');
}

unset($_SESSION["deliveryID"]);
unset($_SESSION["otherUserID"]);
unset($_SESSION["userTypeOrigin"]);

echo "<html lang='en'>";
echo "<head>";
    echo "<meta charset='UTF-8'>";
    echo "<meta name='viewport' content='width=device-width, initial-scale=1.0'>";
    echo "<title>Chat form</title>";
    echo "<link rel='stylesheet' href='../style.css'>";
echo "</head>";

if (isset($_SESSION["message"]) && !empty($_SESSION["message"])) {
    echo "<div id='message' name='message' class='msg'>";
        echo "<p>". $_SESSION["message"] . "</p>";
        echo "<button onclick='closeMessage()'>X</button>";
    echo "</div>";
    unset($_SESSION["message"]);
    echo "<script src='../script.js'></script>";
}

echo "<header>";
    echo "<h1><a href='mainPage.php'>Marketplace</a></h1>";
    echo "<div class='form-mainHeading'>";
        echo "<div class='form-account'>";
            if ($userType == "user") {
                echo "<a href='cart.php'>Return to the cart</a>";
            }
            else {
```

```

        echo "<a href='accountCenter(Deliveries).php'>Return to the Account
Center</a>";
    }
    echo "</div>";
    echo "</div>";
echo "</header>";

mysqli_report(MYSQLI_REPORT_ERROR | MYSQLI_REPORT_STRICT);
$connection = null;

echo "<body>";
    echo "<div class='form-accountDetails'>";
        try{
            $connection = new mysqli("localhost", "root", "", "c2cMarketplacedb");
            $deliveryMessages = mysqli_execute_query($connection, "SELECT * FROM
deliveryMessage WHERE deliveryID = ?", [$deliveryID]);
            $deliveryDetails = mysqli_fetch_array(mysqli_execute_query($connection,
"SELECT * FROM delivery WHERE deliveryID = ?", [$deliveryID]));

            if ($userType == 'user'){
                $otherUserDetails = mysqli_fetch_array(mysqli_execute_query($connection,
"SELECT * FROM vendor WHERE vendorID = ?", [$otherUserID]));
            }
            else{
                $otherUserDetails = mysqli_fetch_array(mysqli_execute_query($connection,
"SELECT * FROM ". $deliveryDetails["userType"] ." WHERE ".
$deliveryDetails["userType"] ."ID = ?", [$otherUserID]));
            }

            echo "<h1>". $otherUserDetails["username"] . "</h1>";
            echo "<hr>";
            echo "<div class='form-cartFlexItemLeft'>";
                if (mysqli_num_rows($deliveryMessages) > 0){
                    while ($deliveryMessage = mysqli_fetch_array($deliveryMessages)){
                        if ($deliveryMessage["userTypeOrigin"] != $userType){
                            echo "<div class='form-leftMessage'>";
                                echo "<p class='form-message '>".
$deliveryMessage["message"] . "</p>";
                                echo "</div>";

                                echo "<div class='form-leftMessage form-leftTimeStamp'>";
                                    echo "<p class='form-message'>".
$deliveryMessage["timeSent"] . "</p><br>";
                                    echo "</div>";
                                }
                            else{
                                echo "<div class='form-rightMessage'>";
                                    echo "<p class='form-message '>".
$deliveryMessage["message"] . "</p>";
                                    echo "</div>";

                                    echo "<div class='form-rightMessage form-rightTimeStamp'>";

```

```

        echo "<p class='form-message'>".
$deliveryMessage["timeSent"] . "</p><br>";
        echo "</div>";
    }
}
}
else{
    echo "<h1>No messages have been sent yet. You can be the first!</h1>";
}
echo "</div>";

    echo "<form action='createMessage.php' method='POST'>";
        echo "<input type='hidden' id='otherUserID' name='otherUserID'
value='$otherUserID'>";
        echo "<input type='hidden' id='userType' name='userType'
value='$userType'>";
        echo "<div class='form-inputAndBtn'>";
            echo "<input type='text' id='message' name='message' placeholder='Type
a message'>";
            echo "<button type='submit' name='formBtn' id='formBtn'
value='$deliveryID'>&rarr;</button>";
            echo "</div>";
        echo "</form>";
    }
    catch(Exception $error){
        echo "<h2>An error occured</h2>";
    }

    if ($connection != null){
        $connection->close();
    }
    echo "<script src='../script.js'></script>";
    echo "</div>";
echo "</body>";
echo "</html>";
?>

```

A php form where the buyer and vendor on a delivery can communicate with each other by sending messages.

createMessage.php

```

<?php
$success = false;
$connectionActive = false;
session_start();

if ($_SERVER["REQUEST_METHOD"] == "POST") {
    $userType = $_POST["userType"];
    $message = $_POST["message"];
    $deliveryID = $_POST["formBtn"];

```

```

$_SESSION["userTypeOrigin"] = $userType;
$_SESSION["otherUserID"] = $_POST["otherUserID"];
$_SESSION["deliveryID"] = $deliveryID;

mysqli_report(MYSQLI_REPORT_ERROR | MYSQLI_REPORT_STRICT);
$connection = null;
try{
    $connection = new mysqli("localhost", "root", "", "c2cMarketplacedb");
    mysqli_autocommit($connection, false);
    mysqli_begin_transaction($connection);
    $connectionActive = true;

    mysqli_execute_query($connection, "INSERT INTO deliveryMessage (message, timeSent,
userTypeOrigin, deliveryID) VALUES (?, ?, ?, ?)", [$message, date("Y-m-d H:i"), $userType,
$deliveryID]);
    mysqli_commit($connection);
    $success = true;
}
catch(Exception $error){
    $_SESSION["message"] = "Error: ". $error->getMessage();
    if ($connectionActive == true){
        mysqli_rollback($connection);
    }
}

if ($connection != null){
    $connection->close();
}
} else {
    $_SESSION["message"] = "Error: POST error occurred";
}

if ($success == true){
    header('Location:deliveryMessages.php');
}
else{
    if ($userType == "user"){
        header('Location:cart.php');
    }
    else{
        header('Location:accountCenter(Deliveries).php');
    }
}
?>

```

(W3Schools, n.d.)

A php file that stores the message sent by the user and when it was sent.

updateDeliveryLocation.php

```

<?php
session_start();
if ($_SERVER["REQUEST_METHOD"] == "POST") {
    $userType = $_POST["userType"];
    $address = $_POST["address"];
    $deliveryID = $_POST["formBtn"];
    $connectionActive = false;
    mysqli_report(MYSQLI_REPORT_ERROR | MYSQLI_REPORT_STRICT);
    $connection = null;

    try{
        $connection = new mysqli("localhost", "root", "", "c2cMarketplacedb");
        mysqli_autocommit($connection, false);
        mysqli_begin_transaction($connection);
        $connectionActive = true;

        mysqli_execute_query($connection, "UPDATE delivery SET deliveryAddress = ? WHERE
deliveryID = ?", [$address, $deliveryID]);
        mysqli_execute_query($connection, "UPDATE delivery SET userApproval = 0 WHERE
deliveryID = ?", [$deliveryID]);
        mysqli_execute_query($connection, "UPDATE delivery SET vendorApproval = 0 WHERE
deliveryID = ?", [$deliveryID]);
        $_SESSION["message"] = "Delivery address has been updated successfully";
        mysqli_commit($connection);
    }
    catch(Exception $error){
        $_SESSION["message"] = "Error: ". $error->getMessage();
        if ($connectionActive == true){
            mysqli_rollback($connection);
        }
    }

    if ($connection != null){
        $connection->close();
    }
} else {
    $_SESSION["message"] = "Error: POST error occured";
}

if ($userType == "user"){
    header('Location:cart.php');
}
else{
    header('Location:accountCenter(Deliveries).php');
}
?>

```

A php file that updates the delivery address of a delivery.

approveDeliveryLocation.php

```

<?php

```

```

$user = "user";
$connectionActive = false;
session_start();

if ($_SERVER["REQUEST_METHOD"] == "POST") {
    $approval = 0;
    if ($_POST["currentApproval"] == 0){
        $approval = 1;
    }
    $deliveryID = $_POST["formBtn"];
    $user = $_POST["userType"];

    try{
        $connection = new mysqli("localhost", "root", "", "c2cMarketplacedb");
        mysqli_autocommit($connection, false);
        mysqli_begin_transaction($connection);
        $connectionActive = true;

        mysqli_execute_query($connection, "UPDATE delivery SET ". $user ."Approval = ?
WHERE ". $user ."ID = ?", [$approval, $_SESSION["userID"]]);
        $_SESSION["message"] = "Delivery location approved successfully";
        mysqli_commit($connection);
    }
    catch(Exception $error){
        $_SESSION["message"] = "Error: ". $error->getMessage();
        if ($connectionActive == true){
            mysqli_rollback($connection);
        }
    }

    if ($connection != null){
        $connection->close();
    }
} else {
    $_SESSION["message"] = "Error: POST error occurred";
}

if ($user == "user"){
    header('Location:cart.php');
}
else{
    header('Location:accountCenter(Deliveries).php');
}
?>

```

A php file that confirms that the buyer or vendor approves of the current delivery address of the delivery.

updateQuantity.php

```
<?php
```

```

session_start();
if ($_SERVER["REQUEST_METHOD"] == "POST") {
    $cartItemID = $_POST["updateBtn"];
    $quantity = $_POST["quantity"];
    $deliveryID = $_POST["deliveryID"];
    $connectionActive = false;
    mysqli_report(MYSQLI_REPORT_ERROR | MYSQLI_REPORT_STRICT);
    $connection = null;

    try{
        $connection = new mysqli("localhost", "root", "", "c2cMarketplacedb");
        mysqli_autocommit($connection, false);
        mysqli_begin_transaction($connection);
        $connectionActive = true;
        $cartItem = mysqli_fetch_array(mysqli_execute_query($connection, "SELECT * FROM
cartItem WHERE cartItemID = ?", [$cartItemID]));
        $listingDetails = mysqli_fetch_array(mysqli_execute_query($connection, "SELECT *
FROM listing WHERE listingID = ?", [$cartItem["listingID"]]));

        if ($quantity != null){
            if ($quantity <= 0){
                $_SESSION["message"] = "Error: Invalid quantity";
            }
            else if ($quantity > $listingDetails["inStock"]){
                $_SESSION["message"] = "Error: Quantity exceeds listing's stock";
            }
        }
        else{
            $_SESSION["message"] = "Error: Quantity is required";
        }

        if (empty($errors)) {
            mysqli_execute_query($connection, "Update cartItem SET quantity = $quantity
WHERE cartItemID = ?", [$cartItemID]);

            $totalCost = 0;
            $deliveryCartItems = mysqli_execute_query($connection, "SELECT * FROM cartItem
WHERE deliveryID = ?", [$deliveryID]);
            while($deliveryCartItem = mysqli_fetch_array($deliveryCartItems)){
                $listingPrice = mysqli_fetch_array(mysqli_execute_query($connection,
"SELECT price FROM listing WHERE listingID = ?", [$deliveryCartItem["listingID"]]));
                $totalCost += ($listingPrice["price"] * $deliveryCartItem["quantity"]);
            }
            mysqli_execute_query($connection, "UPDATE delivery SET totalCost = $totalCost
WHERE deliveryID = ?", [$deliveryID]);

            $_SESSION["message"] = "Product's quantity has been updated successfully";
            mysqli_commit($connection);
        }
    }
    catch(Exception $error){
        $_SESSION["message"] = "Error: ". $error->getMessage();
    }
}

```

```

        if ($connectionActive == true){
            mysqli_rollback($connection);
        }
    }

    if ($connection != null){
        $connection->close();
    }
} else {
    $_SESSION["message"] = "Error: POST error occurred";
}

header('Location:cart.php');
?>

```

A php file that updates the quantity of a listing in a delivery that alters the total price of the delivery.

removeFromCart.php

```

<?php
session_start();
if ($_SERVER["REQUEST_METHOD"] == "POST") {
    $cartItemID = $_POST["removeBtn"];
    $deliveryID = $_POST["deliveryID"];
    $connectionActive = false;
    mysqli_report(MYSQLI_REPORT_ERROR | MYSQLI_REPORT_STRICT);
    $connection = null;

    try {
        $connection = new mysqli("localhost", "root", "", "c2cMarketplacedb");
        mysqli_autocommit($connection, false);
        mysqli_begin_transaction($connection);
        $connectionActive = true;

        $cartItem = mysqli_fetch_array(mysqli_execute_query($connection, "SELECT * FROM
cartItem WHERE cartItemID = ?", [$cartItemID]));
        mysqli_execute_query($connection, "Delete FROM cartItem WHERE cartItemID = ?",
[$cartItemID]);

        $deliveryCartItems = mysqli_execute_query($connection, "SELECT * FROM cartItem
WHERE deliveryID = ?", [$deliveryID]);
        if (mysqli_num_rows($deliveryCartItems) <= 0){
            mysqli_execute_query($connection, "DELETE FROM delivery WHERE deliveryID = ?",
[$deliveryID]);
            mysqli_execute_query($connection, "DELETE FROM deliveryMessage WHERE
deliveryID = ?", [$deliveryID]);
        }
        else{
            $deliveryDetails = mysqli_fetch_array(mysqli_execute_query($connection,
"SELECT * FROM delivery WHERE deliveryID = ?", [$deliveryID]));
            $listingDetails = mysqli_fetch_array(mysqli_execute_query($connection, "SELECT
* FROM listing WHERE listingID = ?", [$cartItem["listingID"]]));

```

```

        mysqli_execute_query($connection, "UPDATE delivery SET totalCost = ? WHERE
deliveryID = ?", [($deliveryDetails["totalCost"] - ($listingDetails["price"] *
$cartItem["quantity"])), $deliveryID]);
    }

    $_SESSION["message"] = "Product removed from cart successfully";
    mysqli_commit($connection);
}
catch(Exception $error) {
    $_SESSION["message"] = "Error: ". $error->getMessage();
    if ($connectionActive == true){
        mysqli_rollback($connection);
    }
}

if ($connection != null){
    $connection->close();
}
} else {
    $_SESSION["message"] = "Error: POST error occurred";
}

header('Location:cart.php');
?>

```

A php file that removes a listing from a delivery and if there are no longer any listings in the delivery then the delivery also gets deleted.

purchaseCartItems.php

```

<?php
if ($_SERVER["REQUEST_METHOD"] == "POST") {
    session_start();
    $success = true;
    $connectionActive = false;
    $total = $_POST["total"];
    $currentAmount = $_POST["currentAmount"];
    $deliveryID = $_POST["paymentBtn"];

    if ($total > $currentAmount){
        $_SESSION["message"] = "You don't have enough funds";
        $success = false;
    }
    mysqli_report(MYSQLI_REPORT_ERROR | MYSQLI_REPORT_STRICT);
    $connection = null;

    if ($success == true) {
        try{
            $connection = new mysqli("localhost", "root", "", "c2cMarketplacedb");
            mysqli_autocommit($connection, false);
            mysqli_begin_transaction($connection);

```

```

        $connectionActive = true;

        $userDetails = mysqli_fetch_array(mysqli_execute_query($connection, "SELECT *
FROM ". $_SESSION["userType"] ." WHERE ". $_SESSION["userType"] ."ID = ?",
[$_SESSION["userID"]]]);
        $deliveryDetails = mysqli_fetch_array(mysqli_execute_query($connection,
"SELECT * FROM delivery WHERE deliveryID = ?", [$deliveryID]));
        $newWallet = $userDetails["eWalletAmount"] - $total;
        mysqli_execute_query($connection, "UPDATE ". $_SESSION["userType"] ." SET
eWalletAmount = ? WHERE $newWallet = ?", [$SESSION["userType"] ."ID",
$_SESSION["userID"]]);
        $cartItems = mysqli_execute_query($connection, "SELECT * FROM cartItem WHERE
deliveryID = ?", [$deliveryID]);

        while($cartItem = mysqli_fetch_array($cartItems)){
            $listingDetails = mysqli_fetch_array(mysqli_execute_query($connection,
"SELECT * FROM listing WHERE listingID = ?", [$cartItem["listingID"]]]);
            $newStock = $listingDetails["inStock"] - $cartItem["quantity"];
            $vendorDetails = mysqli_fetch_array(mysqli_execute_query($connection,
"SELECT * FROM vendor WHERE vendorID = ?", [$listingDetails["vendorID"]]]);
            $newVendorWallet = $vendorDetails["eWalletAmount"] +
($listingDetails["price"] * $cartItem["quantity"]);
            mysqli_execute_query($connection, "UPDATE listing SET inStock = ? WHERE
listingID = ?", [$newStock, $cartItem["listingID"]]);
            mysqli_execute_query($connection, "UPDATE vendor SET eWalletAmount = ?
WHERE vendorID = ?", [$newVendorWallet, $vendorDetails["vendorID"]]);
        }

        mysqli_execute_query($connection, "Delete FROM cartItem WHERE deliveryID = ?",
[$deliveryID]);
        mysqli_execute_query($connection, "Delete FROM delivery WHERE deliveryID = ?",
[$deliveryID]);
        mysqli_execute_query($connection, "Delete FROM deliveryMessage WHERE
deliveryID = ?", [$deliveryID]);
        $ratingHistory = mysqli_execute_query($connection, "SELECT * FROM
userRatingHistory WHERE userType = ? AND userID = ? AND vendorID = ?",
[$deliveryDetails["userType"], $deliveryDetails["userID"], $deliveryDetails["vendorID"]]);
        if (mysqli_num_rows($ratingHistory) <= 0){
            mysqli_execute_query($connection, "INSERT INTO userRatingHistory (rated,
ratingAmount, ratingType, userType, userID, vendorID) VALUES (0, 0, 'buyer', ?, ?, ?)",
[$deliveryDetails["userType"], $deliveryDetails["userID"], $deliveryDetails["vendorID"]]);
            mysqli_execute_query($connection, "INSERT INTO userRatingHistory (rated,
ratingAmount, ratingType, userType, userID, vendorID) VALUES (0, 0, 'vendor', ?, ?, ?)",
[$deliveryDetails["userType"], $deliveryDetails["userID"],
$deliveryDetails["vendorID"]]);
        }

        $_SESSION["message"] = "Purchase was successful";
        mysqli_commit($connection);
    }
    catch(Exception $error){
        $_SESSION["message"] = "Error: ". $error->getMessage();
    }

```

```

        if ($connectionActive == true){
            mysqli_rollback($connection);
        }
    }

    if ($connection != null){
        $connection->close();
    }
} else {
    $_SESSION["message"] = "Error: POST error occurred";
}

header('Location:cart.php');
?>

```

A php file that validates that the user can pay for a delivery, completes the delivery and enables both the buyer and vendor to rate each other if they haven't interacted before.

createListing.php

```

<?php
session_start();
echo "<link rel='stylesheet' href='../style.css'>";
echo "<div class='form-accountDetails'>";
$success = false;
$connectionActive = false;

if ($_SERVER["REQUEST_METHOD"] == "POST") {
    mysqli_report(MYSQLI_REPORT_ERROR | MYSQLI_REPORT_STRICT);
    $connection = null;

    try{
        $connection = new mysqli("localhost", "root", "", "c2cMarketplacedb");
        mysqli_autocommit($connection, false);
        mysqli_begin_transaction($connection);
        $connectionActive = true;

        $title = $_POST["title"];
        $price = $_POST["price"];
        $inStock = $_POST["inStock"];
        $description = $_POST["description"];
        $category = $_POST["productCategories"];
        $picDir = "../images/";
        $picName = basename($_FILES["productVisual"]["name"]);
        $picFilePath = $picDir . $picName;
        $errors = [];

        if (empty($title)) {
            $errors[] = "Username is required";
        }
    }
}

```

```

    if (empty($price)) {
        $errors[] = "Price is required";
    }
    else if ($price < 0){
        $errors[] = "Price can't be negative";
    }

    if (empty($inStock)) {
        $errors[] = "Stock is required";
    }
    else if ($inStock < 1){
        $errors[] = "Stock can't be negative or zero";
    }

    if (empty($errors)) {
        $fileNameFound = false;
        $newFileName = $picFilePath;
        $extraCharacters = "";
        $loopNum = 0;

        do{
            if ($extraCharacters != ""){
                $characterParts = explode('.', $picFilePath);
                $extension = $characterParts[count($characterParts) - 1];
                array_pop($characterParts);
                $newFileName = implode('.', $characterParts) . $extraCharacters .
"." . $extension;
            }

            if (file_exists($newFileName)){
                $fileNameFound = true;
                $loopNum++;
                $extraCharacters = "(" . $loopNum . ")";
            }
            else{
                $fileNameFound = false;
                $picFilePath = $newFileName;
            }
        } while($fileNameFound == true);

        mysqli_execute_query($connection, "INSERT INTO listing (title, price,
description, inStock, category, image, vendorID) VALUES (?, ?, ?, ?, ?, ?, ?)", [$title,
$price, $description, $inStock, $category, $picFilePath, $_SESSION["userID"]]);
        $_SESSION["message"] = "Listing hosted successfully";
        $success = true;

        mysqli_commit($connection);
        move_uploaded_file($_FILES["productVisual"]["tmp_name"], $picFilePath);
    } else {
        echo "<h1>Listing error</h1>";
        echo "<h3>Listing couldn't be registered due to the following error(s):</h3>";
        echo "<div class='form-errorBoard error'>";
    }

```

```

        foreach ($errors as $error) {
            echo "<h4>*$error</h4>";
        }
        echo "</div>";
        echo "<a href='createListing.html' class='form-returnLink'>Return to
listing</a>";
    }
}
catch(Exception $error){
    echo "<h1>Listing error</h1>";
    echo "<h3>Listing couldn't be registered due to the following error(s):</h3>";
    echo "<div class='form-errorBoard error'>";
        echo "<h4>*Error: ". $error->getMessage() . "</h4>";
    echo "</div>";
    echo "<a href='createListing.html' class='form-returnLink'>Return to listing</a>";

    if ($connectionActive == true){
        mysqli_rollback($connection);
    }
}

if ($connection != null){
    $connection->close();
}
} else {
    $success = true;
    $_SESSION["message"] = "Error: POST error occurred";
}
echo "</div>";

if ($success == true){
    header('Location:mainPage.php');
}
?>

```

(Kumar, 2026)

A php file that validates and inserts the listing data into the listing database table.

accountCentre(ProfileDetails).php

```

<?php
session_start();
if (!isset($_SESSION["userID"]) && empty($_SESSION["userID"])){
    header('Location:mainPage.php');
}

$username = $_SESSION["username"];
$firstName = $_SESSION["firstName"];
$lastName = $_SESSION["lastName"];
$phoneNum = $_SESSION["phoneNum"];
$email = $_SESSION["email"];

```

```

$address = $_SESSION["address"];

echo "<html lang='en'>";
echo "<head>";
    echo "<meta charset='UTF-8'>";
    echo "<meta name='viewport' content='width=device-width, initial-scale=1.0'>";
    echo "<title>My Account</title>";
    echo "<link rel='stylesheet' href='../style.css'>";
echo "</head>";

if (isset($_SESSION["message"]) && !empty($_SESSION["message"])){
    echo "<div id='message' name='message' class='msg'>";
        echo "<p>". $_SESSION["message"] . "</p>";
        echo "<button onclick='closeMessage()'>X</button>";
    echo "</div>";
    unset($_SESSION["message"]);
    echo "<script src='../script.js'></script>";
}

echo "<header>";
    echo "<h1><a href='mainPage.php'>Marketplace</a></h1>";
    echo "<div class='form-mainHeading'>";
        echo "<div class='form-account'>";
            echo "<a href='mainPage.php'>Return to the store front</a>";
            echo "</div>";
        echo "</div>";
echo "</header>";

echo "<body>";
    echo "<div class='form-accountCenter'>";
        echo "<div class='form-accountCenterOptions'>";
            echo "<h1 >Account Center</h1>";
            echo "<div class='form-accountCenterOptionsItem form-accCenterOptionPicked'>";
                echo "<a href='accountCenter(ProfileDetails).php'>Profile details</a>";
            echo "</div>";
            echo "<div class='form-accountCenterOptionsItem'>";
                echo "<a href='accountCenter(PassAndSec).php'>Password and security</a>";
            echo "</div>";
            echo "<div class='form-accountCenterOptionsItem'>";
                echo "<a href='accountCenter(UserRatings).php'>User ratings</a>";
            echo "</div>";

            if ($_SESSION["userType"] == "vendor"){
                echo "<div class='form-accountCenterOptionsItem'>";
                    echo "<a href='accountCenter(Deliveries).php'>Ongoing deliveries</a>";
                echo "</div>";
            }
        echo "</div>";

        echo "<div class='form-settingOptions'>";
            echo "<div class='form-settingOptionsSideItems'>";
                echo "<div class='form-inputDetails'>";

```

```

        echo "<label for='username'>Current username:</label><br>";
        echo "<input type='text' name='username' id='username'
value='$username' readonly>";
        echo "</div>";

        echo "<div class='form-inputDetails'>";
        echo "<form action='../accountLogOut.php' method='POST'>";
        echo "<button>Log out</button>";
        echo "</form>";
        echo "</div>";
echo "</div>";

echo "<div class='form-settingOptionsItem'>";
    echo "<div class='form-inputDetails'>";
        echo "<label for='fName'>First name: </label><br>";
        echo "<input type='text' name='fName' id='fName' value='$firstName'
readonly>";
        echo "</div>";
        echo "<div class='form-inputDetails'>";
            echo "<label for='lName'>Last name: </label><br>";
            echo "<input type='text' name='lName' id='lName' value='$lastName'
readonly>";
            echo "</div>";
        echo "</div>";

        echo "<div class='form-settingOptionsItem'>";
        echo "<div class='form-inputDetails'>";
            echo "<label for='phoneNum'>Phone number: </label><br>";
            echo "<input type='tel' name='phoneNum' id='phoneNum'
value='$phoneNum' readonly>";
            echo "</div>";

            echo "<div class='form-inputDetails'>";
                echo "<label for='email'>Email: </label><br>";
                echo "<input type='email' name='email' id='email' value='$email'
readonly>";
                echo "</div>";
            echo "</div>";

            echo "<div class='form-settingOptionsItem'>";
            echo "<div class='form-longInputDetails'>";
                echo "<label for='address'>Address: </label><br>";
                echo "<input type='text' name='address' id='address' value='$address'
readonly>";
                echo "</div>";
            echo "</div>";

            if ($_SESSION["userType"] == "buyer"){
                echo "<div class='form-settingOptionsItem'>";
                echo "<div class='form-inputDetails'>";
                    echo "<form action='vendorRegistration.php' method='POST'>";
                    echo "<button>Become a vendor</button>";

```

```

        echo "</form>";
        echo "</div>";
        echo "</div>";
    }
    mysqli_report(MYSQLI_REPORT_ERROR | MYSQLI_REPORT_STRICT);
    $connection = null;

    try{
        $connection = new mysqli("localhost", "root", "", "c2cMarketplacedb");
        $listingItems = mysqli_execute_query($connection, "SELECT * FROM listing
WHERE vendorID = ?", [$SESSION["userID"]]);

        if(mysqli_num_rows($listingItems) > 0){
            echo "<h1>Current listings</h1>";
            while($recordDetails = mysqli_fetch_array($listingItems)){
                echo "<div class='form-listingList'>";
                echo "<div class='form-cartLeftItem'>";
                echo "<h2>". $recordDetails["title"] . "</h2>";
                echo "<img src='". $recordDetails["image"] . "'
alt='Product.png'>";
                echo "</div>";

                echo "<div class='form-cartRightItem'>";
                echo "<h3>Price: R ". $recordDetails["price"] . "</h3>";
                echo "<form action='updateStock.php' method='POST'>";
                echo "<h4>Stock: <input type='number' name='stock'
id='stock' value='". $recordDetails["inStock"] . "' min=0></h4>";
                echo "<button type='submit' name='updateBtn'
id='updateBtn' value='". $recordDetails["listingID"] . "'>Update stock</button>";
                echo "</form>";

                echo "<form action='../deleteRecord.php' method='POST'>";
                echo "<input type='hidden' value='listing'
name='recordType' id='recordType'>";
                echo "<button type='submit' name='formBtn'
id='formBtn' value='". $recordDetails["listingID"] . "'>Remove listing</button>";
                echo "</form>";
                echo "</div>";
            }
        }
    }
    catch(Exception $error){
        echo "<h2>An error occured</h2>";
    }

    if ($connection != null){
        $connection->close();
    }
    echo "</div>";
    echo "</div>";
    echo "</body>";

```

```
echo "</html>";  
?>
```

A php form that displays the user's information, enables them log out and become a vendor if they are a buyer as well as displays all their listings if they are a vendor.

updateStock.php

```
<?php  
session_start();  
if ($_SERVER["REQUEST_METHOD"] == "POST") {  
    $success = true;  
    $listingID = $_POST["updateBtn"];  
    $stock = $_POST["stock"];  
    $connectionActive = false;  
    mysqli_report(MYSQLI_REPORT_ERROR | MYSQLI_REPORT_STRICT);  
    $connection = null;  
  
    try{  
        $connection = new mysqli("localhost", "root", "", "c2cMarketplacedb");  
        mysqli_autocommit($connection, false);  
        mysqli_begin_transaction($connection);  
        $connectionActive = true;  
  
        if ($stock != null){  
            if ($stock < 1){  
                $_SESSION["message"] = "Error: Invalid stock";  
                $success = false;  
            }  
        }  
        else{  
            $_SESSION["message"] = "Error: stock is required";  
            $success = false;  
        }  
  
        if ($success == true) {  
            mysqli_execute_query($connection, "Update listing SET inStock = $stock WHERE  
listingID = ?", [$listingID]);  
  
            $listingItem = mysqli_fetch_array(mysqli_execute_query($connection, "SELECT *  
FROM listing WHERE listingID = ?", [$listingID]));  
            $cartItems = mysqli_execute_query($connection, "SELECT * FROM cartItem WHERE  
listingID = ?", [$listingID]);  
            while($cartItem = mysqli_fetch_array($cartItems)){  
                if ($cartItem["quantity"] > $listingItem["inStock"]){  
                    $quantityReduction = $cartItem["quantity"] - $listingItem["inStock"];  
                    $totalCost = mysqli_fetch_array(mysqli_execute_query($connection,  
"SELECT totalCost FROM delivery WHERE deliveryID = ?", [$cartItem["deliveryID"]]]);  
                    $newCost = $totalCost["totalCost"] - ($listingItem["price"] *  
$quantityReduction);
```

```

        mysqli_execute_query($connection, "UPDATE cartItem SET quantity = ?
WHERE cartItemID = ?", [($cartItem["quantity"] - $quantityReduction),
$cartItem["cartItemID"]]);
        mysqli_execute_query($connection, "UPDATE delivery SET totalCost = ?
WHERE deliveryID = ?", [$newCost, $cartItem["deliveryID"]]);
    }
}

$_SESSION["message"] = "Listing's stock has been updated successfully";
mysqli_commit($connection);
}
}
catch(Exception $error){
    $_SESSION["message"] = "Error: ". $error->getMessage();
    if ($connectionActive == true){
        mysqli_rollback($connection);
    }
}

if ($connection != null){
    $connection->close();
}
} else {
    $_SESSION["message"] = "Error: POST error occurred";
}

header('Location:accountCenter(ProfileDetails).php');
?>

```

A php file that validates and updates the current stock of a specific listing.

vendorRegistration.php

```

<?php
session_start();

if ($_SERVER["REQUEST_METHOD"] == "POST") {
    $connectionActive = false;
    mysqli_report(MYSQLI_REPORT_ERROR | MYSQLI_REPORT_STRICT);
    $connection = null;

    try{
        $connection = new mysqli("localhost", "root", "", "c2cMarketplacedb");
        mysqli_autocommit($connection, false);
        mysqli_begin_transaction($connection);
        $connectionActive = true;

        $userDetails = mysqli_fetch_array(mysqli_execute_query($connection, "SELECT * FROM
buyer WHERE buyerID = ?", [$_SESSION["userID"]]));
        mysqli_execute_query($connection, "INSERT INTO vendor (firstName, lastName,
username, phoneNum, email, address, password, buyerRating, eWalletAmount, vendorRating,
notify) VALUES (?, ?, ?, ?, ?, ?, ?, ?, ?, 0, ?)", [$userDetails["firstName"],

```

```

$userDetails["lastName"], $userDetails["username"], $userDetails["phoneNum"],
$userDetails["email"], $userDetails["address"], $userDetails["password"],
$userDetails["buyerRating"], $userDetails["eWalletAmount"], $userDetails["notify"]]);
    $vendorDetails = mysqli_fetch_array(mysqli_execute_query($connection, "SELECT
vendorID FROM vendor WHERE username = ?", [$SESSION["username"]]));

    mysqli_execute_query($connection, "Update delivery SET userID = ? WHERE userID = ?
AND userType = 'buyer'", [$vendorDetails["vendorID"], $userDetails["buyerID"]]);
    mysqli_execute_query($connection, "Update delivery SET userType = 'vendor' WHERE
userID = ? AND userType = 'buyer'", [$vendorDetails["vendorID"]]);
    mysqli_execute_query($connection, "Update userRatingHistory SET userID = ? WHERE
userID = ?", [$vendorDetails["vendorID"], $userDetails["buyerID"]]);
    mysqli_execute_query($connection, "Update userRatingHistory SET userType =
'vendor' WHERE userID = ?", [$vendorDetails["vendorID"]]);
    mysqli_execute_query($connection, "Delete FROM buyer WHERE buyerID = ?",
[$userDetails["buyerID"]]);

    $_SESSION["message"] = "Account registered as a vendor successfully";
    mysqli_commit($connection);
    $_SESSION["userID"] = $vendorDetails["vendorID"];
    $_SESSION["userType"] = "vendor";
}
catch(Exception $error){
    $_SESSION["message"] = "Error: ". $error->getMessage();
    if ($connectionActive == true){
        mysqli_rollback($connection);
    }
}

if ($connection != null){
    $connection->close();
}
}
else{
    $_SESSION["message"] = "Error: POST error occurred";
}

header('Location:mainPage.php');
?>

```

A php file that transitions a buyer into a vendor.

accountCentre(PassAndSec).php

```

<?php
session_start();
if (!isset($_SESSION["userID"]) && empty($_SESSION["userID"])){
    header('Location:mainPage.php');
}
echo "<html lang='en'>";

echo "<head>";

```

```

echo "<meta charset='UTF-8'>";
echo "<meta name='viewport' content='width=device-width, initial-scale=1.0'>";
echo "<title>My Account</title>";
echo "<link rel='stylesheet' href='../style.css'>";
echo "</head>";

if (isset($_SESSION["message"]) && !empty($_SESSION["message"])){
    echo "<div id='message' name='message' class='msg'>";
        echo "<p>". $_SESSION["message"] . "</p>";
        echo "<button onclick='closeMessage()'>X</button>";
    echo "</div>";
    unset($_SESSION["message"]);
    echo "<script src='../script.js'></script>";
}

echo "<header>";
    echo "<h1><a href='mainPage.php'>Marketplace</a></h1>";
    echo "<div class='form-mainHeading'>";
        echo "<div class='form-account'>";
            echo "<a href='mainPage.php'>Return to the store front</a>";
            echo "</div>";
        echo "</div>";
echo "</header>";

echo "<body>";
    echo "<div class='form-accountCenter'>";
        echo "<div class='vr form-accountCenterOptions'>";
            echo "<h1 >Account Center</h1>";
            echo "<div class='form-accountCenterOptionsItem'>";
                echo "<a href='accountCenter(ProfileDetails).php'>Profile details</a>";
            echo "</div>";
            echo "<div class='form-accountCenterOptionsItem form-accCenterOptionPicked'>";
                echo "<a href='accountCenter(PassAndSec).php'>Password and security</a>";
            echo "</div>";
            echo "<div class='form-accountCenterOptionsItem'>";
                echo "<a href='accountCenter(UserRatings).php'>User ratings</a>";
            echo "</div>";

            if ($_SESSION["userType"] == "vendor"){
                echo "<div class='form-accountCenterOptionsItem'>";
                    echo "<a href='accountCenter(Deliveries).php'>Ongoing deliveries</a>";
                echo "</div>";
            }
        echo "</div>";

        echo "<div class='form-settingOptions'>";
            echo "<div class='form-settingOptionsSideItems'>";
                echo "<div class='form-inputDetails'>";
                    echo "<form action='../changePasswordForm.php' method='POST'>";
                    echo "<input type='hidden' value='". $_SESSION["userType"] . "'";
name='userType' id='userType'>";

```

```

        echo "<button type='submit' value='". $_SESSION["userID"] ."'
id='formBtn' name='formBtn'>Change password</button>";
        echo "</form>";
        echo "</div>";

        echo "<div class='form-inputDetails'>";
        echo "<form action='../deleteRecord.php' method='POST'>";
        echo "<input type='hidden' value='". $_SESSION["userType"] ."'
name='recordType' id='recordType'>";
        echo "<button type='submit' value='". $_SESSION["userID"] ."'
id='formBtn' name='formBtn'>Delete account</button>";
        echo "</form>";
        echo "</div>";
        echo "</div>";
        echo "</div>";
echo "</body>";

echo "</html>";
?>

```

A php form that enables the user to change their password or delete their account.

changePasswordForm.php

```

<?php
$success = false;
session_start();

if ($_SERVER["REQUEST_METHOD"] == "POST") {
    $userID = $_POST["formBtn"];
    $userType = $_POST["userType"];
    echo "<html lang='en'>";

    echo "<head>";
        echo "<meta charset='UTF-8'>";
        echo "<meta name='viewport' content='width=device-width, initial-scale=1.0'>";
        echo "<title>Change password</title>";
        echo "<link rel='stylesheet' href='style.css'>";
    echo "</head>";

    if ($userType != 'admin'){
        echo "<header>";
            echo "<h1><a href='mainWebsite/mainPage.php'>Marketplace</a></h1>";
            echo "<div class='form-mainHeading'>";
                echo "<div class='form-account'>";
                    echo "<a href='mainWebsite/mainPage.php'>Return to the store
front</a>";
                echo "</div>";
            echo "</div>";
        echo "</header>";
    }
}

```

```

    echo "<body>";
    echo "<div class='form-accountDetails'>";
    echo "<h1>Update password</h1>";
    echo "<form action='changeUserPassword.php' method='POST' onsubmit='return
validatePasswords()'>";
    echo "<input type='hidden' value='$userID' id='userID' name='userID'>";
    echo "<input type='hidden' value='$userType' id='userType'
name='userType'>";

    echo "<div class='form-changePassDetails'>";
    echo "<div class='form-inputDetails'>";
    echo "<label for='password'>Password: </label><br>";
    echo "<div class='form-error'>";
    echo "<input type='password' name='password' id='password'>";
    echo "<small class='error' id='passwordError'></small>";
    echo "</div>";
    echo "</div>";
    echo "</div>";

    echo "<div class='form-changePassDetails'>";
    echo "<div class='form-inputDetails'>";
    echo "<label for='newPassword'>New password: </label><br>";
    echo "<div class='form-error'>";
    echo "<input type='password' name='newPassword'
id='newPassword'>";
    echo "<small class='error' id='newPasswordError'></small>";
    echo "</div>";
    echo "</div>";
    echo "</div>";

    echo "<div class='form-changePassDetails'>";
    echo "<div class='form-inputDetails'>";
    echo "<label for='conPassword'>Confirm new password: </label><br>";
    echo "<div class='form-error'>";
    echo "<input type='password' name='conPassword'
id='conPassword'>";
    echo "<small class='error' id='conPasswordError'></small>";
    echo "</div>";
    echo "</div>";
    echo "</div>";

    echo "<button type='submit' class='form-btnSpace'>Change password</button>";
    echo "</form>";
    echo "<script src='script.js'></script>";
    echo "</div>";
    echo "</body>";
    echo "</html>";
}
echo "</div>";
?>

```

A php form that enables the user to update their password to a new one.

changeUserPassword.php

```
<?php
echo "<link rel='stylesheet' href='style.css'>";
echo "<div class='form-accountDetails'>";
$success = false;
session_start();

if ($_SERVER["REQUEST_METHOD"] == "POST") {
    $userType = $_POST["userType"];
    $connectionActive = false;
    mysqli_report(MYSQLI_REPORT_ERROR | MYSQLI_REPORT_STRICT);
    $connection = null;

    try{
        $connection = new mysqli("localhost", "root", "", "c2cMarketplacedb");
        mysqli_autocommit($connection, false);
        mysqli_begin_transaction($connection);
        $connectionActive = true;

        $userID = $_POST["userID"];
        $password = $_POST["password"];
        $newPassword = $_POST["newPassword"];
        $conPassword = $_POST["conPassword"];
        $errors = [];

        if (empty($password)) {
            $errors[] = "Password is required";
        }
        if (empty($newPassword)) {
            $errors[] = "Password is required";
        }
        else if (strlen($newPassword) < 8){
            $errors[] = "New password must be at least 8 characters or more in length";
        }

        if (empty($conPassword)) {
            $errors[] = "Password is required";
        }
        else if ($conPassword != $newPassword){
            $errors[] = "New password and confirmation password don't match";
        }

        $userDetails = mysqli_fetch_array(mysqli_execute_query($connection, "SELECT * FROM $userType WHERE ". $userType ."ID = ?", [$userID]));
        if (!password_verify($password, $userDetails["password"])){
            $errors[] = "Password is incorrect";
        }

        if (empty($errors)) {
```

```

        mysqli_execute_query($connection, "Update $userType SET password = ? WHERE ".
$userType ."ID = ?", [password_hash($newPassword, PASSWORD_DEFAULT), $userID]);
        $_SESSION["message"] = "Password changed successfully";
        mysqli_commit($connection);
        $success = true;
    }
    else {
        echo "<h1>Account login error</h1>";
        echo "<h3>Password couldn't be changed due to the following error(s):</h3>";
        echo "<div class='form-errorBoard error'>";
        foreach ($errors as $error) {
            echo "<h4>*$error</h4>";
        }
        echo "</div>";

        if ($userType != "admin"){
            echo "<a href='mainWebsite/accountCenter(PassAndSec).php' class='form-
returnLink'>Return to Account Center</a>";
        }
        else{
            echo "<a href='adminWebsite/adminDatabase.php' class='form-
returnLink'>Return to admin database</a>";
        }
    }
}
catch(Exception $error){
    echo "<h1>Account login error</h1>";
    echo "<h3>Password couldn't be changed due to the following error(s):</h3>";
    echo "<div class='form-errorBoard error'>";
        echo "<h4>*Error: ". $error->getMessage() ."</h4>";
    echo "</div>";

    if ($userType != "admin"){
        echo "<a href='mainWebsite/accountCenter(PassAndSec).php' class='form-
returnLink'>Return to Account Center</a>";
    }
    else{
        echo "<a href='adminWebsite/adminDatabase.php' class='form-returnLink'>Return
to admin database</a>";
    }

    if ($connectionActive == true){
        mysqli_rollback($connection);
    }
}

if ($connection != null){
    $connection->close();
}
} else {
    $_SESSION["message"] = "Error: POST error occured";
}

```

```

echo "</div>";

if ($success == true){
    if ($userType != "admin"){
        header('Location:mainWebsite/accountCenter(PassAndSec).php');
    }
    else{
        header('Location:adminWebsite/adminDatabase.php');
    }
}
?>

```

A php file that validates and inserts the user's new password.

accountCentre(UserRatings).php

```

<?php
session_start();
if (!isset($_SESSION["userID"]) && empty($_SESSION["userID"])){
    header('Location:mainPage.php');
}
$username = $_SESSION["username"];

echo "<html lang='en'>";
echo "<head>";
    echo "<meta charset='UTF-8'>";
    echo "<meta name='viewport' content='width=device-width, initial-scale=1.0'>";
    echo "<title>My Account</title>";
    echo "<link rel='stylesheet' href='../style.css'>";
echo "</head>";

if (isset($_SESSION["message"]) && !empty($_SESSION["message"])){
    echo "<div id='message' name='message' class='msg'>";
        echo "<p>". $_SESSION["message"] . "</p>";
        echo "<button onclick='closeMessage()'>X</button>";
    echo "</div>";
    unset($_SESSION["message"]);
    echo "<script src='../script.js'></script>";
}

echo "<header>";
    echo "<h1><a href='mainPage.php'>Marketplace</a></h1>";
    echo "<div class='form-mainHeading'>";
        echo "<div class='form-account'>";
            echo "<a href='mainPage.php'>Return to the store front</a>";
            echo "</div>";
        echo "</div>";
echo "</header>";

echo "<body>";
    echo "<div class='form-accountCenter'>";
        echo "<div class='vr form-accountCenterOptions'>";

```

```

        echo "<h1 >Account Center</h1>";
        echo "<div class='form-accountCenterOptionsItem'>";
            echo "<a href='accountCenter(ProfileDetails).php'>Profile details</a>";
        echo "</div>";
        echo "<div class='form-accountCenterOptionsItem'>";
            echo "<a href='accountCenter(PassAndSec).php'>Password and security</a>";
        echo "</div>";
        echo "<div class='form-accountCenterOptionsItem form-accCenterOptionPicked'>";
            echo "<a href='accountCenter(UserRatings).php'>User ratings</a>";
        echo "</div>";

        if ($_SESSION["userType"] == "vendor"){
            echo "<div class='form-accountCenterOptionsItem'>";
                echo "<a href='accountCenter(Deliveries).php'>Ongoing deliveries</a>";
            echo "</div>";
        }
        echo "</div>";

        echo "<div class='form-settingOptions'>";
            mysqli_report(MYSQLI_REPORT_ERROR | MYSQLI_REPORT_STRICT);
            $connection = null;

            try{
                $connection = new mysqli("localhost", "root", "", "c2cMarketplacedb");
                echo "<div class='form-title'>";
                    $recordFound = false;
                    if ($_SESSION["userType"] == "vendor"){
                        $ratings = mysqli_execute_query($connection, "SELECT * FROM
userRatingHistory WHERE vendorID = ? AND rated = '0' AND ratingType = 'buyer'",
[$_SESSION["userID"]]);

                        if(mysqli_num_rows($ratings) > 0){
                            $recordFound = true;
                            echo "<h2>Provide buyer ratings</h2>";
                            while($ratingDetails = mysqli_fetch_array($ratings)){
                                $vendorDetails =
mysqli_fetch_array(mysqli_execute_query($connection, "SELECT * FROM ".
$ratingDetails["userType"] ." WHERE ". $ratingDetails["userType"] ."ID = ?",
[$ratingDetails["userID"]]));
                                echo "<h2>". $vendorDetails["username"] . "</h2>";
                                echo "<form action='rateUser.php' method='POST'>";
                                    echo "<input type='hidden' id='userType'
name='userType' value='". $ratingDetails["userType"] .'>";
                                    echo "<input type='hidden' id='ratingType'
name='ratingType' value='buyer'>";
                                    echo "<input type='range' min=1 max=5 id='rating'
name='rating'>";
                                    echo "<button type='submit' name='formBtn'
id='formBtn' value='". $ratingDetails["userID"] .'>Confirm rating</button>";
                                    echo "</form>";
                                }
                            }
                        }
                    }
            }
        }
    }

```

```

    }

    $ratings = mysqli_execute_query($connection, "SELECT * FROM
userRatingHistory WHERE userType = ? AND userID = ? AND rated = '0' AND ratingType
='vendor'", [$_SESSION["userType"], $_SESSION["userID"]]);
    if(mysqli_num_rows($ratings) > 0){
        $recordFound = true;
        echo "<h2>Provide vendor ratings</h2>";
        while($ratingDetails = mysqli_fetch_array($ratings)){
            $vendorDetails =
mysqli_fetch_array(mysqli_execute_query($connection, "SELECT * FROM vendor WHERE vendorID
= ?", [$ratingDetails["vendorID"]]));
            echo "<h2>". $vendorDetails["username"] . "</h2>";
            echo "<form action='rateUser.php' method='POST'>";
            echo "<input type='hidden' id='userType' name='userType'
value='vendor'>";
            echo "<input type='hidden' id='ratingType'
name='ratingType' value='vendor'>";
            echo "<input type='range' min=1 max=5 id='rating'
name='rating'>";
            echo "<button type='submit' name='formBtn' id='formBtn'
value='". $ratingDetails["vendorID"] . "'>Confirm rating</button>";
            echo "</form>";
        }
    }

    if ($recordFound == false){
        echo "<h1>* You have no outstanding ratings to give at the moment!
* </h1>";
    }
    echo "</div>";
}
catch(Exception $error){
    echo "<h2>An error occurred</h2>";
}

if ($connection != null){
    $connection->close();
}
echo "<script src='../script.js'></script>";
echo "</div>";
echo "</div>";
echo "</body>";
echo "</html>";
?>

```

A php form that displays and enables users to submit their user ratings.

rateUser.php

```

<?php
session_start();

```

```

if ($_SERVER["REQUEST_METHOD"] == "POST") {
    $userType = $_POST["userType"];
    $rating = $_POST["rating"];
    $ratingType = $_POST["ratingType"];
    $userID = $_POST["formBtn"];
    mysqli_report(MYSQLI_REPORT_ERROR | MYSQLI_REPORT_STRICT);
    $connection = null;
    $connectionActive = false;

    try{
        $connection = new mysqli("localhost", "root", "", "c2cMarketplacedb");
        mysqli_autocommit($connection, false);
        mysqli_begin_transaction($connection);
        $connectionActive = true;

        $totalRating = 0;
        $newRating = 0;
        $allRatings = null;

        if ($ratingType == "buyer"){
            mysqli_execute_query($connection, "UPDATE userRatingHistory SET ratingAmount
= ? WHERE userID = ? AND ratingType = 'buyer'", [$rating, $userID]);
            mysqli_execute_query($connection, "UPDATE userRatingHistory SET rated = 1
WHERE userID = ? AND ratingType = 'buyer'", [$userID]);
            $allRatings = mysqli_execute_query($connection, "SELECT * FROM
userRatingHistory WHERE userID = ? AND rated = '1'", [$userID]);
        }
        else{
            mysqli_execute_query($connection, "UPDATE userRatingHistory SET ratingAmount
= ? WHERE vendorID = ? AND ratingType = 'vendor'", [$rating, $userID]);
            mysqli_execute_query($connection, "UPDATE userRatingHistory SET rated = 1
WHERE vendorID = ? AND ratingType = 'vendor'", [$userID]);
            $allRatings = mysqli_execute_query($connection, "SELECT * FROM
userRatingHistory WHERE vendorID = ? AND rated = '1'", [$userID]);
        }

        while($ratingNum = mysqli_fetch_array($allRatings)){
            $totalRating += $ratingNum["ratingAmount"];
        }
        $newRating = $totalRating / mysqli_num_rows($allRatings);
        mysqli_execute_query($connection, "Update $userType SET ". $ratingType ."Rating = ?
WHERE $newRating = ?", [$userType ."ID", $userID]);

        $_SESSION["message"] = "User rated successfully";
        mysqli_commit($connection);
    }
    catch(Exception $error){
        $_SESSION["message"] = "Error: ". $error->getMessage();
        if ($connectionActive == true){
            mysqli_rollback($connection);
        }
    }
}

```

```

        if ($connection != null){
            $connection->close();
        }
    } else {
        $_SESSION["message"] = "Error: POST error occurred";
    }
}

header('Location:accountCenter(UserRatings).php');
?>

```

A php file that inserts and calculates the rating of a user.

accountCentre(Deliveries).php

```

<?php
session_start();
if (!isset($_SESSION["userID"]) && empty($_SESSION["userID"])){
    header('Location:mainPage.php');
}
$username = $_SESSION["username"];

echo "<html lang='en'>";
echo "<head>";
    echo "<meta charset='UTF-8'>";
    echo "<meta name='viewport' content='width=device-width, initial-scale=1.0'>";
    echo "<title>My Account</title>";
    echo "<link rel='stylesheet' href='../style.css'>";
echo "</head>";

if (isset($_SESSION["message"]) && !empty($_SESSION["message"])){
    echo "<div id='message' name='message' class='msg'>";
        echo "<p>". $_SESSION["message"] . "</p>";
        echo "<button onclick='closeMessage()'>X</button>";
    echo "</div>";
    unset($_SESSION["message"]);
    echo "<script src='../script.js'></script>";
}

echo "<header>";
    echo "<h1><a href='mainPage.php'>Marketplace</a></h1>";
    echo "<div class='form-mainHeading'>";
        echo "<div class='form-account'>";
            echo "<a href='mainPage.php'>Return to the store front</a>";
        echo "</div>";
    echo "</div>";
echo "</header>";

echo "<body>";
    echo "<div class='form-accountCenter'>";
        echo "<div class='vr form-accountCenterOptions'>";
            echo "<h1>Account Center</h1>";

```

```

        echo "<div class='form-accountCenterOptionsItem'>";
        echo "<a href='accountCenter(ProfileDetails).php'>Profile details</a>";
        echo "</div>";
        echo "<div class='form-accountCenterOptionsItem'>";
        echo "<a href='accountCenter(PassAndSec).php'>Password and security</a>";
        echo "</div>";
        echo "<div class='form-accountCenterOptionsItem'>";
        echo "<a href='accountCenter(UserRatings).php'>User ratings</a>";
        echo "</div>";
        echo "<div class='form-accountCenterOptionsItem form-accCenterOptionPicked'>";
        echo "<a href='accountCenter(Deliveries).php'>Ongoing deliveries</a>";
        echo "</div>";
        echo "</div>";

        echo "<div class='form-settingOptions'>";
        mysqli_report(MYSQLI_REPORT_ERROR | MYSQLI_REPORT_STRICT);
        $connection = null;

        try{
            $connection = new mysqli("localhost", "root", "", "c2cMarketplacedb");
            $deliveries = mysqli_execute_query($connection, "SELECT * FROM delivery
WHERE vendorID = ?", [$SESSION["userID"]]);

            if(mysqli_num_rows($deliveries) > 0){
                while($deliveryDetails = mysqli_fetch_array($deliveries)){
                    echo "<div class='form-deliveryForms'>";
                    $spenderDetails =
mysqli_fetch_array(mysqli_execute_query($connection, "SELECT * FROM ".
$deliveryDetails["userType"] ." WHERE ". $deliveryDetails["userType"] ."ID = ?",
[$deliveryDetails["userID"]]));
                    $cartItems = mysqli_execute_query($connection, "SELECT * FROM
cartItem WHERE deliveryID = ?", [$deliveryDetails["deliveryID"]]);
                    echo "<h2>". $spenderDetails["username"] . "</h2>";
                    echo "<h3 class='form-profit'>Total profit: R".
$deliveryDetails["totalCost"] . "</h3>";

                    echo "<div class='form-deliveryFormsAddress'>";
                    echo "<form action='updateDeliveryLocation.php'
method='POST'>";
                    echo "<label for='address'>Delivery address:
</label><br>";
                    echo "<input type='hidden' id='userType'
name='userType' value='vendor'>";
                    echo "<div class='form-deliveryFormsAddressFlex'>";
                    echo "<input type='text' name='address'
id='address' value='". $deliveryDetails["deliveryAddress"] .'>";
                    echo "<button type='submit' name='formBtn'
id='formBtn' value='". $deliveryDetails["deliveryID"] .'>Update delivery
address</button>";

                    echo "</div>";
                    echo "</form>";
                }
            }
        } catch (Exception $e) {
            echo "<div class='form-error'>";
            echo "<p>Error: " . $e->getMessage() . "</p>";
            echo "</div>";
        }
    }
}

```

```

        echo "<div class='form-deliveryFormsAddressFlex'>";
        echo "<label for='vendorApproval'>Your approval:
</label><br>";

        if ($deliveryDetails["vendorApproval"] == 0){
            echo "<input type='checkbox' name='vendorApproval'
id='vendorApproval' disabled>";
        }
        else{
            echo "<input type='checkbox' name='vendorApproval'
id='vendorApproval' checked disabled>";
        }

        echo "<label for='userApproval'>Buyer approval:
</label><br>";

        if ($deliveryDetails["userApproval"] == 0){
            echo "<input type='checkbox' name='userApproval'
id='userApproval' disabled>";
        }
        else{
            echo "<input type='checkbox' name='userApproval'
id='userApproval' checked disabled>";
        }
        echo "</div><br>";

        echo "<form action='approveDeliveryLocation.php'
method='POST'>";

        echo "<input type='hidden' id='userType'
name='userType' value='vendor'>";
        echo "<input type='hidden' id='currentApproval'
name='currentApproval' value='". $deliveryDetails["vendorApproval"] ."'>";
        if ($deliveryDetails["vendorApproval"] == 0){
            echo "<button type='submit' name='formBtn'
id='formBtn' value='". $deliveryDetails["deliveryID"] ."'>Approve delivery
address</button>";
        }
        else{
            echo "<button type='submit' name='formBtn'
id='formBtn' value='". $deliveryDetails["deliveryID"] ."'>Reject delivery
address</button>";
        }
        echo "</form>";
        echo "</div>";

        echo "<div class='form-settingOptionsSideItems'>";
        echo "<form action='deliveryMessages.php' method='POST'>";
        echo "<input type='hidden' id='deliveryID'
name='deliveryID' value='". $deliveryDetails["deliveryID"] ."'>";
        echo "<input type='hidden' id='otherUserID'
name='otherUserID' value='". $spenderDetails[$deliveryDetails["userType"] . "ID"] ."'>";
        echo "<button type='submit' name='formBtn'
id='formBtn' value='vendor'>Chat with customer</button>";
        echo "</form>";

```

```

        echo "<form action='../deleteRecord.php' method='POST'>";
        echo "<input type='hidden' id='userType'
name='userType' value='". $_SESSION["userType"] ."'>";
        echo "<input type='hidden' id='recordType'
name='recordType' value='delivery'>";
        echo "<button type='submit' name='formBtn'
id='formBtn' value='". $deliveryDetails["deliveryID"] ."'>Cancel order delivery</button>";
        echo "</form>";
        echo "</div>";

        echo "<h2>Delivery items</h2><br>";
        while($cartItem = mysqli_fetch_array($cartItems)){
            $productItems =
mysqli_fetch_array(mysqli_execute_query($connection, "SELECT * FROM listing WHERE
listingID = ?", [$cartItem["listingID"]]));

            echo "<div class='form-listingList'>";
            echo "<div class='form-cartLeftItem'>";
            echo "<h3>". $productItems["title"] ."</h3>";
            echo "<img src='". $productItems["image"] ."'
alt='Product.png'>";

            echo "</div>";

            echo "<div class='form-cartRightItem'>";
            echo "<h3 class='form-profit'>Price: R ".
$productItems["price"] * $cartItem["quantity"] ."</h3>";
            echo "<h4>Quantity: <input type='number'
name='quantity' id='quantity' value='". $cartItem["quantity"] ."' min=1 max='".
$productItems["inStock"] ."' readonly></h4>";
            echo "</div>";
            echo "</div>";
        }
        echo "</div>";
    }
}
}
}
catch(Exception $error){
    echo "<h2>An error occured</h2>";
}

if ($connection != null){
    $connection->close();
}
echo "<script src='../script.js'></script>";
echo "</div>";
echo "</div>";
echo "</body>";
echo "</html>";
?>

```

A php form that displays all deliveries associated with the vendor and enables them to update and confirm the delivery address.

searchDatabase.php

```
<?php
session_start();
if ($_SERVER["REQUEST_METHOD"] == "POST") {
    $infoPriority = $_POST["infoPriority"];
    $adPriority = $_POST["adPriority"];
    $orderItem = $_POST["orderItem"];
    $searchBar = strtolower($_POST["searchBar"]);
    $databaseTbl = $_POST["tableBtn"];
    $isAdmin = $_POST["isAdmin"];

    if (trim($searchBar) == ""){
        if ($adPriority == "None"){
            $sqlStatement = "SELECT * FROM ". $databaseTbl ."";
        }
        else{
            if ($databaseTbl == 'listing' && $orderItem == 'username'){
                $sqlStatement = "SELECT * FROM listing INNER JOIN vendor ON
vendor.vendorID = listing.vendorID ORDER BY vendor.username $adPriority";
            }
            else{
                $sqlStatement = "SELECT * FROM ". $databaseTbl ." ORDER BY $orderItem
$adPriority";
            }
        }
    }
    else{
        if ($adPriority == "None"){
            if ($databaseTbl == 'listing' && $infoPriority == 'username'){
                $sqlStatement = "SELECT * FROM ". $databaseTbl ." WHERE vendorID = (SELECT
vendorID FROM vendor WHERE username LIKE '%$searchBar%')";
            }
            else{
                $sqlStatement = "SELECT * FROM ". $databaseTbl ." WHERE $infoPriority LIKE
'%$searchBar%'";
            }
        }
        else{
            if ($databaseTbl == 'listing' && $orderItem == 'username'){
                if ($databaseTbl == 'listing' && $infoPriority == 'username'){
                    $sqlStatement = "SELECT * FROM listing INNER JOIN vendor ON
vendor.vendorID = listing.vendorID WHERE vendor.username LIKE '%$searchBar%' ORDER BY
vendor.username $adPriority";
                }
                else{
                    $sqlStatement = "SELECT * FROM listing INNER JOIN vendor ON
vendor.vendorID = listing.vendorID WHERE $infoPriority LIKE '%$searchBar%' ORDER BY
vendor.username $adPriority";
                }
            }
        }
    }
}
```

```

    }
    else{
        if ($databaseTbl == 'listing' && $infoPriority == 'username'){
            $sqlStatement = "SELECT * FROM ". $databaseTbl ." WHERE vendorID =
(SELECT vendorID FROM vendor WHERE username LIKE '%$searchBar%')";
        }
        else{
            $sqlStatement = "SELECT * FROM ". $databaseTbl ." WHERE $infoPriority
LIKE '%$searchBar%' ORDER BY $orderItem $adPriority";
        }
    }
}
}

$_SESSION["searchStatement"] = $sqlStatement;

if ($isAdmin == "true"){
    header("Location:adminWebsite/". $databaseTbl ."Database.php");
}
else{
    header("Location:mainWebsite/mainPage.php");
}
}
?>

```

A php file that returns a SQL statement based on the search parameters defined by the user.

deleteRecord.php

```

<?php
session_start();
if ($_SERVER["REQUEST_METHOD"] == "POST") {
    $recordID = $_POST["formBtn"];
    $recordType = $_POST["recordType"];
    $userType = $_SESSION["userType"];
    $connectionActive = false;
    mysqli_report(MYSQLI_REPORT_ERROR | MYSQLI_REPORT_STRICT);
    $connection = null;

    if (isset($_POST["userType"])){
        $userType = $_POST["userType"];
    }

    try{
        $connection = new mysqli("localhost", "root", "", "c2cMarketplacedb");
        mysqli_autocommit($connection, false);
        mysqli_begin_transaction($connection);
        $connectionActive = true;

        $_SESSION["message"] = "Account has been deleted successfully";
        if ($recordType != "listing" && $recordType != "delivery"){
            if ($recordType != "admin"){

```

```

        $deliveryID = mysqli_fetch_array(mysqli_execute_query($connection, "SELECT
deliveryID FROM delivery WHERE userID = ?", [$recordID]));
        mysqli_execute_query($connection, "Delete FROM delivery WHERE userID = ?",
[$recordID]);
        mysqli_execute_query($connection, "DELETE FROM deliveryMessage WHERE
deliveryID = ?", [$deliveryID["deliveryID"]]);
        mysqli_execute_query($connection, "Delete FROM cartItem WHERE deliveryID
= ?", [$deliveryID["deliveryID"]]);

        if ($recordType == 'vendor'){
            $listingDetails = mysqli_execute_query($connection, "SELECT * FROM
listing WHERE vendorID = ?", [$recordID]);
            if(mysqli_num_rows($listingDetails) > 0){
                while($recordDetails = mysqli_fetch_array($listingDetails)){
                    unlink(substr($recordDetails["image"], 3));
                    $cartItemDetails =
mysqli_fetch_array(mysqli_execute_query($connection, "SELECT * FROM cartItem WHERE
listingID = ?", [$recordDetails["listingID"]]));
                    mysqli_execute_query($connection, "Delete FROM listing WHERE
listingID = ?", [$recordDetails["listingID"]]);
                    mysqli_execute_query($connection, "Delete FROM cartItem WHERE
listingID = ?", [$recordDetails["listingID"]]);

                    $deliveryCartItems = mysqli_execute_query($connection, "SELECT
* FROM cartItem WHERE deliveryID = ?", [$cartItemDetails["deliveryID"]]);
                    if (mysqli_num_rows($deliveryCartItems) <= 0){
                        mysqli_execute_query($connection, "DELETE FROM delivery
WHERE deliveryID = ?", [$cartItemDetails["deliveryID"]]);
                        mysqli_execute_query($connection, "DELETE FROM
deliveryMessage WHERE deliveryID = ?", [$cartItemDetails["deliveryID"]]);
                    }
                }
            }
        }

        if ($userType != "admin"){
            unset($_SESSION["userID"]);
            unset($_SESSION["firstName"]);
            unset($_SESSION["lastName"]);
            unset($_SESSION["username"]);
            unset($_SESSION["phoneNum"]);
            unset($_SESSION["email"]);
            unset($_SESSION["address"]);
            unset($_SESSION["userType"]);
        }
        $_SESSION["message"] = "Account has been deleted successfully";
    }
}

else if ($recordType == "listing") {
    $listingDetails = mysqli_fetch_array(mysqli_execute_query($connection, "SELECT
* FROM listing WHERE listingID = ?", [$recordID]));

```

```

        $cartItems = mysqli_execute_query($connection, "SELECT * FROM cartItem WHERE
listingID = ?", [$recordID]);
        mysqli_execute_query($connection, "Delete FROM cartItem WHERE listingID = ?",
[$recordID]);

        if(mysqli_num_rows($cartItems) > 0){
            while($cartItemDetails = mysqli_fetch_array($cartItems)){
                $deliveryCartItems = mysqli_execute_query($connection, "SELECT * FROM
cartItem WHERE deliveryID = ?", [$cartItemDetails["deliveryID"]]);
                if (mysqli_num_rows($deliveryCartItems) <= 0){
                    mysqli_execute_query($connection, "DELETE FROM delivery WHERE
deliveryID = ?", [$cartItemDetails["deliveryID"]]);
                    mysqli_execute_query($connection, "DELETE FROM deliveryMessage
WHERE deliveryID = ?", [$cartItemDetails["deliveryID"]]);
                }
                else{
                    $deliveryDetails =
mysqli_fetch_array(mysqli_execute_query($connection, "SELECT * FROM delivery WHERE
deliveryID = ?", [$cartItemDetails["deliveryID"]]));
                    mysqli_execute_query($connection, "UPDATE delivery SET totalCost
= ? WHERE deliveryID = ?", [($deliveryDetails["totalCost"] - ($listingDetails["price"] *
$cartItemDetails["quantity"])), $cartItemDetails["deliveryID"]]);
                }
            }
        }
        else{
            mysqli_execute_query($connection, "Delete FROM cartItem WHERE deliveryID = ?",
[$recordID]);
            $_SESSION["message"] = "Delivery has been cancelled successfully";
        }

        $listingDetails = mysqli_fetch_array(mysqli_execute_query($connection, "SELECT *
FROM listing WHERE listingID = ?", [$recordID]));
        mysqli_execute_query($connection, "Delete FROM $recordType WHERE ".
$recordType ."ID = ?", [$recordID]);
        if ($recordType == "listing") {
            unlink(substr($listingDetails["image"], 3));
            $_SESSION["message"] = "Listing has been deleted successfully";
        }
        else if ($recordType == "delivery"){
            mysqli_execute_query($connection, "DELETE FROM deliveryMessage WHERE
deliveryID = ?", [$recordID]);
        }
        mysqli_commit($connection);
    }
    catch(Exception $error){
        $_SESSION["message"] = "Error: ". $error->getMessage();
        if ($connectionActive == true){
            mysqli_rollback($connection);
        }
    }
}

```

```

        if ($connection != null){
            $connection->close();
        }
    }
else{
    $_SESSION["message"] = "Error: POST error occurred";
}
echo "</div>";

if ($userType != "admin"){
    if ($recordType != "listing" && $recordType != "delivery"){
        header('Location:mainWebsite/mainPage.php');
    }
    else if ($recordType == "listing"){
        header('Location:mainWebsite/accountCenter(ProfileDetails).php');
    }
    else{
        if ($userType == "user"){
            header('Location:mainWebsite/cart.php');
        }
        else{
            header('Location:mainWebsite/accountCenter(Deliveries).php');
        }
    }
}
else{
    header('Location:adminWebsite/'. $recordType .'Database.php');
}
?>

```

A php file that provides a centralised method for database records to be deleted.

adminDatabase.php

```

<?php
session_start();
$sqlStatement = "SELECT * FROM admin";

echo "<html lang='en'>";
echo "<head>";
    echo "<meta charset='UTF-8'>";
    echo "<meta name='viewport' content='width=device-width, initial-scale=1.0'>";
    echo "<title>Users</title>";
    echo "<link rel='stylesheet' href='../style.css'>";
echo "</head>";

if (isset($_SESSION["searchStatement"]) && !empty($_SESSION["searchStatement"])){
    $sqlStatement = $_SESSION["searchStatement"];
    unset($_SESSION["searchStatement"]);
}

```

```

if (isset($_SESSION["message"]) && !empty($_SESSION["message"])){
    echo "<div id='message' name='message' class='msg'>";
    echo "<p>". $_SESSION["message"] . "</p>";
    echo "<button onclick='closeMessage()'>X</button>";
    echo "</div>";
    unset($_SESSION["message"]);
    echo "<script src='../script.js'></script>";
}

echo "<body>";
echo "<div class='form-accountCenter'>";
    echo "<div class='form-accountCenterOptions'>";
        echo "<h1 >Admin center</h1>";
        echo "<div class='form-accountCenterOptionsItem form-accCenterOptionPicked'>";
            echo "<a href='adminDatabase.php'>Admin database</a>";
        echo "</div>";
        echo "<div class='form-accountCenterOptionsItem'>";
            echo "<a href='buyerDatabase.php'>Buyer database</a>";
        echo "</div>";
        echo "<div class='form-accountCenterOptionsItem'>";
            echo "<a href='vendorDatabase.php'>Vendor database</a>";
        echo "</div>";
        echo "<div class='form-accountCenterOptionsItem'>";
            echo "<a href='listingDatabase.php'>Listing database</a>";
        echo "</div>";
    echo "</div>";

    echo "<div class='form-settingOptions'>";
        echo "<div class='form-settingOptionsSideItems'>";
            echo "<h2>Welcome ". $_SESSION["username"] . "</h2>";
            echo "<div class='form-inputDetails'>";
                echo "<form action='../accountLogOut.php' method='POST'>";
                echo "<button>Log out</button>";
            echo "</form>";
            echo "</div>";
        echo "</div>";

        if ($_SESSION["userID"] == 1){
            echo "<a class='form_buttonLink' href='adminCreation.html'><button>Create
a new admin account</button></a>";
        }
        mysqli_report(MYSQLI_REPORT_ERROR | MYSQLI_REPORT_STRICT);
        $connection = null;

        try{
            $connection = new mysqli("localhost", "root", "", "c2cMarketplacedb");
            echo "<div class='form-centeredBar'>";
                echo "<form action='../searchDatabase.php' method='POST'>";
                echo "<div class='form-searchBar'>";
                    echo "<div class='form-searchBarItem'>";
                        echo "<input type='hidden' id='isAdmin' name='isAdmin'
value='true'>";

```

```

        echo "<select id='infoPriority' name='infoPriority'>";
        echo "<option value='adminID'>Search by Admin
ID</option>";

        echo "<option value='username'>Search by
Username</option>";

        echo "</select>";

        echo "<select id='orderItem' name='orderItem'>";
        echo "<option value='adminID'>Order by Admin
ID</option>";

        echo "<option value='username'>Order by
Username</option>";

        echo "</select>";

        echo "<select id='adPriority' name='adPriority'>";
        echo "<option value='ASC'>Order in Ascending</option>";
        echo "<option value='DESC'>Order in
Descending</option>";

        echo "<option value='None'>None</option>";
        echo "</select>";
        echo "</div>";

        echo "<div class='form-inputAndBtn'>";
        echo "<input type='search' id='searchBar'
name='searchBar'>";

        echo "<button type='submit' name='tableBtn' id='tableBtn'
value='listing'>&#160;</button>";
        echo "</div>";
        echo "</div>";
        echo "</form>";
        echo "</div>";

        echo "<div class='form-databaseBlock form-description'>";
        if ($admins = mysqli_execute_query($connection, $sqlStatement)){
            if (mysqli_num_rows($admins) > 0){
                echo "<table>";
                echo "<tr class='form-tableHead'>";
                echo "<th>Admin ID</th>";
                echo "<th>Username</th>";
                if ($_SESSION["userID"] == 1){
                    echo "<th></th>";
                    echo "<th></th>";
                }
                echo "</tr>";

                while($adminDetails = mysqli_fetch_array($admins)){
                    echo "<tr>";
                    echo "<td>". $adminDetails["adminID"] . "</td>";
                    echo "<td>". $adminDetails["username"] . "</td>";

                    if ($_SESSION["userID"] == 1){

```

```

                                echo "<form action='../changePasswordForm.php'
method='POST'>";
                                echo "<input type='hidden' value='admin'
name='userType' id='userType'>";
                                echo "<td><button type='submit' value='".
$adminDetails["adminID"] ."' name='formBtn' id='formBtn'>Update password</button></td>";
                                echo "</form>";

                                if ($adminDetails["adminID"] != 1){
                                    echo "<form action='../deleteRecord.php'
method='POST'>";
                                    echo "<input type='hidden'
value='admin' name='recordType' id='recordType'>";
                                    echo "<td><button type='submit'
value='". $adminDetails["adminID"] ."' name='formBtn' id='formBtn' class='form-
redBtn'>Delete</button></td>";
                                    echo "</form>";
                                }
                            }
                            echo "</tr>";
                        }
                    echo "</table>";
                }
            echo "</div>";
        }
        catch(Exception $error){
            echo "<h2>An error occurred</h2>";
        }

        if ($connection != null){
            $connection->close();
        }
        echo "</div>";
    echo "</div>";
echo "</body>";
echo "</html>";
?>

```

A php form that displays the information of all admin users and enabling them to update admin password and delete other admin users.

adminCreation.php

```

<?php
echo "<link rel='stylesheet' href='../style.css'>";
echo "<div class='form-accountDetails'>";
$success = false;
$connectionActive = false;
session_start();

```

```

if ($_SERVER["REQUEST_METHOD"] == "POST") {
    mysqli_report(MYSQLI_REPORT_ERROR | MYSQLI_REPORT_STRICT);
    $connection = null;

    try{
        $connection = new mysqli("localhost", "root", "", "c2cMarketplacedb");
        mysqli_autocommit($connection, false);
        mysqli_begin_transaction($connection);
        $connectionActive = true;

        $username = $_POST["username"];
        $password = $_POST["password"];
        $conPassword = $_POST["conPassword"];
        $errors = [];

        if (empty($username)) {
            $errors[] = "Username is required";
        }
        else{
            $sqlUsername = mysqli_execute_query($connection, "SELECT * FROM admin WHERE
username = '$username'");
            if (mysqli_num_rows($sqlUsername) > 0){
                $errors[] = "Username already exists";
            }
        }

        if (empty($password)) {
            $errors[] = "Password is required";
        } else if (strlen($password) < 8) {
            $errors[] = "Password must be at least 8 or more characters in length";
        }

        if (empty($conPassword)) {
            $errors[] = "Password must be repeated for confirmation";
        } else if ($password != $conPassword) {
            $errors[] = "Password and confirmation password don't match";
        }

        if (empty($errors)) {
            mysqli_execute_query($connection, "INSERT INTO admin (username, password)
VALUES (?, ?)", [$username, password_hash($password, PASSWORD_DEFAULT)]);
            $success = true;
            $_SESSION["message"] = "Account registered successfully";
            mysqli_commit($connection);
        } else {
            echo "<h1>Admin creation error</h1>";
            echo "<h3>Account couldn't be registered due to the following
errors:</h3>";
            echo "<div class='form-errorBoard error'>";
            foreach ($errors as $error) {
                echo "<h4>$error</h4>";
            }
        }
    }
}

```

```

        echo "</div>";
        echo "<a href='adminCreation.html' class='form-returnLink'>Return to account
creation page</a>";
    }
}
catch(Exception $error){
    echo "<h1>Admin creation error</h1>";
    echo "<h3>Account couldn't be registered due to the following errors:</h3>";
    echo "<div class='form-errorBoard error'>";
        echo "<h4>*Error: ". $error->getMessage() . "</h4>";
    echo "</div>";
    echo "<a href='adminCreation.html' class='form-returnLink'>Return to account
creation page</a>";

    if ($connectionActive == true){
        mysqli_rollback($connection);
    }
}
if ($connection != null){
    $connection->close();
}
} else {
    $_SESSION["message"] = "Error: POST error occured";
}
echo "</div>";

if ($success == true){
    header('Location:adminDatabase.php');
}
?>

```

A php file that validates and inserts the admin details into the admin database table.

buyerDatabase.php

```

<?php
session_start();
$sqlStatement = "SELECT * FROM buyer";

echo "<html lang='en'>";
echo "<head>";
    echo "<meta charset='UTF-8'>";
    echo "<meta name='viewport' content='width=device-width, initial-scale=1.0'>";
    echo "<title>Users</title>";
    echo "<link rel='stylesheet' href='../style.css'>";
echo "</head>";

if (isset($_SESSION["searchStatement"]) && !empty($_SESSION["searchStatement"])){
    $sqlStatement = $_SESSION["searchStatement"];
    unset($_SESSION["searchStatement"]);
}

```

```

if (isset($_SESSION["message"]) && !empty($_SESSION["message"])){
    echo "<div id='message' name='message' class='msg'>";
    echo "<p>". $_SESSION["message"] . "</p>";
    echo "<button onclick='closeMessage()'>X</button>";
    echo "</div>";
    unset($_SESSION["message"]);
    echo "<script src='../script.js'></script>";
}

echo "<body>";
echo "<div class='form-accountCenter'>";
    echo "<div class='form-accountCenterOptions'>";
        echo "<h1 >Admin center</h1>";
        echo "<div class='form-accountCenterOptionsItem'>";
            echo "<a href='adminDatabase.php'>Admin database</a>";
        echo "</div>";
        echo "<div class='form-accountCenterOptionsItem form-accCenterOptionPicked'>";
            echo "<a href='buyerDatabase.php'>Buyer database</a>";
        echo "</div>";
        echo "<div class='form-accountCenterOptionsItem'>";
            echo "<a href='vendorDatabase.php'>Vendor database</a>";
        echo "</div>";
        echo "<div class='form-accountCenterOptionsItem'>";
            echo "<a href='listingDatabase.php'>Listing database</a>";
        echo "</div>";
    echo "</div>";

    echo "<div class='form-settingOptions'>";
        echo "<div class='form-settingOptionsSideItems'>";
            echo "<h2>Welcome ". $_SESSION["username"] . "</h2>";
            echo "<div class='form-inputDetails'>";
                echo "<form action='../accountLogOut.php' method='POST'>";
                echo "<button>Log out</button>";
                echo "</form>";
            echo "</div>";
        echo "</div>";
        echo "<a class='form_buttonLink' href='userCreation.html'><button>Create a new
user account</button></a>";

        mysqli_report(MYSQLI_REPORT_ERROR | MYSQLI_REPORT_STRICT);
        $connection = null;

        try{
            $connection = new mysqli("localhost", "root", "", "c2cMarketplacedb");
            echo "<div class='form-centeredBar'>";
                echo "<form action='../searchDatabase.php' method='POST'>";
                echo "<div class='form-searchBar'>";
                    echo "<div class='form-searchBarItem'>";
                        echo "<input type='hidden' id='isAdmin' name='isAdmin'
value='true'>";
                    echo "<select id='infoPriority' name='infoPriority'>";

```

```

        echo "<option value='buyerID'>Search by Buyer
ID</option>";

        echo "<option value='username'>Search by
Username</option>";

        echo "<option value='email'>Search by Email</option>";
        echo "<option value='address'>Search by
Address</option>";

        echo "<option value='buyerRating'>Search by Buyer
Rating</option>";

        echo "</select>";

        echo "<select id='orderItem' name='orderItem'>";
        echo "<option value='buyerID'>Order by Buyer
ID</option>";

        echo "<option value='username'>Order by
Username</option>";

        echo "<option value='email'>Order by Email</option>";
        echo "<option value='address'>Order by
Address</option>";

        echo "<option value='buyerRating'>Order by Buyer
Rating</option>";

        echo "</select>";

        echo "<select id='adPriority' name='adPriority'>";
        echo "<option value='ASC'>Order in Ascending</option>";
        echo "<option value='DESC'>Order in
Descending</option>";

        echo "<option value='None'>None</option>";
        echo "</select>";
        echo "</div>";

        echo "<div class='form-inputAndBtn'>";
        echo "<input type='search' id='searchBar'
name='searchBar'>";

        echo "<button type='submit' name='tableBtn' id='tableBtn'
value='listing'>&rarr;</button>";
        echo "</div>";
        echo "</div>";
        echo "</form>";
        echo "</div>";

        echo "<div class='form-databaseBlock form-description'>";
        if ($buyers = mysqli_execute_query($connection, $sqlStatement)){
            if (mysqli_num_rows($buyers) > 0){
                echo "<table>";
                echo "<tr class='form-tableHead'>";
                echo "<th>Buyer ID</th>";
                echo "<th>Username</th>";
                echo "<th>First name</th>";
                echo "<th>Last name</th>";
                echo "<th>Phone number</th>";
                echo "<th>Email</th>";
            }
        }
    }
}

```

```

        echo "<th>Address</th>";
        echo "<th>Buyer rating</th>";
        echo "<th></th>";
        echo "<th></th>";
        echo "</tr>";

        while($buyerDetails = mysqli_fetch_array($buyers)){
            echo "<tr>";
            echo "<td>". $buyerDetails["buyerID"] . "</td>";
            echo "<td>". $buyerDetails["username"] . "</td>";
            echo "<td>". $buyerDetails["firstName"] . "</td>";
            echo "<td>". $buyerDetails["lastName"] . "</td>";
            echo "<td>". $buyerDetails["phoneNum"] . "</td>";
            echo "<td>". $buyerDetails["email"] . "</td>";
            echo "<td>". $buyerDetails["address"] . "</td>";
            echo "<td>". $buyerDetails["buyerRating"] . "</td>";

            echo "<form action='changeUserType.php'
method='POST'>";

            echo "<input type='hidden' value='buyer'
name='userType' id='userType'>";

            echo "<td><button type='submit' value='".
$buyerDetails["buyerID"] . "' name='formBtn' id='formBtn'>Transition to
vendor</button></td>";

            echo "</form>";
            echo "<form action='../deleteRecord.php'
method='POST'>";

            echo "<input type='hidden' value='buyer'
name='recordType' id='recordType'>";

            echo "<td><button type='submit' value='".
$buyerDetails["buyerID"] . "' name='formBtn' id='formBtn' class='form-
redBtn'>Delete</button></td>";

            echo "</form>";
            echo "</tr>";
        }
        echo "</table>";
    }
    echo "</div>";
}
catch(Exception $error){
    echo "<h2>An error occured</h2>";
}
echo "</div>";

if ($connection != null){
    $connection->close();
}
echo "</div>";
echo "</body>";
echo "</html>";

```

```
?>
```

A php form that displays all buyer user information, enables the admin to transition buyer users to vendor users and delete user records.

changeUserType.php

```
<?php
if ($_SERVER["REQUEST_METHOD"] == "POST") {
    $connection = new mysqli("localhost", "root", "", "c2cMarketplacedb");

    if ($connection->connect_error) {
        die("Connection has failed:" . $connection->connect_error);
    }
    else {
        mysqli_autocommit($connection, false);
        mysqli_begin_transaction($connection);

        try{
            $userType = $_POST["userType"];
            $userID = $_POST["formBtn"];
            $otherUserType = null;

            if ($userType == "buyer"){
                $otherUserType = "vendor";
            }
            else{
                $otherUserType = "buyer";
            }

            $userDetails = mysqli_fetch_array(mysqli_query($connection, "SELECT * FROM $userType WHERE ". $userType ."ID = '$userID'"));
            if ($otherUserType == "buyer"){
                mysqli_query($connection, "INSERT INTO buyer (firstName, lastName, username, phoneNum, email, address, password, buyerRating, eWalletAmount, notify) VALUES ('. $userDetails["firstName"] ."', '". $userDetails["lastName"] ."', '". $userDetails["username"] ."', '". $userDetails["phoneNum"] ."', '". $userDetails["email"] ."', '". $userDetails["address"] ."', '". $userDetails["password"] ."', '". $userDetails["buyerRating"] ."', '". $userDetails["eWalletAmount"] ."', '". $userDetails["notify"] ."'"));
            }
            else{
                mysqli_query($connection, "INSERT INTO vendor (firstName, lastName, username, phoneNum, email, address, password, buyerRating, eWalletAmount, vendorRating, notify) VALUES ('. $userDetails["firstName"] ."', '". $userDetails["lastName"] ."', '". $userDetails["username"] ."', '". $userDetails["phoneNum"] ."', '". $userDetails["email"] ."', '". $userDetails["address"] ."', '". $userDetails["password"] ."', '". $userDetails["buyerRating"] ."', '". $userDetails["eWalletAmount"] ."', 0, '". $userDetails["notify"] ."'"));
            }
        }
    }
}
```

```

        $newUserDetails = mysqli_fetch_array(mysqli_query($connection, "SELECT * FROM
$otherUserType WHERE username = '". $userDetails["username"] ."'"));
        mysqli_query($connection, "Delete FROM $userType WHERE ". $userType ."ID =
'$userID'");

        $image = null;
        if ($userType == 'vendor'){
            $deliveries = mysqli_fetch_array(mysqli_query($connection, "SELECT * FROM
delivery WHERE vendorID = '". $userDetails["vendorID"] ."'"));
            mysqli_query($connection, "DELETE FROM delivery WHERE deliveryID = '".
$deliveries["deliveryID"] ."'");
            mysqli_query($connection, "DELETE FROM deliveryMessage WHERE deliveryID =
'". $deliveries["deliveryID"] ."'");

            if ($listingDetails = mysqli_query($connection, "SELECT * FROM listing
WHERE vendorID = '". $userDetails["vendorID"] ."'")){
                if(mysqli_num_rows($listingDetails) > 0){
                    while($recordDetails = mysqli_fetch_array($listingDetails)){
                        mysqli_query($connection, "Delete FROM listing WHERE listingID
= '". $recordDetails["listingID"] ."'");

                        $cartItems = mysqli_query($connection, "SELECT * FROM cartItem
WHERE listingID = '". $recordDetails["listingID"] ."'");
                        if (mysqli_num_rows($cartItems) > 0){
                            while ($cartItem = mysqli_fetch_array($cartItems)){
                                mysqli_query($connection, "Delete FROM cartItem WHERE
cartItemID = '". $cartItem["cartItemID"] ."'");
                            }
                        }
                    }
                }
            }
        }
        else{
            mysqli_query($connection, "Update delivery SET userID = '".
$newUserDetails["vendorID"] ."' WHERE userID = '". $userDetails["buyerID"] ."' AND
userType = 'buyer'");
            mysqli_query($connection, "Update delivery SET userType = 'vendor' WHERE
userID = '". $newUserDetails["vendorID"] ."' AND userType = 'buyer'");
            mysqli_query($connection, "Update userRatingHistory SET userID = '".
$newUserDetails["vendorID"] ."' WHERE userID = '". $userDetails["buyerID"] ."'");
            mysqli_query($connection, "Update userRatingHistory SET userType =
'vendor' WHERE userID = '". $newUserDetails["vendorID"] ."'");
        }

        $_SESSION["message"] = "Account transitioned successfully";
        mysqli_commit($connection);
        if ($image != null){
            unlink($image);
        }
    }
}

```

```

        catch(Exception $error){
            $_SESSION["message"] = "Error: ". $error->getMessage();
            mysqli_rollback($connection);
        }
    }
    if ($connection != null){
        $connection->close();
    }
}
else{
    $_SESSION["message"] = "Error: POST error occurred";
}

header('Location:'. $userType .'Database.php');
?>

```

A php file that changes a user from buyer to vendor and vice versa.

userCreation.php

```

<?php
echo "<link rel='stylesheet' href='../style.css'>";
echo "<div class='form-accountDetails'>";
$success = false;
$connectionActive = false;
session_start();

if ($_SERVER["REQUEST_METHOD"] == "POST") {
    mysqli_report(MYSQLI_REPORT_ERROR | MYSQLI_REPORT_STRICT);
    $connection = null;

    try {
        $connection = new mysqli("localhost", "root", "", "c2cMarketplacedb");
        mysqli_autocommit($connection, false);
        mysqli_begin_transaction($connection);
        $connectionActive = true;

        $userType = $_POST["userType"];
        $firstName = $_POST["fName"];
        $lastName = $_POST["lName"];
        $username = $_POST["username"];
        $phoneNum = $_POST["phoneNum"];
        $email = $_POST["email"];
        $address = $_POST["address"];
        $password = $_POST["password"];
        $conPassword = $_POST["conPassword"];
        $notify = isset($_POST["notify"]);

        $errors = [];

        if (empty($firstName)) {
            $errors[] = "First name is required";

```

```

    }
    if (empty($lastName)) {
        $errors[] = "Last name is required";
    }
    if (empty($username)) {
        $errors[] = "Username is required";
    }
    else{
        $sqlUsernameB = mysqli_execute_query($connection, "SELECT username FROM buyer
WHERE username = ?", [$username]);
        if (mysqli_num_rows($sqlUsernameB) > 0){
            $errors[] = "Username already exists";
        }
        else{
            $sqlUsernameV = mysqli_execute_query($connection, "SELECT username FROM
vendor WHERE username = ?", [$username]);
            if (mysqli_num_rows($sqlUsernameV) > 0){
                $errors[] = "Username already exists";
            }
            else{
                $sqlUsernameA = mysqli_execute_query($connection, "SELECT username
FROM admin WHERE username = ?", [$username]);
                if (mysqli_num_rows($sqlUsernameA) > 0){
                    $errors[] = "Username already exists";
                }
            }
        }
    }

    if (empty($phoneNum)) {
        $errors[] = "Phone number is required";
    }

    if (empty($email)) {
        $errors[] = "Email is required";
    } else if (!filter_var($email, FILTER_VALIDATE_EMAIL)) {
        $errors[] = "Invalid email";
    }

    if (empty($address)) {
        $errors[] = "Address is required";
    }

    if (empty($password)) {
        $errors[] = "Password is required";
    } else if (strlen($password) < 8) {
        $errors[] = "Password must be at least 8 or more characters in length";
    }

    if (empty($conPassword)) {
        $errors[] = "Password must be repeated for confirmation";
    } else if ($password != $conPassword) {

```

```

        $errors[] = "Password and confirmation password don't match";
    }

    if (empty($errors)) {
        $notifyChecked = 0;
        if (!empty($notify)){
            $notifyChecked = 1;
        }

        mysqli_execute_query($connection, "INSERT INTO $userType (firstName, lastName,
username, phoneNum, email, address, password, buyerRating, eWalletAmount, notify) VALUES
(?, ?, ?, ?, ?, ?, ?, '0', '0', ?)", [$firstName, $lastName, $username, $phoneNum, $email,
$address, password_hash($password, PASSWORD_DEFAULT), $notifyChecked]);
        mysqli_commit($connection);
        $success = true;
        $_SESSION["message"] = "User created successfully";
    } else {
        echo "<h1>User creation error</h1>";
        echo "<h3>User couldn't be created due to the following error(s):</h3>";
        echo "<div class='form-errorBoard error'>";
            foreach ($errors as $error) {
                echo "<h4>$error</h4>";
            }
        echo "</div>";
        echo "<a href='userCreation.html' class='form-returnLink'>Return to creation
page</a>";
    }
}
catch(Exception $error){
    echo "<h1>User creation error</h1>";
    echo "<h3>User couldn't be created due to the following error(s):</h3>";
    echo "<div class='form-errorBoard error'>";
        echo "<h4>Error: ". $error->getMessage() . "</h4>";
    echo "</div>";
    echo "<a href='userCreation.html' class='form-returnLink'>Return to creation
page</a>";
}

    if ($connectionActive == true){
        mysqli_rollback($connection);
    }
}

    if ($connection != null){
        $connection->close();
    }
} else {
    $success = true;
    $_SESSION["message"] = "Error: POST error occurred";
}
echo "</div>";

if ($success == true){

```

```

    if ($userType == "buyer"){
        header('Location:buyerDatabase.php');
    }
    else{
        header('Location:vendorDatabase.php');
    }
}
?>

```

A php file that validates and inserts the user information as either a buyer or vendor.

vendorDatabase.php

```

<?php
session_start();
$sqlStatement = "SELECT * FROM vendor";

echo "<html lang='en'>";
echo "<head>";
    echo "<meta charset='UTF-8'>";
    echo "<meta name='viewport' content='width=device-width, initial-scale=1.0'>";
    echo "<title>Users</title>";
    echo "<link rel='stylesheet' href='../style.css'>";
echo "</head>";

if (isset($_SESSION["searchStatement"]) && !empty($_SESSION["searchStatement"])){
    $sqlStatement = $_SESSION["searchStatement"];
    unset($_SESSION["searchStatement"]);
}

if (isset($_SESSION["message"]) && !empty($_SESSION["message"])){
    echo "<div id='message' name='message' class='msg'>";
        echo "<p>". $_SESSION["message"] . "</p>";
        echo "<button onclick='closeMessage()'>X</button>";
    echo "</div>";
    unset($_SESSION["message"]);
    echo "<script src='../script.js'></script>";
}

echo "<body>";
    echo "<div class='form-accountCenter'>";
        echo "<div class='form-accountCenterOptions'>";
            echo "<h1 >Admin center</h1>";
            echo "<div class='form-accountCenterOptionsItem'>";
                echo "<a href='adminDatabase.php'>Admin database</a>";
            echo "</div>";
            echo "<div class='form-accountCenterOptionsItem'>";
                echo "<a href='buyerDatabase.php'>Buyer database</a>";
            echo "</div>";
            echo "<div class='form-accountCenterOptionsItem form-accCenterOptionPicked'>";
                echo "<a href='vendorDatabase.php'>Vendor database</a>";
            echo "</div>";
        }
    }

```

```

        echo "<div class='form-accountCenterOptionsItem'>";
        echo "<a href='listingDatabase.php'>Listing database</a>";
        echo "</div>";
    echo "</div>";

    echo "<div class='form-settingOptions'>";
    echo "<div class='form-settingOptionsSideItems'>";
        echo "<h2>Welcome ". $_SESSION["username"] . "</h2>";
        echo "<div class='form-inputDetails'>";
            echo "<form action='../accountLogOut.php' method='POST'>";
                echo "<button>Log out</button>";
            echo "</form>";
        echo "</div>";
    echo "</div>";
    echo "<a class='form_buttonLink' href='userCreation.html'><button>Create a new
user account</button></a>";

    mysqli_report(MYSQLI_REPORT_ERROR | MYSQLI_REPORT_STRICT);
    $connection = null;

    try{
        $connection = new mysqli("localhost", "root", "", "c2cMarketplacedb");
        echo "<div class='form-centeredBar'>";
            echo "<form action='../searchDatabase.php' method='POST'>";
                echo "<div class='form-searchBar'>";
                    echo "<div class='form-searchBarItem'>";
                        echo "<input type='hidden' id='isAdmin' name='isAdmin'
value='true'>";

                        echo "<select id='infoPriority' name='infoPriority'>";
                            echo "<option value='vendorID'>Search by Vendor
ID</option>";

                            echo "<option value='username'>Search by
Username</option>";

                            echo "<option value='email'>Search by Email</option>";
                            echo "<option value='address'>Search by
Address</option>";

                            echo "<option value='buyerRating'>Search by Buyer
Rating</option>";
                            echo "<option value='vendorRating'>Search by Vendor
Rating</option>";

                        echo "</select>";

                        echo "<select id='orderItem' name='orderItem'>";
                            echo "<option value='vendorID'>Order by Vendor
ID</option>";

                            echo "<option value='username'>Order by
Username</option>";

                            echo "<option value='email'>Order by Email</option>";
                            echo "<option value='address'>Order by
Address</option>";

                            echo "<option value='buyerRating'>Order by Buyer
Rating</option>";

```

```

        echo "<option value='vendorRating'>Order by Vendor
Rating</option>";

        echo "</select>";

        echo "<select id='adPriority' name='adPriority'>";
        echo "<option value='ASC'>Order in Ascending</option>";
        echo "<option value='DESC'>Order in
Descending</option>";

        echo "<option value='None'>None</option>";
        echo "</select>";
        echo "</div>";

        echo "<div class='form-inputAndBtn'>";
        echo "<input type='search' id='searchBar'
name='searchBar'>";
        echo "<button type='submit' name='tableBtn' id='tableBtn'
value='listing'>&rarr;</button>";
        echo "</div>";
        echo "</div>";
        echo "</form>";
        echo "</div>";

        echo "<div class='form-databaseBlock form-description'>";
        if ($vendors = mysqli_execute_query($connection, $sqlStatement)){
            if (mysqli_num_rows($vendors) > 0){
                echo "<table>";
                echo "<tr class='form-tableHead'>";
                echo "<th>Vendor ID</th>";
                echo "<th>Username</th>";
                echo "<th>First name</th>";
                echo "<th>Last name</th>";
                echo "<th>Phone number</th>";
                echo "<th>Email</th>";
                echo "<th>Address</th>";
                echo "<th>Buyer rating</th>";
                echo "<th>Vendor rating</th>";
                echo "<th></th>";
                echo "<th></th>";
                echo "</tr>";

                while($vendorDetails = mysqli_fetch_array($vendors)){
                    echo "<tr>";
                    echo "<td>". $vendorDetails["vendorID"] . "</td>";
                    echo "<td>". $vendorDetails["username"] . "</td>";
                    echo "<td>". $vendorDetails["firstName"] . "</td>";
                    echo "<td>". $vendorDetails["lastName"] . "</td>";
                    echo "<td>". $vendorDetails["phoneNum"] . "</td>";
                    echo "<td>". $vendorDetails["email"] . "</td>";
                    echo "<td>". $vendorDetails["address"] . "</td>";
                    echo "<td>".
$vendorDetails["buyerRating"] . "</td>";

```

```

                                echo "<td>".
$vendorDetails["vendorRating"] . "</td>";

                                echo "<form action='changeUserType.php'
method='POST'>";

                                echo "<input type='hidden' value='vendor'
name='userType' id='userType'>";

                                echo "<td><button type='submit' value='".
$vendorDetails["vendorID"] . "' name='formBtn' id='formBtn'>Demote to buyer</button></td>";
                                echo "</form>";
                                echo "<form action='../deleteRecord.php'
method='POST'>";

                                echo "<input type='hidden' value='vendor'
name='recordType' id='recordType'>";

                                echo "<td><button type='submit' value='".
$vendorDetails["vendorID"] . "' name='formBtn' id='formBtn' class='form-
redBtn'>Delete</button></td>";

                                echo "</form>";
                                echo "</tr>";
                                }
                                echo "</table>";
                                }
                                }
                                echo "</div>";
                                }
                                catch(Exception $error){
                                    echo "<h2>An error occurred</h2>";
                                }
                                echo "</div>";

                                if ($connection != null){
                                    $connection->close();
                                }
                                echo "</div>";
echo "</body>";
echo "</html>";
?>

```

A php form that displays vendor information and enables the admin user to change a vendor user into a buyer as well as delete vendor user accounts.

listingDatabase.php

```

<?php
session_start();
$sqlStatement = "SELECT * FROM listing";

echo "<html lang='en'>";
echo "<head>";
    echo "<meta charset='UTF-8'>";
    echo "<meta name='viewport' content='width=device-width, initial-scale=1.0'>";

```

```

    echo "<title>Dashboard</title>";
    echo "<link rel='stylesheet' href='../style.css'>";
echo "</head>";

if (isset($_SESSION["searchStatement"]) && !empty($_SESSION["searchStatement"])){
    $sqlStatement = $_SESSION["searchStatement"];
    unset($_SESSION["searchStatement"]);
}

if (isset($_SESSION["message"]) && !empty($_SESSION["message"])){
    echo "<div id='message' name='message' class='msg'>";
        echo "<p>". $_SESSION["message"] . "</p>";
        echo "<button onclick='closeMessage()'>X</button>";
    echo "</div>";
    unset($_SESSION["message"]);
    echo "<script src='../script.js'></script>";
}

echo "<body>";
    echo "<div class='form-accountCenter'>";
        echo "<div class='form-accountCenterOptions'>";
            echo "<h1 >Admin center</h1>";
            echo "<div class='form-accountCenterOptionsItem'>";
                echo "<a href='adminDatabase.php'>Admin database</a>";
            echo "</div>";
            echo "<div class='form-accountCenterOptionsItem'>";
                echo "<a href='buyerDatabase.php'>Buyer database</a>";
            echo "</div>";
            echo "<div class='form-accountCenterOptionsItem'>";
                echo "<a href='vendorDatabase.php'>Vendor database</a>";
            echo "</div>";
            echo "<div class='form-accountCenterOptionsItem form-accCenterOptionPicked'>";
                echo "<a href='listingDatabase.php'>Listing database</a>";
            echo "</div>";
        echo "</div>";

        echo "<div class='form-settingOptions'>";
            echo "<div class='form-settingOptionsSideItems'>";
                echo "<h2>Welcome ". $_SESSION["username"] . "</h2>";
                echo "<div class='form-inputDetails'>";
                    echo "<form action='../accountLogOut.php' method='POST'>";
                        echo "<button>Log out</button>";
                    echo "</form>";
                echo "</div>";
            echo "</div>";

            mysqli_report(MYSQLI_REPORT_ERROR | MYSQLI_REPORT_STRICT);
            $connection = null;

            try{
                $connection = new mysqli("localhost", "root", "", "c2cMarketplacedb");
                echo "<div class='form-centeredBar'>";
            }

```

```

        echo "<form action='../searchDatabase.php' method='POST'>";
        echo "<div class='form-searchBar'>";
            echo "<div class='form-searchBarItem'>";
                echo "<input type='hidden' id='isAdmin' name='isAdmin'
value='true'>";

                echo "<select id='infoPriority' name='infoPriority'>";
                    echo "<option value='listingID'>Search by Listing
ID</option>";

                    echo "<option value='title'>Search by Title</option>";
                    echo "<option value='price'>Search by Price</option>";
                    echo "<option value='inStock'>Search by
Stock</option>";

                    echo "<option value='category'>Search by
Category</option>";

                    echo "<option value='vendorID'>Search by Vendor
ID</option>";

                    echo "<option value='username'>Search by
username</option>";

                echo "</select>";

                echo "<select id='orderItem' name='orderItem'>";
                    echo "<option value='listingID'>Order by Listing
ID</option>";

                    echo "<option value='title'>Order by Title</option>";
                    echo "<option value='price'>Order by Price</option>";
                    echo "<option value='inStock'>Order by Stock</option>";
                    echo "<option value='category'>Order by
Category</option>";

                    echo "<option value='vendorID'>Order by Vendor
ID</option>";

                    echo "<option value='username'>Order by
username</option>";

                echo "</select>";

                echo "<select id='adPriority' name='adPriority'>";
                    echo "<option value='ASC'>Order in Ascending</option>";
                    echo "<option value='DESC'>Order in
Descending</option>";

                    echo "<option value='None'>None</option>";
                echo "</select>";
            echo "</div>";

            echo "<div class='form-inputAndBtn'>";
                echo "<input type='search' id='searchBar'
name='searchBar'>";

                echo "<button type='submit' name='tableBtn' id='tableBtn'
value='listing'>&rarr;</button>";
                echo "</div>";
            echo "</div>";
        echo "</form>";
    echo "</div>";

```

```

        echo "<div class='form-databaseBlock form-description'>";
        if ($listings = mysqli_execute_query($connection, $sqlStatement)){
            if (mysqli_num_rows($listings) > 0){
                echo "<table>";
                echo "<tr class='form-tableHead'>";
                echo "<th>Listing ID</th>";
                echo "<th>Title</th>";
                echo "<th>Price</th>";
                echo "<th>Description</th>";
                echo "<th>Stock</th>";
                echo "<th>Image</th>";
                echo "<th>Vendor ID</th>";
                echo "<th>Vendor username</th>";
                echo "<th></th>";
                echo "</tr>";

                while($listingDetails = mysqli_fetch_array($listings)){
                    $vendorDetails =
mysqli_fetch_array(mysqli_execute_query($connection, "SELECT * FROM vendor WHERE
vendorID=?", [$listingDetails["vendorID"]]));
                    echo "<tr>";
                    echo "<td>". $listingDetails["listingID"] . "</td>";
                    echo "<td>". $listingDetails["title"] . "</td>";
                    echo "<td>". $listingDetails["price"] . "</td>";
                    echo "<td>".
$listingDetails["description"] . "</td>";
                    echo "<td>". $listingDetails["inStock"] . "</td>";
                    echo "<td><img src='".
$listingDetails["image"] . "' alt='Product.png'></td>";
                    echo "<td>". $vendorDetails["vendorID"] . "</td>";
                    echo "<td>". $vendorDetails["username"] . "</td>";

                    echo "<form action='../deleteRecord.php'
method='POST'>";

                    echo "<input type='hidden' value='listing'
name='recordType' id='recordType'>";

                    echo "<td><button type='submit' value='".
$listingDetails["listingID"] . "' name='formBtn' id='formBtn' class='form-
redBtn'>Delete</button></td>";

                    echo "</form>";
                    echo "</tr>";
                }
                echo "</table>";
            }
        }
        echo "</div>";
    }
    catch(Exception $error){
        echo "<h2>An error occurred</h2>";
    }
    echo "</div>";

```

```

        if ($connection != null){
            $connection->close();
        }
        echo "</div>";
    echo "</body>";
    echo "</html>";
?>

```

A php form that displays all listing information as well as enables the admin user to delete a listing.

c)

accountRegistration.html

```

<html lang="en">

<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Account Registration</title>
    <link rel="stylesheet" href="../style.css">
</head>

<header>
    <h1><a href="mainPage.php">Marketplace</a></h1>
    <div class="form-mainHeading">
        <div class="form-account">
            <a href="mainPage.php">Return to the store front</a>
        </div>
    </div>
</header>

<body>
    <div class="form-accountDetails">
        <h1>Register an account</h1>
        <form action="accountRegistration.php" method="POST" id="accRegistration"
enctype="multipart/form-data" onsubmit="return validateRegistration()">
            <div class="form-accountRegDetails">
                <div class="form-accountRegItem">
                    <div class="form-inputDetails">
                        <label for="fName">First name: </label><br>
                        <div class="form-error">
                            <input type="text" name="fName" id="fName">
                            <small class="error" id="fNameError"></small>
                        </div>
                    </div>
                </div>

                <div class="form-inputDetails">
                    <label for="lName">Last name: </label><br>
                    <div class="form-error">
                        <input type="text" name="lName" id="lName">

```

```

        <small class="error" id="lNameError"></small>
      </div>
    </div>
  </div>
</div>

<div class="form-accountRegDetails">
  <div class="form-accountRegItem">
    <div class="form-inputDetails">
      <label for="username">Username: </label><br>
      <div class="form-error">
        <input type="text" name="username" id="username">
        <small class="error" id="usernameError"></small>
      </div>
    </div>

    <div class="form-inputDetails">
      <label for="phoneNum">Phone Number: </label><br>
      <div class="form-error">
        <input type="tel" name="phoneNum" id="phoneNum">
        <small class="error" id="phoneNumError"></small>
      </div>
    </div>
  </div>
</div>

<div class="form-accountRegDetails">
  <div class="form-accountRegItem">
    <div class="form-inputDetails form-longInputDetails">
      <label for="email">Email: </label><br>
      <div class="form-error">
        <input type="email" name="email" id="email">
        <small class="error" id="emailError"></small>
      </div>
    </div>
  </div>
</div>

<div class="form-accountRegDetails">
  <div class="form-accountRegItem">
    <div class="form-inputDetails form-longInputDetails">
      <label for="address">Address: </label><br>
      <div class="form-error">
        <input type="text" name="address" id="address">
        <small class="error" id="addressError"></small>
      </div>
    </div>
  </div>
</div>

<div class="form-accountRegDetails">
  <div class="form-accountRegItem">

```

```

        <div class="form-inputDetails form-longInputDetails">
            <label for="password">Password: </label><br>
            <div class="form-error">
                <input type="password" name="password" id="password">
                <small class="error" id="passwordError"></small>
            </div>
        </div>
    </div>
</div>

<div class="form-accountRegDetails">
    <div class="form-accountRegItem">
        <div class="form-inputDetails form-longInputDetails">
            <label for="conPassword">Confirm password: </label><br>
            <div class="form-error">
                <input type="password" name="conPassword" id="conPassword">
                <small class="error" id="conPasswordError"></small>
            </div>
        </div>
    </div>
</div>

<div class="form-accountRegDetails">
    <div class="form-accountRegItem">
        <div class="form-inputDetails">
            <input type="checkbox" name="notify" id="notify">
            <label for="notify">Notify me of any promotions on the Marketplace
platform.</label>
        </div>
    </div>
</div>

<div class="form-accountRegDetails">
    <div class="form-accountRegItem">
        <div class="form-inputDetails">
            <input type="checkbox" name="userAgreement" id="userAgreement">
            <label for="userAgreement">You have read and agreed to our user
agreement and privacy
policy.</label>
            <div class="form-error">
                <small class="error" id="userAgreementError"></small>
            </div>
        </div>
    </div>
</div>

<button type="submit">Register</button>
<p>Already have an account? <a href="../accountLogin.html">Log in</a></p>
</form>
<script src="../script.js"></script>
</div>
</body>

```

```
</html>
```

An html form where the user can register a new account that will be categorised as a buyer.

accountLogin.html

```
<html lang="en">

<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Account Login</title>
  <link rel="stylesheet" href="style.css">
</head>

<header>
  <h1><a href="mainWebsite/mainPage.php">Marketplace</a></h1>
  <div class="form-mainHeading">
    <div class="form-account">
      <a href="mainWebsite/mainPage.php">Return to the store front</a>
    </div>
  </div>
</header>

<body>
  <div class="form-accountDetails">
    <h1>Login to an account</h1>
    <form action="accountLogin.php" method="POST" enctype="multipart/form-data"
onsubmit="return validateLogin()">
      <div class="form-accountLogDetails">
        <div class="form-inputDetails">
          <label for="username">Username: </label><br>
          <div class="form-error">
            <input type="text" name="username" id="username">
            <small class="error" id="usernameError"></small>
          </div>
        </div>

        <div class="form-inputDetails">
          <label for="password">Password: </label><br>
          <div class="form-error">
            <input type="password" name="password" id="password">
            <small class="error" id="passwordError"></small>
          </div>
        </div>
      </div>

      <button type="submit" class="form-btnSpace">Login</button>
    </form>
    <script src="script.js"></script>
```

```

    </div>
</body>

</html>

```

(W3Schools, n.d.)

A html form where the user can login into their account, which can be a buyer, vendor or admin account.

createListing.html

```

<html lang="en">

<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Listing creation</title>
    <link rel="stylesheet" href="../style.css">
</head>

<header>
    <h1><a href="mainPage.php">Marketplace</a></h1>
    <div class="form-mainHeading">
        <div class="form-account">
            <a href="mainPage.php">Return to the store front</a>
        </div>
    </div>
</header>

<body>
    <div class="form-accountDetails">
        <h1>Listing details</h1>
        <form action="createListing.php" method="POST" enctype="multipart/form-data"
onsubmit="return validateListing()">
            <div class="form-listingDetails">
                <div class="form-inputDetails">
                    <div class="form-error">
                        <input type="text" name="title" id="title" placeholder="Title">
                        <small class="error" id="titleError"></small>
                    </div>
                </div>

                <div class="form-inputDetails">
                    <div class="form-error">
                        <input type="number" name="price" id="price" placeholder="Price">
                        <small class="error" id="priceError"></small>
                    </div>
                </div>

                <div class="form-inputDetails">
                    <div class="form-error">

```

```

        <input type="number" name="inStock" id="inStock"
placeholder="Stock" min=0>
        <small class="error" id="stockError"></small>
    </div>
</div>

<div class="form-listingDetails">
    <textarea name="description" id="description"
placeholder="Description(Optional)"></textarea>
</div>

<div class="form-listingDetails">
    <div class="form-inputDetails">
        <label for="productCategories">Product categories: </label><br>
        <select name="productCategories" id="productCategories">
            <option value="Home-made">Home-made</option>
            <option value="Grocery">Grocery</option>
            <option value="Health">Health</option>
            <option value="Collectible">Collectible</option>
            <option value="Vehicle">Vehcile</option>
            <option value="Toy">Toy</option>
            <option value="Business">Business</option>
            <option value="Clothing">Clothing</option>
            <option value="Furniture">Furniture</option>
            <option value="Other">Other</option>
        </select>
    </div>

    <div class="form-inputDetails">
        <label for="productVisual">Listing's image: </label><br>
        <div class="form-error">
            <input type="file" name="productVisual" id="productVisual"
accept="image/*">
            <small class="error" id="imageError"></small>
        </div>
    </div>
</div>

    <button type="submit" class="form-btnSpace">Create listing</button>
</form>
<script src="../script.js"></script>
</div>
</footer>
</body>
</html>

```

(W3Schools, n.d.)

A html form that enables vendors to create a new listing associated with them.

adminCreation.html

```
<html lang="en">

<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Admin creation</title>
  <link rel="stylesheet" href="../style.css">
</head>

<body>
  <div class="form-accountDetails">
    <h1>Create an admin account</h1>
    <form action="adminCreation.php" method="POST" enctype="multipart/form-data"
onsubmit="return validateAdminCreation()">
      <div class="form-accountRegDetails">
        <div class="form-inputDetails">
          <label for="username">Username: </label><br>
          <div class="form-error">
            <input type="text" name="username" id="username">
            <small class="error" id="usernameError"></small>
          </div>
        </div>
      </div>

      <div class="form-accountRegDetails">
        <div class="form-inputDetails">
          <label for="password">Password: </label><br>
          <div class="form-error">
            <input type="password" name="password" id="password">
            <small class="error" id="passwordError"></small>
          </div>
        </div>
      </div>

      <div class="form-accountRegDetails">
        <div class="form-inputDetails">
          <label for="password">Confirm password: </label><br>
          <div class="form-error">
            <input type="password" name="conPassword" id="conPassword">
            <small class="error" id="conPasswordError"></small>
          </div>
        </div>
      </div>

      <button type="submit" class="form-btnSpace">Create admin account</button>
    </form>
    <script src="../script.js"></script>
  </div>
</body>
```

```
</html>
```

An html form that enables an admin to create another admin user.

userCreation.html

```
<html lang="en">

<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>User Creation</title>
  <link rel="stylesheet" href="../style.css">
</head>

<body>
  <div class="form-accountDetails">
    <h1>Create a new user</h1>
    <form action="userCreation.php" method="POST" id="accRegistration"
enctype="multipart/form-data" onsubmit="return validateUserCreation()">
      <div class="form-accountRegDetails">
        <div class="form-inputDetails">
          <label for="userType">User Type: </label><br>
          <select name="userType" id="userType">
            <option value="buyer">Buyer</option>
            <option value="vendor">Vendor</option>
          </select>
        </div>
      </div>

      <div class="form-accountRegDetails">
        <div class="form-accountRegItem">
          <div class="form-inputDetails">
            <label for="fName">First name: </label><br>
            <div class="form-error">
              <input type="text" name="fName" id="fName">
              <small class="error" id="fNameError"></small>
            </div>
          </div>

          <div class="form-inputDetails">
            <label for="lName">Last name: </label><br>
            <div class="form-error">
              <input type="text" name="lName" id="lName">
              <small class="error" id="lNameError"></small>
            </div>
          </div>
        </div>

        <div class="form-accountRegDetails">
          <div class="form-accountRegItem">
```

```

        <div class="form-inputDetails">
            <label for="username">Username: </label><br>
            <div class="form-error">
                <input type="text" name="username" id="username">
                <small class="error" id="usernameError"></small>
            </div>
        </div>

        <div class="form-inputDetails">
            <label for="phoneNum">Phone Number: </label><br>
            <div class="form-error">
                <input type="tel" name="phoneNum" id="phoneNum">
                <small class="error" id="phoneNumError"></small>
            </div>
        </div>
    </div>
</div>

<div class="form-accountRegDetails">
    <div class="form-accountRegItem">
        <div class="form-inputDetails form-longInputDetails">
            <label for="email">Email: </label><br>
            <div class="form-error">
                <input type="email" name="email" id="email">
                <small class="error" id="emailError"></small>
            </div>
        </div>
    </div>
</div>

<div class="form-accountRegDetails">
    <div class="form-accountRegItem">
        <div class="form-inputDetails form-longInputDetails">
            <label for="address">Address: </label><br>
            <div class="form-error">
                <input type="text" name="address" id="address">
                <small class="error" id="addressError"></small>
            </div>
        </div>
    </div>
</div>

<div class="form-accountRegDetails">
    <div class="form-accountRegItem">
        <div class="form-inputDetails form-longInputDetails">
            <label for="password">Password: </label><br>
            <div class="form-error">
                <input type="password" name="password" id="password">
                <small class="error" id="passwordError"></small>
            </div>
        </div>
    </div>
</div>

```

```

        </div>

        <div class="form-accountRegDetails">
            <div class="form-accountRegItem">
                <div class="form-inputDetails form-longInputDetails">
                    <label for="conPassword">Confirm password: </label><br>
                    <div class="form-error">
                        <input type="password" name="conPassword" id="conPassword">
                        <small class="error" id="conPasswordError"></small>
                    </div>
                </div>
            </div>
        </div>

        <div class="form-accountRegDetails">
            <div class="form-accountRegItem">
                <div class="form-inputDetails">
                    <input type="checkbox" name="notify" id="notify">
                    <label for="notify">Enable email notifications.</label>
                </div>
            </div>
        </div>

        <button type="submit">Register</button>
    </form>
    <script src="../script.js"></script>
</div>
</body>

</html>

```

An html form that enables an admin user to create either a buyer or vendor user.

d)

script.js

```

const emailPattern = /^[^ ]+@[^ ]+\.[a-z]{2,3}$/;

function validateRegistration() {
    let valid = true;
    const fName = document.getElementById("fName").value.trim();
    const lName = document.getElementById("lName").value.trim();
    const username = document.getElementById("username").value.trim();
    const phoneNum = document.getElementById("phoneNum").value.trim();
    const email = document.getElementById("email").value.trim();
    const address = document.getElementById("address").value.trim();
    const password = document.getElementById("password").value.trim();
    const conPassword = document.getElementById("conPassword").value.trim();
    const userAgreement = document.getElementById("userAgreement").checked;

    document.querySelectorAll(".error").forEach(

```

```

        function (message) {
            message.textContent = "";
        }
    )

    if (fName === "") {
        document.getElementById("fNameError").textContent = "First name is required";
        valid = false;
    }
    if (lName === "") {
        document.getElementById("lNameError").textContent = "Last name is required";
        valid = false;
    }
    if (username === "") {
        document.getElementById("usernameError").textContent = "Username is required";
        valid = false;
    }
    if (phoneNum === "") {
        document.getElementById("phoneNumError").textContent = "Phone number is required";
        valid = false;
    }
    if (email === "") {
        document.getElementById("emailError").textContent = "Email is required";
        valid = false;
    }
    else if (!emailPattern.test(email)) {
        document.getElementById("emailError").textContent = "Email is invalid";
        valid = false;
    }
    if (address === "") {
        document.getElementById("addressError").textContent = "Address is required";
        valid = false;
    }

    if (password === "") {
        document.getElementById("passwordError").textContent = "Password is required";
        valid = false;
    }
    else if (password.length < 8) {
        document.getElementById("passwordError").textContent = "Password must be at least
8 characters or more in length";
        valid = false;
    }
    if (conPassword === "") {
        document.getElementById("conPasswordError").textContent = "Password must be
repeated for confirmation";
        valid = false;
    }
    else if (password !== conPassword) {
        document.getElementById("conPasswordError").textContent = "Password and
confirmation password don't match";
        valid = false;
    }

```

```

    }

    if (userAgreement == false) {
        document.getElementById("userAgreementError").textContent = "You must agree to the
user agreement and privacy policy if you wish to use the marketplace";
        valid = false;
    }

    return valid;
}

function validateLogin(){
    let valid = true;
    const username = document.getElementById("username").value.trim();
    const password = document.getElementById("password").value.trim();

    document.querySelectorAll(".error").forEach(
        function (message) {
            message.textContent = "";
        }
    )

    if (username === "") {
        document.getElementById("usernameError").textContent = "Username is required";
        valid = false;
    }
    if (password === "") {
        document.getElementById("passwordError").textContent = "Password is required";
        valid = false;
    }

    return valid;
}

function validatePasswords(){
    let valid = true;
    const password = document.getElementById("password").value.trim();
    const newPassword = document.getElementById("newPassword").value.trim();
    const conPassword = document.getElementById("conPassword").value.trim();

    document.querySelectorAll(".error").forEach(
        function (message) {
            message.textContent = "";
        }
    )

    if (password === "") {
        document.getElementById("passwordError").textContent = "Password is required";
        valid = false;
    }

    if (newPassword === "") {

```

```

        document.getElementById("newPasswordError").textContent = "New password is
required";
        valid = false;
    }
    else if (newPassword.length < 8) {
        document.getElementById("newPasswordError").textContent = "New password must be at
least 8 characters or more in length";
        valid = false;
    }

    if (conPassword === "") {
        document.getElementById("conPasswordError").textContent = "Password must be
repeated for confirmation";
        valid = false;
    }
    else if (newPassword !== conPassword) {
        document.getElementById("conPasswordError").textContent = "New password and
confirmation password don't match";
        valid = false;
    }

    return valid;
}

function validateListing(){
    let valid = true;
    const title = document.getElementById("title").value.trim();
    const price = document.getElementById("price").value.trim();
    const stock = document.getElementById("inStock").value.trim();
    const productVisual = document.getElementById("productVisual").files;

    document.querySelectorAll(".error").forEach(
        function (message) {
            message.textContent = "";
        }
    )

    if (title === "") {
        document.getElementById("titleError").textContent = "Title is required";
        valid = false;
    }

    if (price === "") {
        document.getElementById("priceError").textContent = "Price is required";
        valid = false;
    }
    else if (price < 0){
        document.getElementById("priceError").textContent = "Price can't be negative";
        valid = false;
    }

    if (stock === "") {

```

```

        document.getElementById("stockError").textContent = "Stock is required";
        valid = false;
    }
    else if(stock < 1){
        document.getElementById("stockError").textContent = "Invalid stock";
        valid = false;
    }

    if (productVisual.length === 0) {
        document.getElementById("imageError").textContent = "Product image is required";
        valid = false;
    }

    return valid;
}

function validateVoucher(){
    let valid = true;
    const fundInc = document.getElementById("fundInc").value.trim();

    document.querySelectorAll(".error").forEach(
        function (message) {
            message.textContent = "";
        }
    )

    if (fundInc === "") {
        document.getElementById("fundError").textContent = "Voucher amount is required";
        valid = false;
    }
    else if(fundInc < 25 || fundInc > 10000){
        document.getElementById("fundError").textContent = "Invalid voucher amount! The
voucher amount can only be between R25 and R10000";
        valid = false;
    }
    return valid;
}

function validateAdminCreation() {
    let valid = true;
    const username = document.getElementById("username").value.trim();
    const password = document.getElementById("password").value.trim();
    const conPassword = document.getElementById("conPassword").value.trim();

    document.querySelectorAll(".error").forEach(
        function (message) {
            message.textContent = "";
        }
    )

    if (username === "") {
        document.getElementById("usernameError").textContent = "Username is required";
    }

```

```

        valid = false;
    }

    if (password === "") {
        document.getElementById("passwordError").textContent = "Password is required";
        valid = false;
    }
    else if (password.length < 8) {
        document.getElementById("passwordError").textContent = "Password must be at least
8 characters or more in length";
        valid = false;
    }

    if (conPassword === "") {
        document.getElementById("conPasswordError").textContent = "Password must be
repeated for confirmation";
        valid = false;
    }
    else if (password !== conPassword) {
        document.getElementById("conPasswordError").textContent = "Password and
confirmation password don't match";
        valid = false;
    }

    return valid;
}

function validateUserCreation(){
    let valid = true;
    const fName = document.getElementById("fName").value.trim();
    const lName = document.getElementById("lName").value.trim();
    const username = document.getElementById("username").value.trim();
    const phoneNum = document.getElementById("phoneNum").value.trim();
    const email = document.getElementById("email").value.trim();
    const address = document.getElementById("address").value.trim();
    const password = document.getElementById("password").value.trim();
    const conPassword = document.getElementById("conPassword").value.trim();

    document.querySelectorAll(".error").forEach(
        function (message) {
            message.textContent = "";
        }
    )

    if (fName === "") {
        document.getElementById("fNameError").textContent = "First name is required";
        valid = false;
    }
    if (lName === "") {
        document.getElementById("lNameError").textContent = "Last name is required";
        valid = false;
    }
}

```

```

if (username === "") {
    document.getElementById("usernameError").textContent = "Username is required";
    valid = false;
}
if (phoneNum === "") {
    document.getElementById("phoneNumError").textContent = "Phone number is required";
    valid = false;
}
if (email === "") {
    document.getElementById("emailError").textContent = "Email is required";
    valid = false;
}
else if (!emailPattern.test(email)) {
    document.getElementById("emailError").textContent = "Email is invalid";
    valid = false;
}
if (address === "") {
    document.getElementById("addressError").textContent = "Address is required";
    valid = false;
}

if (password === "") {
    document.getElementById("passwordError").textContent = "Password is required";
    valid = false;
}
else if (password.length < 8) {
    document.getElementById("passwordError").textContent = "Password must be at least
8 characters or more in length";
    valid = false;
}
if (conPassword === "") {
    document.getElementById("conPasswordError").textContent = "Password must be
repeated for confirmation";
    valid = false;
}
else if (password !== conPassword) {
    document.getElementById("conPasswordError").textContent = "Password and
confirmation password don't match";
    valid = false;
}

return valid;
}

function closeMessage(){
    let msg = document.getElementById("message");
    msg.classList.remove("msg");
    msg.classList.add("hide");
}

```

(RANA, 2026)

A JavaScript file that contains all validation code

e)

style.css

```
h1, h2, h3, h4{
  font-family: 'Times New Roman', Times, serif;
  overflow: auto;
  overflow-wrap: anywhere;
}

label{
  font-family: Arial, Helvetica, sans-serif;
}

header{
  display: flex;
  position: sticky;
  top: 0;
  background-image: linear-gradient(to right, teal, purple);
  width: auto;
  margin-top: 0pt;
  margin-bottom: 0pt;
  align-items: center;

  border-right: 0pt;
  border-left: 0pt;
  border-top: 0pt;
  border-bottom: 2pt;
  border-color: black;
  border-style: solid;
  justify-content: space-between;
}

header h1{
  color: white;
  padding: 10pt;
  margin-top: 0px;
  margin-bottom: 0px;
}

header a{
  font-family: 'Courier New', Courier, monospace;
  font-style: italic;
  color: white;
  text-decoration: none;
  margin-right: 5pt;
}

.form_buttonLink{
  color: white;
  text-decoration: none;
}
```

```

}

.error{
  text-align: center;
  color: red;
  font-family: Arial, Helvetica, sans-serif;
}

.form-error{
  display: grid;
  justify-items: left;
  gap: 2pt;
}

.form-errorBoard{
  background-color: rgb(235, 235, 235);
  border-color: black;
  border-style: solid;
  border-width: 2pt;
  border-radius: 10pt;
  margin-left: 40pt;
  margin-right: 40pt;
  padding: 5pt;
}

button{
  display: block;
  padding: 6px;
  margin-left: 10px;
  border-radius: 10pt;
  border-color: black;
  color: white;
  background-color: darkcyan;
  font-weight: bold;
  font-size: large;
  cursor: pointer;
  font-size: larger;
}

.msg{
  display: flex;
  font-family: 'Courier New', Courier, monospace;
  font-size: large;
  font-weight: bold;
  color: white;
  justify-content: space-between;
  background-color: green;

  border-color: darkgreen;
  border-style: solid;
  border-right: 0pt;
  border-left: 0pt;

```

```

border-top: 0pt;
border-bottom: 2pt;

margin-top: 0pt;
margin-bottom: 0pt;
margin-left: 0pt;
margin-right: 0pt;
}

.msg p{
  margin-left: 4pt;
}

.msg button{
  padding: 12pt;
  font-size: large;
  align-self: center;
  margin-top: 0pt;
  margin-bottom: 0pt;
  margin-right: 4pt;
  background-color: lightgreen;
  width: fit-content;
  height: fit-content;
  border-style: none;
}

.hide{
  display: none;
  visibility: hidden;
  background-color: rebeccapurple;
}

.form-returnLink{
  display: block;
  text-align: center;
  padding: 10pt;
}

.form-mainHeading{
  display: flex;
  font-family: 'Courier New', Courier, monospace;
  padding: 10pt;
  gap: 6pt;
  align-self: stretch;
  align-content: center;
}

.form-mainHeading button{
  padding: 6px;
}

```

```

.form-mainHeading a{
    color: white;
    margin-top: 0px;
    margin-right: 0px;
    margin-left: 0px;
    margin-bottom: 0px;
    transition: color 0.3s;
}

.form-mainHeading a:hover{
    color: yellow;
}

.form-account{
    display: block;
    align-self: stretch;
    align-content: center;
}

.form-quantityBtn{
    display: block;
    justify-items: right;
    justify-content: right;
}

.form-listingList{
    display: flex;
    width: 80%;
    justify-self: center;
    justify-content: space-between;
    background-color: gainsboro;

    border-radius: 10pt;
    border-color: gray;
    border-style: solid;
    margin-bottom: 10pt;
    margin-right: 5pt;
    margin-left: 5pt;
    margin-top: 0pt;
}

.form-listingList img{
    width: 200px;
    height: 200px;
    margin-bottom: 2pt;
    object-fit: contain;
}

.form-listingList input{
    margin-left: 0pt;
    margin-right: 0pt;
    margin-bottom: 4pt;
}

```

```

    margin-top: 4pt;
}

.form-listingList button{
    margin-right: 0pt;
    justify-self: right;
}

.form-cartLeftItem{
    display: block;
    margin-left: 5pt;
}

.form-cartRightItem{
    display: block;
    text-align: right;
    margin-right: 5pt;
    justify-items: right;
    justify-content: right;
    align-content: space-between;
    margin-bottom: 0pt;
    margin-top: 0pt;
}

.form-cartRightItem input{
    width: 60px;
}

body{
    background-image: repeating-linear-gradient(to bottom right, turquoise, orchid,
turquoise);
    border-radius: 10pt;
    margin: 0pt;
    width: auto;
    border-color: black;
}

.form-productsGridContainer{
    display: grid;
    gap: 5px;
    margin-right: 8px;
    margin-left: 8px;
    width: auto;
    grid-template-columns: repeat(auto-fit, minmax(180pt, 1fr));
    align-content: space-evenly;
    justify-items: center;
}

.form-productsGridItem{
    height: fit-content;
    width: 180pt;
    background-image: linear-gradient(to bottom right, white, gray);

```

```

margin-bottom: 8px;
border-radius: 5px;
border-color: black;
border-width: 1pt;
border-style: solid;
box-shadow: 4pt 4pt rgba(128, 128, 128, 0.5);
}

.form-productsGridItem h2{
  font-size: large;
}

.form-productsGridItem h3{
  margin-left: 10px;
  font-size: large;
  margin-bottom: 5px;
}

.form-productsGridItem h4{
  margin-left: 10px;
  font-size: large;
  margin-top: 0px;
  margin-right: 0px;
  margin-bottom: 5px;
}

.form-productsGridItem button{
  margin-right: 10px;
  justify-self: right;
  margin-bottom: 0px;
  margin-top: 0px;
}

.form-productsGridItem img{
  display: block;
  width: 100px;
  height: 100px;
  object-fit: contain;
  border-radius: 10pt;
  padding: 4pt;
}

.form-cost{
  color: darkred;
  font-weight: bold;
}

.form-profit{
  color: darkgreen;
  font-weight: bold;
}

```

```

.form-productsGridItemImage{
    cursor: pointer;
    margin-top: 5px;
    justify-items: center;
}

.form-productsGridItemImage button{
    background-color: rgba(255, 255, 255, 0);
    border-style: none;
}

.form-productsGridVendor{
    display: block;
    margin-right: 8px;
    margin-left: 8px;
    justify-items: right;
}

.form-productsGridVendor h2{
    margin-top: 0px;
    margin-bottom: 8px;
}

.form-productsGridVendor h3{
    margin-top: 0px;
    margin-bottom: 8px;
    font-size: 14pt;
}

.form-accountDetails{
    background-image: linear-gradient(to bottom right, lightgray, wheat);
    width: auto;
    margin-top: 10pt;
    margin-left: 10pt;
    margin-right: 10pt;
    margin-bottom: 10pt;

    border-radius: 10pt;
    box-shadow: 4pt 4pt rgba(128, 128, 128, 0.5);
    text-align: left;
    padding: 5pt;
}

.form-accountDetails h1{
    text-align: center;
    margin-top: 5pt;
}

.form-accountDetails h3{
    text-align: center;
    margin-top: 5pt;
    margin-bottom: 1pt;
}

```

```

}

.form-accountDetails p{
    padding: 10pt;
    font-size: 15pt;
    text-align: center;
    margin-bottom: 5pt;
}

.form-accountDetails a{
    color: blue;
}

.form-accountDetails button{
    margin-top: 10pt;
    margin-left: 10pt;
    margin-bottom: 10pt;
    justify-self: left;
}

.form-accountRegDetails{
    display: flex;
    align-items: center;
    text-align: left;
    justify-items: right;
    margin-left: 10pt;
    margin-right: 10pt;
}

.form-accountRegItem{
    display: flex;
    margin-bottom: 25pt;
    gap: 10pt;
}

.form-accountLogDetails{
    display: flex;
    align-items: center;
    padding: 1pt;
    justify-content: space-evenly;
}

.form-changePassDetails{
    align-items: center;
    padding: 15pt;
}

.form-listingDetails{
    display: flex;
    align-items: start;
    justify-content: center;
    padding: 10pt;
}

```

```

    gap: 10pt;
}

.form-listingDetails input{
    width: 100%;
}

.form-listingDetails textarea{
    font-size: large;
}

label{
    justify-self: left;
    margin-bottom: 2pt;
}

input{
    border-radius: 4pt;
    padding: 5pt;
    border-color: black;
    font-size: 12pt;
}

textarea{
    box-sizing: border-box;
    resize: none;
    font-size: 12pt;
    width: 75%;
}

.form-inputDetails{
    display: inline-block;
    justify-self: center;
}

.form-longInputDetails input{
    width: 204%;
}

.form-btnSpace{
    margin-bottom: 10pt;
    margin-top: 0pt;
}

hr{
    height: 1pt;
    background-color: gray;
    margin-top: 0px;
    margin-right: 0px;
    margin-left: 0px;
    margin-bottom: 0px;
}

```

```

.vr{
  border-right: 2pt solid gray;
  height: auto;
  margin-top: 0px;
  margin-right: 0px;
  margin-left: 0px;
  margin-bottom: 0px;
}

.form-title{
  display: block;
  justify-self: center;
}

.form-accountCenter{
  display: flex;
  height: fit-content;
  gap: 5pt;
  margin-top: 10pt;
  margin-left: 10pt;
  margin-right: 10pt;
  margin-bottom: 10pt;
}

.form-accountCenterOptions{
  display: block;
  flex: 0.5;
  background-color: lightgray;
  justify-items: center;
  height: fit-content;

  padding: 6pt;
  border-style: none;
  border-radius: 10px;
  border-top-right-radius: 0px;
  border-bottom-right-radius: 0px;
}

.form-accountCenterOptions h1{
  overflow-wrap: normal
}

.form-accountCenterOptionsItem{
  display: block;
  margin-top: 5pt;
  margin-bottom: 5pt;
  width: 100%;
  text-align: center;
}

.form-accountCenterOptionsItem a{

```

```

display: block;
color: black;
width: 100%;
text-decoration: none;
background-color: white;
border-style: none;
border-radius: 10px;
padding: 10px 0px;
transition: color 0.1s;
}

.form-accountCenterOptionsItem a:hover{
background-color: rgb(60, 60, 60);
color: white;
}

.form-accCenterOptionPicked a{
background-color: rgb(100, 100, 100);
color: white;
}

.form-accountCenter img{
width: 150pt;
height: 150pt;
border-color: black;
object-fit: cover;
}

.form-settingOptions{
display: block;
flex: 2.5;
height: fit-content;
width: auto;
background-image: linear-gradient(to bottom right, lightgray, wheat);
justify-items: left;

padding: 6pt;
border-style: none;
border-radius: 10px;
border-top-left-radius: 0px;
}

.form-settingOptions h1{
margin-left: 5pt;
}

.form-settingOptionsItem{
display: flex;
gap: 5pt;
padding: 5pt;
}

```

```

.form-settingOptionsItem button{
    margin-bottom: 0pt;
}

.form-settingOptionsSideItems{
    display: flex;
    box-sizing: border-box;
    width: 98%;
    justify-content: space-between;
    justify-items: left;
    margin-top: 4pt;
    padding: 0pt;
    margin-left: 5pt;
    margin-right: 5pt;
}

.form-deliveryForms{
    display: grid;
    width: auto;
    justify-self: center;
    justify-items: left;
    background-color: whitesmoke;

    border-radius: 10pt;
    border-color: black;
    border-style: solid;
    border-width: 1px;
    margin-bottom: 10pt;
    margin-top: 0pt;
    padding: 5pt;
}

.form-deliveryForms h2{
    margin-top: 2pt;
    margin-bottom: 0pt;
}

.form-deliveryForms h3{
    margin-bottom: 5pt;
}

.form-deliveryFormsAddress{
    display: block;
    border-radius: 5pt;
    background-color: gainsboro;
    padding: 5pt;
    margin-bottom: 0pt;
}

.form-deliveryForms button{
    margin-top: 0pt;
    margin-bottom: 0pt;
}

```

```

        justify-self: left;
    }

    .form-deliveryFormsAddress input{
        margin-right: 8pt;
    }

    .form-deliveryFormsAddress input[type='text']{
        width: 400px;
    }

    .form-deliveryFormsAddressFlex{
        display: flex;
        align-items: center;
    }

    .form-adminDetails{
        background-color: white;
        border-radius: 10pt;
        margin-left: 200pt;
        margin-right: 200pt;
        box-shadow: 4pt 4pt rgba(128, 128, 128, 0.5);
        padding: 5pt;
    }

    .form-adminDetails h1{
        text-align: center;
        margin-top: 5pt;
    }

    .form-adminDetails h3{
        text-align: center;
        margin-top: 5pt;
        margin-bottom: 1pt;
    }

    .form-adminDetails button{
        margin-top: 10pt;
        margin-bottom: 10pt;
        justify-self: center;
    }

    .form-adminLogDetails{
        display: flex;
        align-items: center;
        padding: 1pt;
        justify-content: space-evenly;
    }

    table{
        display: block;
        max-width: 100%;
    }

```

```

border-collapse: collapse;
table-layout: auto;
overflow-x: auto;
}

th{
  position: sticky;
}

th, td{
  font-size: medium;
  padding: 4pt;
}

table, th, td{
  border-width: 1pt;
  border-color: black;
  border-style: solid;
  text-align: center;
}

.form-databaseBlock{
  display: block;
  justify-self: center;
  justify-items: center;
  box-sizing: border-box;
  width: 800px;
}

.form-tableHead{
  background-color: gray;
  color: white;
}

.form-redBtn{
  background-color: red;
}

.form-permissionTable{
  display: block;
  background-color: green;
  justify-items: center;
  gap: 10pt;
  padding: 10pt;
  margin-bottom: 10pt;
}

.form-permissionTableItem{
  display: block;
  background-color: antiquewhite;
}

```

```

.form-cartFlex{
  display: flex;
  flex-wrap: nowrap;
  padding: 5pt;
  gap: 5pt;
  justify-content: space-between;
}

.form-cartFlexItemLeft{
  flex: 2.75;
}

.form-cartFlexItemRight{
  flex: 1.25;
}

.form-checkout{
  width: auto;
  justify-items: left;
  background-color: white;
  border-radius: 10pt;
  box-shadow: 4pt 4pt rgba(128, 128, 128, 0.5);
  padding: 5pt;
  margin-bottom: 5pt;
}

.form-checkout h2{
  font-size: large;
  overflow: auto;
  overflow-wrap: anywhere;
}

.form-checkout h3{
  font-size: large;
  justify-self: center;
  margin-bottom: 4pt;
}

.form-checkout label{
  font-size: small;
}

.form-checkout input{
  font-size: small;
  margin-top: 2pt;
}

.form-checkout button{
  font-size: medium;
  margin-left: 0px;
  margin-right: 0px;
}

```

```

.form-centeredBar{
  display: block;
  width: 100%;
  justify-items: center;
}

.form-searchBar{
  display: flex;
  width: auto;
  background-image: linear-gradient(to bottom right, white, gray);
  padding: 4pt;
  justify-content: space-between;
  align-items: center;

  border-radius: 10pt;
  margin-top: 10pt;
  margin-left: 5pt;
  margin-right: 5pt;
}

.form-searchBarItem{
  display: flex;
  flex: 1.4;
  align-self: center;
  justify-content: left;
  width: auto;
}

.form-searchBarItem select{
  margin-left: 5pt;
  margin-right: 10pt;
  font-size: 16px;
}

.form-searchBar input{
  width: auto;
}

.form-inputAndBtn{
  display: flex;
  flex: 1.6;
  align-items: center;
}

.form-inputAndBtn input[type=search]{
  width: 100%;
  border-radius: 0px;
}

.form-inputAndBtn input[type=text]{
  width: 100%;
}

```

```

    height: 34px;
    border-radius: 0px;
}

.form-inputAndBtn button{
    margin-left: 0px;
    border-radius: 0px;
    border-style: none;
}

.form-productPage{
    display: flex;
    background-color: white;
    margin-right: 8px;
    margin-left: 8px;
    margin-top: 5px;
    margin-bottom: 0px;
    justify-content: space-between;

    border-radius: 8pt;
    border-style: none;
    width: auto;
}

.form-productPageLeft{
    display: block;
    justify-items: center;
    align-items: start;
    width: 60%;
    margin-left: 5pt;
    margin-right: 5pt;
    margin-top: 5pt;
    margin-bottom: 5pt;
}

.form-productPageLeft img{
    display: block;
    object-fit: contain;
    border-radius: 5pt;
    border-color: gray;
    border-width: 2pt;
    border-style: solid;
    width: 350px;
    height: 350px;
}

.form-productPageRight{
    display: block;
    justify-items: left;
    width: 100%;
    overflow: auto;
    overflow-wrap: anywhere;
}

```

```

margin-left: 5pt;
margin-right: 5pt;
margin-top: 2pt;
}

.form-productPageRight h3{
display: block;
margin-bottom: 0px;
margin-top: 4px;
}

.form-productPageRight p{
display: block;
font-size: medium;
margin-left: 25px;
font-family: Arial, Helvetica, sans-serif;
}

.form-moveLeft{
margin-left: 15px;
}

.form-description{
margin-bottom: 2px;
}

.form-description td{
overflow-wrap: break-word;
}

.form-productPageMainHeading{
display: flex;
width: 100%;
justify-content: space-between;
}

.form-message{
display: block;
text-align: left;
width: auto;
overflow: auto;
overflow-wrap: anywhere;
color: white;

padding: 5pt;
margin-left: 5pt;
margin-right: 5pt;
margin-top: 5pt;
margin-bottom: 5pt;
}

```

```

.form-leftMessage p{
  font-size: larger;
  background-color: gray;
  justify-self: left;
}

.form-leftTimeStamp p{
  font-size: small;
}

.form-rightMessage p{
  font-size: larger;
  background-color: green;
  justify-self: right;
}

.form-rightTimeStamp p{
  font-size: small;
}

@media only screen and (max-width: 600px){
  header{
    font-size: small;
  }

  .form-mainHeading{
    font-size: small;
  }

  .form-mainHeading button{
    font-size: small;
  }

  .form-mainHeading a{
    font-size: smaller;
  }

  .form-inputDetails{
    font-size: small;
  }

  .form-inputDetails input{
    font-size: smaller;
    padding: 2px;
  }

  .form-longInputDetails{
    font-size: small;
  }

  .form-longInputDetails input{
    font-size: smaller;
  }
}

```

```

padding: 2px;
}

.form-settingOptions input{
margin-right: 0pt;
width: 100%;
}

.form-settingOptionsItem button{
font-size: smaller;
}

.form-accountCenter{
display: flex;
height: fit-content;
gap: 5pt;
margin-top: 10pt;
margin-left: 10pt;
margin-right: 10pt;
margin-bottom: 10pt;
}

.form-accountCenterOptions{
font-size: x-small;
}

.form-settingOptions{
font-size: xx-small;
}

.form-listingDetails textarea{
font-size: small;
}

.form-accountDetails h1{
font-size: medium;
}

.form-accountDetails h3{
font-size: small;
}

.form-accountDetails p{
font-size: small;
}

.button{
font-size: medium;
}

table, td, th{
font-size: xx-small;
}

```

```

}

.form-databaseBlock{
    width: 250px;
}

.msg{
    font-size: smaller;
}

.msg button{
    font-size: smaller;
}

.form-deliveryForms button{
    font-size: x-small;
}

.form-deliveryForms h2{
    font-size: small;
}

.form-deliveryForms h3{
    font-size: x-small;
}

.form-deliveryFormsAddress{
    font-size: x-small;
}

.form-deliveryFormsAddress input{
    padding: 0px;
}

.form-deliveryFormsAddress input[type='text']{
    width: 150px;
    font-size: xx-small;
    height: 25px;
}

.form-quantityBtn{
    font-size: xx-small;
}

.form-listingList input{
    width : 40px;
}

.form-listingList img{
    width: 75px;
    height: 75px;
}

```

```
.form-listingList button{
    font-size: x-small;
}

.form-listingList input{
    font-size: x-small;
}

.form-checkout{
    font-size: x-small;
}

.form-checkout h2{
    font-size: x-small;
}

.form-checkout h3{
    font-size: x-small;
}

.form-checkout label{
    font-size: xx-small;
}

.form-checkout input{
    font-size: xx-small;
}

.form-checkout button{
    font-size: x-small;
}

.form-productPage{
    font-size: x-small;
}

.form-productPageLeft img{
    width: 120px;
    height: 120px;
}

.form-productPageRight p{
    font-size: small;
}

.form-searchBar{
    font-size: x-small;
}

.form-searchBarItem{
    display: block;
}
```

```

    flex: 0.8;
}

.form-searchBarItem select{
    margin-bottom: 3pt;
    margin-top: 3pt;
    width: 100px;
    font-size: small;
}

.form-inputAndBtn input{
    font-size: xx-small;
}

.form-productsGridContainer{
    grid-template-columns: repeat(auto-fit, minmax(90pt, 1fr));
}

.form-productsGridItem{
    width: 90pt;
    font-size: x-small;
}

.form-productsGridItem h2{
    font-size: small;
}

.form-productsGridItem h3{
    font-size: small;
}

.form-productsGridItem h4{
    font-size: x-small;
}

.form-productsGridItem button{
    font-size: small;
}

.form-productsGridItem img{
    width: 50px;
    height: 50px;
}

.form-leftMessage p{
    font-size: medium;
    background-color: gray;
    justify-self: left;
}

.form-leftTimeStamp p{
    font-size: xx-small;
}

```

```

}

.form-rightMessage p{
    font-size: medium;
    background-color: green;
    justify-self: right;
}

.form-rightTimeStamp p{
    font-size: xx-small;
}
}

@media only screen and (min-width: 600px) and (max-width: 900px){
    header{
        font-size: medium;
    }

    .form-mainHeading{
        font-size: medium;
    }

    .form-mainHeading button{
        font-size: medium;
    }

    .form-mainHeading a{
        font-size: small;
    }

    .form-inputDetails{
        font-size: medium;
    }

    .form-inputDetails input{
        font-size: small;
        padding: 2px;
    }

    .form-longInputDetails{
        font-size: medium;
    }

    .form-longInputDetails input{
        font-size: small;
        padding: 2px;
    }

    .form-settingOptions input{
        margin-right: 0pt;
        width: 100%;
    }
}

```

```

.form-settingOptionsItem button{
    font-size: small;
}

.form-listingDetails textarea{
    font-size: medium;
}

.msg{
    font-size: medium;
}

.msg button{
    font-size: medium;
}

table, td, th{
    font-size: small;
}

.form-databaseBlock{
    width: 650px;
}

.form-deliveryForms button{
    font-size: smaller;
}

.form-deliveryForms h2{
    font-size: medium;
}

.form-deliveryForms h3{
    font-size: smaller;
}

.form-deliveryFormsAddress{
    font-size: smaller;
}

.form-deliveryFormsAddress input{
    padding: 4px;
}

.form-deliveryFormsAddress input[type='text']{
    width: 220px;
    font-size: x-small;
    height: 25px;
}

.form-quantityBtn{

```

```

        font-size: x-small;
    }

    .form-listingList input{
        width : 30px;
    }

    .form-listingList img{
        width: 130px;
        height: 130px;
    }

    .form-listingList button{
        font-size: smaller;
    }

    .form-listingList input{
        font-size: smaller;
    }

    .form-checkout{
        font-size: smaller;
    }

    .form-checkout h3{
        font-size: smaller;
    }

    .form-checkout label{
        font-size: x-small;
    }

    .form-checkout input{
        font-size: x-small;
    }

    .form-checkout button{
        font-size: smaller;
    }

    .form-accountDetails h1{
        font-size: large;
    }

    .form-accountDetails h3{
        font-size: medium;
    }

    .form-accountDetails p{
        font-size: medium;
    }

```

```

.button{
    font-size: large;
}

.form-productPage{
    font-size: small;
}

.form-productPageLeft img{
    width: 250px;
    height: 250px;
}

.form-productPageRight p{
    font-size: medium;
}

.form-searchBar{
    font-size: small;
}

.form-searchBarItem{
    display: block;
    flex: 1.4;
}

.form-searchBarItem select{
    margin-bottom: 3pt;
    margin-top: 3pt;
    width: 100px;
    font-size: medium;
}

.form-inputAndBtn input{
    font-size: small;
}

.form-productsGridContainer{
    grid-template-columns: repeat(auto-fit, minmax(120pt, 1fr));
}

.form-productsGridItem{
    width: 120pt;
    font-size: small;
}

.form-productsGridItem h2{
    font-size: medium;
}

.form-productsGridItem h3{
    font-size: medium;
}

```

```

}

.form-productsGridItem h4{
    font-size: small;
}

.form-productsGridItem button{
    font-size: medium;
}

.form-productsGridItem img{
    width: 75px;
    height: 75px;
}

.form-leftMessage p{
    font-size: large;
    background-color: gray;
    justify-self: left;
}

.form-leftTimeStamp p{
    font-size: x-small;
}

.form-rightMessage p{
    font-size: large;
    background-color: green;
    justify-self: right;
}

.form-rightTimeStamp p{
    font-size: x-small;
}
}

@media only screen and (min-width: 768px) and (max-width: 992px){
    .form-databaseBlock{
        width: 400px;
    }
}

```

(W3Schools, n.d.) (W3Schools, n.d.)

A css file that specifies the style of all files.

f)

Buyer database table

Server: 127.0.0.1 » Database: c2marketplacedb » Table: buyer

Showing rows 0 - 1 (2 total, Query took 0.0039 seconds.)

SELECT * FROM `buyer`

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Extra options

	buyerID	firstName	lastName	username	phoneNum	email	address	password	buyerRating	eWalletAmount	notify
☐ Edit ☐ Copy ☐ Delete	56	Keenan	Batista	CoolGuy123	0432287628	keenan@gmail.com	43 Night Avenue Durbanville	\$2y\$10\$JouAYKVL30SejMaBle9q.M9LDusC8HwBRGJ1HLx.0...	0	0	1
☐ Edit ☐ Copy ☐ Delete	60	Wendy	Gomez	GnomeLady	0298321758	wendy@gmail.com	5 Park Street Tyger Valley	\$2y\$10\$/ZLKFG2kWYybcHW5g0CGQ2JUyIWHIM5g/hPmUzaSw...	0	4500	1

Check all | With selected: Edit Copy Delete Export

Vendor database table

Server: 127.0.0.1 » Database: c2marketplacedb » Table: vendor

Showing rows 0 - 2 (3 total, Query took 0.0087 seconds.)

SELECT * FROM `vendor`

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Extra options

	vendorID	firstName	lastName	username	phoneNum	email	address	password	buyerRating	vendorRating	eWal
☐ Edit ☐ Copy ☐ Delete	39	Randy	Johnson	Rad.Johny	0228994121	johnrad@gmail.com	5 Bumpy Road Pretoria	\$2y\$10\$BUo6TuSauJEC4SKbu6qHegGUIQyUIRID4VO9Mwdh2G...	0	0	
☐ Edit ☐ Copy ☐ Delete	40	Sarah	Dickinson	sarah_D	0954879964	sarah@gmail.com	32 Curvy Road Durbanville	\$2y\$10\$M.dskyyict1jnGjpPsRJORH7JfQckyJy6dPG.Xi5y...	0	0	
☐ Edit ☐ Copy ☐ Delete	41	Marco	Downey	ElectronicMasters	0341785456	electronicmasters@gmail.com	13 Dream Avenue Tyger Valley	\$2y\$10\$mQsraSZuLM1G7HLrLvv3uoB4FJbOeAUQBF1EDgTEAa...	0	0	

Check all | With selected: Edit Copy Delete Export

Admin database table

Server: 127.0.0.1 » Database: c2marketplacedb » Table: admin

Showing rows 0 - 1 (2 total, Query took 0.0230 seconds.)

SELECT * FROM `admin`

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Extra options

	adminID	username	password
☐ Edit ☐ Copy ☐ Delete	1	admin1	\$2y\$10\$Li/UOsOWBebjOKp8jolmV.0twrf38ULWfDZ0aEY3ON7...
☐ Edit ☐ Copy ☐ Delete	5	admin2	\$2y\$10\$RaiKu95/rNnRoZ7rB.h2ke.E.9MD5IPBDRi2nwQrqaG...

Check all | With selected: Edit Copy Delete Export

Listing database table

Server: 127.0.0.1 » Database: c2cmarketplacedb » Table: listing

Showing rows 0 - 5 (6 total, Query took 0.0505 seconds.)

SELECT * FROM `listing`

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Extra options

	listingID	title	price	description	in Stock	category	image	vendorID
☐ Edit ☐ Copy ☐ Delete	101	Radical cookies	20	Chocolate chip cookies for radical people. Freshly...	12	Home-made	../images/best-chocolate-chip-cookies.jpg	39
☐ Edit ☐ Copy ☐ Delete	102	Dining chair	375	A well-made, antique, dining chair.	5	Furniture	../images/lois-solid-wood-with-marble-top-dining-c...	40
☐ Edit ☐ Copy ☐ Delete	103	Baked beans	25	Healthy, canned, baked beans.	17	Grocery	../images/Koo Product Beans in Tomato Sauce2x.png	40
☐ Edit ☐ Copy ☐ Delete	104	Cabbage	20	Fresh cabbages grown in a home garden.	3	Grocery	../images/235.jpg	40
☐ Edit ☐ Copy ☐ Delete	105	Toy Car	60	A classic, red car for children.	6	Toy	../images/eng_pm_Beetle-car-metal-toy-car-opening-...	41
☐ Edit ☐ Copy ☐ Delete	106	Cup	300	A home-made cup.	10	Home-made	../images/mug-raccoon-family-1-1024x768.jpg	41

Check all | With selected: Edit Copy Delete Export

Delivery database table

Server: 127.0.0.1 » Database: c2cmarketplacedb » Table: delivery

Showing rows 0 - 1 (2 total, Query took 0.0112 seconds.)

SELECT * FROM `delivery`

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Extra options

	deliveryID	totalCost	deliveryAddress	userApproval	vendorApproval	userType	userID	vendorID
☐ Edit ☐ Copy ☐ Delete	73	45		0	0	vendor	41	40
☐ Edit ☐ Copy ☐ Delete	75	400		1	0	buyer	60	40

Check all | With selected: Edit Copy Delete Export

DeliveryMessage database table

Server: 127.0.0.1 » Database: c2cmarketplacedb » Table: deliverymessage

MySQL returned an empty result set (i.e. zero rows). (Query took 0.0065 seconds.)

SELECT * FROM `deliverymessage`

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

messageID message timeSent userTypeOrigin deliveryID

Query results operations

Create view

CartItem database table

Server: 127.0.0.1 » Database: c2cmarketplacedb » Table: cartitem

Browse Structure SQL Search Insert Export Import Privileges

Showing rows 0 - 3 (4 total, Query took 0.0222 seconds.)

```
SELECT * FROM `cartitem`
```

☐ Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

☐ Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Extra options

		cartItemID	quantity	listingID	deliveryID
☐	Edit Copy Delete	131	1	104	73
☐	Edit Copy Delete	132	1	103	73
☐	Edit Copy Delete	134	1	102	75
☐	Edit Copy Delete	135	1	103	75

☐ Check all | With selected: Edit Copy Delete Export

UserRatingHistory database table

Server: 127.0.0.1 » Database: c2cmarketplacedb » Table: userratinghistory

Browse Structure SQL Search Insert Export Import Privileges

Showing rows 0 - 1 (2 total, Query took 0.0231 seconds.)

```
SELECT * FROM `userratinghistory`
```

☐ Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

☐ Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Extra options

		ratingId	rated	ratingAmount	ratingType	userType	userID	vendorID
☐	Edit Copy Delete	6	0	0	buyer	buyer	60	39
☐	Edit Copy Delete	7	0	0	vendor	buyer	60	39

☐ Check all | With selected: Edit Copy Delete Export

GitHub repository link: <https://github.com/Keenan-Batista/ITECA-Website>

2.5) Conclusion

The main website will provide most of its functionality after users register an account, which will give them access to altering their profile details, viewing listings and buying deliveries after communication with the vendor. The buyer will also have the option to transition into a vendor role where they can create listings and negotiate with buyers.

Additionally, admin users are separate from the main website and have access to the admin website where they can perform administrative tasks, such as updating and deleting database records.

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